

5G RAN

Carrier Aggregation Feature Parameter Description

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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added measurement of operator-level numbers of CA-capable UEs, with new related counters subject to the "optimized measurement of the numbers of CA-capable UEs in cells" function. For details, see 4.8.2 Optimized Measurement of the Numbers of CA-capable UEs in Cells .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite

- Editorial changes

Added descriptions of the CA UE data split stop optimization function. For details, see [4.7 CA Data Split](#), [5.1.2 Network Analysis](#), [5.1.4.1 Data Configuration](#), [6.1.2 Network Analysis](#), and [6.1.4.1 Data Configuration](#).

Deleted the mutually exclusive relationship between inter-FR CA and downlink experience satisfaction guarantee from [7.3.2.2 Mutually Exclusive Functions](#).

Revised descriptions of the impact relationships of the "proactive avoidance of remote interference" function with "uplink intra-band CA" and "uplink intra-FR inter-band CA". For details, see [5.2.3.2.3 Function Impacts](#) and [6.2.3.2.3 Function Impacts](#).

Added descriptions of the impact relationships of "downlink intra-band CA", "uplink intra-band CA", "downlink intra-FR inter-band CA", and "uplink intra-FR inter-band CA" with "user-rate-based dynamic network slice resource management" and "user-rate-based dynamic QCI resource management". For details, see [5.1.3.2.3 Function Impacts](#), [5.2.3.2.3 Function Impacts](#), [6.1.3.2.3 Function Impacts](#), and [6.2.3.2.3 Function Impacts](#).

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed. Examples of such changes include changes in the names of reference documents.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 05 (2025-11-07).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added C-RNTI conflict detection and conflict-triggered reconfiguration. For details, see: • 4.6.1.1 Common Procedure for SCell Configuration • 4.6.1.5 C-RNTI Conflict Detection and Conflict-triggered Reconfiguration • 4.6.2 SCell Change • 5.1.2.1 Benefits • 5.1.2.2 Impacts • 5.1.4.1.1 Data Preparation • 5.1.4.1.2 Using MML Commands • 6.1.2.1 Benefits • 6.1.2.2 Impacts • 6.1.4.1.1 Data Preparation • 6.1.4.1.2 Using MML Commands	Modified parameters: • Added the INTRA_GNB_CRNTI_CON_DETECT_SW option to the gNodeBCarrMgmt.CaEnhancedAlgoSwitch parameter. • Added the CRNTI_CONFLICT_RECONFIG_SW option to the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter.	Low-frequency TDD High-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added UE-number-based SCell configuration control. For details, see: • 4.6.1.6 UE-Number-based SCell Configuration Control • 5.1.2.1 Benefits • 5.1.2.2 Impacts • 5.1.4.1.1 Data Preparation • 5.1.4.1.2 Using MML Commands • 6.1.2.1 Benefits • 6.1.2.2 Impacts • 6.1.4.1.1 Data Preparation • 6.1.4.1.2 Using MML Commands	Added parameters: • NRCellCaMgmtConfig.Forbid2ccUeRatioThld • NRCellCaMgmtConfig.Forbid3ccUeRatioThld Modified parameter: Added the UE_NUM_BASED_SCELL_CONFIG_SW option to the gNodeBCarrMgmt.CaEnhancedAlgoSwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added non-CQI-based SCell deactivation for CA UEs performing downlink large-packet services. For details, see: • 4.6.4 SCell Deactivation • 5.1.2.1 Benefits • 5.1.4.1.1 Data Preparation • 5.1.4.1.2 Using MML Commands • 6.1.2.1 Benefits • 6.1.4.1.1 Data Preparation • 6.1.4.1.2 Using MML Commands	Modified parameter: Added the NOT_DEACT_SCC_BASED_CQI_SW option to the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added a method for observing the activation and running state of intra-band CA and intra-FR inter-band CA. For details, see: • 5.1.4.2 Activation Verification • 5.2.4.2 Activation Verification • 6.1.4.2 Activation Verification • 6.2.4.2 Activation Verification	None	Low-frequency TDD High-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added the impact relationships of time-domain power saving BWP (low-frequency TDD) with intra-band CA, inter-FR CA, and intra-FR inter-band CA. For details, see: • 5.1.3.2.3 Function Impacts • 5.2.3.2.3 Function Impacts • 6.1.3.2.3 Function Impacts • 6.2.3.2.3 Function Impacts • 7.3.2.3 Function Impacts	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added the SRS and PUSCH/PUCCH concurrency compatibility function. For details, see: • 5.2.1.5 SRS and PUSCH/PUCCH Concurrency Compatibility • 5.2.2.1 Benefits • 5.2.3.2.1 Prerequisite Functions • 5.2.3.2.2 Mutually Exclusive Functions • 5.2.4.1.1 Data Preparation • 5.2.4.1.2 Using MML Commands • 6.2.1.4 SRS and PUSCH/PUCCH Concurrency Compatibility • 6.2.2.1 Benefits • 6.2.3.2.1 Prerequisite Functions • 6.2.3.2.2 Mutually Exclusive Functions • 6.2.4.1.1 Data Preparation • 6.2.4.1.2 Using MML Commands	Modified parameter: Added the SRS_PUSCH_CONCURRENCE_COMPAT_SWITCH option to the NRDUCellCarrMgmt.CaCompatibilitySwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added uplink enhancement for smart carrier selection. For details, see: • 4.6.1.4 Service-based Optimal Carrier Selection Policies • 5.2.2.1 Benefits • 5.2.3.2.1 Prerequisite Functions • 5.2.4.1.1 Data Preparation • 5.2.4.1.2 Using MML Commands • 6.2.2.1 Benefits • 6.2.3.2.1 Prerequisite Functions • 6.2.4.1.1 Data Preparation • 6.2.4.1.2 Using MML Commands	Modified parameter: Added the SMART_CARR_SEL_UL_ENH_SW option to the NRDUCellCollabServ.MultiCarrMgmtSw parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Replaced the reserved parameter controlling the "multi-carrier combination selection based on multiple bandwidth levels" function with a formal one: Added the RSVDSWPARAM0_BIT31 option of NRCeIIRsvd.RsvdSwParam0 to the disuse list and replaced it with the MULTI_CARR_MULTI_BWP_OPT_SW option of NRCeIIaMgmtConfig.CaStrategySwitch. For details, see: • 4.6.1.3 CA Capability Combination Selection Policies • 5.1.4.1.1 Data Preparation • 5.1.4.1.2 Using MML Commands • 6.1.4.1.1 Data Preparation • 6.1.4.1.2 Using MML Commands	Modified parameter: Added the MULTI_CARR_MULTI_BWP_OPT_SW option to the NRCeIIaMgmtConfig.CaStrategySwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Deleted the mutually exclusive relationship between uplink intra-band CA and PUSCH scheduling in self-contained slots. For details, see 5.2.3.2.2 Mutually Exclusive Functions.	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added a cell-level method for observing the number of CA UEs and SCell configuration. For details, see: • 4.8.2 Optimized Measurement of the Numbers of CA-capable UEs in Cells • 5.2.4.2 Activation Verification • 6.2.4.2 Activation Verification	None	Low-frequency TDD High-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added adaptive CSI-RS beam sweeping. For details, see: • 5.1.1.3 SCell CSI-RS Beam Sweeping • 5.1.2.1 Benefits • 5.1.2.2 Impacts • 5.1.4.1.1 Data Preparation • 5.1.4.1.2 Using MML Commands • 6.1.1.2 SCell CSI-RS Beam Sweeping • 6.1.2.1 Benefits • 6.1.2.2 Impacts • 6.1.4.1.1 Data Preparation • 6.1.4.1.2 Using MML Commands	Modified parameter: Added the CA_CSIRS_BM_SWEEP_ADPT_SW option to the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added functions for improving the SCell activation and deactivation success rates. For details, see: • 4.6.3 SCell Activation • 4.6.4 SCell Deactivation • 5.1.2.1 Benefits • 5.1.4.1.1 Data Preparation • 5.1.4.1.2 Using MML Commands • 6.1.2.1 Benefits • 6.1.4.1.1 Data Preparation • 6.1.4.1.2 Using MML Commands	Activated the following: • RSVDSWPARAM3_BIT32 of the NRDUCellRsvdExt01.RsvdSwParam3 parameter • NRDUCellRsvdExt01.RsvdParam15	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added the impact relationships of RF module fast deep dormancy with downlink intra-band CA and downlink intra-FR inter-band CA. For details, see: • 5.1.3.2.3 Function Impacts • 6.1.3.2.3 Function Impacts	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the mutually exclusive relationship between inter-FR CA and downlink experience satisfaction guarantee. For details, see 7.3.2.2 Mutually Exclusive Functions.	None	Low-frequency TDD High-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the impact relationship between inter-FR CA and PUCCH receive channel adaptation. For details, see 7.3.2.3 Function Impacts.	None	High-frequency TDD	5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added support for flexible configuration of the SINR threshold for uplink data split to SCells in CA. For details, see: • 4.7 CA Data Split • 5.2.4.1.1 Data Preparation • 5.2.4.1.2 Using MML Commands • 6.2.4.1.1 Data Preparation • 6.2.4.1.2 Using MML Commands	Added the NRDUCellCarrMgmt.UlCaSccSplitSinrThld parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added SCC removal due to consecutive non-split in uplink CA. For details, see: • 4.7 CA Data Split • 5.2.2.1 Benefits • 5.2.2.2 Impacts • 5.2.4.1.1 Data Preparation • 5.2.4.1.2 Using MML Commands • 6.2.2.1 Benefits • 6.2.2.2 Impacts • 6.2.4.1.1 Data Preparation • 6.2.4.1.2 Using MML Commands	Added the NRDUCellCarrMgmt.UlCaContNonSplitSccDelThld parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added the impact relationships between flexible experience differentiation and the following functions: • Downlink intra-band CA: See 5.1.3.2.3 Function Impacts. • Downlink intra-FR inter-band CA: See 6.1.3.2.3 Function Impacts. • Inter-FR CA (trial): See 7.3.2.3 Function Impacts.	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added the "FR2 primary-SCell-inclusive activation and deactivation" function. For details, see: • 7.1.4 FR2 Primary-SCell-Inclusive Activation and Deactivation • 7.2.1 Benefits • 7.2.2 Impacts • 7.3.2.1 Prerequisite Functions • 7.3.3 Hardware • 7.3.6 Others • 7.4.1.1 Data Preparation • 7.4.1.2 Using MML Commands	Modified parameter: Added the FR2_SCELL_ACTIV_DEACTV_SW and FR2_SCELL_FAST_ACTIV_DEACTV_SW options to the NRDUCellCarrMgmt.InterFrCaEnhSw parameter. Added the NRDUCellCarrMgmt.Fr2ScellDeactvProhibitTmr and NRDUCellCarrMgmt.Fr2ScellActvInterval parameters.	Low-frequency TDD High-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added support for a maximum of six high-frequency CCs to be aggregated in the downlink for intra-band CA in the case of 200 MHz per high-frequency cell. For details, see: • 4.4 CA Band +Bandwidth Combinations • 5.1.4.1.1 Data Preparation • 5.1.4.1.2 Using MML Commands	Added the gNodeBCarrMgmt.Fr2IntraBandCa200MccMaxNum parameter.	High-frequency TDD	5900 series base stations (macro base stations)
Added support for a maximum of six high-frequency CCs to be aggregated in the downlink for inter-FR CA in the case of 200 MHz per high-frequency cell. For details, see: • 4.4 CA Band +Bandwidth Combinations • 7.3.2.3 Function Impacts • 7.4.1.1 Data Preparation • 7.4.1.2 Using MML Commands	Added the NRDUCellCarrMgmt.InterFrCa200MFr2CcMaxNum parameter.	Low-frequency TDD High-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added support for the 2:3 slot configuration (2_3_DSUUU) by the gNodeB in high frequency scenarios. The corresponding changes in this document are as follows: - Added support for the 2:3 + 2:3 slot configuration combination by intra-band CA. For details, see 4.3 Slot Configuration Combinations. - Added the NRDUCellCarrMgmt.Fr2CaCarrMgmtPolicy configuration requirement under the 2:3 slot configuration (2_3_DSUUU). For details, see 5.3.3.4 Cells.	None	High-frequency TDD	5900 series base stations (macro base stations)
Added the CSI-RS beam sweeping optimization. For details, see: 5.1.1.3 SCell CSI-RS Beam Sweeping 5.1.2.1 Benefits 5.1.4.1.1 Data Preparation 5.1.4.1.2 Using MML Commands 6.1.1.2 SCell CSI-RS Beam Sweeping 6.1.2.1 Benefits 6.1.4.1.1 Data Preparation 6.1.4.1.2 Using MML Commands	Modified parameter: Added the CSI_BEAM_SWEEPING_OPT_SW option to the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added the impact relationship between inter-FR CA and mmWave PUSCH MU-MIMO. For details, see 7.3.2.3 Function Impacts .	None	Low-frequency TDD High-frequency TDD	5900 series base stations (macro base stations)
Added the impact relationship between inter-FR CA and intelligent beam selection. For details, see 7.3.2.3 Function Impacts .	None	High-frequency TDD	5900 series base stations (macro base stations)
Added support for transfer of downlink CA UEs by the "UE-number-based and spectral-efficiency-based connected mode MLB" function and revised descriptions of the impact relationships of this function with downlink intra-band CA and downlink intra-FR inter-band CA. For details, see: • 5.1.3.2.3 Function Impacts • 6.1.3.2.3 Function Impacts	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added the impact relationships of network slicing with downlink intra-band CA (for high-frequency TDD) and inter-FR CA. For details, see 5.1.3.2.3 Function Impacts and 7.3.2.3 Function Impacts .	None	High-frequency TDD	5900 series base stations (macro base stations)

- Editorial changes
Added descriptions of the uplink CA data split threshold optimization function. For details, see:

- [4.7 CA Data Split](#)
- [5.2.2.2 Impacts](#)
- [5.2.4.1.1 Data Preparation](#)
- [5.2.4.1.2 Using MML Commands](#)
- [6.2.2.2 Impacts](#)
- [6.2.4.1.1 Data Preparation](#)
- [6.2.4.1.2 Using MML Commands](#)

Added the impact relationships of uplink intra-band CA and uplink intra-FR inter-band CA with UE-number-based and spectral-efficiency-based connected mode MLB. For details, see [5.2.3.2.3 Function Impacts](#) and [6.2.3.2.3 Function Impacts](#).

Revised descriptions of the impact relationships of network slicing with downlink intra-band CA and downlink intra-FR inter-band CA. For details, see [5.1.3.2.3 Function Impacts](#) and [6.1.3.2.3 Function Impacts](#).

Added descriptions of the impact relationships of "UL and DL Decoupling" with "downlink intra-band CA" and "downlink intra-FR inter-band CA". For details, see [5.1.3.2.3 Function Impacts](#) and [6.1.3.2.3 Function Impacts](#).

Revised the switch of the UL and DL Decoupling function in the impact relationship of this function with inter-FR CA. For details, see [7.3.2.3 Function Impacts](#).

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed. Examples of such changes include changes in the names of reference documents.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FOFD-020205	Intra-Band CA	5.1 Downlink Intra-Band CA 5.2 Uplink Intra-Band CA
FOFD-031201	Intra-FR Inter-Band CA	6.1 Downlink Intra-FR Inter-Band CA 6.2 Uplink Intra-FR Inter-Band CA
FOFD-070206	Inter-FR CA (Trial)	7 Inter-FR Inter-Band CA

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Downlink intra-band CA	None	5.1 Downlink Intra-Band CA
Uplink intra-band CA	None	5.2 Uplink Intra-Band CA
Flexible carrier management (high-frequency TDD)	None	5.3 Flexible Carrier Management (High-Frequency TDD)
Downlink intra-FR inter-band CA	None	6.1 Downlink Intra-FR Inter-Band CA
Uplink intra-FR inter-band CA	None	6.2 Uplink Intra-FR Inter-Band CA
Inter-FR inter-band CA	Supported only in SA networking	7 Inter-FR Inter-Band CA

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high

frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Downlink intra-band CA	Supported in both low and high frequency bands, with the following differences: • Low-frequency CA and high-frequency CA differ in the supported bands and bandwidth combinations. • In high-frequency CA, SCells can be configured, activated, deactivated, or removed (removal triggered by SCell unavailability or dynamically) but cannot be changed. In low-frequency CA, SCells can be configured, changed, activated, deactivated, or removed (triggered by either event A2 or SCell unavailability). • In high-frequency CA, SCells can be configured only in a blind manner, and the NRCellRelation MO, the NRCellFreqRelation MO, and the SCC_BLIND_CONFIG_SW option of the NRCellCaMgmtConfig.CaStrategySwitch parameter do not need to be configured. In low-frequency CA, SCells can be configured either in a blind manner or based on measurements. • CA data split is supported only in low frequency bands. • High-frequency CA can be performed only in a single baseband processing unit, while low-frequency CA can be performed in a single baseband processing unit or between baseband processing units. • The NRCellRelation MO does not need to be configured for SCell configuration in high-frequency CA. • Operator-defined CA band+bandwidth combinations are supported only in low frequency bands. • SCC PMI-based weight enhancement is supported only in low frequency bands. • SCell CSI-RS beam sweeping is supported only in low frequency bands.	5.1 Downlink Intra-Band CA

Function Name	Difference	Chapter/Section
Uplink intra-band CA	Supported in both low and high frequency bands, with the following differences: • Low-frequency CA and high-frequency CA differ in the supported bands and bandwidth combinations. • In high-frequency CA, SCells can be configured, activated, deactivated, or removed (removal triggered by SCell unavailability or dynamically) but cannot be changed. In low-frequency CA, SCells can be configured, changed, activated, deactivated, or removed (triggered by either event A2 or SCell unavailability). • In high-frequency CA, SCells can be configured only in a blind manner, and the NRCellRelation MO, the NRCellFreqRelation MO, and the SCC_BLIND_CONFIG_SW option of the NRCellCaMgmtConfig.CaStrategySwitch parameter do not need to be configured. In low-frequency CA, SCells can be configured either in a blind manner or based on measurements. • CA data split is supported only in low frequency bands. • High-frequency CA can be performed only in a single baseband processing unit, while low-frequency CA can be performed in a single baseband processing unit or between baseband processing units. • The NRCellRelation MO does not need to be configured for SCell configuration in high-frequency CA. • Operator-defined CA band+bandwidth combinations are supported only in low frequency bands. • SCell uplink timing works only in low-frequency non-contiguous CA.	5.2 Uplink Intra-Band CA
Flexible carrier management (high-frequency TDD)	Supported only in high frequency bands	5.3 Flexible Carrier Management (High-Frequency TDD)

Function Name	Difference	Chapter/Section
Downlink intra-FR inter-band CA	Supported only in low frequency bands	6.1 Downlink Intra-FR Inter-Band CA
Uplink intra-FR inter-band CA	Supported only in low frequency bands	6.2 Uplink Intra-FR Inter-Band CA
Inter-FR inter-band CA	Supported in both low and high frequency bands, with the following constraint: PCells must be low-frequency cells, and SCells must be high-frequency cells.	7 Inter-FR Inter-Band CA

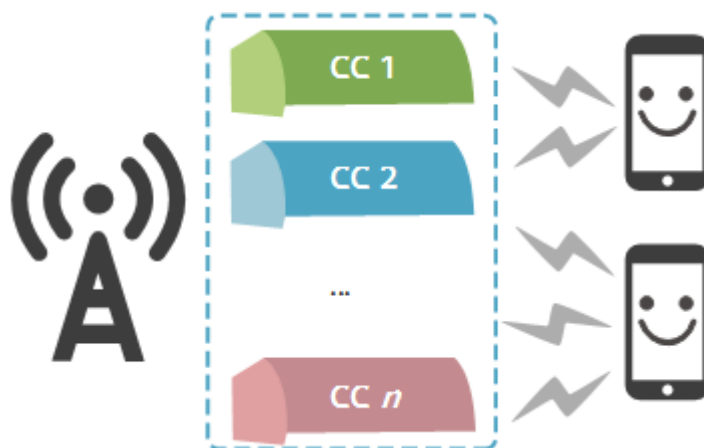
3 Overview

3.1 Definition

3GPP Release 15 requires that individual New Radio (NR) UEs work with up to 1 GHz of bandwidth so that users can enjoy services at higher rates. **However, operators' spectrum resources may not be contiguous or may exceed the single-carrier bandwidth capability defined in 3GPP specifications. Against this backdrop, 3GPP introduced carrier aggregation (CA), which allows for aggregation of multiple contiguous or non-contiguous component carriers (CCs) to expand bandwidth to the 3GPP-required value of 1 GHz and increase users' peak data rates in the uplink and downlink.**

Figure 3-1 illustrates an example of CA.

Figure 3-1 CA for UEs



For details about the protocol stack with CA enabled, see section 6.7 "Carrier Aggregation" in 3GPP TS 38.300 V15.6.0.

This document explains the basic principles and scenarios of CA. For details about functions related to CA experience improvement, see *CA Experience Improvement*.

3.2 Classification

NR CA is classified into intra-FR CA and inter-FR CA based on the frequency ranges (FRs) that CCs belong to.

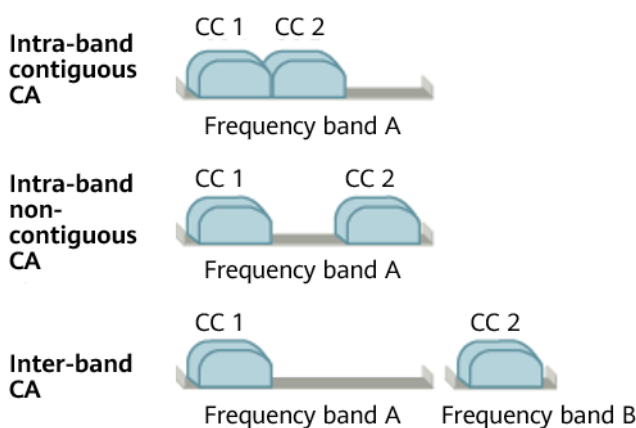
Intra-FR CA aggregates CCs that belong to the same FR (that is, the low FR or high FR), for example, CCs in bands n41 and n77.

Intra-FR CA can be classified into intra-band CA and inter-band CA. Intra-band CA can be further classified into intra-band contiguous CA and intra-band non-contiguous CA. For details about how to determine intra-band contiguous CA and intra-band non-contiguous CA, see [Frequency](#) and [Frequency](#).

- Intra-band CA: CCs that belong to the same frequency band are aggregated.
 - Intra-band contiguous CA: CCs that belong to the same frequency band and are contiguous in the frequency domain are aggregated.
 - Intra-band non-contiguous CA: CCs that belong to the same frequency band and are non-contiguous in the frequency domain are aggregated.
- Inter-band CA: CCs that belong to different frequency bands are aggregated.

[Figure 3-2](#) uses 2CC aggregation to illustrate intra-band contiguous CA, intra-band non-contiguous CA, and inter-band CA.

Figure 3-2 Classification of intra-FR CA



Inter-FR CA aggregates CCs that belong to different FRs (that is, some CCs in the low FR and the others in the high FR), for example, CCs in bands n38 and n257.

NOTE

CA can also be classified into downlink CA and uplink CA.

3.3 CA Application Scenarios

CA can be used between intra- or inter-gNodeB cells, involving both intra-FR CA (including intra-band CA and inter-band CA) and inter-FR CA.

Intra-gNodeB CA Scenarios

CA works in three scenarios: intra-gNodeB co-coverage carriers (scenario 1), intra-gNodeB different-coverage carriers (scenario 2), and intra-gNodeB carriers with one providing edge coverage for another (scenario 3). Intra-band CA mainly works in scenarios 1 and 3. Inter-band CA mainly works in scenarios 2 and 3. [Figure 3-3](#), [Figure 3-4](#), and [Figure 3-5](#) show the CA scenarios, which use 2CC aggregation as an example. In the figures, **F1** and **F2** denote two carrier frequencies.

Figure 3-3 Intra-gNodeB co-coverage carriers

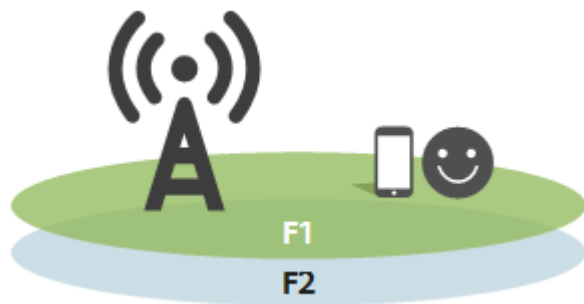


Figure 3-4 Intra-gNodeB different-coverage carriers

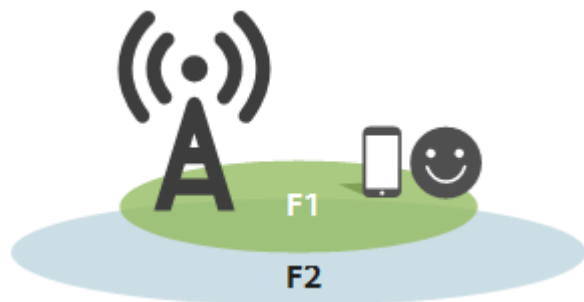


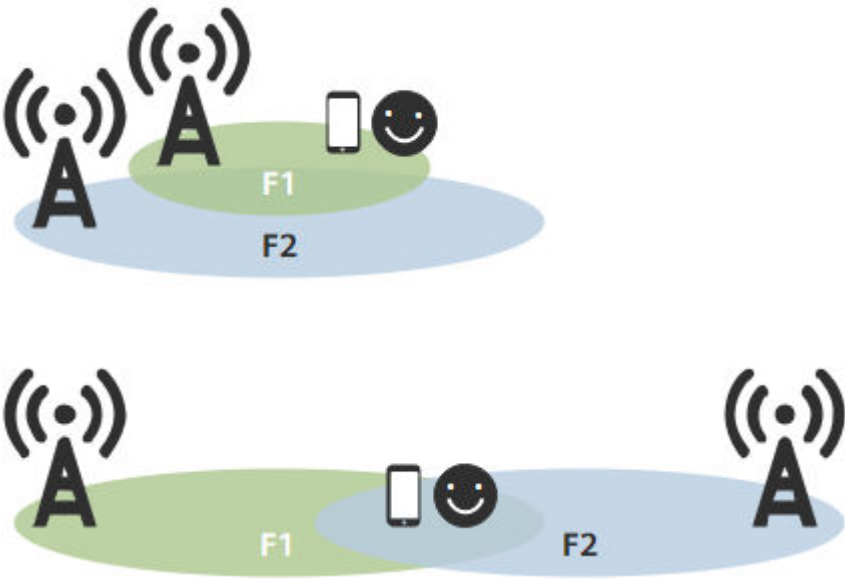
Figure 3-5 Intra-gNodeB carriers with one providing edge coverage for another



Inter-gNodeB CA Scenarios

[Figure 3-6](#) shows the inter-gNodeB CA scenarios. In the figure, **F1** and **F2** denote two carrier frequencies.

Figure 3-6 Inter-gNodeB CA scenarios



3.4 Overview of CA Functions

The gNodeB offers various CA functions to CA UEs according to their CA capabilities and the carrier management principles. These functions allow CA UEs to have larger bandwidths for data transmission. [Table 3-1](#) and [Table 3-2](#) provide an overview of these functions.

NOTE

- UEs report their CA capabilities to the gNodeB through the *BandCombination* IE in the *UECapabilityInformation* message. This IE contains the UE-supported band combinations.
- For details about carrier management principles, see [4 General Principles](#).

Table 3-1 Overview of CA functions (1)

Function Name	Intra-Band	Inter-Band	Intra-gNodeB	Inter-gNodeB	Uplink	Downlink
Downlink intra-gNodeB intra-band CA	√	-	√	-	-	√
Uplink intra-gNodeB intra-band CA	√	-	√	-	√	-

Function Name	Intra-Band	Inter-Band	Intra-gNodeB	Inter-gNodeB	Uplink	Downlink
Flexible carrier management (high-frequency TDD)	√	-	√	-	√	√
Downlink intra-gNodeB intra-FR inter-band CA	-	√	√	-	-	√
Uplink intra-gNodeB intra-FR inter-band CA	-	√	√	-	√	-
Inter-FR inter-band CA	-	√	√	√	√	√
CA SRS carrier switching	√	√	√	√	-	√
Inter-gNodeB CA	√	√	-	√	-	√
3CC aggregation	√	√	√	√	-	√
Fast CA	√	√	√	√	-	√
Massive CA	-	√	√	√	-	√
CA-UE-specific frequency-priority-based inter-frequency handover ^a	√	√	√	√	√	√

Function Name	Intra-Band	Inter-Band	Intra-gNodeB	Inter-gNodeB	Uplink	Downlink
a: CA-UE-specific frequency-priority-based inter-frequency handover is an inter-frequency handover procedure specific to CA UEs that is performed to change the PCell for each CA UE. This function works in all the scenarios listed in this table.						

 NOTE

In the preceding table, "√" indicates that the function is supported in the specified scenario, while "-" indicates that the function is not supported.

Table 3-2 Overview of CA functions (2)

Function Name	Function Switch	Reference
Downlink intra-gNodeB intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	5.1 Downlink Intra-Band CA
Uplink intra-gNodeB intra-band CA	INTRA_BAND_UL_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	5.2 Uplink Intra-Band CA
Flexible carrier management (high-frequency TDD)	NRDUCellCarrMgmt.Fr2CaCarrMgmtPolicy set to USER_TRAFFIC_BASED	5.3 Flexible Carrier Management (High-Frequency TDD)
Downlink intra-gNodeB intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	6.1 Downlink Intra-FR Inter-Band CA
Uplink intra-gNodeB intra-FR inter-band CA	INTRA_FR_INTER_BAND_UL_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	6.2 Uplink Intra-FR Inter-Band CA
Inter-FR inter-band CA	INTER_FR_FR1TOFR2_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	7 Inter-FR Inter-Band CA
CA SRS carrier switching	SRS_CARRIER_SWITCHING_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter	<i>CA Experience Improvement</i>

Function Name	Function Switch	Reference
Inter-gNodeB CA	INTER_GNODEB_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>CA Experience Improvement</i>
3CC aggregation	NRDUCellCarrMgmt.CaDLMaxCcNum set to DL3CC	<i>CA Experience Improvement</i>
Fast CA	Fast intra-gNodeB CA: CA_ULTRA_LOW_LATENCY_SPLIT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter Fast inter-gNodeB CA: INTER_GNB_ULTRALOW_LAT_SPLIT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter	<i>CA Experience Improvement</i>
Massive CA	4CC aggregation: NRDUCellCarrMgmt.CaDLMaxCcNum set to DL4CC 5CC aggregation: NRDUCellCarrMgmt.CaDLMaxCcNum set to DL5CC	<i>CA Experience Improvement</i>
CA-UE-specific frequency-priority-based inter-frequency handover	CA_FREQ_PRI_BASED_INTERF_HO_SW option of the NRCCellAlgoSwitch.InterFreqHoSwitch parameter	<i>CA Experience Improvement</i>

4 General Principles

The basic concepts of CA and SCell-related procedures apply to both this document and *CA Experience Improvement*.

4.1 Related Concepts

- PCell
A primary cell (PCell) is the cell that a CA UE camps on. The CA UE works in the same way in the PCell as it does in a single-carrier cell.
- SCell
A secondary cell (SCell) is a cell that a base station configures for a CA UE through an RRC message. This cell provides the CA UE with additional radio resources. In an SCell, there can be either downlink transmission alone, or both downlink and uplink transmission.
- CC
Component carriers (CCs) are the carriers that are aggregated for a CA UE.
- PCC
The primary component carrier (PCC) is the carrier of the PCell.
- SCC
A secondary component carrier (SCC) is the carrier of an SCell.
There are two types of SCCs:
 - Uplink+downlink SCC: Both PUSCH and sounding reference signal (SRS) resources are available. That is, such an SCC is configured for both the downlink and uplink.
 - Downlink-only SCC: Neither PUSCH nor SRS resources are available. That is, such an SCC is configured only for the downlink.
- Symmetric CA and asymmetric CA
 - Symmetric CA: The number of CCs in the uplink is equal to the number of CCs in the downlink.
 - Asymmetric CA: The number of CCs in the uplink is different from the number of CCs in the downlink. Generally, a CA UE uses more CCs in the downlink than in the uplink.

Whether symmetric or asymmetric CA can be used for a CA UE depends on the UE's capabilities. Certain UEs do not support symmetric CA.

4.2 Related Events

- Event A5

The signal quality of the PCell drops below threshold 1 and the signal quality of a neighboring cell exceeds threshold 2.

Triggering condition for event A5: $(Ms + Hys < Thresh1)$ and $(Mn + Ofn + Ocn - Hys > Thresh2)$ are true throughout *TimeToTrig*.

In this condition:

- *Ms* and *Mn* are the measurement results of the serving cell and a neighboring cell, respectively.
- *Hys* is the hysteresis for the measurement result.
- *Thresh1* and *Thresh2* are thresholds. For CA event A5, *Thresh1* is fixed at –30 dBm, while *Thresh2* is determined collectively by the **NRCellCaMgmtConfig.CaA5RsrpThld2** and **gNBCaFrequency.CaSccA5RsrpThld2Offset** parameters.
- *Ofn* is the frequency-specific offset for the neighboring cell.
- *Ocn* is the cell individual offset (CIO) for the neighboring cell.
- *TimeToTrig* is the duration throughout which the entering condition for an event must be met before the event can be triggered. The time-to-trigger for CA event A5 is specified by the **NRCellCaMgmtConfig.CaA5TimeToTrig** parameter.

 NOTE

In the following sections, the threshold for event A5 refers to threshold 2.

Event A5 is used for measurement-based SCell configuration. For details, see [4.6.1 SCell Configuration](#).

- Event A1

The signal quality of the serving cell exceeds a specific threshold. If a cell has been configured as a downlink SCell, the gNodeB uses event A1 for measurement-based uplink SCell configuration. For details, see [4.6.1 SCell Configuration](#).

Triggering condition for event A1: $(Ms - Hys > Thresh)$ is true throughout *TimeToTrig*.

In this condition:

- *Ms* is the measurement result of the serving cell.
- *Hys* is the hysteresis for the measurement result. The value of this variable is always 2 dBm.
- *Thresh* is the threshold. *Thresh* for CA event A1 is the same as *Thresh2* for CA event A5. For details, see [Event A5](#).
- *TimeToTrig* is the duration throughout which the entering condition for an event must be met before the event can be triggered. The time-to-trigger for CA event A1 is the same as that for CA event A5. For details, see [Event A5](#).

- Event A6

The signal quality of an SCell's intra-frequency neighboring cell becomes offset better than that of the SCell. After receiving an event A6 report, the gNodeB changes the SCell to a new one while keeping the PCell unchanged. For details, see [4.6.2 SCell Change](#).

Triggering condition for event A6: $(Mn + Ocn - Hys > Ms + Ocs + Off)$ is true throughout *TimeToTrig*.

In this condition:

- *Mn* is the reference signal received power (RSRP) measurement result of a neighboring cell.
- *Ocn* is the cell-specific offset specified by the **NRCellRelation.CellIndividualOffset** parameter for the neighboring cell. This parameter is configured on the PCell side.
- *Hys* is the hysteresis for event A6. The value of this variable is always 1 dB.
- *Ms* is the RSRP measurement result of the SCell.
- *Ocs* is the cell-specific offset for the SCell. The value of this variable is always 0 dB.
- *Off* is the offset for event A6. The offset is specified by the **NRCellCaMgmtConfig.CaA6Offset** parameter.
- *TimeToTrig* is the duration throughout which the entering condition for an event must be met before the event can be triggered. The time-to-trigger for CA event A6 is specified by the **NRCellCaMgmtConfig.CaA6TimeToTrig** parameter.

- Event A2

The signal quality of an SCell drops below a specific threshold. The threshold for CA event A2 is determined by the **NRCellCaMgmtConfig.CaA2RsrpThld**, **NRCellCaMgmtConfig.UlCaA2RsrpThld**, and **gNBCaFrequency.CaA2RsrpThldOffset** parameters. Event A2 is used for removal of SCells with poor signal quality. For details, see [4.6.5 SCell Removal](#).

4.3 Slot Configuration Combinations

The gNodeB works with the slot configuration combinations listed in [Table 4-1/ Table 4-2/ Table 4-3](#).

Low-Frequency CA

In particular, the compatibility with single-period 8:2 and dual-period 8:2 slot configuration combinations is controlled by the switch specified by the **CA_SLOT_CONFIG_EXT_SW** option of the **NRCellCaMgmtConfig.CaStrategySwitch** parameter. Downlink CA is compatible with these combinations only if this switch is turned on for the PCell.

Table 4-1 Slot configuration combinations supported for low-frequency CA

CA Type	Slot Configuration Combination
Intra-band CA (TDD +TDD)	• 4:1 + 4:1 • Single-period 8:2 + Single-period 8:2 • Dual-period 8:2 + Dual-period 8:2 NOTE Intra-band CA requires all CCs to have the same uplink-downlink slot configuration, according to section 4.2 "Applicability of minimum requirements" in 3GPP TS 38.101-1 V15.6.0, 3GPP TS 38.101-2 V15.6.0, and 3GPP TS 38.101-3 V15.6.0.
Intra-FR inter-band CA (TDD+TDD)	• Same slot configuration: – 4:1 + 4:1 – Single-period 8:2 + Single-period 8:2 – Dual-period 8:2 + Dual-period 8:2 • Different slot configurations: – 4:1 + Single-period 8:2 – 4:1 + Dual-period 8:2 – Single-period 8:2 + Dual-period 8:2
3CC aggregation (TDD+TDD+TDD)	Intra-band 3CC aggregation: • 4:1 + 4:1 + 4:1 • Single-period 8:2 + Single-period 8:2 + Single-period 8:2 • Dual-period 8:2 + Dual-period 8:2 + Dual-period 8:2 Intra-FR inter-band 3CC aggregation: • Same slot configuration: – 4:1 + 4:1 + 4:1 – Single-period 8:2 + Single-period 8:2 + Single-period 8:2 – Dual-period 8:2 + Dual-period 8:2 + Dual-period 8:2 • Different slot configurations: – 4:1 + 4:1 + Single-period 8:2, or 4:1 + Single-period 8:2 + Single-period 8:2 – 4:1 + 4:1 + Dual-period 8:2, or 4:1 + Dual-period 8:2 + Dual-period 8:2 – Single-period 8:2 + Single-period 8:2 + Dual-period 8:2, or Dual-period 8:2 + Dual-period 8:2 + Single-period 8:2

CA Type	Slot Configuration Combination
Massive CA	The supported slot configuration combinations are the same as those supported for 3CC aggregation. NOTE Massive CA allows for aggregation of at most five intra-FR inter-band carriers and is compatible with slot configuration combinations in which there are at most two different types of TDD slot configurations.
SRS carrier switching enhancement in TDD-only 3CC aggregation scenarios	Single-period 8:2 + Single-period 8:2 + Single-period 8:2

High-Frequency CA

Table 4-2 Slot configuration combinations supported for high-frequency CA

CA Type	Slot Configuration Combination
Intra-band CA ($n \times$ TDD) (For details about the value of n , see 4.4 CA Band +Bandwidth Combinations .)	$n \times 4:1$ $n \times 2:3$ NOTE Intra-band CA requires all CCs to have the same uplink-downlink slot configuration, according to section 4.2 "Applicability of minimum requirements" in 3GPP TS 38.101-1 V15.6.0, 3GPP TS 38.101-2 V15.6.0, and 3GPP TS 38.101-3 V15.6.0.

Inter-FR CA

Table 4-3 Slot configuration combinations supported for inter-FR CA

CA Type	Slot Configuration Combination
Inter-FR CA	Low-frequency 1CC: • Low-frequency 4:1 + High-frequency 4:1 • Low-frequency single-period 8:2 + High-frequency 4:1 Low-frequency 2CC: • Low-frequency 4:1 + Low-frequency 4:1 + High-frequency 4:1 • Low-frequency 4:1 + Low-frequency single-period 8:2 + High-frequency 4:1 • Low-frequency single-period 8:2 + Low-frequency single-period 8:2 + High-frequency 4:1 NOTE All high-frequency CCs must have the same uplink-downlink slot configuration.

4.4 CA Band+Bandwidth Combinations

Low-Frequency CA

Intra-band CA and intra-FR inter-band CA both work in low frequency bands.

- For downlink intra-band CA, a maximum of three contiguous or non-contiguous CCs can be aggregated. 3CC aggregation requires that the **NRDUCellCarrMgmt.CaDlMaxCcNum** parameter be set to **DL3CC**. (For more details, see *CA Experience Improvement*.) This requirement does not apply to 2CC aggregation.
- For downlink intra-FR inter-band CA, a maximum of five CCs can be aggregated.
 - 3CC aggregation requires that the **NRDUCellCarrMgmt.CaDlMaxCcNum** parameter be set to **DL3CC**.
 - 4CC aggregation requires that the **NRDUCellCarrMgmt.CaDlMaxCcNum** parameter be set to **DL4CC**.
 - 5CC aggregation requires that the **NRDUCellCarrMgmt.CaDlMaxCcNum** parameter be set to **DL5CC**.

For more details, see *CA Experience Improvement*. These requirements do not apply to 2CC aggregation.

- For uplink intra-band CA and uplink intra-FR inter-band CA, a maximum of two contiguous or non-contiguous CCs can be aggregated.

Both default and customized (not protocol-defined) band+bandwidth combinations are compatible with intra-band CA and intra-FR inter-band CA in low frequency bands:

- Default combinations: **Table 4-4** provides the tables in 3GPP TS 38.101-1 that list the band+bandwidth combinations for intra-band CA and intra-FR inter-band CA.

Table 4-4 Band+bandwidth combinations for low-frequency CA

CA Type	Band+Bandwidth Combination
Intra-band CA	Same as protocol-defined ones. For details, see Table 5.5A.1-1 (for contiguous CA) and Table 5.5A.2-1 (for non-contiguous CA) in 3GPP TS 38.101-1 V18.6.0.
Intra-FR inter-band CA	Same as protocol-defined ones. For details, see Table 5.5A.3.1-1 (2CC), Table 5.5A.3.2-1 (3CC), Table 5.5A.3.3-1 (4CC), and Table 5.5A.3.4-1 (5CC) in 3GPP TS 38.101-1 V18.6.0.
Note 1: The preceding band+bandwidth combinations can be used only after cells with the specific bandwidths have been set up in the corresponding frequency bands. Note 2: A CC can be configured for uplink CA only if it is also configured for downlink CA.	

- Customized combinations: In addition to the preceding combinations, which are supported by default, operators can customize band+bandwidth combinations for CA. After a customized combination is defined with the frequency bands and bandwidths specified, gNodeBs can aggregate carriers in the combination.

CA band+bandwidth combinations can be defined as follows:

- Set the ID of each customized CA band+bandwidth combination using the **gNBCustCaBandComb.CustCaBandCombId** parameter.
- Specify customized CA bandwidth combination sets by using the **gNBCustCaBandComb.BandwidthCombSet** parameter. It is recommended that this parameter be set to **0** to maximize compatibility with UE capabilities.
- Set the following parameters to specify the bands and bandwidths in the customized combinations:
 - **gNBCustCaBandComb.FrequencyBand1** + **gNBCustCaBandComb.FrequencyBandwidth1**
 - **gNBCustCaBandComb.FrequencyBand2** + **gNBCustCaBandComb.FrequencyBandwidth2**
 - **gNBCustCaBandComb.FrequencyBand3** + **gNBCustCaBandComb.FrequencyBandwidth3**
 - **gNBCustCaBandComb.FrequencyBand4** + **gNBCustCaBandComb.FrequencyBandwidth4**
 - **gNBCustCaBandComb.FrequencyBand5** + **gNBCustCaBandComb.FrequencyBandwidth5**

 NOTE

According to the definition of fallback band combination in section 3.1 "Definitions" of 3GPP TS 38.306 V17.3.0, a fallback band combination supports the same channel bandwidth for each carrier as its parent band combination. Therefore, in a UE capability fallback case of CA, CA works after the fallback only if the parent band+bandwidth combinations are completely configured.

High-Frequency CA

Only intra-band CA works in high-frequency NR TDD.

- In the downlink, at most six contiguous CCs can be aggregated for intra-band CA. Non-contiguous intra-band CA does not work.
- In the uplink, at most four contiguous CCs can be aggregated for intra-band CA. Non-contiguous intra-band CA does not work.

Only default band+bandwidth combinations are compatible with intra-band CA in high frequency bands. The following lists the combinations for intra-band CA according to 3GPP TS 38.101-2 V16.6.0.

- Supported band combinations: n257/n258/n261
- Supported bandwidth combinations:
 - In the case of 100 MHz per high-frequency cell: $m \times 100$ MHz in the downlink and $n \times 100$ MHz in the uplink ($1 \leq n \leq m \leq 4$)
 - In the case of 200 MHz per high-frequency cell (SA scenario):
 - If the **gNodeBCarrMgmt.Fr2IntraBandCa200MccMaxNum** parameter is set to **DL6CC_UL4CC**: $m \times 200$ MHz in the downlink and $n \times 200$ MHz in the uplink ($1 \leq m \leq 6; 1 \leq n \leq 4; n \leq m$)
 - If the **gNodeBCarrMgmt.Fr2IntraBandCa200MccMaxNum** parameter is set to **DL4CC_UL4CC**: $m \times 200$ MHz in the downlink and $n \times 200$ MHz in the uplink ($1 \leq n \leq m \leq 4$).
 - In the case of 200 MHz per high-frequency cell (NSA scenario): $m \times 200$ MHz in the downlink and $n \times 200$ MHz in the uplink ($1 \leq n \leq m \leq 4$)

 NOTE

The preceding bandwidth combinations can be used only when cells with the specific bandwidths have been set up in the corresponding frequency bands. A CC can be configured for uplink CA only if it is also configured for downlink CA.

Inter-FR CA

Based on the number of low-frequency CCs, inter-FR CA is classified into the following types:

- Inter-FR CA with one low-frequency CC: The one low-frequency CC acts as the PCC, and all high-frequency CCs act as SCCs. This type of CA works in both the uplink and the downlink.
- Inter-FR CA with two low-frequency CCs: The two low-frequency CCs are a PCC and an SCC, and all high-frequency CCs act as SCCs. This type of CA works in both the uplink and the downlink. However, uplink CA does not work between the two low-frequency CCs.

Inter-FR CA requires that all the high-frequency SCCs be in the same frequency band and contiguously spread in the frequency domain.

- Inter-FR CA with two low-frequency CCs is compatible with the following combinations:
 - Low-frequency CCs in bands n41 and n79 + high-frequency CCs in band n258
 - Low-frequency CCs in band n78 + high-frequency CCs in band n258
 - Low-frequency CCs in bands n40 and n78 + high-frequency CCs in band n257 or n258
 - Low-frequency CCs in bands n77 and n79 + high-frequency CCs in band n258

The following are the bandwidth combinations supported in low and high frequency bands:

- Bandwidth combinations of low-frequency cells: They must meet the requirements stipulated in section 5.3.5 "UE channel bandwidth per operating band" of 3GPP TS 38.101-1 V18.6.0 and the requirements stipulated in section 5.5A.1 "Inter-band CA configurations between FR1 and FR2" of 3GPP TS 38.101-3 V17.7.0.
- Bandwidth combinations of high-frequency cells:
 - In the case of 100 MHz per high-frequency cell: $m \times 100$ MHz in the downlink and $n \times 100$ MHz in the uplink ($1 \leq n \leq m \leq 4$)
 - In the case of 200 MHz per high-frequency cell:
 - If the **NRDUCellCarrMgmt.InterFrCa200MFr2CcMaxNum** parameter is set to **DL6CC_UL4CC**: $m \times 200$ MHz in the downlink and $n \times 200$ MHz in the uplink ($1 \leq m \leq 6$; $1 \leq n \leq 4$; $n \leq m$)
 - If the **NRDUCellCarrMgmt.InterFrCa200MFr2CcMaxNum** parameter is set to **DL4CC_UL4CC**: $m \times 200$ MHz in the downlink and $n \times 200$ MHz in the uplink ($1 \leq n \leq m \leq 4$).

The **NRDUCellCarrMgmt.InterFrCa200MFr2CcMaxNum** parameter is configured on the PCell.

- Inter-FR CA with one low-frequency CC is compatible with the following combinations: low-frequency CCs in band n38, n40, n41, n77, n78, or n79 + high-frequency CCs in band n257, n258, or n261.

The following are the bandwidth combinations supported in low and high frequency bands:

- Bandwidth combinations of low-frequency cells: all cell bandwidths supported by low-frequency NR TDD cells. For details, see *Scalable Bandwidth*.
- Bandwidth combinations of high-frequency cells:
 - In the case of 100 MHz per high-frequency cell: $m \times 100$ MHz in the downlink and $n \times 100$ MHz in the uplink ($1 \leq n \leq m \leq 4$)
 - In the case of 200 MHz per high-frequency cell:
 - If the **NRDUCellCarrMgmt.InterFrCa200MFr2CcMaxNum** parameter is set to **DL6CC_UL4CC**: $m \times 200$ MHz in the downlink and $n \times 200$ MHz in the uplink ($1 \leq m \leq 6$; $1 \leq n \leq 4$; $n \leq m$)

- If the **NRDUCellCarrMgmt.InterFrCa200MFr2CcMaxNum** parameter is set to **DL4CC_UL4CC**: $m \times 200$ MHz in the downlink and $n \times 200$ MHz in the uplink ($1 \leq n \leq m \leq 4$).

The **NRDUCellCarrMgmt.InterFrCa200MFr2CcMaxNum** parameter is configured on the PCell.

NOTE

Channel bandwidths for carriers supported by certain UEs for CA are sometimes less than or equal to the bandwidths supported by single carriers. In addition, supported bandwidth combinations for CA sometimes vary between UEs provided by different vendors.

For example, an operator has two frequencies in band n78. On each frequency, a cell operates with a bandwidth of 100 MHz. Three UEs on the network support different bandwidth combinations in band n78: UE1 supports 100 MHz+40 MHz, UE2 supports 100 MHz+20 MHz, and UE3 supports 100 MHz+100 MHz.

- If both **NRDUCellBwp.UlMinCarrierBw** and **NRDUCellBwp.DlMinCarrierBw** are set to **20M** for the two cells: Only 100 MHz+20 MHz can take effect for UE1, despite its support for 100 MHz+40 MHz; 100 MHz+20 MHz can take effect for UE2; 100 MHz+100 MHz can take effect for UE3, whose bandwidth capability is the same as the cell bandwidth capability.
- If both **NRDUCellBwp.UlMinCarrierBw** and **NRDUCellBwp.DlMinCarrierBw** are set to **40M** for the two cells: 100 MHz+40 MHz can take effect for UE1; CA cannot be performed for UE2; 100 MHz+100 MHz can take effect for UE3, whose bandwidth capability is the same as the cell bandwidth capability.

To resolve the issue occurring when the preceding types of UEs are present on the network, enable UE bandwidth adaptation, with which the UE-reported band+bandwidth combinations take effect for CA of the UEs. UE bandwidth adaptation is controlled by the **UE_BW_ADAPTIVE_SW** option of the **NRDUCellBwp.BwpConfigSwitch** parameter. For details about this function, see *Scalable Bandwidth*.

4.5 PCell Management

PCell management includes the "prohibition of acting as the PCell" function ([Prohibition of Acting as the PCell](#)). The PCell management policy configuration can distinguish between NSA and SA networking ([NSA/SA-specific PCell Management Policy Configuration](#)).

Prohibition of Acting as the PCell

The "prohibition of acting as the PCell" function is controlled by the **PCELL_PROHIBIT_SW** option of the **NRCaMgmtConfig.CaStrategySwitch** parameter: If this option is selected, the cell cannot act as the PCell for CA and the SCell configuration procedure will not be initiated. For UEs that newly access the cell and UEs camping on the cell but not in the CA state, CA will not take effect. For UEs in the CA state with the cell acting as their PCell, selecting the function option does not affect their existing SCells; the gNodeB can remove the SCells but cannot add new SCells or change SCells for these UEs.

NSA/SA-specific PCell Management Policy Configuration

This function is controlled by the **NRCaMgmtConfig.SpecUePcellPolicy** parameter.

- If this parameter is set to **NSA_SA**, the PCell management policy can take effect for both NSA and SA UEs.

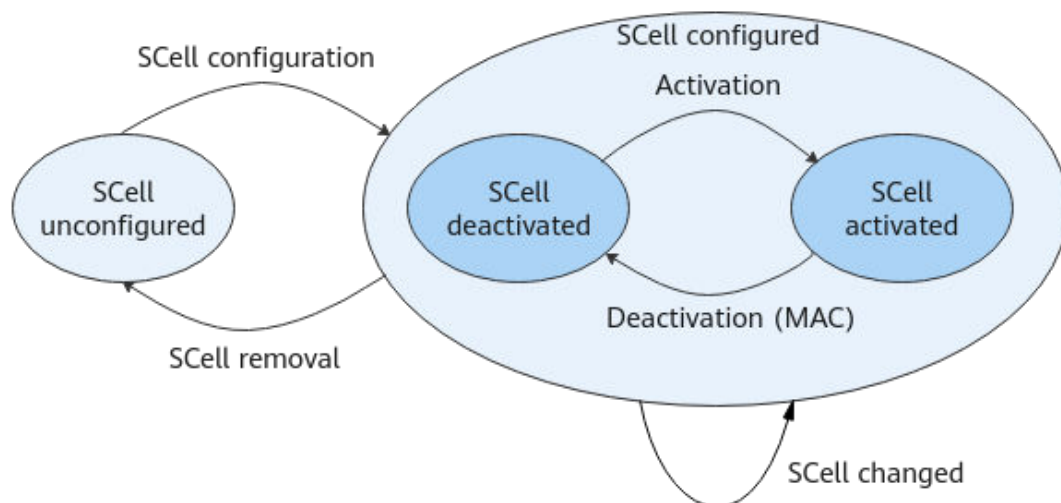
- If this parameter is set to **NSA**, the PCell management policy can take effect only for NSA UEs.
- If this parameter is set to **SA**, the PCell management policy can take effect only for SA UEs.

After this function takes effect, the PCell management policy configuration can distinguish between NSA and SA networking. This flexible configuration ensures optimal performance in both networking.

4.6 SCell Management

With CA enabled, SCell configuration is triggered for a CA UE during initial access, an incoming handover, or an incoming RRC connection reestablishment. After an SCell is configured, it can be changed, activated, deactivated, or removed. These SCell management activities are performed by the gNodeB that the PCell belongs to. **CA takes effect only after an SCell is configured and activated for a CA UE.** Figure 4-1 shows the transitions between the SCell states. In low-frequency CA, SCells can be configured, changed, activated, deactivated, or removed. In high-frequency CA, SCells can be configured, activated, or deactivated, or can be removed dynamically or based on SCell unavailability, but cannot be removed based on event A2 or changed.

Figure 4-1 SCell state transitions



4.6.1 SCell Configuration

4.6.1.1 Common Procedure for SCell Configuration

A gNodeB starts SCell configuration for a CA UE during initial access, an incoming handover, or an incoming RRC connection reestablishment.

SCells are configured in a blind manner or based on measurements.

- Blind SCell configuration: Cells that meet certain conditions are directly configured as SCells without being measured.

- Measurement-based SCell configuration: Cells that meet the threshold for event A5 are configured as SCells.

 NOTE

The gNodeB checks whether the blind SCell configuration conditions are met. If the conditions are met, the gNodeB performs blind SCell configuration. If the conditions are not met or blind SCell configuration fails, measurement-based SCell configuration is performed.

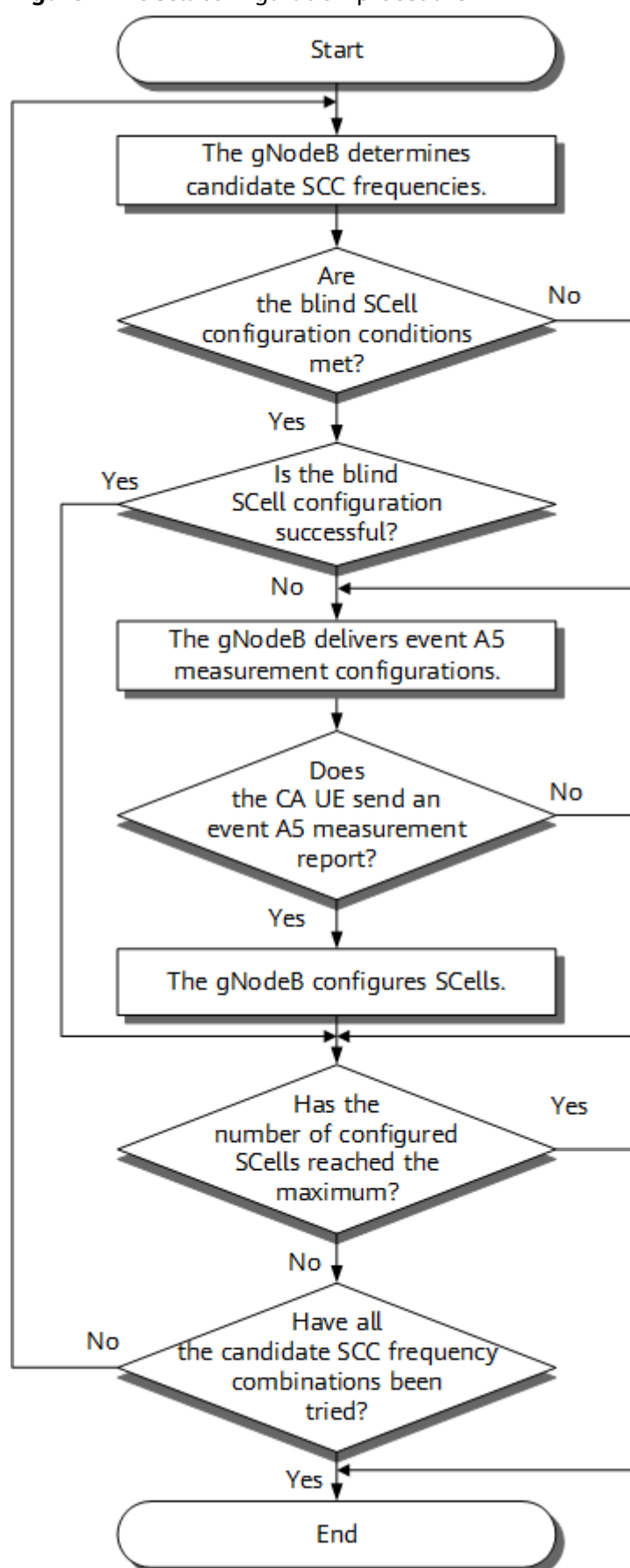
In low-frequency CA, intra-gNodeB SCells can be configured in a blind manner or based on measurements, while inter-gNodeB SCells can be configured only based on measurements.

In high-frequency CA, intra-gNodeB SCells can be configured only in a blind manner.

SCell Configuration Procedure in Low-Frequency Scenarios

Figure 4-2 illustrates the SCell configuration procedure in low-frequency scenarios.

Figure 4-2 SCell configuration procedure



The SCell configuration procedure varies depending on the setting of the smart carrier selection switch (the **SMART_CARR_SEL_SW** option of the **NRDUCellCollabServ.MultiCarrMgmtSw** parameter):

- The smart carrier selection switch is on.
 - a. The gNodeB selects candidate frequencies, which are included in a CA frequency group, have not been allocated to the CA UE, and belong to the same operator as the UE. A CA frequency group is configured using the **gNBCaFrequency** MO. A frequency belongs to an operator if the operator is configured using the **gNBCaFrequency.OperatorIdList** parameter.
 - b. The gNodeB checks whether the blind SCell configuration conditions are met. If the conditions are met, the procedure goes to **c**. Otherwise, the procedure goes to **d**.

The blind SCell configuration conditions are as follows:

- The **SCC_BLIND_CONFIG_SW** option of the **NRCellCaMgmtConfig.CaStrategySwitch** parameter is selected.
- A blind-configurable candidate SCell (with **NRCellRelation.BlindScellConfigFlag** set to **TRUE**) operates on a candidate SCC frequency.
- c. The gNodeB sends an RRCReconfiguration message to configure appropriate cells (cells with **NRCellRelation.BlindScellConfigFlag** set to **TRUE** on the candidate SCC frequencies and PLMN IDs overlapping those of the PCell) as SCells in a blind manner. (Cells mentioned in **c** all refer to blind-configurable candidate SCells.)

If multiple frequencies serve cells with

NRCellRelation.BlindScellConfigFlag set to **TRUE**, the gNodeB selects a number of optimal cells from those cells as the targets for blind SCell configuration following a certain policy. (The number of cells selected is equal to the number of SCells to be configured. For example, a maximum of one cell is selected in the case of 2CC aggregation, and a maximum of two cells are selected in the case of 3CC aggregation.) The policy depends on the setting of the service-aware carrier selection switch (the **SERVICE_AWARE_CARR_SEL_SW** option of the **NRDUCellCollabServ.MultiCarrMgmtSw** parameter).

- If the service-aware carrier selection switch is off, the gNodeB follows a policy according to **4.6.1.3 CA Capability Combination Selection Policies** to select the optimal CA capability combination for the UE.
- If the service-aware carrier selection switch is on, the gNodeB follows a policy according to **4.6.1.4 Service-based Optimal Carrier Selection Policies** to select the optimal CA capability combination for the UE.

If the blind SCell configuration succeeds, the procedure goes to **g**. If it fails, the procedure goes to **d** for measurement-based configuration.

- d. The gNodeB delivers event A5 measurement configurations to the CA UE. The event A5 threshold is the sum of the values of **NRCellCaMgmtConfig.CaA5RsrpThld2** and

gNBCaFrequency.CaSccA5RsrpThld2Offset. When there are multiple candidate frequencies, the gNodeB delivers the event A5 measurement configurations related to all the candidate frequencies supported by the UE.

- e. The gNodeB checks whether it has received an A5 measurement report from the CA UE.
 - If the gNodeB has received an A5 measurement report before the A5 measurement waiting timer (specified by the **gNodeBAlgo.MultiCarrierMeasWaitTimer** parameter) expires, the procedure goes to **f**. There are two cases. In the first case, the gNodeB has received A5 measurement reports related to all candidate frequencies before the timer expires. In this case, the gNodeB immediately proceeds to **f** and stops the timer. In the second case, the gNodeB has received A5 measurement reports related to (not necessarily all) candidate frequencies when the timer expires. In this case, the gNodeB proceeds to **f** upon timer expiry.
 - If the gNodeB has not received A5 measurement reports on any candidate frequency before the A5 measurement waiting timer (specified by the **gNodeBAlgo.MultiCarrierMeasWaitTimer** parameter) expires, the procedure goes to **g**.
- f. The gNodeB configures SCells for the CA UE.

The gNodeB selects a number of optimal cells (each with the highest RSRP value on its operating frequency and PLMN IDs overlapping those of the PCell) as SCells following a certain policy. (The number of cells selected is equal to the number of SCells to be configured. For example, a maximum of one cell is selected in the case of 2CC aggregation, and a maximum of two cells are selected in the case of 3CC aggregation.) The policy depends on the setting of the service-aware carrier selection switch:

- If the service-aware carrier selection switch is off, the gNodeB follows a policy according to **4.6.1.3 CA Capability Combination Selection Policies** to select the optimal CA capability combination for the UE.
- If the service-aware carrier selection switch is on, the gNodeB follows a policy according to **4.6.1.4 Service-based Optimal Carrier Selection Policies** to select the optimal CA capability combination for the UE.

After an SCell is selected, the gNodeB checks whether this SCell can be configured according to **4.6.1.5 C-RNTI Conflict Detection and Conflict-triggered Reconfiguration**.

- g. The gNodeB automatically checks whether the number of successfully configured SCells has reached the maximum.
 - i. If it has, the SCell configuration procedure ends.
 - ii. If it has not, the procedure goes to **h**.
- h. The gNodeB checks whether all candidate frequency combinations have been tried.
 - i. If they have, the configuration procedure ends.

- ii. If they have not, the next candidate frequency combination is tried.
- The smart carrier selection switch is off.
 - a. The gNodeB selects candidate frequencies, which are included in a CA frequency group, have not been allocated to the CA UE, and belong to the same operator as the UE. A CA frequency group is configured using the **gNBCaFrequency** MO. A frequency belongs to an operator if the operator is configured using the **gNBCaFrequency.OperatorIdList** parameter.
 - b. The gNodeB checks whether the blind SCell configuration conditions are met. If the conditions are met, the procedure goes to **c**. Otherwise, the procedure goes to **d**.

The blind SCell configuration conditions are as follows:

 - The **SCC_BLIND_CONFIG_SW** option of the **NRCellCaMgmtConfig.CaStrategySwitch** parameter is selected.
 - A blind-configurable candidate SCell (with **NRCellRelation.BlindScellConfigFlag** set to **TRUE**) operates on a candidate SCC frequency.
 - c. The gNodeB sends an RRCReconfiguration message to configure appropriate cells (cells with **NRCellRelation.BlindScellConfigFlag** set to **TRUE** on the candidate SCC frequencies and PLMN IDs overlapping those of the PCell) as SCells in a blind manner. (Cells mentioned in **c** all refer to blind-configurable candidate SCells.)

If multiple frequencies serve cells with **NRCellRelation.BlindScellConfigFlag** set to **TRUE**, the gNodeB selects a number of optimal cells from those cells as SCells according to **4.6.1.3 CA Capability Combination Selection Policies**. The number of cells selected is equal to the number of SCells to be configured. For example, a maximum of one cell is selected in the case of 2CC aggregation, and a maximum of two cells are selected in the case of 3CC aggregation.

If the blind SCell configuration succeeds, the procedure goes to **g**. If it fails, the procedure goes to **d** for measurement-based configuration.
 - d. The gNodeB delivers event A5 measurement configurations to the CA UE. The threshold for event A5 is the sum of the values of **NRCellCaMgmtConfig.CaA5RsrpThld2** and **gNBCaFrequency.CaSccA5RsrpThld2Offset**. When there are multiple candidate SCC frequencies, the gNodeB selects a number of candidate SCC frequencies in the way described in **4.6.1.3 CA Capability Combination Selection Policies** and delivers the event A5 measurement configurations related to the selected frequencies. The number of frequencies selected is equal to the number of SCells to be configured. For example, a maximum of one frequency is selected in the case of 2CC aggregation, and a maximum of two frequencies are selected in the case of 3CC aggregation.
 - e. The gNodeB checks whether it has received an A5 measurement report from the CA UE.
 - If the gNodeB has received an A5 measurement report before the A5 measurement waiting timer (specified by the **gNodeBALgo.MultiCarrierMeasWaitTimer** parameter) expires, the

procedure goes to [f](#). There are two cases. In the first case, the gNodeB has received A5 measurement reports related to all candidate frequencies before the timer expires. In this case, the gNodeB immediately proceeds to [f](#) and stops the timer. In the second case, the gNodeB has received A5 measurement reports related to (not necessarily all) candidate frequencies when the timer expires. In this case, the gNodeB proceeds to [f](#) upon timer expiry.

- If the gNodeB has not received A5 measurement reports on any candidate frequency before the A5 measurement waiting timer (specified by the **gNodeBAlgo.MultiCarrierMeasWaitTimer** parameter) expires, the procedure goes to [g](#).
- f. The gNodeB configures SCells for the CA UE.
The gNodeB selects available cells with the highest RSRP values and PLMN IDs overlapping those of the PCell as SCells and selects the optimal CA capability combination for the UE in the way described in [4.6.1.3 CA Capability Combination Selection Policies](#).
After an SCell is selected, the gNodeB checks whether this SCell can be configured according to [4.6.1.5 C-RNTI Conflict Detection and Conflict-triggered Reconfiguration](#).
- g. The gNodeB automatically checks whether the number of successfully configured SCells has reached the maximum.
 - i. If it has, the SCell configuration procedure ends.
 - ii. If it has not, the procedure goes to [h](#).
- h. The gNodeB checks whether all candidate frequency combinations have been tried.
 - i. If they have, the configuration procedure ends.
 - ii. If they have not, the next candidate frequency combination is tried.

If all the candidate SCC frequency combinations have been tried but the number of successfully configured SCells is still less than the maximum, the gNodeB periodically checks whether the uplink or downlink traffic volume of the CA UE meets the SCell activation conditions described in [4.6.3 SCell Activation](#).

- If the SCell activation conditions are met, another SCell configuration procedure (measurement-based SCell configuration only) will be triggered to configure SCells again. If the number of configured uplink SCells does not reach the maximum, the gNodeB delivers A1 measurement configurations to a cell that has been configured as an SCell in the downlink. In other scenarios, the gNodeB delivers A5 measurement configurations.
- If the SCell activation conditions are not met, another SCell configuration procedure will not be triggered.

This period is determined as follows:

- If the **SMART_CARR_SEL_SW** option of the **NRDUCellCollabServ.MultiCarrMgmtSw** parameter is deselected, the gNodeB takes 15 seconds as a period.
- If the **SMART_CARR_SEL_SW** option of the **NRDUCellCollabServ.MultiCarrMgmtSw** parameter is selected, the gNodeB takes the value of the **NRDUCellCarrMgmt.SccCfgInterval** parameter as a period. This parameter specifies the SCC configuration interval.

 NOTE

- In NSA networking, a low-frequency gNodeB cannot initiate CA combination capability renegotiation. Instead, the gNodeB only supports the CA combinations that are selected according to UE capabilities when the eNodeB adds or changes a secondary cell group (SCG).
- In low-frequency scenarios, if a measurement report indicates the PCI of an extended neighboring cell (for which the **NRCellRelation.NCellClassLabel** parameter is set to **EXTENDED**), the handling varies as follows:
 - If the **EXTD_NCELL_MULTICARR_COMPAT_SW** option of the **NRCellAlgoSwitch.AnrSwitch** parameter is selected, a CGI measurement on that cell is started. If CGI measurement reporting is successful, that cell is configured as an SCell. If CGI measurement reporting fails, that cell is not configured as an SCell this time.
 - If the **EXTD_NCELL_MULTICARR_COMPAT_SW** option of the **NRCellAlgoSwitch.AnrSwitch** parameter is deselected, no CGI measurement on that cell is started and that cell is not configured as an SCell.
- Flow control against multi-carrier operations at the gNodeB applies to the SCell configuration procedure. The flow control level is specified by the **gNodeBAlgo.MultiCarrierFlowCtrlLevel** parameter. Different flow control levels correspond to different CPU load thresholds. After the CPU usage reaches the corresponding threshold, SCells will no longer be configured for newly admitted UEs. For the network impacts of flow control against multi-carrier operations at the gNodeB, see [5.1.2.2 Impacts](#) and [6.1.2.2 Impacts](#).

SCell Configuration Procedure in High-Frequency Scenarios

In high-frequency scenarios, the gNodeB configures a cell that meets all of the following conditions as an SCell in a blind manner through an RRCReconfiguration message:

- The cell and the PCell are intra-gNodeB cells served by the same AAU.
- The downlink frequencies of the cell and the PCell belong to the same CA frequency group.
- The PLMN IDs of the cell and the PCell overlap.

Flexible carrier management can be used in high-frequency scenarios. This function is controlled by the **NRDUCellCarrMgmt.Fr2CaCarrMgmtPolicy** parameter.

- When this parameter is set to **USER_TRAFFIC_BASED**, dynamic SCell configuration is performed based on services.
- When this parameter is set to **USER_EXP_BASED**, the full SCell configuration policy is used. With this policy, the gNodeB performs an SCell configuration procedure at 15-second intervals if the number of successfully configured SCells has not reached the maximum.

For details about flexible carrier management, see [5.3 Flexible Carrier Management \(High-Frequency TDD\)](#).

 NOTE

In NSA networking, a high-frequency gNodeB can initiate CA combination capability renegotiation.

4.6.1.2 SCell Configuration Procedure Upon PCell Handovers

When triggered by a PCell handover, the SCell configuration procedure differs from the procedure described in [4.6.1.1 Common Procedure for SCell Configuration](#) in that:

1. The source cell sends a handover request that contains the measurement results of neighboring cells on the operating frequencies of the configured SCells.
2. The target cell sends the source cell a handover request acknowledgment that contains the information about new SCells.
3. The source cell sends a handover command to configure the new SCells for the UE and release the old SCells.

As indicated by the preceding procedure, SCells remain configured throughout the PCell handover. This prevents SCell configuration from being interrupted by the handover and ensures user experience of the CA UE during the handover. However, this procedure prolongs the handover delay, which therefore cannot meet the requirement in certain scenarios. To address this issue, SCell configuration has been optimized for PCell handovers. Optimized SCell configuration during PCell handovers involves the following functions:

- Prohibition of SCell configuration inheritance during incoming handovers: This function can be enabled by selecting the **HO_IN_SCC_CFG_INHERIT_POL_SW** option of the **NRCeCellCaMgmtConfig.CaStrategySwitch** parameter for a low-frequency PCell. With this function, the target cell of an incoming handover no longer configures SCells based on the information sent from the source cell. This function shortens the handover delay but results in a period of time in which SCell configuration is interrupted.
- Prohibition of blind SCell configuration during incoming handovers: This function can be enabled by selecting the **HO_IN_SCC_BLIND_CONFIG_POL_SW** option of the **NRCeCellCaMgmtConfig.CaStrategySwitch** parameter for a low-frequency PCell. With this function, the target cell configures SCells based on A5 measurements rather than in a blind manner.

NOTE

In inter-FR CA scenarios, prohibition of blind SCell configuration during incoming handovers takes effect only for low-frequency SCells, but not for high-frequency SCells.

Smart Carrier Selection Optimization

If the **SMART_CARR_SEL_SW** option of the **NRDUCellCollabServ.MultiCarrMgmtSw** parameter is selected, the following issues exist for SCell configuration during PCell handovers:

- In certain scenarios, SCell configuration is interrupted during handovers.
- The old and new SCells must operate on the same frequency. SCells cannot be changed to their inter-frequency cells during handovers, even if their inter-frequency cells deliver better quality.

To address these issues, smart carrier selection optimization is introduced. It is controlled by the **SMART_CARR_SEL_OPT_SW** option of the

NRCellCaMgmtConfig.CaStrategySwitch parameter. This function initiates an SCell selection after a PCell handover, in order to:

- Reduce the duration in which SCell configuration is interrupted.
- Increase the probability that SCells are changed to better-quality cells and the uplink and downlink throughputs of the CA UE rise after the handover, thanks to the new rule that the old and new SCells can be inter-frequency cells.

4.6.1.3 CA Capability Combination Selection Policies

Basic Functions of CA Capability Combination Selection Policies

In low-frequency scenarios, for better UE performance, the gNodeB selects the optimal combination of UE capabilities indicated by the reported *FeatureSetDownlinkPerCC* and *FeatureSetUplinkPerCC* IEs, including supported bandwidths, multiple-input multiple-output (MIMO) capabilities, and supported modulation orders. The policy for selecting optimal UE capability combinations is determined by the **NRCellCaMgmtConfig.CaBandCombSelectionStrat** parameter.

- If the **NRCellCaMgmtConfig.CaBandCombSelectionStrat** parameter is set to **BANDWIDTH_FIRST**, the gNodeB selects the optimal capability combination by evaluating the following multi-carrier factors in sequence: downlink bandwidth, theoretical downlink peak rate, uplink bandwidth, and theoretical uplink peak rate.
- If the **NRCellCaMgmtConfig.CaBandCombSelectionStrat** parameter is set to **PEAK_RATE_FIRST**, the gNodeB selects the optimal capability combination by evaluating the following multi-carrier factors in sequence: theoretical downlink peak rate, downlink bandwidth, theoretical uplink peak rate, and uplink bandwidth.
- If the **NRCellCaMgmtConfig.CaBandCombSelectionStrat** parameter is set to **UL_PEAK_RATE_FIRST**, the gNodeB selects the optimal capability combination by evaluating the following multi-carrier factors in sequence: theoretical uplink peak rate, uplink bandwidth, theoretical downlink peak rate, and downlink bandwidth.
- If the **NRCellCaMgmtConfig.CaBandCombSelectionStrat** parameter is set to **DEFAULT**, the gNodeB selects the optimal capability combination by evaluating the following multi-carrier factors in sequence: number of CCs to be aggregated in the uplink and theoretical uplink peak rate.

NOTE

When evaluating CA capability combinations, the gNodeB adheres to the following rule: If multiple CA capability combinations have the same priority, the gNodeB preferentially selects the CA frequency group with the smallest value of the **gNBCaFrequency.CaFreqGroupId** parameter. If multiple frequencies are available in a CA frequency group, the gNodeB preferentially selects the frequency with the smallest value of the **gNBCaFrequency.SsbFreqPos** parameter.

Optimized CA Capability Combination Selection Policies with SRS-based Weights Considered

In low-frequency scenarios, the gNodeB also automatically considers the gains from SRS-based weights if the **SRS_WT_BASED_CARR_SEL_SW** option of the

NRCellCaMgmtConfig.CaCompatibilitySwitch parameter is selected and the **NRCellCaMgmtConfig.CaBandCombSelectionStrat** parameter is set to **BANDWIDTH_FIRST**, **PEAK_RATE_FIRST**, or **UL_PEAK_RATE_FIRST**.

Multi-Carrier Combination Selection Based on Multiple Bandwidth Levels

When a cell is configured with minimum carrier bandwidths (**NRDUCellBwp.DlMinCarrierBw** and **NRDUCellBwp.UlMinCarrierBw**), UEs in the cell can operate on these bandwidths in addition to the bandwidth of the cell. The CA capabilities for UEs vary with the bandwidth.

Prior to introducing multi-carrier combination selection based on multiple bandwidth levels, the gNodeB considers only the cell bandwidth when following a CA capability combination selection policy to select an optimal multi-carrier capability combination. This approach may fail to identify the optimal combination because some UEs exhibit better CA capabilities (for example, more layers) on the minimum carrier bandwidths than on the cell bandwidth. To address this, multi-carrier combination selection based on multiple bandwidth levels is introduced in low frequency bands. This function is controlled by the **MULTI_CARR_MULTI_BWP_OPT_SW** option of the **NRCellCaMgmtConfig.CaStrategySwitch** parameter. If this option is selected, this function is enabled and the gNodeB considers both the UE-supported bandwidths of the PCell and SCells as well as minimum carrier bandwidths to select an optimal CA capability combination.

- PCell: If the bandwidth of the PCell in the selected optimal combination differs from the bandwidth in use for a UE, the gNodeB instructs the UE to use the optimal bandwidth. If they are the same, no change is made.
- SCell: During SCell configuration, the gNodeB instructs the UE to use the bandwidth of the SCell in the selected optimal combination.

For details about the minimum carrier bandwidths, see *Scalable Bandwidth*.

4.6.1.4 Service-based Optimal Carrier Selection Policies

In low-frequency scenarios, service-based optimal carrier selection is available. This function allows carrier selection policies to be changed flexibly based on UE traffic requirements given a specified CA capability combination selection policy. UEs therefore can achieve user-perceived rates that most fully meet their requirements.

Service-based optimal carrier selection is controlled by the **SERVICE_AWARE_CARR_SEL_SW** option of the **NRDUCellCollabServ.MultiCarrMgmtSw** parameter. A procedure of this function is triggered once upon a change in the traffic volume model of a UE, which is listed in [Table 4-5](#). [Table 4-5](#) describes how this function works when different CA capability combination policies are configured. In this table, **CA Capability Combination Selection Policy** lists policies specified by **NRCellCaMgmtConfig.CaBandCombSelectionStrat**, and **Actual CA Capability Combination Selection Policy** lists policies selected using this function based on both traffic and a specified policy.

Table 4-5 CA capability combination selection policies after service-based optimal carrier selection is enabled

Traffic Volume Model	CA Capability Combination Selection Policy	Actual CA Capability Combination Selection Policy
Uplink: small Downlink: large	Preference based on the downlink bandwidth	Preference based on the downlink bandwidth
	Preference based on the downlink peak rate	Preference based on the downlink peak rate
	Preference based on the uplink peak rate	Preference based on the downlink peak rate
Uplink: small Downlink: small	Preference based on the downlink bandwidth	Preference based on the downlink bandwidth
	Preference based on the downlink peak rate	Preference based on the downlink peak rate
	Preference based on the uplink peak rate	Preference based on the uplink peak rate
Uplink: large Downlink: small	Preference based on the downlink bandwidth	Preference based on the uplink peak rate
	Preference based on the downlink peak rate	Preference based on the uplink peak rate
	Preference based on the uplink peak rate	Preference based on the uplink peak rate
Uplink: large Downlink: large	Preference based on the downlink bandwidth	Preference based on the downlink bandwidth
	Preference based on the downlink peak rate	Preference based on the downlink peak rate
	Preference based on the uplink peak rate	Preference based on the uplink peak rate

Service-based optimal carrier selection has the following limitations: When the traffic volume model changes for a UE capable of 3CC or massive CA, preference based on the uplink peak rate cannot take effect in the CA capability combination selection policy in scenarios where the optimal uplink and downlink CA combinations are not in the same combination. As a result, uplink CA performance cannot be ensured. In this case, uplink enhancement for smart carrier selection is introduced. This function is controlled by the **SMART_CARR_SEL_UL_ENH_SW** option of the **NRDUCellCollabServ.MultiCarrMgmtSw** parameter. After this enhancement function is enabled and the traffic volume model changes for a UE capable of 3CC or massive CA, preference based on the uplink peak rate can take effect in the CA capability combination selection policy in scenarios where the optimal uplink and downlink CA combinations are not in the same combination. As a result, uplink CA performance can be ensured.

4.6.1.5 C-RNTI Conflict Detection and Conflict-triggered Reconfiguration

A cell radio network temporary identifier (C-RNTI) is a temporary ID of a UE in connected mode in a cell. The gNodeB uses different C-RNTIs for the UEs in uplink and downlink scheduling. The C-RNTIs of UEs in a cell must be different to avoid inter-UE interference.

C-RNTI Conflict Detection

Without C-RNTI conflict detection, the C-RNTI of a UE in an SCell to be configured may have been used by another UE in this SCell. In this case, a C-RNTI conflict occurs, which intensifies inter-UE interference and affects UE performance. As such, the C-RNTI conflict detection function is introduced. This function is controlled by the **INTRA_GNB_CRNTI_CON_DETECT_SW** option of the **gNodeBCarrMgmt.CaEnhancedAlgoSwitch** parameter. Note that the preceding parameter-controlled C-RNTI conflict detection function is an intra-gNodeB function, because inter-gNodeB C-RNTI conflict detection takes effect by default and does not need to be configured.

After C-RNTI conflict detection takes effect, the gNodeB checks whether the C-RNTI of a UE is unique within an SCell to be configured.

- If so, the SCell is configured.
- If not, the gNodeB does not configure the SCell.

This function takes effect only in measurement-based SCell configuration, but not blind SCell configuration.

C-RNTI Conflict-triggered Reconfiguration

When intra- or inter-gNodeB C-RNTI conflict detection takes effect and the gNodeB determines that the C-RNTI of a UE in an SCell to be configured has been used by another UE within the SCell, the C-RNTI conflict-triggered reconfiguration function is introduced to ensure normal SCell configuration for the UE and guarantee UE performance. This function is controlled by the **CRNTI_CONFLICT_RECONFIG_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoSwitch** parameter. With this function, the gNodeB re-allocates C-RNTIs to UEs for which C-RNTI conflicts are detected, and attempts to reconfigure the SCell. This reduces the risk of SCell configuration failures caused by C-RNTI conflicts.

The preceding option controls both intra-gNodeB and inter-gNodeB C-RNTI conflict-triggered reconfiguration. In intra-gNodeB scenarios, C-RNTI conflict-triggered reconfiguration takes effect only when C-RNTI conflict detection is enabled.

This function takes effect only in measurement-based SCell configuration, but not blind SCell configuration.

 NOTE

Consider an inter-FR CA scenario where (1) two low-frequency 2CCs and at least one high-frequency SCC have been configured; (2) During low-frequency SCC change, a C-RNTI conflict is detected in the SCC to be configured. In this scenario, the gNodeB will delete all configured high-frequency SCCs during C-RNTI reconfiguration. The gNodeB will subsequently reconfigure high-frequency SCCs based on the high-frequency SCC configuration procedure.

4.6.1.6 UE-Number-based SCell Configuration Control

CA UEs count towards the RRC_CONNECTED UE number limits for both the PCell and SCells. In scenarios where these limits are likely to be reached (for example, during major sports events), non-CA UEs may struggle to connect to the network. As such, UE-number-based SCell configuration control is introduced in low frequency bands. This function is controlled by the **UE_NUM_BASED_SCELL_CONFIG_SW** option of the **gNodeBCarrMgmt.CaEnhancedAlgoSwitch** parameter.

This function includes two subfunctions: UE-number-based 2CC aggregation prohibition and UE-number-based 3CC aggregation prohibition. After the **UE_NUM_BASED_SCELL_CONFIG_SW** option of the **gNodeBCarrMgmt.CaEnhancedAlgoSwitch** parameter is selected, the related UE proportion thresholds must be configured for the subfunctions to take effect. UE-number-based SCell configuration control works when either of the subfunctions takes effect.

- UE-number-based 2CC aggregation prohibition: takes effect when the **NRCaMgmtConfig.Forbid2ccUeRatioThld** parameter is set to a value other than **100**. After this function takes effect in a cell, the gNodeB periodically calculates the proportion of UEs in the cell. If this proportion is greater than the value of this parameter, CA with two or more CCs aggregated is disallowed for UEs that treat this cell as their PCell. Also, this cell is prohibited from being configured as an SCell for UEs in other cells.
- UE-number-based 3CC aggregation prohibition: takes effect when the **NRCaMgmtConfig.Forbid3ccUeRatioThld** parameter is set to a value other than **100**. After this function takes effect in a cell, the gNodeB periodically calculates the proportion of UEs in the cell. If this proportion is greater than the value of this parameter, CA with three or more CCs aggregated is disallowed for UEs that treat this cell as their PCell.

Note the following: Proportion of UEs in a cell = (Number of UEs in a cell / Maximum number of RRC_CONNECTED UEs in a cell) x 100%, where the number of UEs in a cell includes the number of RRC_CONNECTED UEs in a cell and the number of UEs that treat the local cell as their SCell (including SUL UEs). For details about the maximum number of RRC_CONNECTED UEs in a cell, see the UE number specifications of the corresponding RAT in "NR Technical Specifications of Baseband Processing Units" in "BBU Technical Specifications" in *3900 & 5900 Series Base Station Product Documentation*.

NOTE

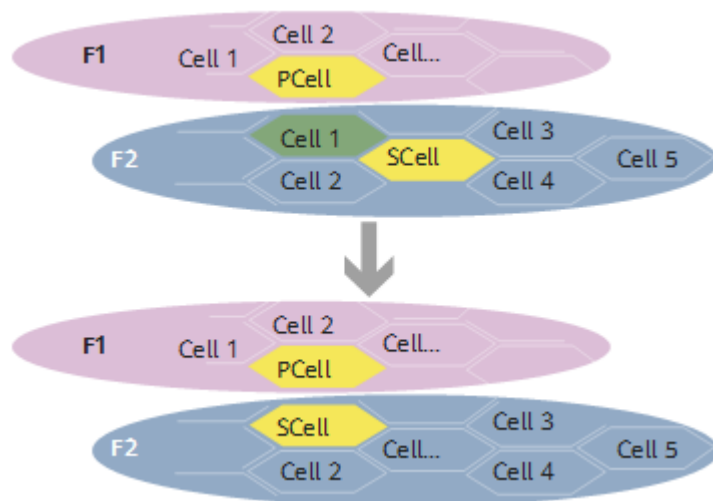
When the value of **NRCaMgmtConfig.Forbid2ccUeRatioThld** or **NRCaMgmtConfig.Forbid3ccUeRatioThld** is changed (for example, from **100** to a value other than **100** to enable UE-number-based SCell configuration control), the latest setting does not take effect immediately for CA UEs in the RRC_CONNECTED state. Instead, the latest setting takes effect only when an SCell addition or change procedure is triggered.

4.6.2 SCell Change

If a CA UE with SCells configured receives better signal quality from an intra-frequency neighboring cell of an SCell than from that SCell, the serving gNodeB of the PCell can change the SCell while keeping the PCell unchanged.

Figure 4-3 illustrates this function.

Figure 4-3 SCell change



1. The gNodeB sends an RRCReconfiguration message to a CA UE after configuring SCells for the UE. The message contains event A6 measurement configurations, in which the offset for event A6 is specified by the **NRCaMgmtConfig.CaA6Offset** parameter configured on the PCell side.
2. After receiving A6 measurement reports from the CA UE, the gNodeB tries the reported candidate SCells in descending order of RSRP for an SCell change. If the change can be performed, the gNodeB sends an RRCReconfiguration message to change the SCell to the new cell for the CA UE.

NOTE

SCell change is supported only in low frequency bands, but not in high frequency bands. As adding a SCell during SCell change is considered SCell configuration, the SCell change procedure will also trigger C-RNTI conflict detection and conflict-triggered reconfiguration. For details about the functions, see [4.6.1.5 C-RNTI Conflict Detection and Conflict-triggered Reconfiguration](#).

4.6.3 SCell Activation

The gNodeB sends either a MAC control element (CE) or an RRC reconfiguration message to activate SCells, as described in [SCell Activation Through a MAC CE](#) or [SCell Activation Through an RRC Reconfiguration Message](#) respectively.

For low-frequency NR TDD SCells, precise SCell activation can be used to quickly distribute data to the SCells, improving user experience of CA UEs. For details, see [Precise SCell Activation](#).

After sending the MAC CE or RRC reconfiguration message, the gNodeB checks whether the SCells have been successfully activated. For details, see [SCell Activation Check](#).

The functions described in [SCell Activation Success Rate Improvement](#) can be used to increase the SCell activation success rate.

SCell Activation Through a MAC CE

For traffic volume to trigger SCell activation, the gNodeB first evaluates whether the UE meets the uplink or downlink conditions for SCell activation listed in [Table 4-6](#).

Table 4-6 Conditions for traffic volume–triggered SCell activation

Traffic Volume Type	Condition
Uplink	The gNodeB determines that the uplink data size indicated in the buffer status report (BSR) is greater than or equal to the uplink threshold for SCell activation. The threshold is determined by the current PCell air interface capability and the NRDUCellCarrMgmt.CaUIScellActThldSlotNum parameter value. Given the same air interface capability of the PCell, a smaller value of the NRDUCellCarrMgmt.CaUIScellActThldSlotNum parameter results in a higher probability of activating SCells, and a larger value results in a lower probability.

Traffic Volume Type	Condition
Downlink	(High frequency bands) The gNodeB determines that downlink data volume buffered at the RLC layer for the UE is greater than or equal to the downlink threshold for SCell activation. The threshold is equal to $\min\{y, \text{NRDUCellCarrMgmt.DlStateTransitBufVolThld}\}$, where y is calculated based on the current PCell air interface capability and the NRDUCellCarrMgmt.CaDlScellActThldSlotNum parameter value. The smaller the values of the NRDUCellCarrMgmt.CaDlScellActThldSlotNum and NRDUCellCarrMgmt.DlStateTransitBufVolThld parameters, the higher the probability of activating SCells. The larger the values, the lower the probability. (Low frequency bands) The gNodeB determines that downlink data volume buffered at the RLC layer for the UE is greater than or equal to the downlink threshold for SCell activation. The threshold is equal to $\min\{y, \text{Buffered traffic volume threshold for downlink state transitions}\}$. In this expression: y is calculated based on the current PCell air interface capability and the NRDUCellCarrMgmt.CaDlScellActThldSlotNum parameter value. The buffered traffic volume threshold for downlink state transitions is specified by the NRDUCellCarrMgmt.DlStateTransitBufVolThld parameter or the NRDUCellQciBearer.QciDlStTransitBufVolThld parameter. The former is a cell-level threshold, and the latter is a QCI-level one. The NRDUCellQciBearer.QciDlStTransitBufVolThld parameter is valid only if it is set to a value other than 65535 and the QCI_CA_BWP_SWITCH_THLD_CTRL_SW option of the NRDUCellQciBearer.ServiceDiffAlgoSwitch parameter is selected. If the NRDUCellQciBearer.QciDlStTransitBufVolThld parameter setting is valid for the QCI of the QoS flows to which services of a UE are mapped, the QCI-level threshold takes effect; otherwise, the cell-level threshold takes effect. If multiple QCIs are involved in this case, the setting specific to the QCI with the highest priority (specified by the gNBQciBearer.PriorityLevel parameter) applies. The smaller the values of the NRDUCellCarrMgmt.CaDlScellActThldSlotNum, NRDUCellCarrMgmt.DlStateTransitBufVolThld, and NRDUCellQciBearer.QciDlStTransitBufVolThld parameters, the higher the probability of activating SCells. The larger the values, the lower the probability.

Traffic Volume Type	Condition
Note: The uplink data size indicated in the BSR and the downlink data volume buffered at the RLC layer are filtered using the coefficient specified by the NRDUCellCarrMgmt.RlcBufferFilterCoeff parameter.	

When the preceding activation conditions are met, the gNodeB performs the following operations at every transmission time interval (TTI):

- If the uplink condition for SCell activation is met, the gNodeB sends a MAC CE to activate SCells configured for the uplink, but not those configured only for the downlink.
- If the downlink condition for SCell activation is met, the gNodeB sends a MAC CE to activate all SCells.

For details about SCell activation, see section 6.1.3.10 "SCell Activation/Deactivation MAC CEs" in 3GPP TS 38.321 V15.6.0.

NOTE

In low-frequency NSA networking, only data that can be transmitted within 10 ms is distributed to NR when **gNBpdcpparamGroup.DlDataPdcpsplitMode** (which specifies the downlink data split mode at the PDCP layer) is set to **SCG_AND_MCG**, **MCG_PREFERRED** or **ADAPT_PRIMARY_PATH**. That is, the downlink data volume buffered at the RLC layer equals at most the amount allowed to be transmitted within 10 ms. The CA SCell activation rate in NSA networking may be lower than that in SA networking. For the CA SCell activation rate in NSA networking to reach that in SA networking:

- If the PCell is an NR TDD cell, it is recommended that the **NRDUCellCarrMgmt.CaDlScellActThldSlotNum** parameter be set to a value less than or equal to 13 and the **NRDUCellCarrMgmt.DlStateTransitBufVolThld** and **NRDUCellQciBearer.QciDlStTransitBufVolThld** parameters be set to values less than 10 KB.

When **NRDUCellCarrMgmt.CaUlScellActThldSlotNum** or **NRDUCellCarrMgmt.CaDlScellActThldSlotNum** is set to 0, SCells can be activated immediately after UE access regardless of whether the conditions for traffic volume-triggered SCell activation are met. In addition, the SCells will not be deactivated based on the traffic volume. This way, CA is likely to take effect. For details about the network impacts, see [5.1.2.2 Impacts](#), [5.2.2.2 Impacts](#), [6.1.2.2 Impacts](#), and [6.2.2.2 Impacts](#).

SCell Activation Through an RRC Reconfiguration Message

3GPP Release 16 introduces the function of SCell activation through an RRC reconfiguration message. When **NRDUCellCarrMgmt.CaUlScellActThldSlotNum** or **NRDUCellCarrMgmt.CaDlScellActThldSlotNum** is set to 0, no MAC CE is required for SCell activation for UEs that have the capabilities indicated by the *directMCG-ScellActivation-r16* or *directSCG-ScellActivation-r16* IE. Instead, the gNodeB activates SCells by including the *sCellState-r16* IE in an RRC reconfiguration message for SCell configuration. If both **NRDUCellCarrMgmt.CaUlScellActThldSlotNum** and **NRDUCellCarrMgmt.CaDlScellActThldSlotNum** are set to non-zero values, the gNodeB sends a MAC CE to activate SCells.

Precise SCell Activation

After receiving a MAC CE or an RRC reconfiguration message for SCell activation, the UE responds with HARQ information. It also measures periodic CSI-RSs of the SCells and reports the measurement results. Data distribution to an SCell cannot be started until the SCell receives the CSI-RS measurement results.

If the SCell is a low-frequency NR TDD cell, precise SCell activation can be used to quickly distribute data to the SCell, improving user experience of the UE. With this function, the gNodeB proactively checks whether the SCell is available, after receiving HARQ information from the UE. If the SCell is available, the gNodeB directly starts CA data split.

Precise SCell activation is enabled by selecting the **PRECISE_SCELL_ACTIVATION_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch** parameter for both the PCell and SCells. If this option is selected for the PCell and some of the TDD SCells, precise SCell activation takes effect only for the SCells with this option selected.

For the impact of precise SCell activation on the network, see [5.1.2 Network Analysis](#) and [6.1.2 Network Analysis](#).

Precise SCell activation does not take effect when multiple timing advances (MTA) is used for uplink timing. For details about MTA, see [5.2.1.2 CA Uplink Timing](#) and [6.2.1.2 CA Uplink Timing](#).

Precise SCell activation does not take effect when the uplink CCE allocation failure rate is greater than 30%. This failure rate is equal to $(1 - (\text{N.CCE.UL.AggLvl16Num} + \text{N.CCE.UL.AggLvl1Num} + \text{N.CCE.UL.AggLvl2Num} + \text{N.CCE.UL.AggLvl4Num} + \text{N.CCE.UL.AggLvl8Num}) / \text{N.CCE.UL.AllocReq.Num}) \times 100\%$.

SCell Activation Check

After sending a MAC CE or RRC reconfiguration message for SCell activation, the gNodeB checks whether the SCells have been successfully activated. An SCell is considered to be successfully activated if one of the following conditions is met:

- The SCell receives valid CQI reports.
- The SCell receives valid CSI-RS beam sweeping reports. This condition applies when the early data split function is enabled (by selecting the **EARLY_SPLIT_SW** option of the **NRDUCellCarrMgmt.CaCompatibilitySwitch** parameter).
- The SCell is available according to the result of a proactive check by the gNodeB. This condition applies when the SCell is a low-frequency NR TDD cell and precise SCell activation is enabled.

For details about the network impacts of selecting the **EARLY_SPLIT_SW** option of the **NRDUCellCarrMgmt.CaCompatibilitySwitch** parameter, see [5.1.2.2 Impacts](#) and [6.1.2.2 Impacts](#).

SCell Activation Success Rate Improvement

The SCell activation success rate is subject to multiple factors. The following two functions improve the SCell activation success rate from the perspectives of DRX status consistency and coverage.

- SCell activation and deactivation suspension during DRX wakeup protection: When DRX is in effect, there may be false scheduling request (SR) detection. That is, the gNodeB determines that a UE is in the DRX active time, but the UE is actually in the DRX sleep time. When false SR detection occurs in the PCell, the SCell activation success rate is affected. As such, the "SCell activation and deactivation suspension during DRX wakeup protection" function is introduced. With this function in effect, the gNodeB suspends SCell activation and deactivation for a UE during the DRX wakeup protection period (a period where false SR detection may occur); and resumes SCell activation and deactivation when this period elapses. This function is controlled by the **RSVDSWPARAM3_BIT32** option of the **NRDUCellRsvdExt01.RsvdSwParam3** parameter. For details about DRX, see *DRX*.
- SCell activation prohibition under weak coverage: The SCell activation success rate is low for a UE under weak coverage in the PCell. As such, the SCell activation prohibition under weak coverage function is introduced. This function is controlled by **NRDUCellRsvdExt01.RsvdParam15**.
 - When this parameter ranges from **1** to **126**, this function takes effect. If the SRS RSRP of a UE in the PCell is less than or equal to the parameter value, the gNodeB considers the UE to be under weak coverage and does not activate any SCell for the UE.
 - If this parameter is set to **0** or a value greater than **126**, this function does not take effect.

The actual value of **NRDUCellRsvdExt01.RsvdParam15** is the configured value minus 157. For example, if this parameter is set to **1**, the actual value is -156 dBm.

4.6.4 SCell Deactivation

SCell deactivation by the gNodeB can be triggered by any of the following factors:

- Traffic volume of a CA UE. For details, see [Traffic Volume-triggered SCell Deactivation](#).
- Channel quality in an SCell. For details, see [Channel Quality-triggered SCell Deactivation](#).
- Consecutive residual block errors in an SCell. For details, see [Consecutive Residual Block Error-triggered SCell Deactivation](#).
- Volume of data distributed from the PCell to an SCell. For details, see [Distributed Data Volume-triggered SCell Deactivation](#).

The gNodeB also supports delayed deactivation and deactivation protection. For details about these functions, see [Delayed SCell Deactivation](#) and [SCell Deactivation Protection](#), respectively.

The functions described in [SCell Deactivation Success Rate Improvement](#) can be used to increase the SCell deactivation success rate.

NOTE

For details about SCell deactivation, see section 6.1.3.10 "SCell Activation/Deactivation MAC CEs" in 3GPP TS 38.321 V15.6.0.

Traffic Volume–triggered SCell Deactivation

For traffic volume to trigger SCell deactivation, the gNodeB first evaluates whether the UE meets the uplink or downlink conditions for SCell deactivation listed in [Table 4-7](#).

Table 4-7 Conditions for traffic volume–triggered SCell deactivation

Uplink/Downlink	Condition
Uplink	Both of the following conditions are met: • The gNodeB determines that the uplink data size indicated in the BSR is less than the uplink data volume threshold for SCell deactivation. The threshold is determined by the current PCell air interface capability and the NRDUCellCarrMgmt.CaUIScellDeactThldSlotNum parameter value. • The uplink UE throughput is less than min{x, uplink RLC throughput threshold}. x indicates the uplink throughput of the UE in the PCell calculated based on the current radio channel quality, and the uplink RLC throughput threshold is specified by the NRDUCellCarrMgmt.RlcULThroughputThld parameter. NOTE When NRDUCellCarrMgmt.CaUIScellDeactThldSlotNum or NRDUCellCarrMgmt.RlcULThroughputThld is set to 0, traffic volume–triggered SCell deactivation does not take effect.

Uplink/Downlink	Condition
Downlink	(High frequency bands) Both of the following conditions are met: • The gNodeB determines that the downlink data volume buffered at the RLC layer for the UE is less than the downlink data volume threshold for SCell deactivation. The threshold is equal to $\min\{y, \text{NRDUCellCarrMgmt.DlStateTransitBufVolThld} \times 0.5\}$, where y is calculated based on the current PCell air interface capability and the NRDUCellCarrMgmt.CaDlScellDeactThldSlotNum parameter value. • The downlink UE throughput is less than $\min\{z, \text{Downlink RLC throughput threshold}\}$. z indicates the downlink throughput of the UE in the PCell calculated based on the current radio channel quality, and the downlink RLC throughput threshold is specified by the NRDUCellCarrMgmt.RlcDlThroughputThld parameter. NOTE When NRDUCellCarrMgmt.CaDlScellDeactThldSlotNum or NRDUCellCarrMgmt.RlcDlThroughputThld is set to 0, traffic volume-triggered SCell deactivation does not take effect. (Low frequency bands) Both of the following conditions are met: • The gNodeB determines that the downlink data volume buffered at the RLC layer for the UE is less than the downlink data volume threshold for SCell deactivation. The threshold is equal to $\min\{y, \text{Buffered traffic volume threshold for downlink state transitions} \times 0.5\}$. In this expression: y is calculated based on the current PCell air interface capability and the NRDUCellCarrMgmt.CaDlScellDeactThldSlotNum parameter value. The buffered traffic volume threshold for downlink state transitions is specified by the NRDUCellCarrMgmt.DlStateTransitBufVolThld parameter or the NRDUCellQciBearer.QciDlStTransitBufVolThld parameter. The former is a cell-level threshold, and the latter is a QCI-level one. The NRDUCellQciBearer.QciDlStTransitBufVolThld parameter is valid only if it is set to a value other than 65535 and the QCI_CA_BWP_SWITCH_THLD_CTRL_SW option of the NRDUCellQciBearer.ServiceDiffAlgoSwitch parameter is selected. If the NRDUCellQciBearer-

Uplink/Downlink	Condition
	er.QciDlStTransitBufVolThld parameter setting is valid for the QCI of the QoS flows to which services of a UE are mapped, the QCI-level threshold takes effect; otherwise, the cell-level threshold takes effect. If multiple QCIs are involved in this case, the setting specific to the QCI with the highest priority (specified by the gNBQciBearer.PriorityLevel parameter) applies. The downlink UE throughput is less than $\min\{z, \text{Downlink RLC throughput threshold}\}$. z indicates the downlink throughput of the UE in the PCell calculated based on the current radio channel quality, and the downlink RLC throughput threshold is specified by the NRDUCellCarrMgmt.RlcDlThroughputThld parameter. NOTE When NRDUCellCarrMgmt.CaDlScellDeactThldSlotNum or NRDUCellCarrMgmt.RlcDlThroughputThld is set to 0, traffic volume-triggered SCell deactivation does not take effect.
Note: The downlink data volume buffered at the RLC layer and the uplink data size indicated in the BSR are filtered using the coefficient specified by the NRDUCellCarrMgmt.RlcBufferFilterCoeff parameter.	

When the preceding deactivation conditions are met, the gNodeB performs the following operations every second:

- If both the uplink and downlink conditions for SCell deactivation are met, the gNodeB delivers a MAC CE to deactivate all the SCells of the CA UE.
- If only the downlink conditions for SCell deactivation are met, the gNodeB delivers a MAC CE to deactivate all SCells of the CA UE that are configured only for the downlink.
- If only the uplink conditions for SCell deactivation are met, the gNodeB does not deactivate any SCells.

Channel Quality-triggered SCell Deactivation

When the spectral efficiency corresponding to the CQI of an SCell reported by the CA UE is lower than the spectral efficiency corresponding to **NRDUCellCarrMgmt.SccDeactvCqiThld** configured for the SCell, the gNodeB delivers a MAC CE to deactivate the SCell. If the **NRDUCellCarrMgmt.SccDeactvCqiThld** parameter is set to **0** for an SCell, channel quality-triggered SCell deactivation does not take effect for the SCell.

It is recommended that channel quality-triggered SCell deactivation be enabled to prevent the gNodeB from scheduling data in SCells with poor channel quality.

An SCell with low spectral efficiency (namely poor channel quality) as indicated by its CQI, can still deliver certain throughput for a CA UE performing downlink large-

packet services. In this case, if the SCell is deactivated based on channel quality, the throughput of the CA UE will be affected. As such, non-CQI-based SCell deactivation for CA UEs performing downlink large-packet services has been introduced. This function is controlled by the **NOT_DEACT_SCC_BASED_CQI_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoSwitch** parameter. If this option is selected, channel-quality-triggered SCell deactivation does not take effect for CA UEs performing downlink large-packet services, regardless of whether **NRDUCellCarrMgmt.SccDeactvCqiThld** is set to **0**. Non-CQI-based SCell deactivation for CA UEs performing downlink large-packet services is supported only in low frequency bands.

Consecutive Residual Block Error-triggered SCell Deactivation

(Low-frequency scenarios) If the **BIT_ERROR_SCC_DEACTV_SW** option of the **NRDUCellCarrMgmt.CaCompatibilitySwitch** parameter is selected, consecutive residual block error-triggered SCell deactivation is enabled. When the gNodeB detects consecutive downlink residual block errors in an SCell of a CA UE, the gNodeB sends a MAC CE to deactivate the SCell.

(High-frequency scenarios) If the gNodeB detects consecutive uplink or downlink residual block errors in an SCell of a CA UE, the gNodeB delivers a MAC CE to deactivate the SCell.

Distributed Data Volume-triggered SCell Deactivation

(Low-frequency scenarios) If the **SPLIT_VOL_BASED_SCELL_DEACT_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch** parameter is selected, distributed data volume-triggered SCell deactivation is enabled. When the gNodeB detects that the volume of data distributed from the PCell to an SCell keeps 0 for 30 consecutive seconds, the gNodeB delivers a MAC CE to deactivate the SCell.

Delayed SCell Deactivation

Delayed SCell deactivation is controlled by the SCell deactivation waiting timer, which is specified by the **NRDUCellCarrMgmt.ScellDeactvWaitTimer** parameter.

- If this parameter is set to **0**, delayed SCell deactivation does not take effect. The gNodeB initiates SCell deactivation for a UE immediately when it detects that the UE meets the deactivation conditions.
- If this parameter is set to a value other than **0**, delayed SCell deactivation takes effect. The gNodeB initiates SCell deactivation when it detects that a UE continuously meets the deactivation conditions throughout a period longer than or equal to the value of this parameter.

SCell Deactivation Protection

There is a possibility that the gNodeB sends a MAC CE to deactivate an SCell, but the UE fails to receive the MAC CE and therefore does not have the SCell deactivated. In such a case, however, the gNodeB considers that the SCell has been deactivated. Such inconsistency of the SCell deactivation status between the gNodeB and the UE affects the reliability of the CSI reported by the UE, which in turn affects the data rate of the UE. To address this issue, SCell deactivation protection is introduced. This function is controlled by the

SCell_DEACT_PROT_SW option of the **NRDUCellCarrMgmt.CaCompatibilitySwitch** parameter. It works only in low frequency bands.

With this function, if the gNodeB detects that the CSI is continuously unreliable after delivery of a MAC CE for deactivation, it sends a MAC CE to deactivate the SCell again.

SCell Deactivation Success Rate Improvement

The SCell deactivation success rate can be increased through the SCell activation and deactivation suspension during DRX wakeup protection function. For details about this function, see [SCell Activation Success Rate Improvement](#).

4.6.5 SCell Removal

SCells can be removed dynamically or as triggered by event A2 or SCell unavailability.

NOTE

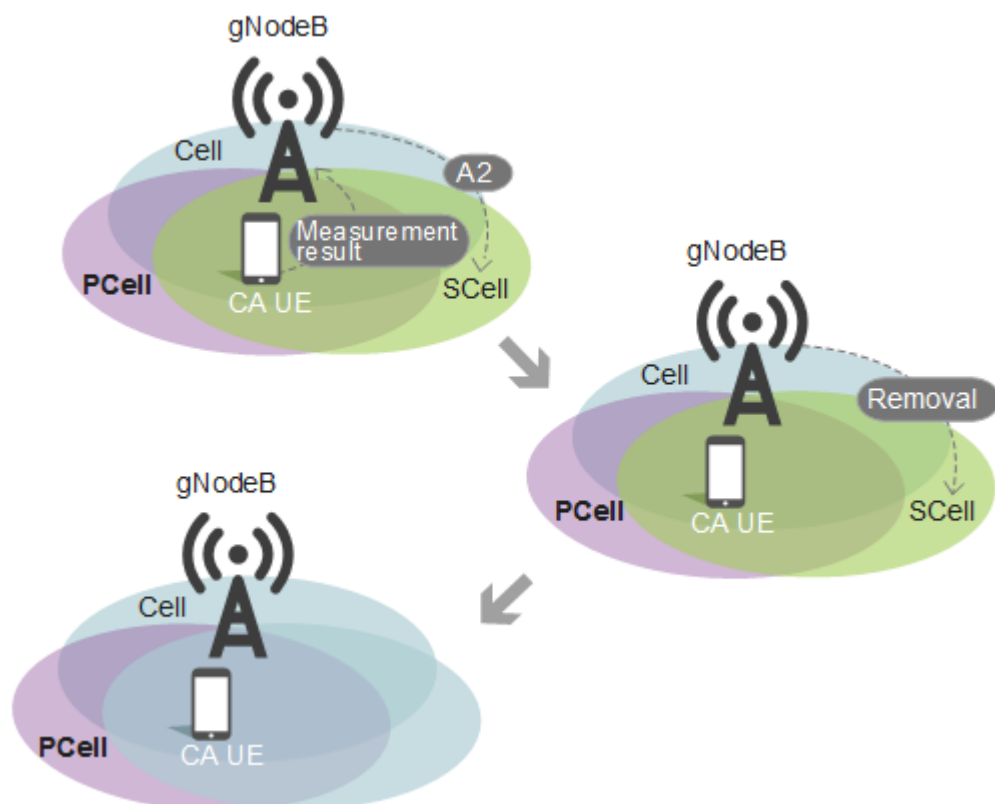
SCell removal triggered by event A2 is supported only in low frequency bands, not in high frequency bands.

Dynamic SCell removal is supported only in high frequency bands, not in low frequency bands.

SCell Removal Triggered by Event A2

If a CA UE reports event A2 for an SCell, the gNodeB removes the SCell as shown in [Figure 4-4](#).

Figure 4-4 SCell removal



1. The gNodeB delivers to a CA UE the event A2 measurement configuration related to its SCells that have been configured.
2. If the RSRP of an SCell is lower than the threshold for uplink or downlink CA event A2, the CA UE sends an uplink or downlink CA event A2 measurement report that contains the SCell, respectively.
3. The gNodeB sends the CA UE an RRCReconfiguration message to remove the SCell. If the report is an uplink event report, the gNodeB will further configure the cell as an SCell again but only for the downlink.

The threshold for downlink CA event A2 equals the sum of **NRCellCaMgmtConfig.CaA2RsrpThld** and **gNBCaFrequency.CaA2RsrpThldOffset**. The threshold for uplink CA event A2 equals the sum of **NRCellCaMgmtConfig.UlCaA2RsrpThld** and **gNBCaFrequency.CaA2RsrpThldOffset**.

It is recommended that the threshold for uplink CA event A2 be greater than that for downlink CA event A2. If this condition is not met and the reported RSRP of an SCell is greater than the threshold for uplink CA event A2 and less than that for downlink CA event A2, the gNodeB removes the SCell in both the uplink and downlink.

It is recommended that the thresholds for CA event A2 be less than the threshold for CA event A5. If this condition is not met and the reported RSRP of an SCell is greater than the threshold for CA event A5 and less than the thresholds for CA event A2, the CA UE reports event A2 immediately after the gNodeB configures the SCell based on event A5 measurements, causing the SCell to be removed. Once this happens and if the traffic volume of the CA UE meets the SCell

activation conditions for a long period of time, the gNodeB will periodically initiate SCell configuration and removal for the CA UE.

SCell Removal Triggered by SCell Unavailability

If an SCell becomes unavailable due to faults, worsened inter-gNodeB transmission, or other reasons, the gNodeB sends the CA UE an RRCReconfiguration message to remove the SCell.

Dynamic SCell Removal

In high-frequency scenarios, flexible carrier management can be used when the **NRDUCellCarrMgmt.Fr2CaCarrMgmtPolicy** parameter is set to **USER_TRAFFIC_BASED**. With this setting, SCells can be removed dynamically.

For details about flexible carrier management, see [5.3 Flexible Carrier Management \(High-Frequency TDD\)](#).

4.7 CA Data Split

CA data split allows a CA UE to be simultaneously scheduled in its PCell and SCells. Data split involves an interaction between the PCell and SCells, with this interaction inevitably causing delays. For services with stringent requirements on delay, such as video services, data split can be replaced with scheduling only in the PCell.

In high-frequency scenarios, after SCells are activated for a UE, the serving gNodeB of the PCell distributes part of the data buffered at the RLC layer to the SCells.

In low-frequency scenarios, CA data split is available only in downlink intra-gNodeB CA, uplink intra-gNodeB CA, and downlink inter-gNodeB CA. For details, see [Downlink Intra-gNodeB CA](#), [Uplink Intra-gNodeB CA](#), and [Downlink Inter-gNodeB CA](#).

Downlink Intra-gNodeB CA

In downlink intra-gNodeB CA, after SCells are activated for a UE, the serving gNodeB of the PCell distributes part of the data buffered at the RLC layer to the SCells. This occurs only when the buffered RLC data volume is greater than or equal to the downlink threshold for a CA data split, which is specified by the **NRDUCellCarrMgmt.IntraGnbCaDlSplitThld** parameter on the SCell side. If this parameter is set to **ADAPTIVE**, the gNodeB calculates the threshold based on the data rates of historical scheduling for the UE. If this parameter is set to another value, the configured value is taken as the threshold.

For more accurate CA data split, real-time data split by RLC for downlink intra-gNodeB CA can be used. This function takes effect when the **NRDUCellCarrMgmt.DlIntraGnbCaSplitPolicy** parameter is set to **REAL_TIME**. Once this function takes effect, the gNodeB serving the PCell uses an optimized data split algorithm to distribute data buffered at the RLC layer to SCells. For the network impacts of real-time data split by RLC for downlink intra-gNodeB CA, see [5.1.2.2 Impacts](#) and [6.1.2.2 Impacts](#).

To improve downlink experience of CA UEs, CA data split stop based on the scheduling capability ratio is introduced. This function takes effect when the ratio of the scheduling capability of a higher-throughput carrier to that of a lower-throughput carrier is greater than the threshold specified by the **NRDUCellCarrMgmt.SplitStopSchCapbRatioThld** parameter (the value of this parameter must be greater than zero). Data distribution to the carrier with the lower scheduling capability is stopped. For the network impacts of this function, see [5.1.2.1 Benefits](#) and [6.1.2.1 Benefits](#).

In some scenarios, the gNodeB does not distribute data of specific CA UEs to their SCCs. Without the CA UE data split stop optimization function, the PCC data split weight (proportion of the data volume distributed to the PCC to the total data volume) is not the optimal value when data split to SCCs is stopped for a CA UE. This increases the data scheduling duration and affects user experience. As such, the CA UE data split stop optimization function is introduced. This function is controlled by the **SPLIT_STOP_OPT_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoSwitch** parameter. After this function takes effect, the gNodeB optimizes the PCC data split weight when data split to SCCs is stopped for a CA UE. This shortens the data scheduling duration and guarantees user experience.

Uplink Intra-gNodeB CA

In uplink intra-gNodeB CA, after SCells are activated for a UE, the gNodeB evaluates both the channel quality of the carriers serving the UE and the amount of to-be-scheduled data for the UE. A data split is triggered only when both criteria are met:

- Qualified carrier channel quality: The gNodeB assesses channel quality across multiple dimensions, with a necessary condition being that the SINR of the carriers must be greater than or equal to the SINR threshold for uplink data split to SCells (specified by the **NRDUCellCarrMgmt.UlCaSccSplitSinrThld** parameter).
- Sufficient amount of to-be-scheduled data: The amount must be greater than or equal to the SCell uplink data split threshold (a scenario-specific threshold internally calculated by the gNodeB).

The total power of a UE may decrease after SCC configuration. If there is consecutive non-split due to unqualified carrier channel quality, the UE cannot benefit from CA. However, the UE rate on a single carrier (PCC) may decrease due to the decrease in the total power. To mitigate this, SCC removal due to consecutive non-split in uplink CA is introduced. This function is controlled by the **NRDUCellCarrMgmt.UlCaContNonSplitSccDelThld** parameter.

- If this parameter is set to **255**, SCC removal due to consecutive non-split in uplink CA does not take effect.
- If the parameter value is not **255** and the count of consecutive non-split events due to unqualified carrier channel quality exceeds this value, the gNodeB removes the corresponding uplink SCC to ensure that the UE rate on the PCC is not affected.

In low-frequency scenarios, the following functions can be used to improve the accuracy of data volume distributed to SCells in the uplink.

- Multi-frequency actual-PUSCH-based inner-loop power control: For details, see *Power Control*. This function is controlled by the

MF_ACTUAL_PUSCH_ILPC_SW option of the **NRDUCellUIPcConfig.UlPwrCtrlAlgoExtSwitch** parameter.

- Uplink CA data split threshold optimization: refines the SCell uplink data split threshold calculation for improved stability and accuracy. This function is controlled by the **UL_CA_SPLIT_THLD_OPT_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoSwitch** parameter.

Downlink Inter-gNodeB CA

In downlink inter-gNodeB CA, after SCells are activated for a UE, the serving gNodeB of the PCell distributes part of the data buffered at the RLC layer to the SCells. This occurs only when the buffered RLC data volume is greater than or equal to the downlink threshold for a CA data split. The threshold is calculated by the gNodeB based on the amount of data that can be transmitted within the length of the inter-gNodeB delay (the maximum inter-gNodeB delay is specified by the **NRDUCellCarrMgmt.InterGnodebDelay** parameter).

In some scenarios, the gNodeB does not distribute data of specific CA UEs to their SCCs. Without the CA UE data split stop optimization function, the PCC data split weight (proportion of the data volume distributed to the PCC to the total data volume) is not the optimal value when data split to SCCs is stopped for a CA UE. This increases the data scheduling duration and affects user experience. As such, the CA UE data split stop optimization function is introduced. This function is controlled by the **SPLIT_STOP_OPT_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoSwitch** parameter. After this function takes effect, the gNodeB optimizes the PCC data split weight when data split to SCCs is stopped for a CA UE. This shortens the data scheduling duration and guarantees user experience.

4.8 Performance Measurement Optimization

4.8.1 Optimized Measurement of Downlink CA Throughput

Without optimization, the average throughput of SCell-activated UEs is measured with tail data excluded, as stipulated in section "Average DL UE throughput in gNB" in 3GPP TS 28.552. Tail data is the data transmitted in the slot when the downlink buffer is emptied. The proportion of tail data may rise because of the improved CA UE scheduling capability after CA is enabled on a commercial network with a small-packet traffic model. This may hinder the average throughput of SCell-activated UEs from increasing and therefore prevent the throughput from accurately reflecting the air interface capabilities of CA UEs.

The average throughput of SCell-activated UEs is classified into the following types:

- Average throughput of SCell-activated CA UEs, which is equal to $(N.CA.SccActiveUser.ThpVol.DL - N.CA.ThpVol.DL.LastSlot) / N.CA.ThpTime.DL.RmvLastSlot$
- Average throughput of SCell-activated inter-gNodeB CA UEs, which is equal to $(N.IntergNB.CA.SccActiveUser.ThpVol.DL - N.IntergNB.CA.ThpVol.DL.LastSlot) / N.IntergNB.CA.ThpTime.DL.RmvLastSlot$

- Average throughput of SCell-activated CA UEs in the 3CC aggregation state, which is equal to $(N.3CC.CA.ThpVol.DL - N.3CC.CA.ThpVol.DL.LastSlot) / N.3CC.CA.ThpTime.DL.RmvLastSlot$
- Average throughput of SCell-activated massive CA UEs, which is equal to $(N.Massive.CA.ThpVol.DL - N.Massive.CA.ThpVol.DL.LastSlot) / N.Massive.CA.ThpTime.DL.RmvLastSlot$

Take the average throughput of SCell-activated CA UEs as an example. The **N.CA.SccActiveUser.ThpVol.DL** counter indicates the total downlink data volume for SCell-activated UEs, the **N.CA.ThpVol.DL.LastSlot** counter indicates the tail data volume, and the **N.CA.ThpTime.DL.RmvLastSlot** counter indicates the duration in which the data excluding tail data is transmitted.

To enable the average throughput of SCell-activated UEs to more accurately reflect the air interface capabilities of CA UEs, optimized measurement of downlink CA throughput is introduced. This function is controlled by the **DL_CA_THPT_L_SLOT_STATS_POL_SW** option of the **NRDUCellOamParam.StatisticsStrategySwitch** parameter. It optimizes the method for measuring the average throughput of SCell-activated UEs.

Still take the average throughput of SCell-activated CA UEs as an example. In the numerator, the value of the **N.CA.ThpVol.DL.LastSlot** counter is reduced to 0. In the denominator, the tail data transmission duration is converted and added to the **N.CA.ThpTime.DL.RmvLastSlot** counter. (The duration is converted with the estimated TB size in large-packet scenarios as the denominator and the tail data volume as the numerator. The converted duration is shorter than one slot.)

With the optimization, the preceding counters will change as follows:

- The values of the following counters are reduced to 0:
N.CA.ThpVol.DL.LastSlot, **N.IntergNB.CA.ThpVol.DL.LastSlot**, **N.3CC.CA.ThpVol.DL.LastSlot**, and **N.Massive.CA.ThpVol.DL.LastSlot**.
- The values of the following counters increase:
N.CA.ThpTime.DL.RmvLastSlot, **N.IntergNB.CA.ThpTime.DL.RmvLastSlot**, **N.3CC.CA.ThpTime.DL.RmvLastSlot**, and **N.Massive.CA.ThpTime.DL.RmvLastSlot**.

This function increases the average throughput of SCell-activated UEs. The amount of the increase depends on the proportion of tail data and the spectral efficiency of tail data scheduling. The higher the proportion and the higher the average spectral efficiency, the greater the increase.

4.8.2 Optimized Measurement of the Numbers of CA-capable UEs in Cells

The following counters measure the average and maximum numbers of CA-capable UEs and those of 3CC-capable UEs in a cell.

- **N.User.CA.Capability.Avg**
- **N.User.CA.Capability.Max**
- **N.User.3CC.CA.Capability.Avg**
- **N.User.3CC.CA.Capability.Max**
- **N.User.UL.CA.Capability.Avg**

- **N.User.UL.CA.Capability.Avg.PLMN**
- **N.User.3CC.CA.Capability.Avg.PLMN**

Such a number is not necessarily equal to the number of CA-capable or 3CC-capable UEs that can actually work in the capability-indicated CA state with the present cell configurations. To enable the preceding counters to measure only CA-capable and 3CC-capable UEs that can actually work in the capability-indicated CA states with the present cell configurations, select the **CA_CAP_USER_STATS_POL_SW** option of the **gNodeBAlgo.NrSmartMultiCarrSw** parameter for the cell. After this option is selected, the values of the preceding counters may decrease.

The following uses the **N.User.3CC.CA.Capability.Avg** and **N.User.3CC.CA.Capability.Max** counters as an example to describe the differences in counter measurement before and after this optimized measurement function is enabled:

- With this function disabled
 - SA networking: If a UE is handed over from a cell where 3CC aggregation is applicable to a cell where 3CC aggregation is not applicable, the UE is counted into the **N.User.3CC.CA.Capability.Max** and **N.User.3CC.CA.Capability.Avg** counters in the target cell.
 - NSA networking:
 - If a UE is handed over from a cell where 3CC aggregation is applicable to a cell where 3CC aggregation is not applicable, the UE is counted into the **N.User.3CC.CA.Capability.Max** and **N.User.3CC.CA.Capability.Avg** counters in the target cell.
 - If a UE is capable of 3CC aggregation in SA networking but not in NSA networking, the UE is counted into the **N.User.3CC.CA.Capability.Max** and **N.User.3CC.CA.Capability.Avg** counters.
- With this function enabled:
 - SA networking: If a UE is handed over from a cell where 3CC aggregation is applicable to a cell where 3CC aggregation is not applicable, the UE is no longer counted into the **N.User.3CC.CA.Capability.Max** or **N.User.3CC.CA.Capability.Avg** counter in the target cell.
 - NSA networking:
 - If a UE is handed over from a cell where 3CC aggregation is applicable to a cell where the UE capability indicates that the UE is not capable of 3CC aggregation in NSA networking, the UE is no longer counted into the **N.User.3CC.CA.Capability.Max** or **N.User.3CC.CA.Capability.Avg** counter in the target cell.
 - If a UE is capable of 3CC aggregation in SA networking but not in NSA networking, the UE is no longer counted into the **N.User.3CC.CA.Capability.Max** or **N.User.3CC.CA.Capability.Avg** counter.

This function brings similar changes to the measurement of the other counters related to the number of CA-capable UEs.

5 Intra-Band CA

5.1 Downlink Intra-Band CA

5.1.1 Principles

5.1.1.1 Overview

With downlink intra-band CA enabled, SCell configuration is triggered when a CA UE initiates RRC connection setup during initial access, an incoming handover, or an incoming RRC connection reestablishment. After SCells are configured and activated, the CA UE can benefit from downlink CA.

In low-frequency scenarios, downlink intra-band CA can be enabled by turning on the downlink intra-band CA switch, which is the **INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch.CaAlgoSwitch** parameter. 2CC aggregation is supported by default. For details about how to configure aggregation of more than two carriers, see *CA Experience Improvement*.

In high-frequency NR TDD, downlink intra-band CA takes effect only when both the downlink intra-band CA switch and the uplink intra-band CA switch (the **INTRA_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch.CaAlgoSwitch** parameter) are turned on.

For the band+bandwidth combinations for downlink intra-band CA, see [4.4 CA Band+Bandwidth Combinations](#).

5.1.1.2 SCC PMI-based Weight Enhancement

There are two types of SCCs: uplink+downlink SCCs and downlink-only SCCs. SCells on uplink+downlink SCCs (uplink+downlink SCells) can use SRS-based weights, and therefore downlink beamforming performance in these SCells is the same as that in PCells. SCells on downlink-only SCCs (downlink-only SCells) can use SRS-based weights only if SRS carrier switching is enabled, which enables these SCells to obtain SRS resources. However, if SRS carrier switching is disabled or if it is enabled but does not take effect for UEs, only PMI-based weights can be used in these SCells. As a result, downlink beamforming performance in the TDD

downlink-only SCells is worse than that when SRS-based weights are used. For details about SRS carrier switching, see *CA Experience Improvement*.

To improve downlink beamforming performance of these TDD SCells, SCC PMI-based weight enhancement has been introduced so that SRSs in the PCell can be used to calculate PMI-based weights for SCells. This function applies when a TDD downlink-only SCell and the PCell are intra-gNodeB cells served by the same RF module in intra-band contiguous CA in low frequency bands. This function is controlled by the **SCC_PMI_ENH_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch** parameter. It is supported only in low frequency bands, but not in high frequency bands.

5.1.1.3 SCell CSI-RS Beam Sweeping

Basic Principles

For downlink CA to take effect, the gNodeB must obtain the directions of CSI-RS beams for a UE in the PCell and SCells.

In asymmetric CA, some SCells are configured only for the downlink and therefore the gNodeB cannot obtain the beam directions for the UE in these SCells based on the uplink. As a result, these SCells cannot be used. When all the following conditions are met, however, the beam directions for the UE in the PCell and SCells are the same, and the gNodeB can prevent the preceding issue with the use of the CSI-RS beam direction in the PCell:

- The PCell and SCells operate in the same band.
- The PCell and SCells are served by the same RF module.
- The spacing between the center frequencies of the PCell and each SCell is less than 500 MHz. The center frequencies of these cells are specified by the **NRDUCell.UINarfcn** and **NRDUCell.DINarfcn** parameters.

If not all the preceding conditions are met, the beam direction in the PCell cannot be used for SCells. As a result, the SCells cannot be used.

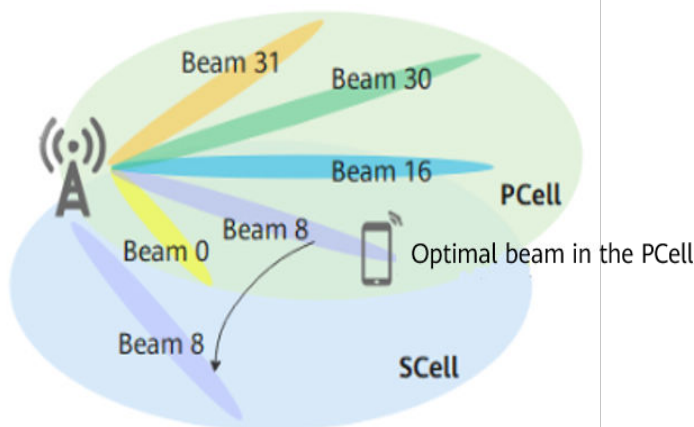
To address this issue, the SCell CSI-RS beam sweeping function is introduced. This function is controlled by the **CARR_CSI_BEAM_SWEEPING_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoSwitch** parameter. This function must be enabled when the beam directions in the PCell and SCells differ. Otherwise, SCells cannot be configured and UE throughput in the PCell decreases due to frequent gap-assisted measurements for SCell configuration.

Figure 5-1 illustrates the SCell CSI-RS beam sweeping function, which works as follows:

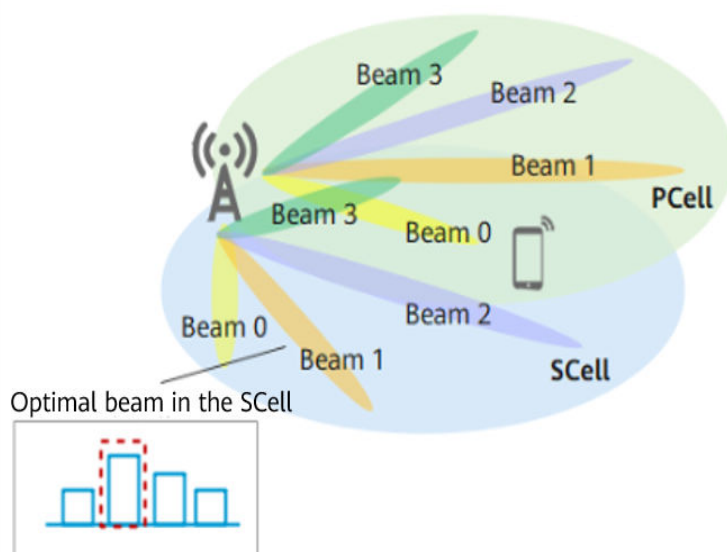
1. The gNodeB sends CSI-RS for beam measurement (CSI-RS for BM) in an SCell on CSI-RS beams in different directions in turn, so as to obtain the optimal beam for the UE in the SCell. Then the gNodeB sends CSI-RS for channel measurement (CSI-RS for CM) on this optimal beam to the UE.
2. The UE performs measurements for CQI, PMI, and rank indication (RI) based on the CSI-RS for CM sent by the SCell.
3. The UE reports the CQI, RI, and PMI measurement results of the SCell on the PUCCH or PUSCH in the PCell.

Figure 5-1 SCell CSI-RS beam sweeping

CSI-RS beam directions are the same in the PCell and SCell.



CSI-RS beam directions are different in the PCell and SCell.



NOTE

In symmetric CA scenarios, the gNodeB obtains the CSI-RS beam directions for a UE in SCells based on the uplink. The SCell CSI-RS beam sweeping function does not take effect even if it is enabled.

SCell CSI-RS beam sweeping needs to be enabled only when the SCell has multiple CSI-RS beams. It does not need to be enabled when the SCell has only one CSI-RS beam. For details about the number of CSI-RS beams in a cell, see *Beam Management*.

CSI-RS Beam Sweeping Optimization

CSI-RS beam sweeping described in [Basic Principles](#) is performed in either periodic or aperiodic mode. After the sweeping succeeds, the UE reports the optimal beam in an SCell. Resources must be reserved for such information

reporting in both periodic and aperiodic sweeping scenarios (or only the supported sweeping scenario if the UE does not support both). The resources available for UEs to report information are limited. If resources are insufficient, other features that require UE information reporting for CA carriers may fail to take effect. Considering that periodic CSI-RS beam sweeping has a high success rate, CSI-RS beam sweeping optimization is introduced to reduce the resource consumption by UE information reporting. CSI-RS beam sweeping optimization is controlled by the **CSI_BEAM_SWEEPING_OPT_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoSwitch** parameter.

- If this option is selected:
 - If a UE supports periodic sweeping, such sweeping is performed and resources are reserved only for UE information reporting for periodic sweeping.
 - If a UE does not support periodic sweeping, aperiodic sweeping is performed and resources are reserved only for UE information reporting for aperiodic sweeping.
- If this option is deselected, CSI-RS beam sweeping optimization does not take effect.

Adaptive CSI-RS Beam Sweeping

The **maxNumberSSB-CSI-RS-ResourceOneTx** field is carried in the capability information that a UE reports to the gNodeB. This field indicates a maximum total quantity of specific reference signal resources (including CSI-RS signal resources) that can be used by the UE to simultaneously measure L1-RSRP in all serving cells in a slot. Without adaptive CSI-RS beam sweeping, the gNodeB does not check whether the total number of CSI-RS resources required by all CCs exceeds the total number of CSI-RS resources supported by a UE during CA configuration. As a result, the total number of CSI-RS resources required by all CCs may exceed the UE capability, and some UEs may experience RRC connection reestablishments or service drops. As such, adaptive CSI-RS beam sweeping is introduced, which is controlled by the **CA_CSIRS_BM_SWEEP_ADPT_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoSwitch** parameter. After this function takes effect, the gNodeB checks whether the CSI-RS resources required by all CCs exceed the total number of CSI-RS resources supported by a UE. If this is true, the gNodeB either avoids SCell configuration or reduces the number of SCCs to prevent issues such as reestablishment or service drops of UEs.

5.1.1.4 CA PDCCH Capacity Optimization

Assume that a UE receives an uplink grant and then a downlink grant in a single slot and sends HARQ feedback on the PUSCH. For certain such UEs, the downlink block error rate (BLER) increases. This issue can be prevented when the start PDCCH symbol for CA UEs is **fixed** at the first available PDCCH symbol in the cells serving these UEs. However, PDCCH resources in the cells are allocated less flexibly for CA UEs using this method. Data transmission for CA UEs may even be affected in the case of insufficient PDCCH resources in heavy-load scenarios.

CA PDCCH capacity optimization is therefore introduced to improve data transmission for CA UEs in heavy-load scenarios. This function allows the start symbol of the PDCCH to be flexibly selected for CA UEs. In addition, the scheduling timing is internally optimized to prevent the downlink BLER from

increasing. This function is controlled by the **CA_PDCCH_CAPACITY_OPT_SWITCH** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch** parameter. This function is supported only in low frequency bands, not in high frequency bands.

5.1.1.5 CA PDCCH Aggregation Level Optimization

CA PDCCH aggregation level optimization has been introduced to improve the downlink data transmission performance of CA UEs. This function applies an upper limit (specified by the **NRDUCellPdcchAlgo.PdcchOuterLoopAdjUpLimit** parameter) to the accumulated outer-loop adjustment amount in PDCCH aggregation level adaptation for CA UEs. Outer-loop adjustment for the PDCCH can be performed when the accumulated outer-loop adjustment amount is lower than the upper limit, but it can no longer be performed when this accumulated amount is greater than or equal to the upper limit. A smaller value of the **NRDUCellPdcchAlgo.PdcchOuterLoopAdjUpLimit** parameter indicates a lower upper limit and results in a higher PDCCH aggregation level that is adaptively selected. A larger value results in a lower PDCCH aggregation level. This function is supported in low frequency bands, but not in high frequency bands.

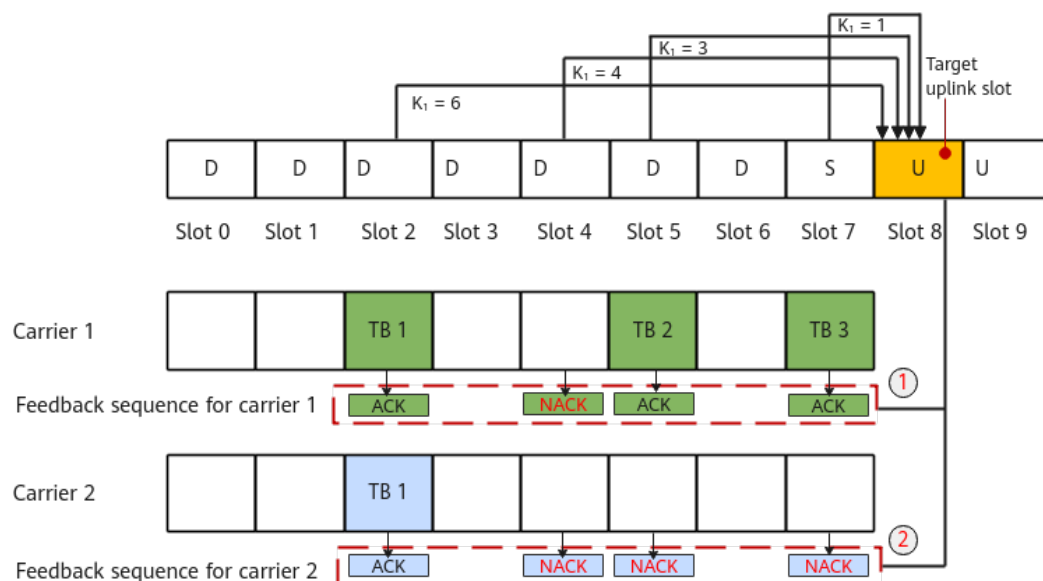
5.1.1.6 Dynamic-Codebook-based HARQ Feedback over the PUCCH

3GPP TS 38.331 has defined two HARQ-ACK codebook types: type 1 and type 2.

- Type-1 codebooks are semi-static HARQ-ACK codebooks. Once K_1 is configured by the higher layer, the UE provides, in each uplink slot, the HARQ feedback about all the slots that are K_1 slots prior to the current feedback slot. In those slots, data transmission is possible.
- Type-2 codebooks are dynamic HARQ-ACK codebooks. Unlike a UE using a semi-static codebook, a UE using a dynamic codebook sends HARQ feedback only about the resources where data was transmitted.

Figure 5-2 provides an example. In this example, the UE can simultaneously receive data over at most two carriers, and it correctly receives data over both carriers. K_1 received by the UE is {1, 3, 4, 6}, which is configured in the PCell in multi-carrier cases.

- If a semi-static HARQ-ACK codebook is used, the UE sends the HARQ feedback about slots 2, 4, 5, and 7 of both carriers in the target HARQ feedback slot 8. In this situation, eight bits are required for the feedback, regardless of whether the UE has received data in those slots.
- If a dynamic HARQ-ACK codebook is used, the UE sends the HARQ feedback only about slots with data transmission (that is, slots 2, 5, and 7 of carrier 1 and slot 2 of carrier 2) in the target HARQ feedback slot 8. In this situation, only four bits are required for the feedback.

Figure 5-2 Example of data transmission in the 2CC aggregation state


As shown in this example, the use of a semi-static HARQ-ACK codebook consumes more resources than the use of a dynamic HARQ-ACK codebook. Currently, semi-static HARQ-ACK codebooks are always used for the PUCCH, resulting in a limited number of CA UEs that can be scheduled in a cell. To address this issue, dynamic HARQ-ACK codebooks can now be used for the PUCCH. This usage is controlled by the **PUCCH_DYN_HARQ_ACK_CB_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch** parameter.

- If this option is selected and all the following conditions are met, the gNodeB sets the *PDSCH-HARQ-ACK-Codebook* field to **dynamic** when starting SCell configuration for a CA UE during initial access, an incoming handover, or an incoming RRC connection reestablishment, and the UE then uses dynamic HARQ-ACK codebooks for the PUCCH:
 - The number of SCell-configured UEs is greater than the value of the **NRDUCellCarrMgmt.DynCodebookCaUeNumThld** parameter, or the number of RRC_CONNECTED UEs in the PCell is greater than 800.
 - The UE is not a low-frequency inter-gNodeB CA UE, for which low-frequency inter-gNodeB SCells have been configured.
- If this option is deselected, UEs use semi-static HARQ-ACK codebooks.

In medium- and heavy-load scenarios (where the PRB usage is greater than 70%), there will be limited specifications of dynamic HARQ-ACK codebooks over the PUCCH, leading to a limited number of CA UEs that can be scheduled in a cell. As such, dynamic PUCCH codebook specification improvement for CA UEs is introduced. This function is controlled by the **PUCCH_DYN_CB_SPECS_IMP_SW** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter. After this function is enabled, there are more available dynamic HARQ-ACK codebooks over the PUCCH for CA UEs, and the number of CA UEs that can be scheduled in a cell increases.

5.1.2 Network Analysis

5.1.2.1 Benefits

The downlink peak data rates of CA UEs increase.

Table 5-1 lists the ranges of peak data rates that a CA UE can achieve using downlink 2CC aggregation in low frequency bands.

Table 5-1 Ranges of peak data rates achieved using downlink 2CC aggregation in low frequency bands

SCell Weight Type	Peak Data Rate Range
SRS-based	$(A \times 98\% + B \times 98\%)$ to $(A \times 100\% + B \times 100\%)$
Precoding matrix indication (PMI)-based	$(A \times 98\% + B \times 95\%)$ to $(A \times 100\% + B \times 100\%)$
Note 1: The SCell weight type depends on CA UE capabilities. If a CA UE transmits SRSs in an SCell, SRS-based weights are used in the SCell. Otherwise, PMI-based weights are used in the SCell. Note 2: A is the downlink data rate that the UE in the single-carrier state can achieve in a cell that is later configured as the PCell for the UE. Note 3: B is the downlink data rate that the UE in the single-carrier state can achieve in a cell that is later configured as an SCell and uses SRS- or PMI-based weights.	

For the ranges of peak data rates that a CA UE can achieve using aggregation of more than two CCs in low frequency bands in the downlink, see *CA Experience Improvement*.

The peak data rate that a CA UE can achieve using downlink contiguous 4CC aggregation in high-frequency NR TDD equals $(A + B + C + D) \times 100\%$. A , B , C , and D are the downlink data rates that the UE in the single-carrier state can achieve in individual cells before the cells are configured as the PCell and SCells for the UE.

NOTE

UEs cannot achieve the preceding maximum downlink peak data rates because section 4.1 "Supported max data rate" in 3GPP TS 38.306 V15.6.0 places a limit on the maximum downlink peak data rates of UEs. The following methods can be used to remove the limit:

- For certain UEs powered by chips with strong processing capabilities: Use device-pipe synergy for gNodeBs to identify UEs that support this synergy function. After such UEs are identified, the gNodeBs will remove their limit. In this way, these UEs can achieve higher downlink peak data rates as their chips allow for rates beyond the limit. To enable this function, select both the **SPEC_UE_IDENTIFY_SW** and **FR1_DL_TBS_LIMIT_CANCEL_SW** options of the **NRDUCellUeCoop.SpecUeCooperationSwitch** parameter in low-frequency scenarios and select both the **SPEC_UE_IDENTIFY_SW** and **FR2_DL_TBS_LIMIT_CANCEL_SW** options of the **NRDUCellUeCoop.SpecUeCooperationSwitch** parameter in high-frequency scenarios.
- For all UEs: Reduce the value of the parameter *OH* (indicating overhead) used in the formula for calculating the downlink transport block size (TBS). The parameter *OH* is specified by **NRDUCellCarrMgmt.Fr1CaDlTbsParamOh** in low-frequency scenarios and by **NRDUCellCarrMgmt.Fr2CaDlTbsParamOh** in high-frequency NR TDD.

After CA is enabled, downlink UE throughput increases. In addition, first packet delay decreases, which equals **N.Traffic.DL.RlcFirstPktDelay.Time** divided by **N.Qos.DL.RlcFirstPktDelay.Num**.

The CA enhancements provide the following benefits:

- SCC PMI-based weight enhancement increases downlink throughput in TDD SCells by 1% to 4%.
- SCell CSI-RS beam sweeping can increase the **User Downlink Average Throughput (DU)**. CSI-RS beam sweeping optimization in SCell CSI-RS beam sweeping further optimizes the allocation of CA UE reporting capabilities among multiple carriers, improving CA UE performance. After adaptive CSI-RS beam sweeping takes effect, the values of the following indicators decrease if the number of CSI resources for UEs on the network is less than the number of CSI resources required by all CCs: (1) **Service Call Drop Rate (CU, Inactive)**; (2) RRC reestablishment rate ($\frac{\text{N.RRC.ReEst.Att} + (\text{N.RRC.SetupReq.Succ} + \text{N.RRC.ResumeReq.Succ} + \text{N.HO.InterRAT.E2N.ExecSuccIn}) \times 100\%}{\text{N.RRC.ReEst.Att} + (\text{N.RRC.SetupReq.Succ} + \text{N.RRC.ResumeReq.Succ} + \text{N.HO.InterRAT.E2N.ExecSuccIn})}$).
- CA PDCCH capacity optimization, when enabled in heavy-load scenarios, increases both the **User Downlink Average Throughput (DU)** and **Cell Downlink Average Throughput (DU)**.
- Dynamic-codebook-based HARQ feedback over the PUCCH reduces the number of bits used for HARQ feedback and increases the probability of using PUCCH format 1 (with two or fewer bits for HARQ feedback), improving PUCCH resource efficiency and increasing the number of CA UEs that can be scheduled in a cell.
- Dynamic PUCCH codebook specification improvement for CA UEs further increases the number of CA UEs that can be scheduled in a cell. This function improves the downlink throughput of CA UEs in medium- and heavy-load scenarios (where the PRB usage is greater than 70%).
- CA data split stop based on the scheduling capability ratio enables data distribution to the carriers with lower scheduling capabilities to be stopped. This improves CA UE experience and possibly increases the **User Downlink Average Throughput (DU)** when some carriers serving individual CA UEs suffer severe interference.

- CA UE data split stop optimization may shorten the data scheduling delay of CA UEs for which data split is stopped.
- C-RNTI conflict detection prevents interference caused by C-RNTI conflicts, thereby increasing the average throughput of SCell-activated CA UEs in scenarios (such as heavy-load scenarios) where C-RNTI conflicts are likely to occur.
 - Average throughput of SCell-activated downlink CA UEs =
$$\frac{(N.CA.SccActiveUser.ThpVol.DL - N.CA.ThpVol.DL.LastSlot)}{N.CA.ThpTime.DL.RmvLastSlot}$$
 - Average throughput of SCell-activated uplink CA UEs =
$$\frac{(N.CA.SccActiveUser.ThpVol.UL - N.CA.ThpVol.UL.SmallPkt)}{N.CA.ThpTime.UL.RmvSmallPkt}$$
- C-RNTI conflict-triggered reconfiguration reduces the risk of SCell configuration failures due to C-RNTI conflicts, thereby increasing the average throughput of SCell-activated CA UEs in scenarios (such as heavy-load scenarios) where C-RNTI conflicts are likely to occur.
 - Average throughput of SCell-activated downlink CA UEs =
$$\frac{(N.CA.SccActiveUser.ThpVol.DL - N.CA.ThpVol.DL.LastSlot)}{N.CA.ThpTime.DL.RmvLastSlot}$$
 - Average throughput of SCell-activated uplink CA UEs =
$$\frac{(N.CA.SccActiveUser.ThpVol.UL - N.CA.ThpVol.UL.SmallPkt)}{N.CA.ThpTime.UL.RmvSmallPkt}$$
- UE-number-based SCell configuration control limits the use of CA to fewer RRC_CONNECTED UEs in scenarios where the number of RRC_CONNECTED UEs may reach the upper limit (for example, during major sports events). This ensures that more UEs can access the network.
- Non-CQI-based SCell deactivation for CA UEs performing downlink large-packet services increases the average throughput of SCell-activated downlink CA UEs, which is calculated using the following formula:
$$\frac{(N.CA.SccActiveUser.ThpVol.DL - N.CA.ThpVol.DL.LastSlot)}{N.CA.ThpTime.DL.RmvLastSlot}$$
- Delayed SCell deactivation:
 - A larger value of the SCell deactivation waiting timer results in a longer time for UEs to stay in the SCell-activated state and therefore higher average throughput of SCell-activated CA UEs, but it also leads to higher UE power consumption. If the timer value is too large and flow control is triggered, user experience is affected.
 - A smaller value of the SCell deactivation waiting timer results in a shorter time for UEs to stay in the SCell-activated state and therefore lower average throughput of SCell-activated CA UEs, but it also leads to lower UE power consumption.

Average throughput of SCell-activated CA UEs =
$$\frac{(N.CA.SccActiveUser.ThpVol.DL - N.CA.ThpVol.DL.LastSlot)}{N.CA.ThpTime.DL.RmvLastSlot}$$
- SCell deactivation protection increases the values of the following indicators in the case of inconsistent SCell deactivation status between the base station and UEs:
 - Average throughput of CA UEs configured with SCells:
$$(N.CA.SccConfigUser.ThpVol.DL -$$

- $\frac{\text{N.CA.SccConfigUser.ThpVol.DL.LastSlot}}{\text{N.CA.SccConfigUser.ThpTime.DL.RmvLastSlot}}$
- **User Downlink Average Throughput (DU)**
 - **User Uplink Average Throughput (DU)**
 - Precise SCell activation enables data to be distributed to SCells earlier, increasing the average throughput of CA UEs with the SCells activated. The more frequently SCells are activated and deactivated, the higher the gains. This function does not offer benefits when SCells remain in the activated or deactivated state without changing their activation states.
 - SCell activation and deactivation suspension during DRX wakeup protection: After this function takes effect, the number of UEs for which SCells can be activated and deactivated may decrease in the PCell, the SCell activation success rate ($\text{N.CA.SCell.Act.Succ}/\text{N.CA.SCell.Act.Att}$) and SCell deactivation success rate ($\text{N.CA.SCell.Deact.Succ}/\text{N.CA.SCell.Deact.Att}$) may increase, but the number of SCell activation times ($\text{N.CA.SCell.Act.Att}$) and the number of SCell deactivation times ($\text{N.CA.SCell.Deact.Att}$) may decrease.
 - SCell activation prohibition under weak coverage: When this function is in effect, the following impacts may occur depending on the value of **NRDUCellRsvdExt01.RsvdParam15**:
 - A smaller value of **NRDUCellRsvdExt01.RsvdParam15** results in more UEs for which SCells can be activated in the PCell, a lower SCell activation success rate ($\text{N.CA.SCell.Act.Succ}/\text{N.CA.SCell.Act.Att}$), more SCell activation times ($\text{N.CA.SCell.Act.Att}$), and a higher average throughput of SCell-configured CA UEs ($\frac{\text{N.CA.SccConfigUser.ThpVol.DL} - \text{N.CA.SccConfigUser.ThpVol.DL.LastSlot}}{\text{N.CA.SccConfigUser.ThpTime.DL.RmvLastSlot}}$).
 - A larger value of **NRDUCellRsvdExt01.RsvdParam15** results in fewer UEs for which SCells can be activated in the PCell, a higher SCell activation success rate ($\text{N.CA.SCell.Act.Succ}/\text{N.CA.SCell.Act.Att}$), fewer SCell activation times ($\text{N.CA.SCell.Act.Att}$), and a lower average throughput of SCell-configured CA UEs ($\frac{\text{N.CA.SccConfigUser.ThpVol.DL} - \text{N.CA.SccConfigUser.ThpVol.DL.LastSlot}}{\text{N.CA.SccConfigUser.ThpTime.DL.RmvLastSlot}}$).
 - Prohibition of blind SCell configuration during incoming handovers, a sub-function of optimized SCell configuration during PCell handovers, allows cells measured in the source cell of a handover to be preferentially configured as SCells when the SCell coverage on the target cell side is poorer.
 - Multi-carrier combination selection based on multiple bandwidth levels increases **User Downlink Average Throughput (DU)** and **User Uplink Average Throughput (DU)** of UEs if a relatively large number of UEs on the network exhibit better CA capabilities (for example, more layers) on the minimum carrier bandwidths.

 NOTE

There may be zero gains or even negative impacts (3% at most) for a UE at certain positions. For example, this may occur if the UE receives the CSI-RS wave at its crest from the downlink-only SCell or if there are excessively many multipath signals.

5.1.2.2 Impacts

The impacts between this function and other functions are described in [5.1.3.2.3 Function Impacts](#).

- System capacity
A CA UE with SCells configured has only one RRC connection to the network. Such a UE consumes one sales unit of the license for the number of RRC_CONNECTED UEs. However, the CA UE consumes one hardware resource unit in the PCell and SCells separately. With n CC aggregation for all UEs on the network, the maximum allowed number of RRC_CONNECTED UEs decreases to $1/n$ (n is an integer).
- Resource usage
 - Overall physical resource block (PRB) usage of the network
After CA is enabled, cell load can be rapidly balanced through carrier management and scheduling to help utilize idle resources and increase the overall PRB usage of the network.
 - Physical uplink control channel (PUCCH) and physical uplink shared channel (PUSCH) overheads
Each CA UE sends the acknowledgment/negative acknowledgment (ACK/NACK) and CSI related to its SCells in its PCell with the PUCCH. When the PUSCH is not scheduled, the UE sends the information over the PUCCH. When the PUSCH is scheduled, the UE sends the information over the PUSCH. Therefore, the PUCCH overhead or the signaling overhead on the PUSCH increases.
 - CPU usage of baseband processing units (BBPs)
After downlink intra-band CA is enabled, the CPU usage of the BBP where the PCell is configured increases with the amount of data forwarded from the PCell to SCells at the RLC layer. If the CPU usage is too high, the peak data rate of the UE decreases.
- Overall throughput in the entire network
CA does not directly affect network capacity. However, when the resources on a network have not been exhausted, CA increases the overall resource usage and network throughput.
- Key performance indicator (KPI) fluctuation
 - Channel quality of UEs that treat a cell as their PCell differs from that of UEs that treat the cell as their SCell. Therefore, CA will cause CQIs and MCS indexes to change in a cell configured as an SCell for UEs.
 - After CA is enabled, PUCCH format 3 and format 4 are used more often than before. Therefore, the average value of counters **N.UL.SINR.PUCCH.Index0** to **N.UL.SINR.PUCCH.Index7** decreases slightly.
 - Before CA is enabled, all UEs are in the single-carrier state, have SRS resources, and can use SRS-based weights. After CA is enabled, some UEs have to use only PMI-based weights in their SCells where no SRS resources are available. PMI-based weights result in lower MCS indexes being selected for UEs than SRS-based weights. Therefore, with all other factors constant, the average of counters **N.ChMeas.PDSCH.MCS.0** through **N.ChMeas.PDSCH.MCS.28** will decrease and the usage of

control channel elements (CCEs) on the PDCCH will increase in an SCell after CA is enabled if UEs have no SRS resources in this SCell.

- After CA is enabled, SCells are not activated if the network traffic is light, which is indicated by the value of the **N.User.CA.PCell.DL.SCellAct.Avg** counter being 0. However, the PDCCH CCE usage on each carrier decreases. The reason is that although SCells are not activated for a CA UE, the gNodeB has configured the semi-static HARQ-ACK codebook for the UE, which leads to a lower proportion of discontinuous transmissions (DTXs) than that when the dynamic HARQ-ACK codebook was used before the SCells were configured.
- When CA is in effect, the PUCCH exists only in the PCell. As a result, fewer DTXs are detected, the DTX-based PDCCH aggregation level adjustment in the PCell and SCells is performed less frequently, and the PDCCH aggregation level adjustment slows down. Consequently, the BLER (including the **IBLER** and **RBLER**) may increase.
- After CA is enabled in NSA networking, the values of the **N.NsaDc.SgNB.Mod.Req.Succ** and **N.NsaDc.SgNB.Mod.Req.Att** counters increase because the gNodeB needs to configure SCells through RRCReconfiguration messages.
- After CA is enabled in NSA networking, the values of the **N.NsaDc.SgNB.Mod.Required.Succ** and **N.NsaDc.SgNB.Mod.Required.Att** counters increase because the gNodeB needs to deliver measurement configurations to CA UEs through RRCReconfiguration messages.
- If no cell measured by a UE at the cell edge meets the threshold for event A5, the gNodeB cannot configure an SCell for the UE. In this case, the gNodeB delivers event A5 measurement configurations to the UE every 15s if the UE traffic volume is constantly high. Since these periodic inter-frequency measurements require measurement gaps, the downlink throughput of the UE in the PCell decreases.
- Initial scheduling begins after an SCell is activated. This scheduling affects the average user-perceived data rates of CA UEs to an extent depending on the scheduling duration. (The duration is related to factors such as target weights, rank adjustment period, and PCell air interface quality.) As a result, the average user-perceived data rates of CA UEs decrease in non-full buffer scenarios with frequent SCell deactivation and activation.
- After flow control against multi-carrier operations at the gNodeB is triggered, SCells will no longer be configured for newly admitted UEs. As a result, the **User Uplink Average Throughput (DU)** and **User Downlink Average Throughput (DU)** may decrease. In particular, if this level of flow control is high (indicated by the value **HIGH** of the **gNodeBAlgo.MultiCarrierFlowCtrlLevel** parameter), the **Service Call Drop Rate (CU, Inactive)** in the cell may rise.
- As data is distributed to SCells, scheduling capabilities for CA UEs are stronger. As a result, the proportion of the last slots during each of which the downlink buffer becomes empty in the total data transmission duration may increase and the average user-perceived rate may decrease. (The proportion equals $(\text{N.ThpTime.DL} - \text{N.ThpTime.DL.RmvLastSlot}) / \text{N.ThpTime.DL}$).
- During optimal SCC frequency combination selection, event A5 inter-frequency measurement configurations are delivered to obtain the RSRP

of each frequency. As inter-frequency measurement requires measurement gaps, the downlink throughput of UEs in their PCells decreases.

- Parameters related to SCell activation and deactivation have the following impacts on the network:
 - With the **NRDUCellCarrMgmt.CaDIScellActThldSlotNum** parameter set to a smaller value, the following impacts are more likely to occur:
 - There will be a higher probability of SCell activation, a longer SCell-activated duration, higher average throughput of SCell-activated CA UEs, and higher UE power consumption.
 - The values of the following counters increase:
 - **N.CA.SCell.Act.Att**
 - **N.CA.SCell.Act.Succ**
 - **N.CA.SCell.Deact.Att**
 - **N.CA.SCell.Deact.Succ**
 - With the **NRDUCellCarrMgmt.CaDIScellDeactThldSlotNum** or **NRDUCellCarrMgmt.RlcDlThroughputThld** parameter set to a smaller value, the following impacts are more likely to occur:
 - There will be a lower probability of SCell deactivation, a longer SCell-activated duration, higher average throughput of SCell-activated CA UEs, and higher UE power consumption.
 - The values of the following counters decrease:
 - **N.CA.SCell.Deact.Att**
 - **N.CA.SCell.Deact.Succ**
 - **N.CA.SCell.Act.Att**
 - **N.CA.SCell.Act.Succ**
 - Reconfiguring **NRDUCellCarrMgmt.DlStateTransitBufVolThld** or **NRDUCellQciBearer.QciDlStTransitBufVolThld**, which controls both SCell activation and SCell deactivation, causes the values of the following counters to fluctuate (the fluctuation is dependent on the threshold setting and the traffic volume of CA UEs on the live network):
 - **N.User.CA.PCell.DL.SCellAct.Avg**
 - **N.CA.PCell.ThpVol.Dl.Cell**
 - **N.CA.SCell.ThpVol.Dl.Cell**
 - **N.CA.SCell.Deact.Att**
 - **N.CA.SCell.Deact.Succ**
 - **N.CA.SCell.Act.Att**
 - **N.CA.SCell.Act.Succ**

- With channel quality-triggered SCell deactivation, consecutive residual block error-triggered SCell deactivation, and distributed data volume-triggered SCell deactivation enabled:
 - SCell deactivation is triggered in more scenarios. This results in a shorter SCell-activated duration, lower average throughput of SCell-activated CA UEs, and lower UE power consumption.
 - The values of the following counters increase:
 - **N.CA.SCell.Deact.Att**
 - **N.CA.SCell.Deact.Succ**
 - **N.CA.SCell.Act.Att**
 - **N.CA.SCell.Act.Succ**
- SCell deactivation protection will change performance counters as follows:
 - The value of the **N.CA.SCell.Deact.Att** counter increases.
 - The value of the **N.CA.SCell.Deact.Succ** counter increases.
 - The value of the **N.CA.SCell.Act.Att** counter changes.
 - The value of the **N.CA.SCell.Act.Succ** counter changes.

Average throughput of SCell-activated CA UEs =

$$\frac{(\text{N.CA.SccActiveUser.ThpVol.DL} - \text{N.CA.ThpVol.DL.LastSlot})}{\text{N.CA.ThpTime.DL.RmvLastSlot}}$$

- Certain functions have the following additional network impacts:
 - Precise SCell activation: The downlink IBLER increases during the period in which the gNodeB checks SCell availability. During this period as well as the period from when the gNodeB starts data split to when it receives CSI-RS measurement results, the average downlink MCS index and average downlink rank may decrease, and the **Downlink Resource Block Utilizing Rate (DU)** and average downlink CCE aggregation level increase.
 - Early data split: SCells are occasionally activated for CA to take effect for CA UEs in advance. As a result, the CQIs and MCS indexes of the cells acting as SCells fluctuate.
 - SCell CSI-RS beam sweeping: After its subfunction "adaptive CSI-RS beam sweeping" takes effect, the values of counters related to the number of CA UEs (for example, **N.User.CA.SCell.DL.Avg**) may decrease if the number of CSI resources for UEs on the network is less than the number of CSI resources required by all CCs.
 - CA PDCCH aggregation level optimization: A smaller value of the **NRDUCellPdcchAlgo.PdcchOuterLoopAdjUpLimit** parameter results in a better PDCCH demodulation capability but possibly a smaller number of UEs that can be scheduled in the cell.
 - Dynamic-codebook-based HARQ feedback over the PUCCH: This function causes the downlink initial block error rate (IBLER) to increase, because of its handling on missing DCI detection.

Downlink IBLER = $(\text{N.DL.SCH.QPSK.ErrTB.Ibler} + \text{N.DL.SCH.16QAM.ErrTB.Ibler} + \text{N.DL.SCH.64QAM.ErrTB.Ibler} +$

$$\frac{N.DL.SCH.256QAM.ErrTB.Ibler}{(N.DL.SCH.QPSK.TB + N.DL.SCH.16QAM.TB + N.DL.SCH.64QAM.TB + N.DL.SCH.256QAM.TB)} \times 100\%$$

- Real-time data split by RLC for downlink intra-gNodeB CA: The **User Downlink Average Throughput (DU)** fluctuates when the load is extremely heavy.
- Prohibition of acting as the PCell: CA will not take effect for UEs that newly access the cell or UEs camping on the cell but not in the CA state. These UEs cannot benefit from CA and therefore optimal user experience cannot be reached. For UEs in the CA state with the cell acting as their PCell, selecting the function option does not affect their existing SCells; the gNodeB can remove the SCells but cannot add new SCells or change SCells for these UEs. The UEs mentioned here refer to those with the "prohibition of acting as the PCell" function in effect. Whether this function is in effect for UEs in different networking depends not only on the cell's function switch setting but also on NSA/SA-specific PCell management policy configuration (detailed in [NSA/SA-specific PCell Management Policy Configuration](#)).
- After CA is enabled, especially when the **SCC_BLIND_CONFIG_SW** option of the **NRCellCaMgmtConfig.CaStrategySwitch** parameter is selected, cell-edge user experience may improve. Cell-edge UEs that would otherwise enter idle or inactive mode due to poor experience may remain in connected mode. As a result, cell-edge UEs may stay online for a longer time, and the proportion of cell-edge UEs may increase. Consequently, the service drop rate of PCells sometimes rises, as cell-edge UEs are more likely to experience service drops.
- CA UE data split stop optimization: If data split is stopped for some CA UEs, more data is scheduled on the PCC due to the optimized PCC data split weight. This may decrease the user-perceived rate of non-CA UEs on the PCC.
- C-RNTI conflict-triggered reconfiguration: The RRC reconfiguration times and service drop rate vary, depending on the number of CA capability combinations selected during SCell configuration. Selecting more CA capability combinations increases the likelihood of fewer RRC reconfiguration times and a lower service drop rate. Selecting fewer CA capability combinations increases the likelihood of more RRC reconfiguration times and a higher service drop rate. In addition, this function increases the value of the following counters:
 - **N.CA.SCell.Add.Att**
 - **N.CA.SCell.Add.Succ**
 - **N.CA.SCell.Mod.Att**
 - **N.CA.SCell.Mod.Succ**
- UE-number-based SCell configuration: The proportion of UEs for which CA takes effect and the network-wide UE throughput (including the uplink and downlink throughputs) change, depending on the configured thresholds expressed as the proportion of UEs for prohibiting 2CC or 3CC aggregation. Higher thresholds indicate higher proportions of CA UEs and higher network-wide UE throughput. Lower thresholds indicate lower

proportions of CA UEs and lower network-wide UE throughput. In addition, after the **UE_NUM_BASED_SCELL_CONFIG_SW** option of the **gNodeBCarrMgmt.CaEnhancedAlgoSwitch** parameter is selected, regardless of whether UE-number-based 2CC aggregation prohibition and UE-number-based 3CC aggregation prohibition take effect, more cell load information will be exchanged over the eXn interface, leading to increased length of a single message and possible more message fragments in some scenarios. As a result, the values of counters related to receive and transmit traffic in the SCTP Link Measurement subset may increase.

- Prohibition of SCell configuration inheritance during incoming handovers, a sub-function of optimized SCell configuration during PCell handovers, shortens the handover delay for CA UEs but results in a period of time in which SCell configuration is interrupted. Therefore, the uplink and downlink throughputs of the CA UEs decrease before new SCells are configured for the UEs.
- Multi-carrier combination selection based on multiple bandwidth levels may increase the CPU usage of the main control board, which may decrease the number of CA UEs in heavy-load scenarios. The following counters are involved:
 - **N.User.CA.PCell.DL.Avg** (average number of downlink CA UEs that use the local cell as their PCells)
 - **N.User.CA.SCell.DL.Avg** (average number of downlink CA UEs that use the local cell as their SCells)
 - **N.User.CA.PCell.UL.Avg** (average number of uplink CA UEs that use the local cell as their PCells)
 - **N.User.CA.SCell.UL.Avg** (average number of uplink CA UEs that use the local cell as their SCells)

5.1.3 Requirements

5.1.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-020205	Intra-Band CA	NR0S00IBCA00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.1.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Smart carrier selection	SMART_CARRIER_SEL_SW option of the NRDUCellConfig.NRMultiCarrierMgmtSw parameter	<i>Carrier Aggregation</i>	Smart carrier selection must be enabled before smart carrier selection optimization can be enabled.
Low-frequency TDD	Prohibition of acting as the PCell	PCELL_PROHIBIT_SW option of the NRCellConfig.CellStrategySwitch parameter	<i>Carrier Aggregation</i>	The NSA/SA-specific PCell management policy configuration function takes effect only when the "prohibition of acting as the PCell" function is enabled.
Low-frequency TDD	Dynamic-codebook-based HARQ feedback over the PUCCH	PUCCH_DYNAMIC_HARQ_ACK_CB_SW option of the NRDUCellConfig.EnhancedAlgorithmExtSwitch parameter	<i>Carrier Aggregation</i>	The "dynamic PUCCH codebook specification improvement for CA UEs" function takes effect only when the "dynamic-codebook-based HARQ feedback over the PUCCH" function is enabled.
Low-frequency TDD	Downlink Minimum Carrier Bandwidth	NRDUCellBwp.DlMinCarrierBw	<i>Scalable Bandwidth</i>	Multi-carrier combination selection based on multiple bandwidth levels requires that the NRDUCellBwp.DlMinCarrierBw parameter be set to a value other than NOT_CONFIG .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink Minimum Carrier Bandwidth	NRDUCellBwp.UlMinCarrierBw	<i>Scalable Bandwidth</i>	Multi-carrier combination selection based on multiple bandwidth levels requires that NRDUCellBwp.UlMinCarrierBw be set to a value other than NOT_CONFIG .
Low-frequency TDD	Smart carrier selection	SMART_CARRIER_SEL_SW option of the NRDUCellConfig.SmartCarrierMgmtSw parameter	<i>Carrier Aggregation</i>	Multi-carrier combination selection based on multiple bandwidth levels takes effect only when smart carrier selection is enabled.

5.1.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	For an NR TDD high-speed cell (with NRDUCell.HighSpeedFlag set to HIGH_SPEED), downlink intra-band CA can be enabled only in the following three scenarios: - Hyper cells, for which NRDUCell.NrDualCellNetworkingMode is set to HYPER_CELL - Combined cells in basic or DPS transmission mode, for which NRDUCell.NrDualCellNetworkingMode is set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTRP.DataTransMode is set to BASIC_MODE or DPS_MODE - Common cells, for which NRDUCell.NrDualCellNetworkingMode is set to NORMAL_CELL
Low-frequency TDD	Compensation for SAR decrease (low-frequency TDD)	REDUCE_SAR_COMPENSATION_SW option of the NRDUCellUeCoop.SpecUeCooperationSwitch parameter	<i>Device-Pipe Synergy</i>	Intra-band CA does not work with compensation for specific absorption rate (SAR) decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellIgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	Intra-band CA does not work with the function of low latency and high reliability.
Low-frequency TDD	Optimization of non-contention-based RA beyond cell radius	CONTENTION_FREE_RA_OPT_SW option of the NRDUCellPrach.RachAlgoOptSwitch parameter	<i>Random Access Control</i>	Downlink intra-band CA does not work with optimization of non-contention-based RA beyond cell radius.
Low-frequency TDD	Inter-FR CA	INTER_FR_FR1TOFR2_CARRIER_SWITCH option of the NRDUCellIgoSwitch.CarrierAlgoSwitch parameter	<i>Carrier Aggregation</i>	When service-aware carrier selection is enabled (by selecting the SERVICE_AWARE_CARRIER_SEL_SW option of the NRDUCellCollabService.MultiCarrierMgmtSwitch parameter), inter-FR CA cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	gNodeB Adaptive Fast Processing Switch	GNB_ADAPT_FAST_PROC ESS_SW option of the gNodeBParam.ServiceDiffProcSwitch parameter	None	Dynamic-codebook-based HARQ feedback over the PUCCH (controlled by the PUCCH_DYN_HARQ_ACK_CB_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgorithmSwitch parameter) cannot be enabled when the gNodeB adaptive fast processing switch is on. This switch cannot be turned on when this function is enabled.
Low-frequency TDD	Distributed massive MIMO cell	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgorithmSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	SCell CSI-RS beam sweeping cannot be enabled in distributed massive MIMO cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	One piece of cell resource for CSI-RS	NRDUCellCsirs.CsirsCellResourceNum set to 1_RESOURCE	<i>Beam Management</i>	SCell CSI-RS beam sweeping must be disabled in a cell where one piece of cell resource is used for CSI-RS and either of the following conditions is not met: - The NRDUCellTrp.TxRxMode parameter is set to 64T64R or 128T128R in at least one NRDUCellTrp MO associated with this cell through the NRDUCellCsirs.NrDuCellId parameter. - The NRDUCellCsirs.FR1MaxCellCsirsPortNum parameter is set to 32PORT.
Low-frequency TDD	Experience-based smart carrier selection	MULTI_FREQ_SMART_SEL_SW option of the NRCelAlgoSwitch.MultiFreqAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	If the NRCelCaMgmtConfig.CaBandCombSelectionStrat parameter (specifying the policy on multi-carrier capability combination selection) is set to DEFAULT, experience-based smart carrier selection cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell deployment in IAB mode	NRDUCell.DeployPosition set to DEPLOY_IAB	None	NRDUCell.DeployPosition can be set to DEPLOY_IAB only when downlink intra-band CA is disabled.
Low-frequency TDD	Downlink 8-layer transmission for a single FWA user	DL_SU_EIGHT_LAYER_SW option of the NRDUCellDLRank.DlRankSelAlgoSw parameter	<i>FWA</i>	Dynamic-codebook-based HARQ feedback over the PUCCH does not work with downlink 8-layer transmission for a single FWA user.
High-frequency TDD	None	None	None	None

5.1.3.2.3 Function Impacts

- Impact relationships with basic functions under RAN functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Cell-level PUCCH time-frequency format	None	<i>Channel Management</i>	The NRDUCellPucch.St ructureType parameter must be set to LONG_STRUCTURE for a low-frequency cell. If it is set to SHORT_STRUCTURE, intra-band CA does not take effect. For a high-frequency cell, intra-band CA still takes effect when the NRDUCellPucch.St ructureType parameter is set to SHORT_STRUCTURE.
Low-frequency TDD	SRS period	NRDUCellSrs.SrsPeriod	<i>Channel Management</i>	When the SRS period (specified by the NRDUCellSrs.SrsPeriod parameter) in the PCell of an antenna-selection UE is set to 5 ms or 10 ms, SRS resources may fail to be allocated to the UE in its SCells. If this is the case, user experience is affected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink enhanced scheduling of aperiodic CSI reports	APER_CSI_RPT_ENHANCE_SW option of the NRDUCellConfig.CsiSwitch parameter	<i>Uplink Scheduling</i>	The status (enabled or disabled) of uplink enhanced scheduling of aperiodic CSI reports must be consistent between the PCell and all SCells of a CA UE. Status inconsistency affects the downlink throughput of the UE.
Low-frequency TDD	PDCCH rate matching	PDCCH_RATEMATCH_SW option of the NRDUCellPDSCH.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	PDCCH rate matching does not take effect when SCells of CA UEs are transmitting the common PDCCH. It is recommended that common PDCCH rate matching in SCells (controlled by the SCC_COMMON_PDCCH_RM_SW option of the NRDUCellCarrierManagement.CaEnhancedAlgorithmExtSwitch parameter) be enabled, to increase the peak data rates in the SCells.
High-frequency TDD	DL 256QAM	NRDUCellAdaptiveSwitch.DL256QAMSwitch	<i>Modulation Schemes</i>	With DL 256QAM enabled in high-frequency TDD scenarios, a larger value of the NRDUCellCarrierManagement.DLStateTransitBufferVolThld parameter results in a lower probability of SCell activation.

- Impact relationships with functions related to UCN under RAN functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UL CoMP	INTRA_GNB_UL_COMP_SW option of the NRDUCellA lgoSwitch.CompSwitch parameter	<i>CoMP</i>	Uplink coordinated multipoint reception (UL CoMP) cannot be performed for CA UEs in their SCells.
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellA lgoSwitch.CompSwitch parameter	<i>CoMP</i>	When CA is enabled, intra-base-station joint transmission cannot be performed for CA UEs in their SCells.
Low-frequency TDD	Joint Transmission Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE and NRDUCellM ultiTrp.DataTransMode set to JOINT_MODE	<i>Cell Combination</i>	In low-speed scenarios, downlink intra-band 2CC aggregation is supported by combined cells in joint transmission mode (that is, cells with Joint Transmission Cell Combination enabled) that are served by UBBPg or UBBPfw series baseband processing units. If downlink intra-band CA is enabled for combined cells in joint transmission mode, candidate SCells must also be combined cells in joint transmission mode.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	When downlink intra-band CA is enabled, Co-MM cannot be performed for CA UEs in their SCells.
Low-frequency TDD	Hyper Cell Cell Combination	NRDUCell.NrDuCellNetworkingMode	<i>Hyper Cell Cell Combination</i>	The cells involved in downlink intra-band CA cannot include both hyper cells and combined cells. (Combined cells are those with NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTrp.DataTransMode set to BASIC_MODE or DPS_MODE .)

- Impact relationships with functions related to energy saving

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Power saving BWP (low-frequency TDD)	BWP2_SWITCH option of the NRDUCellUePwrSaving . BwpPwrSavingSw parameter	<i>UE Power Saving</i>	BWP2, which provides a narrow bandwidth, is not configured for CA UEs in their SCells. When power saving BWP or "QCI-level threshold control for BWP switching" is enabled together with CA, there are the following impacts: • A larger value of NRDUCellCarrMgmt.CaDlScellActivateThldSlotNum results in a higher traffic volume threshold for UE switching from BWP2 to BWP1 and therefore a lower probability of the switching, which decreases UE power consumption. • A larger value of NRDUCellCarrMgmt.CaDlScellDeactivateThldSlotNum results in a higher traffic volume threshold for UE switching from BWP1 to BWP2 and therefore a higher probability of the switching, which decreases UE power consumption. • A larger value of NRDUCellCarrM

RAT	Function Name	Function Switch	Reference	Description
				gmt.DlStateTransitionBufVolThld or NRDUCellQciBearer.QciDlStateTransitionBufVolThld results in a lower probability of UE switching from BWP2 to BWP1 and a higher probability of UE switching from BWP1 to BWP2. As a result, UEs stay in BWP2 for a longer time and their power consumption decreases. • A smaller value of NRDUCellCarrierMgmt.DlStateTransitionBufVolThld or NRDUCellQciBearer.QciDlStateTransitionBufVolThld results in a higher probability of UE switching from BWP2 to BWP1 and a lower probability of UE switching from BWP1 to BWP2. As a result, UEs stay in BWP2 for a shorter time and their power consumption increases. BWP switching and RRC reconfiguration may occur more

RAT	Function Name	Function Switch	Reference	Description
				frequently. For cell-edge UEs under weak coverage, the service drop rate may increase. • SCells are more likely to be deactivated if the NRDUCellCarrMgmt.RlcDLThroughputThld parameter value is less than the NRDUCellUePwrSaving.Bwp1ToBwp2DLThptThld or NRDUCellQciBearer.QciBwp1ToBwp2DLThptThld parameter value. If NRDUCellCarrMgmt.RlcDLThroughputThld is set to 0, SCells will not be deactivated and UEs will not switch to BWP2.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Time-domain power saving BWP (low-frequency TDD)	TDM_BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	BWP2 is not configured for CA UEs in their SCells. When CA and time-domain power saving BWP are both enabled, there are the following impacts: • A larger value of NRDUCellCarrMgmt.CaDisCellActThldSlotNum results in a higher traffic volume threshold for UE switching from BWP2 to BWP1 and therefore a lower probability of the switching, which decreases UE power consumption. • A larger value of NRDUCellCarrMgmt.CaDisCellDeactThldSlotNum results in a higher traffic volume threshold for UE switching from BWP1 to BWP2 and therefore a higher probability of the switching, which decreases UE power consumption. • A larger value of NRDUCellCarrMgmt.DlStateTransitBufVolThld results in a lower probability

RAT	Function Name	Function Switch	Reference	Description
				of UE switching from BWP2 to BWP1 and a higher probability of UE switching from BWP1 to BWP2. As a result, UEs stay in BWP2 for a longer time and their power consumption decreases. • A smaller value of NRDUCellCarrMgmt.DlStateTransitBufVolThld results in a higher probability of UE switching from BWP2 to BWP1 and a lower probability of UE switching from BWP1 to BWP2. As a result, UEs stay in BWP2 for a shorter time and their power consumption increases. BWP switching and RRC reconfiguration may occur more frequently. For cell-edge UEs under weak coverage, the service drop rate may increase. • SCells are more likely to be deactivated if the NRDUCellCarrM

RAT	Function Name	Function Switch	Reference	Description
				gmt.RlcDlThroughputThld parameter value is less than the NRDUCellUePowerSaving.Bwp1ToBwp2DlThptThld parameter value. If NRDUCellCarrMgmt.RlcDlThroughputThld is set to 0 , SCells will not be deactivated and UEs will not switch to BWP2.
Low-frequency TDD	Intelligent carrier shutdown	INTRA_GNB_MULTI_CA_RR_SD_SW or INTER_GNB_MULTI_CA_RR_SD_SW option of the NRDUCellAgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When intelligent carrier shutdown takes effect, the capacity-layer cells that have been shut down cannot be configured as SCells for UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DRX	NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch	<i>DRX</i>	Data split for a UE is affected if the UE is in different discontinuous reception (DRX) states in its PCell and SCells. When intra-band CA and DRX are both enabled, it is recommended that collaboration between CA and DRX be enabled. The collaboration is controlled by the CA_AND_DRX_STATUS_COLLAB_SWITCH option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter.
Low-frequency TDD	Maximum-bandwidth reduction	OVERHEATING_ASSISTANCE_INFO_SW and UE_BW_REDUCE_SW options of the NRCelAlgoSwitch.UeAssistanceInfoSwitch parameter	<i>UE Power Saving</i>	When a UE requests maximum-bandwidth reduction, the base station deactivates all SCells of the UE in multi-carrier scenarios.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH skipping	PDCCH_SKIP_SW option of the NRDUCellUePwrSaving.SchPwrSavingSw parameter	<i>UE Power Saving</i>	With both downlink intra-band CA and PDCCH skipping enabled: • If the CA_UE_PDCCH_SKIP_SW option of the NRDUCellUePwrSaving.SchPwrSavingSw parameter is deselected, PDCCH skipping does not take effect for UEs with SCells configured. • If the CA_UE_PDCCH_SKIP_SW option of the NRDUCellUePwrSaving.SchPwrSavingSw parameter is selected, PDCCH skipping can take effect for CA UEs with SCells not activated, but not for CA UEs with SCells activated.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SSSG switching	UE_PWR_SAVING_ENHANCE_SW and SSSG_SWITCHING_SW options of the NRDUCellUePwrSaving . <i>UePowerSavingSwitch</i> parameter	<i>UE Power Saving</i>	With both downlink intra-band CA and search space set group (SSSG) switching enabled, a sparse SSSG can be configured for PCCs but not for SCCs. In this case, SCell activation for CA requires switching from a sparse SSSG to a dense SSSG and therefore is delayed.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast carrier shutdown	Capacity-layer cells: INTRA_GNB_MULTICARRIER_SD_SWITCH option (intra-base-station multi-carrier scenarios) or INTER_GNB_MULTICARRIER_SD_SWITCH option (inter-base-station multi-carrier scenarios) of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter FAST_CARRIER_SHUTDOWN_SWITCH option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter Basic-layer cells:	<i>Energy Conservation and Emission Reduction</i>	After fast carrier shutdown takes effect, the capacity-layer cells that have been shut down cannot be configured as SCells for UEs.

RAT	Function Name	Function Switch	Reference	Description
		FAST_CARR_SHUT_SERV_ASSUR_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter		
Low-frequency TDD	Symbol power saving	NRDUCellAlgoSwitch.SymbolShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	After dynamic PUCCH codebook specification improvement for CA UEs takes effect, the number of scheduling times for CA UEs increases. This shortens the symbol power saving duration of a cell (N.PowerSaving.SymbolShutdown.Duration).
Low-frequency TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATS parameter	<i>Multi-RAT Coordinated Energy Saving</i>	After RF module fast deep dormancy takes effect, capacity-layer cells that have been shut down cannot be configured as SCells for UEs.

- Impact relationships with functions related to RAN services

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS field strength-based positioning	• SRS_RSRP_LOCATION_SW option of the NRCellAlgSwitch.LcsSwitch parameter • gNBRSFPConfig.UserServiceType set to LCS	<i>Positioning</i>	Downlink intra-band CA does not take effect for UEs whose UE service type is LCS when SRS field strength-based positioning is enabled and the COMMON_USER_LCS_SW option of the NRDUCellPosAlg.LcsMeasureAlgSwitch parameter is deselected.
Low-frequency TDD	Precise positioning (LampSite NR)	• UTDOA_SW option of the NRCellAlgSwitch.LcsSwitch parameter • gNBRSFPConfig.UserServiceType set to LCS	<i>Positioning</i>	Downlink intra-band CA does not take effect for UEs whose UE service type is LCS when precise positioning (LampSite NR) is enabled and the COMMON_USER_LCS_SW option of the NRDUCellPosAlg.LcsMeasureAlgSwitch parameter is deselected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Precise positioning (NR)	• AOA_UTDOA_HYBRID_LOCATION_SWITCH option of the NRCellAIgoSwitch.LcsSwitch parameter • gNBRfspConfig.UserServiceType set to LCS	<i>Positioning</i>	Downlink intra-band CA does not take effect for UEs whose UE service type is LCS when precise positioning (NR) is enabled and the COMMON_USER_LCS_SW option of the NRDUCellPosAlgo.LcsMeasureAlgoSw parameter is deselected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-base-station CA downlink data split for Cloud X services	CLOUD_X_INTRA_GNB_CA_ENH_S W option of the NRDUCellA lgoSwitch.C loudVrSchS witch parameter	<i>Ultimate Cloud X Experience</i>	When both intra-base-station CA downlink data split for Cloud X services and distributed data volume-triggered SCell deactivation (controlled by the SPLIT_VOL_BASED_SCELL_DEACT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter) are enabled: - A larger value of the gNBQciBearer.CloudServSplitCoefficient parameter results in a higher CA downlink data split threshold for Cloud X UEs. If the buffered data volume is lower than the data split threshold for a long time, SCells will be deactivated and the values of the N.CA.SCell.Deact.Att and N.CA.SCell.Deact.Succ counters will increase. - A smaller value of the gNBQciBearer.CloudServSplitCoefficient parameter results in a lower CA

RAT	Function Name	Function Switch	Reference	Description
				downlink data split threshold for Cloud X UEs, a longer SCell-activated duration, and higher UE power consumption.
Low-frequency TDD	CA prohibition for VoNR UEs	VONR_NOT_CONFIG_CA_SW option of the NRCellCaMgmtConfig.CaStrategySwitch parameter	<i>VoNR</i>	When CA prohibition for VoNR UEs is enabled, CA does not take effect for VoNR UEs.

- Impact relationships with functions related to carrier management under RAN functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Experience-based smart carrier selection	MULTI_FREQ_SMART_SELECTION_SW option of the NRCellAlgoSwitch.MultiFreqAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	If both smart carrier selection (controlled by the SMART_CARR_SELECTION_SW option of the NRDUCellCollabServ.MultiCarrMgmtSw parameter) and experience-based smart carrier selection are enabled, inter-frequency measurement configurations related to event A5 for experience-based smart carrier selection are preferentially delivered.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink multi-frequency coordination	MULTI_FREQ_UL_COORD_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter	<i>Multi-Frequency Smart Selection</i>	The gNodeB does not select UEs for which downlink CA has taken effect.
Low-frequency TDD	Inter-FR CA	INTER_FR_FR1TOFR2_CA_SW option of the NRDUCellAlgoSwitch.CarrierAggregationSwitch parameter	<i>Carrier Aggregation</i>	For a UE in the intra-band CA state: • If the band +bandwidth combination for CA at the UE meets the band +bandwidth requirements of inter-FR CA, inter-FR CA can simultaneously take effect for the UE. If this is the case, downlink CA instead of uplink CA can take effect between the two low-frequency cells. • Otherwise, inter-FR CA cannot simultaneously take effect for the UE.
Low-frequency TDD	Multi-frequency beam coordination	MULTI_FREQ_BEAM_COORD_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter	<i>Multi-Frequency Smart Selection</i>	The gNodeB does not select downlink CA UEs whose SCells perform SRS measurement.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CCE-based data split stop for CA UEs	CCE_BASED_CA_SPLIT_STOP_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	CCE-based data split stop for CA UEs takes effect only for UEs in the downlink CA state. After this function takes effect, data scheduling for CA UEs with SCells activated may accelerate and traffic volume-triggered SCell deactivation is more likely to take effect.
Low-frequency TDD	SRS carrier switching	SRS_CARRIER_SWITCHING_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter	<i>CA Experience Improvement</i>	Dynamic-codebook-based HARQ feedback over the PUCCH does not take effect for UEs for which SRS carrier switching has taken effect and from which <i>switchingTimeUL</i> with a value other than 0 μ s is reported.
Low-frequency TDD	Massive CA	NRDUCellCarrMgmt.CaDLMaxCNum set to DL4CC or DL5CC	<i>CA Experience Improvement</i>	Dynamic-codebook-based HARQ feedback over the PUCCH does not take effect for UEs in the massive CA state.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-FR CA coverage enhancement	PUCCH_GROUP_ADAPT_SW option of the NRDUCellCarrierMgmt.InterFrCaEnhSw parameter	<i>Carrier Aggregation</i>	Dynamic-codebook-based HARQ feedback over the PUCCH does not take effect for inter-FR CA UEs for each of which a single set of PUCCH-based feedback mechanisms is used.
Low-frequency TDD	Experience-based smart carrier selection	MULTI_FREQ_SMART_SELECTOR_SW option of the NRCellAlgoSwitch.MultiFreqAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	Assume that both service-aware carrier selection (controlled by the SERVICE_AWARE_CARR_SEL_SW option of the NRDUCellCollabServ.MultiCarrMgmtSw parameter) and experience-based smart carrier selection are enabled. Experience-based smart carrier selection can take effect to trigger carrier changes for a UE only if the UE has not been considered as a target UE for service-aware carrier selection.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SCC coverage expansion	- SA networking: SA_SCC_COV_THLD_ADAPT_SW option of the NRCellAlgoSwitch.MultiFreqReqAlgoSwitch parameter - NSA networking: NSA_SCC_COV_THLD_ADAPT_SW option of the NRCellAlgoSwitch.MultiFreqReqAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	When both selection of the optimal combination of multi-carrier frequencies (controlled by the SMART_CARR_SELECTION_SW option of the NRDUCellCollabServ.MultiCarrMgmtSw parameter) and SCC coverage expansion are enabled, the inter-frequency A5 measurement configurations related to the former function are preferentially delivered.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SCC coverage expansion	- SA networking: SA_SCC_COV_THLD_ADAPT_SW option of the NRCellAlgSwitch.MultiFreqAlgSwitch parameter - NSA networking: NSA_SCC_COV_THLD_ADAPT_SW option of the NRCellAlgSwitch.MultiFreqAlgSwitch parameter	<i>Multi-Frequency Smart Selection</i>	For CA UEs for which SCell coverage expansion has taken effect, the threshold for event A2 related to SCell coverage expansion is used in A2-triggered SCell removal.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	RSRP prediction model optimization	NR_TO_NR_COV_GRID_SW option of the gNodeBAlgo.NrSmartMultiCarrSw parameter	<i>Virtual Grid-based Multi-Frequency Coordination</i>	If the NR_TO_NR_COV_GRID_SW option of the gNodeBAlgo.NrSmartMultiCarrSw parameter is selected, data distribution to SCells is delayed. Therefore, the average downlink throughput of CA UEs, which is equal to $(N.CA.SccActiveUser.ThpVol.DL - N.CA.ThpVol.DL.LastSlot) / N.CA.ThpTime.DL.RmvLastSlot$, slightly decreases in burst service scenarios.

- Impact relationships with functions related to mobility management under RAN functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE-number-based connected mode MLB	• INTER_FREQ_LOAD_BALANCING_CONNECTED_MODE_MLB_SWITCH option of the NRCellInfoMlb parameter • NRCellInfoMlb.TriggerMode set to CELL_USER_NUM	<i>Mobility Load Balancing in Connected Mode</i>	During the procedure of UE-number-based connected mode MLB, the gNodeB can select CA UEs as the target UEs to be handed over only when NRCellInfoMlb.TriggerMode is set to DL_CA , and can only transfer downlink CA UEs for the purpose of uplink MLB. After a CA UE is handed over to the target cell, the CA configuration may no longer be applicable, resulting in reduced uplink and downlink user-perceived rates of the UE.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Spectral-efficiency-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SWITCH option of the NRCellInfo.MlbAlgorithmSwitch parameter NRCellInfo.MlbTriggerMode set to SPECTRUM_EFFECTIVE_USER_NUMBER	<i>Mobility Load Balancing in Connected Mode</i>	During the procedure of spectral-efficiency-based connected mode MLB, the gNodeB can select CA UEs as the target UEs to be handed over only when NRCellInfo.MlbTransferCA is set to DL_CA , and can only transfer downlink CA UEs for the purpose of uplink MLB. After a CA UE is handed over to the target cell, the CA configuration may no longer be applicable, resulting in reduced uplink and downlink user-perceived rates of the UE.
Low-frequency TDD	5QI-specific UE-number-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SWITCH option of the NRCellInfo.MlbAlgorithmSwitch parameter NRCellInfo.MlbTriggerMode set to SPECIFIED_5QI_USER_NUMBER	<i>Mobility Load Balancing in Connected Mode</i>	When selecting UEs to be handed over, the gNodeB does not select UEs on which CA has taken effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Radio-resource-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SWITCH option of the NRCellAlgoSwitch.MlbAlgoSwitch parameter selected, and NRCellMlb.MlbTriggerMode set to RADIO_RESOURCE	<i>Mobility Load Balancing in Connected Mode</i>	The gNodeB does not select UEs that treat the local cell as their PCell or SCell for handovers.
Low-frequency TDD High-frequency TDD	Service-based handover to SCC frequencies	SERV_BASED_HO_TO_SCC_FREQ_SWITCH option of the gNBMobilityCommParam.MobilityAlgoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	CA UEs are allowed to be handed over to cells on their SCC frequencies through service-based inter-frequency handovers only when service-based handover to SCC frequencies is enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NSA/SA selection based on downlink traffic volume	• MOBILITY_TO_EUTRAN_SWITCH option of the NRCellAIgoSwitch.InterRatServiceMobilitySwitch parameter • NSA_SA_DL_SELECTION_OPTION_SWITCH option of the gNodeBParam.NRworkingOptionOptSwitch parameter	<i>NSA-SA Selection Based on User Experience</i>	The gNodeB includes the selected EN-DC combination in the <i>selectedbandCombinationInfoEN-DC-v1540</i> IE of the handover request message. The eNodeB then configures the EN-DC combination for NSA networking. This EN-DC combination remains unchanged before the SCG added by NSA/SA selection based on downlink traffic volume is released. Therefore, the eNodeB cannot add SCells based on the selection by downlink CA.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NSA/SA selection based on uplink coverage	MOBILITY_TO_EU_TRANS_W option of the NRCellAIgoSwitch.InterRatServiceMobilitySw parameter NSA_SA_UL_SEL_OPT_SW option of the gNodeBParam.NetworkinOptionOptSw parameter	<i>NSA-SA Selection Based on User Experience</i>	The gNodeB includes the selected EN-DC combination in the <i>selectedbandCombinationInfoEN-DC-v1540</i> IE of the handover request message. The eNodeB then configures the EN-DC combination for NSA DC. This EN-DC combination remains unchanged before the SCG added by NSA/SA selection based on uplink coverage is released. Therefore, the eNodeB cannot add SCells based on the selection by downlink CA.

- Impact relationships with functions related to interference mitigation under RAN functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Enhanced intra-frequency SSB interference avoidance	INTRA_FREQ_SSB_AVOID_ENH_SW option of the NRDUCellAIgoSwitch.CommChnlInterfAvoidSwitch parameter	<i>3D Networking Experience Improvement</i>	When downlink intra-band CA is enabled, enhanced intra-frequency SSB interference avoidance does not take effect for CA UEs in their SCells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-frequency CSI-RS interference avoidance	INTRA_FREQ_CSIRS_AVOID_SW option of the NRDUCellAsgoSwitch.CommChnlInterfAvoidSwitch parameter	<i>3D Networking Experience Improvement</i>	When downlink intra-band CA is enabled, intra-frequency CSI-RS interference avoidance does not take effect for CA UEs in their SCells.
Low-frequency TDD	Enhanced inter-frequency SSB interference avoidance	INTER_FREQ_SSB_AVOID_ENH_SW option of the NRDUCellAsgoSwitch.CommChnlInterfAvoidSwitch parameter	<i>3D Networking Experience Improvement</i>	When downlink intra-band CA is enabled, enhanced inter-frequency SSB interference avoidance does not take effect for CA UEs in their SCells.
Low-frequency TDD	Inter-frequency CSI-RS interference avoidance	INTER_FREQ_CSIRS_AVOID_SW option of the NRDUCellAsgoSwitch.CommChnlInterfAvoidSwitch parameter	<i>3D Networking Experience Improvement</i>	When downlink intra-band CA is enabled, inter-frequency CSI-RS interference avoidance does not take effect for CA UEs in their SCells.
Low-frequency TDD	Proactive avoidance of remote interference	SRS_RIM_INTERF_AVOID_SW and SRS_INTERF_COORD_S_SLOT_SW options of the NRDUCellSrsSrsAlgoExtSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	2CC aggregation takes effect for fewer UEs when proactive avoidance of remote interference is enabled.

- Impact relationships with functions related to MIMO under RAN functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDSCH MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAlgoSwitch . MuMimoSwitch parameter	<i>MIMO (TDD)</i>	If CA UEs do not transmit SRSs in their SCells, they cannot be paired for multi-user MIMO (MU-MIMO) in the SCells. As a result, the downlink cell capacity decreases.
Low-frequency TDD	Service-based adaptive scheduling	NRDUCellPdsch.DLSchOptTimeThld	<i>MIMO (TDD)</i> <i>Massive MIMO AHR (Low-Frequency TDD)</i>	For UEs in their SCells of downlink intra-band CA, the activation of service-based adaptive scheduling may be affected by the mechanism of data distribution to the SCells.

- Impact relationships with other functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-RAT ANR	NR_NR_ANR_SW option of the NRCelAlgoSwitch . AnrSwitch parameter	<i>ANR</i>	If intra-band CA is disabled, measurement-based SCell configuration does not take effect; therefore, intra-RAT automatic neighbor relation (ANR) will not be triggered by CA. If intra-band CA is enabled but SCells are configured in a blind manner, intra-RAT ANR will not be triggered in this scenario either.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Light-load time sequence optimization	LIGHT_LOAD_TIME_SEQUENCE_OPT_SW option of the NRDUCellAlgorithmSwitch.LowLatencySwitch parameter	None	Light-load time sequence optimization does not take effect for CA UEs.
Low-frequency TDD High-frequency TDD	Network slicing	Multiple parameters are required for network slicing deployment. For details, see <i>Network Slicing</i> .	<i>Network Slicing</i>	The network slices configured for an SCell must include UEs' network slices.
Low-frequency TDD	User-rate-based dynamic network slice resource management	DYNAMIC_NETWORK_SLICE_SW option of the NRDUCellCommonSwitch.ServiceDiffAlgoExtSwitch parameter	<i>Network Slicing</i>	When user-rate-based dynamic network slice resource management is enabled, the NRDUCellQciBearer.DynResDLTargetRate parameter value configured for the PCell also applies to the SCells.
Low-frequency TDD	User-rate-based dynamic QCI resource management	DYNAMIC_QCI_GROUP_SW option of the NRDUCellFeatureSwitch.QciBasedDiffServExpSw parameter	<i>QCI-based Resource Control</i>	When user-rate-based dynamic QCI resource management is enabled, the NRDUCellQciBearer.DynResDLTargetRate parameter value configured for the PCell also applies to the SCells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	QCI-based uplink basic preallocation	UL_PREALLOCATION_QCI_SW option of the gNBDUSchParamGroup.ScheduleOptSwitch parameter	<i>Service-differentiated Experience Guarantee</i>	For CA UEs, QCI-based uplink basic preallocation is performed in their PCells but not SCells.
Low-frequency TDD	Remote Interference Avoidance for SRS Sw	SRS_RIM_INTERF_AVOID_SW option of the NRDUCellSrsSrsAlgoExtSwitch parameter	None	2CC aggregation takes effect for fewer UEs when the SRS remote interference avoidance switch is on.
Low-frequency TDD	UE bandwidth adaptation	UE_BW_ADAPTIVE_SW option of the NRDUCellBwp.BwpConfigSwitch parameter	<i>Scalable Bandwidth</i>	Consider a scenario where UE bandwidth adaptation is enabled and minimum carrier bandwidths (NRDUCellBwp.DlMinCarrierBw and NRDUCellBwp.UlMinCarrierBw) are configured. After multi-carrier combination selection based on multiple bandwidth levels takes effect, if the bandwidth of the PCell in the optimal combination selected based on the CA capability combination policy differs from the bandwidth in use for a UE, the gNodeB instructs the UE to use the optimal bandwidth.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Directional outmigration of cell-center UEs from an extended-range cell	VIRTUAL_D EDICATED_ NET_SW option of the NRCellAlgo Switch . <i>ExtendedCellRangePolSw</i> parameter	<i>Extended Cell Range</i>	Directional outmigration of cell-center UEs from an extended-range cell can take effect for CA PCells but not SCells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-base-station CA downlink data split	CLOUD_X_INTRA_GNB_CA_ENH_S option of the NRDUCellA lgoSwitch.C loudVrSchS witch parameter	<i>Layered QoS</i>	When both intra-base-station CA downlink data split and distributed data volume-triggered SCell deactivation are enabled: - A larger value of the gNBQciBearer.CloudServSplitCoefficient parameter results in a higher CA downlink data split threshold for R18 PDU Set UEs. If the buffered data volume is lower than the data split threshold for a long time, SCells will be deactivated and the values of the N.CA.SCell.Deactivate.Att and N.CA.SCell.Deactivate.Succ counters will increase. - A smaller value of the gNBQciBearer.CloudServSplitCoefficient parameter results in a lower CA downlink data split threshold for R18 PDU Set UEs, a longer SCell-activated duration, and higher UE power consumption.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Flexible experience differentiation	FLEX_EXP_DIFF_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter	<i>Service-differentiated Experience Guarantee</i>	After both "CA" and "flexible experience differentiation" are enabled, the base station estimates the uplink or downlink peak data rate of each carrier of a UE. After both "CA" and "flexible experience differentiation" are enabled, the following rules also apply: (1) If the SCell activation threshold is set to a small value, the estimated uplink or downlink peak UE data rate is more accurate, but the UE power consumption increases. (2) If an SCell cannot be activated because the SCell activation threshold is set to a large value, the estimated uplink or downlink peak UE data rate may be inaccurate. It is recommended that the NRDUCellCarrMgmt.CaDIScellActThldSlotNum and NRDUCellCarrMgmt.CaUIScellActThldSlotNum parameters be set to 0 in this scenario.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UL and DL Decoupling	NRDUCellA lgoSwitch.U lDlDecouplingSwitch	<i>UL and DL Decoupling</i>	For UEs that support both UL and DL Decoupling and intra-band CA, either UL and DL Decoupling or intra-band CA can take effect at a time.

5.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

NOTE

One prerequisite for CA is that the corresponding cells have been set up. The following requirements for base station models, boards, and RF modules are based on this prerequisite.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

In low frequency bands, CA can be performed in one baseband processing unit or between baseband processing units. In high frequency bands, CA can be performed only in one baseband processing unit.

RF Modules

Downlink intra-band CA:

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

SCell PMI-based weight enhancement:

All AAUs that each have 32T32R or more TX/RX channels and can each serve at least two NR TDD carriers support this function. For details, see the technical

specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

SCell CSI-RS beam scheme:

Low-frequency RF modules with at least 16T16R support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.1.3.4 Networking

Frequency

There must be at least two frequencies available on the live network.

(Low-frequency scenarios) The **NRCellFreqRelation** MO must be already configured.

The spacing between the center frequencies of CCs for contiguous intra-band CA must meet the following requirements:

For low-frequency intra-band contiguous CA to take effect, the spacing must be less than or equal to the nominal channel spacing required by section 5.4A.1 "Channel spacing for CA" in 3GPP TS 38.101-1 V15.6.0. Otherwise, the CCs are aggregated for intra-band non-contiguous CA. When being less than the nominal channel spacing, the center frequency spacing must be a multiple of the least common multiple of the channel raster and SCS. The nominal channel spacing is calculated using different formulas depending on the channel raster (defined for each frequency band in Table 5.4.2.3-1 "Applicable NR-ARFCN per operating band" in 3GPP TS 38.101-1 V15.6.0):

- When the channel raster for the frequency band is 100 kHz, the nominal channel spacing is calculated using the following formula 1:

$$\text{Nominal channel spacing} = \left\lceil \frac{BW_{\text{Channel}(1)} + BW_{\text{Channel}(2)} - 2 \lfloor GB_{\text{Channel}(1)} - GB_{\text{Channel}(2)} \rfloor}{0.6} \right\rceil \times 0.3$$

- When the channel raster for the frequency band is not 100 kHz, the nominal channel spacing is calculated using the following formula 2:

$$\text{Nominal channel spacing} = \left\lceil \frac{BW_{\text{Channel}(1)} + BW_{\text{Channel}(2)} - 2 \lfloor GB_{\text{Channel}(1)} - GB_{\text{Channel}(2)} \rfloor}{0.015 \times 2^{n+1}} \right\rceil \times 0.015 \times 2^n, n = \mu_0$$

where

- $BW_{\text{Channel}(1)}$ and $BW_{\text{Channel}(2)}$ are the bandwidths of two CCs, CC 1 and CC 2, respectively.
- μ_0 is the μ value corresponding to the maximum subcarrier spacing (SCS) that is supported by both CC 1 and CC 2.
- $GB_{\text{Channel}(1)}$ and $GB_{\text{Channel}(2)}$ are the minimum guard bands corresponding to μ_0 for CC 1 and CC 2, respectively.

Assume that CC 1 (bandwidth: 60 MHz) and CC 2 (bandwidth: 100 MHz) belong to n41. Their nominal channel spacing is calculated as follows:

1. Determine the formula to be used: Formula 2 is to be used because the channel raster for n41 is not 100 kHz, according to Table 5.4.2.3-1 "Applicable NR-ARFCN per operating band" in 3GPP TS 38.101-1 V15.6.0.

- Determine μ_0 : Both CC 1 and CC 2 support 30 kHz SCS and 60 kHz SCS according to [Figure 5-3](#), which lists the SCSs supported by each bandwidth in n41 (the figure is adapted from Table 5.3.5-1 "Channel bandwidths for each NR band" in 3GPP TS 38.101-1 V15.6.0). The maximum SCS that both CCs support is 60 kHz, which corresponds to the μ value of 2. Therefore, μ_0 equals 2. For the μ value corresponding to each SCS, see Table 4.2-1 "Supported transmission numerologies" in 3GPP TS 38.211 V15.6.0.

Figure 5-3 SCSs for n41

NR Band	SCS (kHz)	UE Channel Bandwidth													
		5 MHz	10 MHz	15 MHz	20 MHz	25 MHz	30 MHz	40 MHz	50 MHz	60 MHz	70 MHz	80 MHz	90 MHz	100 MHz	
n41	15	N/A	Yes	Yes	Yes	N/A	Yes	Yes	Yes	N/A	N/A	N/A	N/A	N/A	
n41	30	N/A	Yes	Yes	Yes	N/A	Yes	Yes	Yes	Yes	N/A	Yes	Yes	Yes	
n41	60	N/A	Yes	Yes	Yes	N/A	Yes	Yes	Yes	Yes	N/A	Yes	Yes	Yes	

- Determine $G_{B_{\text{Channel}(1)}}$ and $G_{B_{\text{Channel}(2)}}$: $G_{B_{\text{Channel}(1)}}$ equals 1530 kHz and $G_{B_{\text{Channel}(2)}}$ equals 1370 kHz according to [Figure 5-4](#), which is adapted from Table 5.3.3-1 "Minimum guardband for each UE channel bandwidth and SCS (kHz)" in 3GPP TS 38.101-1 V15.6.0. (In the case of formula 1, the minimum guard bands for two CCs are also determined using the maximum one of the SCSs supported by both CCs.)

Figure 5-4 Minimum guardband for each UE channel bandwidth and SCS (kHz)

SCS (kHz)	UE Channel Bandwidth													
	5 MHz	10 MHz	15 MHz	20 MHz	25 MHz	30 MHz	40 MHz	50 MHz	60 MHz	70 MHz	80 MHz	90 MHz	100 MHz	
15	242.5	312.5	382.5	452.5	522.5	592.5	552.5	692.5	N/A	N/A	N/A	N/A	N/A	
30	505	665	645	805	785	945	905	1045	825	965	925	885	845	
60	N/A	1010	990	1330	1310	1290	1610	1570	1530	1490	1450	1410	1370	

- Calculate the nominal channel spacing using the formula with $BW_{\text{Channel}(1)}$, $BW_{\text{Channel}(2)}$, $G_{B_{\text{Channel}(1)}}$, $G_{B_{\text{Channel}(2)}}$, and μ_0 determined. The result is 79.8 MHz.

In high-frequency NR TDD, the spacing between the center frequencies of CCs must be less than or equal to the nominal channel spacing required by section 5.4A.1 "Channel spacing for CA" in 3GPP TS 38.101-2 V15.7.0. Otherwise, the CCs are aggregated for intra-band non-contiguous CA. When being less than the nominal channel spacing, the center frequency spacing must be a multiple of the least common multiple of the channel raster and SCS. Use the same method to determine μ_0 for calculating the nominal channel spacing as in low frequency bands.

SCC PMI-based weight enhancement takes effect only in downlink intra-band contiguous CA.

NOTE

If the spacing between the center frequencies of CCs to be aggregated is too small, there will be a relatively large proportion of bandwidth overlap between the PCell and the SCell. This results in UE compatibility risks. To mitigate these risks, a spacing equal to the nominal channel spacing is recommended.

Coverage

The coverage areas provided by the cells involved in CA must overlap. The smaller the overlapping range, the smaller the area where intra-band CA takes effect.

SCC PMI-based weight enhancement takes effect only when SCells are served by the same gNodeB and RF module as PCells.

Cell

The inter-frequency NR cell relationships must be already configured (using the **NRCeIlRelatIOn** MO) in low-frequency scenarios.

For intra-frequency cells, the network must be planned to prevent physical cell identifier (PCI) conflicts.

Downlink intra-band CA does not take effect in NR supplementary uplink (SUL) cells, for which the **NRDUCell.DuplexMode** parameter is set to **CELL_SUL**.

The **NRDUCell.HighSpeedFlag** parameter, if configured, of the cells to be engaged in downlink intra-band CA must meet one of the following requirements before downlink intra-band CA can take effect:

- If the **LOW_HIGH_SPEED_CELL_SPT_CA_SW** option of the **NRCeIlCaMgmtConfig.CaStrategySwitch** parameter is deselected, the **NRDUCell.HighSpeedFlag** parameter must be set to the same value for all these cells.
- If the **LOW_HIGH_SPEED_CELL_SPT_CA_SW** option of the **NRCeIlCaMgmtConfig.CaStrategySwitch** parameter is selected, the **NRDUCell.HighSpeedFlag** parameter can be set to either the same value or different values for these cells. If the values are different between the cells, the value must be **LOW_SPEED** for the PCell, whereas the value **HIGH_SPEED** can be set for some or all SCells. After this option is selected, the number of CA UEs increases, and the **User Downlink Average Throughput (DU)** increases.

Downlink intra-band CA does not take effect for low-frequency cells configured with short PUCCH. This restriction does not apply to high-frequency cells.

In high-frequency scenarios, the PCell and SCells must be served by the same RF module. In low-frequency scenarios, there is no such requirement.

Dynamic-codebook-based HARQ feedback over the PUCCH requires that the PCell be a low-frequency NR TDD cell.

SCell CSI-RS beam sweeping is supported only when the **NRDUCELL.DuplexMode** parameter is set to **CELL_TDD**.

SCell CSI-RS beam sweeping is supported only when the **NRDUCellTrp.TxRxMode** parameter is set to **16T16R** or higher.

5.1.3.5 Others

- For this function to deliver a theoretical peak data rate to SA CA UEs, the maximum bit rate that each CA UE subscribes to in the 5G core network (5GC) cannot be lower than this theoretical value.

- For this NR CA function to deliver a theoretical peak data rate to NSA UEs, the maximum bit rate that each CA UE subscribes to in the evolved packet core (EPC) cannot be lower than this theoretical value.
- UE requirements:
 - UEs must support CA. During initial access, an incoming handover, or an incoming RRC connection reestablishment, a UE reports its CA capabilities (through the *BandCombination* IE in the *UECapabilityInformation* message) to the gNodeB.
 - UEs must support semi-static codebooks (as indicated by *semiStaticHARQ-ACK-Codebook* in the *phy-Parameters* IE) for feedback over the PUCCH.

5.1.4 Operation and Maintenance

5.1.4.1 Data Configuration

5.1.4.1.1 Data Preparation

Before activating this function in low-frequency scenarios, ensure that the **NRCellRelation** and **NRCellFreqRelation** MOs have been configured. For details about the configuration, see *SA Mobility Management in Connected Mode* and *NSA Mobility Management*.

[Table 5-2](#) and [Table 5-3](#) describe the parameters used for function activation and optimization, respectively. The following parameters do not take effect for SUL cells, for which the **NRDUCell.DuplexMode** parameter is set to **CELL_SUL**.

Table 5-2 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
CA Algorithm Switch	NRDUCellAlgoSwitch.CaAlgoSwitch	INTRA_BAND_CA_SW	Select this option.
CA Algorithm Switch	NRDUCellAlgoSwitch.CaAlgoSwitch	INTRA_BAND_UL_CA_SW	In high-frequency NR TDD, select both the INTRA_BAND_CA_SW and INTRA_BAND_UL_CA_SW options of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter.
NR Cell ID	NRCellCaMgmtConfig.NrCellId	N/A	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
CA Strategy Switch	NRCaMgmtConfig.CaStrategySwitch	SCC_BLIND_CONFIG_SW	Select this option if blind SCell configuration is required in low-frequency scenarios. This option does not need to be set in high-frequency scenarios.
CA Frequency Group ID	gNBaFrequency.CaFreqGroupId	N/A	Set this parameter based on the network plan. It is recommended that all frequencies planned for CA be configured in the same frequency group.
Downlink NARFCN	gNBaFrequency.DlNarfcn	N/A	Set this parameter based on the network plan.
SSB Frequency Position	gNBaFrequency.SsbFreqPos	N/A	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
Operator ID List	gNBCaFrequency.OperatorIdList	N/A	Set this parameter based on the network plan. The value of this parameter is expressed in decimal format. The gNodeB converts the decimal number into a 32-bit binary number. If the number is less than 32 bits, the most significant bits are 0s by default. The bits from right to left have a one-to-one relationship with the local operator indexes (specified by the gNBOperator.OperatorId parameter) in ascending order. For example, bit 0 at the rightmost place corresponds to the operator with the gNBOperator.OperatorId parameter set to 0. If a bit is set to 1, the corresponding operator can use the frequency. If a bit is set to 0, the corresponding operator cannot use the frequency. Take the value 5 of the gNBCaFrequency.OperatorIdList parameter as an example. The binary format of this value is 101, indicating that the operators with gNBOperator.OperatorId set to 0 and 2 can use the CA frequency and the operators with gNBOperator.OperatorId set to 1 and values greater than 2 cannot use the CA frequency.

Parameter Name	Parameter ID	Option	Setting Notes
Blind SCell Configuration Flag	NRCellRelation.BlindScellConfigFlag	N/A	To enable blind SCell configuration in low-frequency scenarios, set this parameter to TRUE for each blind-configurable candidate SCell. This option does not need to be set in high-frequency scenarios. You are advised to specify only one cell as a blind-configurable candidate SCell on each candidate SCC frequency. If multiple blind-configurable candidate SCells are specified on a candidate SCC frequency, only one of these cells can be configured as an SCell and this cell is uncertain.
CA A5 RSRP Threshold 2	NRCellCaMgmtConfig.CaA5RsrpThld2	N/A	To trigger SCell configuration only when the SCells provide better service quality, increase the value of this parameter. The value of this parameter must be greater than that of NRCellCaMgmtConfig.CaA2RsrpThld.
CA SCC A5 RSRP Threshold 2 Offset	gNBCaFrequency.CaScCA5RsrpThld2Offset	N/A	To adjust the CA event A5 threshold for different SCCs, set this parameter according to operators' policies.
CA A5 Time-to-Trigger	NRCellCaMgmtConfig.CaA5TimeToTrigger	N/A	Set this parameter based on the network plan. A smaller value of this parameter results in a shorter time required before event A5 can be reported and a greater impact of RSRP fluctuations after the SCell coverage meets the entering conditions for event A5. A larger value of this parameter results in the opposite effects.

Parameter Name	Parameter ID	Option	Setting Notes
CA A6 Time-to-Trigger	NRCellCaMgmtConfig.CaA6TimeToTrigger	N/A	Set this parameter based on the network plan. A smaller value of this parameter results in a shorter time required before intra-frequency event A6 can be reported and a greater impact of RSRP fluctuations after the intra-frequency neighboring cell meets the entering conditions for event A6. A larger value of this parameter results in the opposite effects.
Multi-Carrier Measurement Wait Timer	gNodeBAlgo.MultiCarrierMeasWaitTimer	N/A	Retain the default value. The smaller the value of this parameter, the shorter the measurement waiting time. In scenarios with a large number of candidate frequencies, however, if the value is too small, SCells may not be configured because no measurement reports are received, or better SCells may not be configured because no measurement reports on better candidates are received. As a result, the user-perceived rates of CA UEs decrease. A larger value of this parameter indicates a longer measurement waiting time and results in a higher probability of configuring better SCells but a longer time taken for SCell configuration.
CA A6 Offset	NRCellCaMgmtConfig.CaA6Offset	N/A	To trigger SCell changes only when the neighboring cells of the SCells provide better service quality, increase the value of this parameter.

Parameter Name	Parameter ID	Option	Setting Notes
SCC Deactivation CQI Threshold	NRDUCellCar rMgmt.SccDe activCqiThld	N/A	It is recommended that this parameter be set to 5 . To ensure that SCells are activated only when the channel quality of CA UEs is high, increase the value of this parameter.
CA A2 RSRP Threshold	NRCeCellCaMg mtConfig.Ca A2RsrpThld	N/A	To trigger SCell removals only when the SCells provide poorer service quality, reduce the value of this parameter.
CA A2 RSRP Threshold Offset	gNBCaFreque ncy.CaA2Rsrp ThldOffset	N/A	To adjust the CA event A2 threshold for different SCCs, set this parameter according to operators' policies.
Multi-carrier Flow Control Level	gNodeBAlgo. MultiCarrierF lowCtrlLevel	N/A	Use the recommended value.
Intra-gNodeB CA DL Data Split Threshold	NRDUCellCar rMgmt.Intra GnbCaDlSplit Thld	N/A	The value ADAPTIVE is recommended. Set this parameter to a large value if it is expected that data is distributed to SCells only when the volume of buffered RLC data is large.
DL Intra-gNodeB CA Split Policy	NRDUCellCar rMgmt.DlIntr aGnbCaSplitP olicy	N/A	The value REAL_TIME is recommended.
Split Stop Sch Capb Ratio Threshold	NRDUCellCar rMgmt.SplitS topSchCapbR atioThld	N/A	The value 50 is recommended.
CA Enhanced Algorithm Switch	NRDUCellCar rMgmt.CaEn hancedAlgoS witch	SPLIT_STOP_ OPT_SW	You are advised to select this option.
Customized CA Band Combination ID	gNBCustCaB andComb.Cus tCaBandCom bld	N/A	This parameter identifies an operator-defined CA band +bandwidth combination configured on the base station. Set this parameter as required.

Parameter Name	Parameter ID	Option	Setting Notes
Bandwidth Combination Set	gNBCustCaB andComb.BandwidthCombSet	N/A	Set this parameter as required.
Frequency Band 1	gNBCustCaB andComb.FrequencyBand1	N/A	This parameter specifies CA band 1. Set this parameter as required.
Frequency Bandwidth 1	gNBCustCaB andComb.FrequencyBandwidth1	N/A	This parameter specifies the bandwidth of CA band 1. Set this parameter as required.
Frequency Band 2	gNBCustCaB andComb.FrequencyBand2	N/A	This parameter specifies CA band 2. Set this parameter as required.
Frequency Bandwidth 2	gNBCustCaB andComb.FrequencyBandwidth2	N/A	This parameter specifies the bandwidth of CA band 2. Set this parameter as required.
CA Enhanced Algorithm Extended Switch	NRDUCellCar rMgmt.CaEnhancedAlgorithmSwitch	SCC_PMI_ENH_SW	It is recommended that this option be selected for intra-band contiguous CA.
FR1 CA Downlink TBS Parameter OH	NRDUCellCar rMgmt.Fr1CaDLTbsParamOh	N/A	The value 14 is recommended. To increase the maximum downlink TBS, set this parameter to a smaller value.
FR2 CA Downlink TBS Parameter OH	NRDUCellCar rMgmt.Fr2CaDLTbsParamOh	N/A	The value 18 is recommended. To increase the maximum downlink TBS, set this parameter to a smaller value.

Parameter Name	Parameter ID	Option	Setting Notes
CA Activate and BWP Switch DL Thld Slot Num	NRDUCellCarrMgmt.CaDIScellActThldSlotNum	N/A	The value 16 is recommended for low-frequency NR TDD cells. The value 64 is recommended for high-frequency NR TDD cells. A larger value of this parameter results in a higher downlink threshold for SCell activation and a lower probability of SCell activation. The value of this parameter must be greater than one half of the NRDUCellCarrMgmt.CaDIScellDeactThldSlotNum parameter value. In low-frequency NSA networking, to ensure NR SCell activation when downlink dynamic data split mode is used at the PDCP layer, reduce the value of the NRDUCellCarrMgmt.CaDIScellActThldSlotNum parameter as follows: In the case of the PCell being an NR TDD cell, set the NRDUCellCarrMgmt.CaDIScellActThldSlotNum parameter to a value less than or equal to 20. NOTE Downlink dynamic data split mode is used at the PDCP layer when data is configured as follows: - In Option 3, the NsaDcQciParamGroup.DlDataPdcpsplitMode parameter is set to SCG_AND_MCG, SCG_PREFERRED, or ADAPT_PRIMARY_PATH. - In Option 3x, the gNBpdcpsplitMode parameter is set to SCG_AND_MCG, MCG_PREFERRED, or ADAPT_PRIMARY_PATH.

Parameter Name	Parameter ID	Option	Setting Notes
CA Deactivate and BWP Switch DL Thld Slot Num	NRDUCellCar rMgmt.CaDIS cellDeactThl dSlotNum	N/A	The value 4 is recommended for low-frequency cells. The value 16 is recommended for high-frequency cells. A larger value of this parameter results in a higher downlink threshold for SCell deactivation and a higher probability of SCell deactivation.
DL State Transition Buffer Volume Threshold	NRDUCellCar rMgmt.DIS taTransitBuf VolThld	N/A	- For both NSA and SA networking: The value 20 is recommended for TDD cells. - For the CA activation rate in NSA networking to reach that in SA networking: A value less than 20 is recommended for TDD cells.
Service Differentiation Algorithm Switch	NRDUCellQci Bearer.Servic eDiffAlgoSwi tch	QCI_CA_BWP_ SWITCH_THL D_CTRL_SW	Select this option if the buffered traffic volume threshold for downlink state transitions needs to be set on a per QCI basis.
QCI DL State Transit Buffer Volume Threshold	NRDUCellQci Bearer.QciDL StTransitBuf VolThld	N/A	- For both NSA and SA networking: The value 20 is recommended for TDD cells. - For the CA activation rate in NSA networking to reach that in SA networking: A value less than 20 is recommended for TDD cells.
RLC Downlink Throughput Threshold	NRDUCellCar rMgmt.RlcDL ThroughputThld	N/A	The value 2000 is recommended. A larger value of this parameter results in a higher downlink threshold for SCell deactivation and a higher probability of SCell deactivation.

Parameter Name	Parameter ID	Option	Setting Notes
RLC Buffer Filtering Coefficient	NRDUCellCar rMgmt.RlcBuf ferFilterCoeff	N/A	Use the recommended value. A larger value of this parameter results in a higher level of sensitivity of CA UEs to the instantaneous change in the buffered data volume.
CA SCell Config PRB Usage Threshold	NRCeIlCaMg mtConfig.CaS cellConfigPrb UsageThld	N/A	To prevent cells with high PRB usages from being configured as SCells for CA, reduce the value of this parameter.
CA Enhanced Algorithm Extended Switch	NRDUCellCar rMgmt.CaEn hancedAlgoE xtSwitch	CA_PDCCH_C APACITY_OPT _SWITCH	Use the recommended value.
CA Enhanced Algorithm Extended Switch	NRDUCellCar rMgmt.CaEn hancedAlgoE xtSwitch	SPLIT_VOL_B ASED_SCELL_ DEACT_SW	To further reduce UE power consumption in low-frequency scenarios, it is recommended that this option be selected.
CA Enhanced Algorithm Switch	NRDUCellCar rMgmt.CaEn hancedAlgoS witch	NOT_DEACT_ SCC_BASED_C QI_SW	You are advised to select this option.
CA Compatibility Switch	NRDUCellCar rMgmt.CaCo mpatibilityS witch	BIT_ERROR_S CC_DEACTV_S W	It is recommended that this option be selected in low-frequency scenarios.
PDCCH Outer-Loop Adjustment Upper Limit	NRDUCellPdc chAlgo.Pdcch OuterLoopAd jUpLimit	None	Use the recommended value. When this parameter is set, the NRDUCellPdcch.UeSpecificPd cchAggLvl parameter must be set to ADAPTIVE .
SCC Configuration Interval	NRDUCellCar rMgmt.SccCf gInterval	None	You are advised to use the recommended value.
Dynamic Codebook CA UE Number Threshold	NRDUCellCar rMgmt.DynC odebookCaU eNumThld	None	Set this parameter as required.
CA Enhanced Algorithm Extended Switch	NRDUCellCar rMgmt.CaEn hancedAlgoE xtSwitch	PUCCH_DYN_ HARQ_ACK_C B_SW	You are advised to select this option.

Parameter Name	Parameter ID	Option	Setting Notes
PUCCH Algorithm Switch	NRDUCellPucch.PucchAlgoSwitch	PUCCH_DYN_CB_SPECS_IM_P_SW	You are advised to select this option in medium- and heavy-load scenarios.
SCell Deactivation Wait Timer	NRDUCellCarMgmt.ScellDeactvWaitTimer	None	Set this parameter as required.
CA Compatibility Switch	NRDUCellCarMgmt.CaCompatibilitySwitch	SCELL_DEACT_PROT_SW	Select this option if SCell deactivation protection is required.
CA Strategy Switch	NRCeCellCaMgmtConfig.CaStrategySwitch	HO_IN_SCC_CFG_INHERIT_POL_SW	Set this option as required.
CA Strategy Switch	NRCeCellCaMgmtConfig.CaStrategySwitch	HO_IN_SCC_B_LIND_CONFIG_POL_SW	Set this option as required.
CA Strategy Switch	NRCeCellCaMgmtConfig.CaStrategySwitch	MULTI_CARR_MULTI_BWP_OPT_SW	You are advised to select this option if a relatively large number of UEs on the network exhibit better CA capabilities (for example, more layers) on the minimum carrier bandwidths. It is recommended that this option be deselected in other scenarios.
CA Enhanced Algorithm Extended Switch	NRDUCELLCARRMGMT.CaEnhancedAlgoExtSwitch	PRECISE_SCELL_ACTIVATION_SW	You are advised to select this option.
Reserved Switch Parameter 3	NRDUCellRsvdExt01.RsvdSwParam3	RSVDSWPARAM3_BIT32	Select this option as required if DRX is enabled and the SCell activation or deactivation success rate is relatively low.
Reserved Parameter 15	NRDUCellRsvdExt01.RsvdParam15	None	Set this parameter as required when a large number of UEs on the network are under weak coverage and the SCell activation success rate is relatively low.

Parameter Name	Parameter ID	Option	Setting Notes
CA Enhanced Algorithm Switch	NRDUCellCar rMgmt.CaEn hancedAlgoS witch	CARR_CSI_BE AM_SWEEPIN G_SW	Set this option as required.
CA Enhanced Algorithm Switch	NRDUCellCar rMgmt.CaEn hancedAlgoS witch	CSI_BEAM_S WEEPING_OP T_SW	You are advised to select this option.
CA Enhanced Algorithm Switch	NRDUCellCar rMgmt.CaEn hancedAlgoS witch	CA_CSIRS_BM _SWEEP_AD P_T_SW	You are advised to select this option.
CA Strategy Switch	NRCeIlCaMg mtConfig.CaS trategySwitc h	SMART_CARR _SEL_OPT_SW	Retain the default value.
Statistics Strategy Switch	NRDUCellOa mParam.Stati sticsStrategy Switch	DL_CA_THPT_ L_SLOT_STAT S_POL_SW	Select this option if the average throughput of SCell-activated UEs is expected to more accurately reflect the actual air interface capability of CA UEs.
NR Smart Multi-Carrier Switch	gNodeBAlgo. NrSmartMult iCarrSw	CA_CAP_USER _STATS_POL_ SW	Select this option if it is expected that CA-capable UE counters measure only the numbers of UEs that can actually work in the capability-indicated CA state with the present cell configurations.
CA Strategy Switch	NRCeIlCaMg mtConfig.CaS trategySwitc h	LOW_HIGH_S PEED_CELL_S PT_CA_SW	You are advised to select this option.

Parameter Name	Parameter ID	Option	Setting Notes
FR2 Intra-band CA 200M CC Max Num	gNodeBCarr Mgmt.Fr2Intra BandCa200 MCCMaxNum	None	- When there are four or fewer contiguous high-frequency carriers with a bandwidth of 200 MHz each, it is recommended that this parameter be set to DL4CC_UL4CC. - When there are five or six contiguous high-frequency carriers with a bandwidth of 200 MHz each, it is recommended that this parameter be set to DL6CC_UL4CC. This parameter takes effect only in SA networking.
CA Enhanced Algorithm Switch	gNodeBCarr Mgmt.CaEnhancedAlgoSwitch	UE_NUM_BASED_SCELL_CONFIG_SWITCH	Select this option if the number of UEs that can access the network may be limited (for example, during major sports events).
Forbid 2CC UE Ratio Threshold	NRCaCellMgmtConfig.Forbid2ccUeRatioThld	None	Set this parameter as required. If the number of UEs that can access the network may be limited (for example, during major sports events), this parameter can be set to a value other than 100. If both NRCaCellMgmtConfig.Forbid2ccUeRatioThld and NRCaCellMgmtConfig.Forbid3ccUeRatioThld are not set to 100, the value of NRCaCellMgmtConfig.Forbid2ccUeRatioThld must be greater than that of NRCaCellMgmtConfig.Forbid3ccUeRatioThld.

Parameter Name	Parameter ID	Option	Setting Notes
Forbid 3CC UE Ratio Threshold	NRCeMgmtConfig.FoRbid3ccUeRatioThld	None	Set this parameter as required. If the number of UEs that can access the network may be limited (for example, during major sports events), this parameter can be set to a value other than 100 . If both NRCeMgmtConfig.FoRbid2ccUeRatioThld and NRCeMgmtConfig.FoRbid3ccUeRatioThld are not set to 100 , the value of NRCeMgmtConfig.FoRbid2ccUeRatioThld must be greater than that of NRCeMgmtConfig.FoRbid3ccUeRatioThld .
CA Enhanced Algorithm Switch	gNodeBCarrMgmt.CaEnhancedAlgoSwitch	INTRA_GNB_C RNTI_CON_D TECT_SW	You are advised to select this option.
CA Enhanced Algorithm Switch	NRDUCellCarMgmt.CaEnhancedAlgoSwitch	CRNTI_CONFL ICT_RECONF G_SW	You are advised to select this option.

Table 5-3 Parameters used for optimization

Parameter Name	Parameter ID	Option	Setting Notes
Specified UE Cooperation Switch	NRDUCellUeCoop.SpecUeCooperationSwitch	SPEC_UE_IDE NTIFY_SW	Set this option as required.
Specified UE Cooperation Switch	NRDUCellUeCoop.SpecUeCooperationSwitch	FR1_DL_TBS_ LIMIT_CANCE L_SW	Set this option as required.
Specified UE Cooperation Switch	NRDUCellUeCoop.SpecUeCooperationSwitch	FR2_DL_TBS_ LIMIT_CANCE L_SW	Set this option as required.

Parameter Name	Parameter ID	Option	Setting Notes
CA Enhanced Algorithm Extended Switch	NRDUCellCar rMgmt.CaEn hancedAlgoE xtSwitch	CA_AND_DRX _STATUS_COL LAB_SW	It is recommended that this option be selected when both intra-band CA and DRX are enabled in low-frequency scenarios.
Multi-Carrier Capb Comb Select Strat	NRCeCellCaMg mtConfig.Ca BandCombSe lectionStrat	N/A	Set this parameter as required.
Multi-Carrier Compatibility Switch	NRCeCellCaMg mtConfig.Ca Compatibility Switch	SRS_WT_BASE D_CARR_SEL_ SW	It is recommended that this option be selected only when a UE reports its capability of multiple CA band+bandwidth combinations that differ only in the SRS carrier switching capability.
Multi-Carrier Management Switch	NRDUCellCol labServ.Multi CarrMgmtSw	SMART_CARR _SEL_SW	You are advised to select this option if the number of an operator's frequencies is greater than the number of frequencies that can be aggregated for a UE.
Multi-Carrier Management Switch	NRDUCellCol labServ.Multi CarrMgmtSw	SERVICE_AWA RE_CARR_SEL_ SW	You are advised to select this option if UEs in a cell have various requirements and the number of an operator's frequencies is greater than the number of frequencies that can be aggregated for a UE.
CA Compatibility Switch	NRDUCellCar rMgmt.CaCo mpatibilityS witch	EARLY_SPLIT_ SW	You are advised to select this option.
CA Enhanced Algorithm Extended Switch	NRDUCellCar rMgmt.CaEn hancedAlgoE xtSwitch	SCC_COMMO N_PDCCH_RM _SW	It is recommended that this option be selected when both intra-band CA and PDCCH rate matching are enabled in low-frequency scenarios.
CA Strategy Switch	NRCeCellCaMg mtConfig.CaS trategySwitc h	PCELL_PROHI BIT_SW	You are advised to use the recommended value.

Parameter Name	Parameter ID	Option	Setting Notes
Specific UE PCell Policy	NRCeCellCaMg mtConfig.Spe cUePcellPolic y	None	Set this parameter based on the network plan.

5.1.4.1.2 Using MML Commands

Before using MML commands, refer to [5.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the downlink intra-band CA algorithm switch in low-frequency scenarios
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CaAlgoSwitch=INTRA_BAND_CA_SW-1;
//Turning on both the downlink intra-band CA switch and the uplink intra-band CA switch in high-
frequency NR TDD
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
CaAlgoSwitch=INTRA_BAND_CA_SW-1&INTRA_BAND_UL_CA_SW-1;
//(Using low-frequency 2CC aggregation as an example) Configuring the CA frequency group and its
frequencies, the offsets for event A2 and event A5, and the operators allowed to use the frequencies (all
operators in this example). The DfNarfcn and SsbFreqPos parameter values must be the same as those in
the NRDUCell MO.
ADD GNBFAFREQUENCY: CaFreqGroupId=0, DfNarfcn=630000, SsbFreqPos=7812, CaA2RsrpThldOffset=0,
CaSccA5RsrpThld2Offset=0, OperatorIdList=4294967295;
ADD GNBFAFREQUENCY: CaFreqGroupId=0, DfNarfcn=636664, SsbFreqPos=7881, CaA2RsrpThldOffset=0,
CaSccA5RsrpThld2Offset=0, OperatorIdList=4294967295;
//(Using high-frequency 4CC aggregation as an example) Configuring the CA frequency group and its
frequencies, the offsets for event A2 and event A5, and the operators allowed to use the frequencies (all
operators in this example). The DfNarfcn and SsbFreqPos parameter values must be the same as those in
the NRDUCell MO.
ADD GNBFAFREQUENCY: CaFreqGroupId=1, DfNarfcn=2069165, SsbFreqPos=22435, CaA2RsrpThldOffset=0,
CaSccA5RsrpThld2Offset=0, OperatorIdList=4294967295;
ADD GNBFAFREQUENCY: CaFreqGroupId=1, DfNarfcn=2072497, SsbFreqPos=22447, CaA2RsrpThldOffset=0,
CaSccA5RsrpThld2Offset=0, OperatorIdList=4294967295;
ADD GNBFAFREQUENCY: CaFreqGroupId=1, DfNarfcn=2075829, SsbFreqPos=22458, CaA2RsrpThldOffset=0,
CaSccA5RsrpThld2Offset=0, OperatorIdList=4294967295;
ADD GNBFAFREQUENCY: CaFreqGroupId=1, DfNarfcn=2079161, SsbFreqPos=22470, CaA2RsrpThldOffset=0,
CaSccA5RsrpThld2Offset=0, OperatorIdList=4294967295;
//Setting the measurement waiting timer for multi-carrier operations
MOD GNODEBALGO: MultiCarrierMeasWaitTimer=30;
//Setting CA-related parameters
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaA2RsrpThld=-109, CaA5RsrpThld2=-105,
CaA5TimeToTrig=640MS, CaA6Offset=2, CaA6TimeToTrig=640MS;
//(Optional for low-frequency cells; these parameters do not need to be set for high-frequency cells, where
blind SCell configuration is the only SCell configuration function and takes effect by default) Setting
parameters related to blind SCell configuration
//Turning on the switch of blind SCell configuration
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=SCC_BLIND_CONFIG_SW-1;
//Setting BlindScellConfigFlag to TRUE
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1234, CellId=1,
```

```

BlindScellConfigFlag=TRUE;
//Setting the SCC configuration interval
MOD NRDUCELLCARRMGMT: NrDuCellId=0, SccCfgInterval=8;
//Setting the level of flow control against multi-carrier operations at the gNodeB
MOD GNODEBALGO: MultiCarrierFlowCtrlLevel=MEDIUM;
//Setting the CQI threshold for SCell deactivation
MOD NRDUCELLCARRMGMT: NrDuCellId=0, SccDeactvCqiThld=5;
//Enabling CSI-RS beam sweeping
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=CARR_CSI_BEAM_SWEEPING_SW-1;
//((Optional) Enabling CSI-RS beam sweeping optimization and adaptive CSI-RS beam sweeping
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
CaEnhancedAlgoSwitch=CSI_BEAM_SWEEPING_OPT_SW-1&CA_CSIRS_BM_SWEEP_ADPT_SW-1;
//Setting the downlink data split threshold for intra-gNodeB CA
MOD NRDUCELLCARRMGMT: NrDuCellId=0, IntraGnbCaDlSplitThld=ADAPTIVE;
//Enabling real-time data split by RLC for downlink intra-gNodeB CA
MOD NRDUCELLCARRMGMT: NrDuCellId=0, DlIntraGnbCaSplitPolicy=REAL_TIME;
//Enabling CA data split stop based on the scheduling capability ratio
MOD NRDUCELLCARRMGMT: NrDuCellId=0, SplitStopSchCapbRatioThld=50;
//((Optional) Enabling CA UE data split stop optimization
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=SPLIT_STOP_OPT_SW-1;
//((Optional; TDD) Configuring an operator-defined CA band+bandwidth combination
ADD GNBCUSTCABANDCOMB: CustCaBandCombId=1, BandwidthCombSet=0, FrequencyBand1=N77,
FrequencyBandwidth1=CELL_BW_100M, FrequencyBand2=N77, FrequencyBandwidth2=CELL_BW_90M;
//((Optional) Enabling non-CQI-based SCell deactivation for CA UEs performing downlink large-packet
services
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=NOT_DEACT_SCC_BASED_CQI_SW-1;
//Enabling consecutive residual block error-triggered SCell deactivation in low-frequency scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaCompatibilitySwitch=BIT_ERROR_SCC_DEACTV_SW-1;
//Enabling distributed data volume-triggered SCell deactivation in low-frequency scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
CaEnhancedAlgoExtSwitch=SPLIT_VOL_BASED_SCELL_DEACT_SW-1;
//Setting the SCell deactivation waiting timer in low-frequency scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=0, SCellDeactvWaitTimer=10;
//((Optional) Enabling SCell deactivation protection in low-frequency scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaCompatibilitySwitch=SCELL_DEACT_PROT_SW-1;
//Enabling SCC PMI-based weight enhancement
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoExtSwitch=SCC_PMI_ENH_SW-1;
//Setting the downlink traffic volume thresholds for SCell activation and deactivation in low-frequency
scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaDlScellActThldSlotNum=16, CaDlScellDeactThldSlotNum=4;
//Setting the downlink traffic volume thresholds for SCell activation and deactivation in high-frequency NR
TDD
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaDlScellActThldSlotNum=64, CaDlScellDeactThldSlotNum=16;
//Setting the cell-level threshold of buffered traffic volume for downlink state transitions related to CA
SCells in NR TDD scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=0, DlStateTransitBufVolThld=20;
//Turning on QCI_CA_BWP_SWITCH_THLD_CTRL_SW in low-frequency NR TDD scenarios if the buffered
traffic volume threshold for downlink state transitions needs to be set on a per QCI basis
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=80,
ServiceDiffAlgoSwitch=QCI_CA_BWP_SWITCH_THLD_CTRL_SW-1;
//Setting the QCI-level threshold of buffered traffic volume for downlink state transitions related to CA
SCells in low-frequency NR TDD scenarios
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=80, QciDlStTransitBufVolThld=20;
//Setting the RlcBufferFilterCoeff parameter
MOD NRDUCELLCARRMGMT: NrDuCellId=0, RlcBufferFilterCoeff=1;
//Setting the downlink throughput threshold for SCell deactivation
MOD NRDUCELLCARRMGMT: NrDuCellId=0, RlcDlThroughputThld=2000;
//Setting the parameter OH for downlink TBS in low-frequency CA
MOD NRDUCELLCARRMGMT: NrDuCellId=0, Fr1CaDlTbsParamOh=14;
//Setting the parameter OH for downlink TBS in high-frequency CA
MOD NRDUCELLCARRMGMT: NrDuCellId=0, Fr2CaDlTbsParamOh=18;
//Setting the PRB usage threshold for CA SCell configuration
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaScellConfigPrbUsageThld=100;
//Turning on the CA PDCCH capacity optimization switch
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
CaEnhancedAlgoExtSwitch=CA_PDCCH_CAPACITY_OPT_SWITCH-1;
//Setting the PdcchOuterLoopAdjUpLimit parameter
MOD NRDUCELLPDCCH: NrDuCellId=0, UeSpecificPdcchAggLvl=ADAPTIVE;
MOD NRDUCELLPDCCHALGO: NrDuCellId=0, PdcchOuterLoopAdjUpLimit=40;

```

```
//Setting DynCodebookCaUeNumThld (take 120 as an example)
MOD NRDUCELLCARRMGMT: NrDuCellId=0, DynCodebookCaUeNumThld=120;
//Enabling dynamic-codebook-based HARQ feedback over the PUCCH
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoExtSwitch=PUCCH_DYN_HARQ_ACK_CB_SW-1;
//((Optional; recommended in medium- or heavy-load scenarios) Enabling dynamic PUCCH codebook
specification improvement for CA UEs
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=PUCCH_DYN_CB_SPECS_IMP_SW-1;
//Enabling precise SCell activation (The switch must be turned on for both the PCell and SCells.)
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoExtSwitch=PRECISE_SCELL_ACTIVATION_SW-1;
//((Optional) Turning on the switch for SCell activation and deactivation suspension during DRX wakeup
protection if the SCell activation or deactivation success rate is relatively low when DRX is enabled
MOD NRDUCELLRSVDEXT01: NrDuCellId=0, RsvdSwParam3=RSVDSWPARAM3_BIT32-1;
//((Optional) Turning on the switch for SCell activation prohibition under weak coverage if a large number
of UEs on the network are under weak coverage and the SCell activation success rate is relatively low
MOD NRDUCELLRSVDEXT01: NrDuCellId=0, RsvdParam15=27;
//((Optional) Turning on HO_IN_SCC_CFG_INHERIT_POL_SW to enable prohibition of SCell configuration
inheritance during incoming handovers
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=HO_IN_SCC_CFG_INHERIT_POL_SW-1;
//((Optional) Turning on HO_IN_SCC_BLIND_CONFIG_POL_SW to enable prohibition of blind SCell
configuration during incoming handovers
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=HO_IN_SCC_BLIND_CONFIG_POL_SW-1;
//((Optional) Turning on SMART_CARR_SEL_OPT_SW to enable smart carrier selection optimization
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=SMART_CARR_SEL_OPT_SW-1;
//((Optional) Selecting the MULTI_CARR_MULTI_BWP_OPT_SW option (recommended only when a relatively
large number of UEs on the network exhibit better CA capabilities, for example, more layers, on the
minimum carrier bandwidths; not recommended in other scenarios)
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=MULTI_CARR_MULTI_BWP_OPT_SW-1;
//((Optional) Enabling optimized measurement of downlink CA throughput so that the average throughput
of SCell-activated UEs can more accurately reflect the actual air interface capability of CA UEs
MOD NRDUCELLLOAMPARAM: NrDuCellId=0,
StatisticsStrategySwitch=DL_CA_THPT_L_SLOT_STATS_POL_SW-1;
//((Optional) Enabling optimized measurement of the numbers of CA-capable UEs in cells so that CA-
capable UE counters measure only the numbers of UEs that can actually work in the capability-indicated
CA state with the present cell configurations
MOD GNODEBALGO: NrSmartMultiCarrSw=CA_CAP_USER_STATS_POL_SW-1;
//((Optional; recommended) Enabling CA between low- and high-speed cells
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=LOW_HIGH_SPEED_CELL_SPT_CA_SW-1;
//((Optional) Setting Fr2IntraBandCa200MccMaxNum to DL6CC_UL4CC if there are five or six contiguous
high-frequency carriers with a bandwidth of 200 MHz each
MOD GNODEBCARRMGMT: Fr2IntraBandCa200MccMaxNum=DL6CC_UL4CC;
//((Optional) Enabling UE-number-based SCell configuration control if the number of UEs that can access
the network may be limited (for example, during major sports events)
MOD GNODEBCARRMGMT: CaEnhancedAlgoSwitch=UE_NUM_BASED_SCELL_CONFIG_SW-1;
MOD NRCELLCAMGMTCONFIG: NrCellId=0, Forbid2ccUeRatioThld=80, Forbid3ccUeRatioThld=70;
//((Optional; recommended) Turning on the intra-gNodeB C-RNTI conflict detection switch
MOD GNODEBCARRMGMT: CaEnhancedAlgoSwitch=INTRA_GNB_CRNTI_CON_DETECT_SW-1;
//((Optional) Turning on the C-RNTI conflict-triggered reconfiguration switch
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=CRNTI_CONFLICT_RECONFIG_SW-1;
```

Optimization Command Examples

```
//Enabling device-pipe synergy to remove the limit on the maximum downlink peak data rates of UEs
MOD NRDUCELLUECOOP: NrDuCellId=0,
SpecUeCooperationSwitch=SPEC_UE_IDENTIFY_SW-1&FR1_DL_TBS_LIMIT_CANCEL_SW-1&FR2_DL_TBS_LIMI
T_CANCEL_SW-1;
//Enabling collaboration between CA and DRX when both intra-band CA and DRX are enabled
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoExtSwitch=
CA_AND_DRX_STATUS_COLLAB_SW-1;
//Enabling preferential selection of a UE-reported CA capability combination based on the peak data rate
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaBandCombSelectionStrat=PEAK_RATE_FIRST;
//Enabling CA capability combination selection with SRS-based weights considered
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaCompatibilitySwitch=SRS_WT_BASED_CARR_SEL_SW-1;
//Turning on the smart carrier selection switch
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, MultiCarrMgmtSw=SMART_CARR_SEL_SW-1;
//Turning on the service-aware carrier selection switch
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, MultiCarrMgmtSw=SERVICE_AWARE_CARR_SEL_SW-1;
//Turning on the early data split switch
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaCompatibilitySwitch=EARLY_SPLIT_SW-1;
//Turning on the switch for common PDCCH rate matching in SCells in low-frequency scenarios
```

```
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoExtSwitch= SCC_COMMON_PDCCH_RM_SW-1;
//Turning on PCELL_PROHIBIT_SW and configuring the UE-specific PCell policy (If PCELL_PROHIBIT_SW is
turned on for a cell, the cell cannot act as the PCell for CA. Determine the switch setting based on the
network plan.)
MOD NRCELLCAMGMTCONFIG: NrCellId=0,
CaStrategySwitch=PCELL_PROHIBIT_SW-1,SpecUePcellPolicy=NSA_SA;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the downlink intra-band CA algorithm switch in low-frequency scenarios
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CaAlgoSwitch=INTRA_BAND_CA_SW-0;
//Turning off both the downlink intra-band CA switch and the uplink intra-band CA switch in high-
frequency NR TDD
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
CaAlgoSwitch=INTRA_BAND_CA_SW-0&INTRA_BAND_UL_CA_SW-0;
//Turning off the blind SCell configuration switch
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=SCC_BLIND_CONFIG_SW-0;
//Disabling CSI-RS beam sweeping optimization and adaptive CSI-RS beam sweeping
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
CaEnhancedAlgoSwitch=CSI_BEAM_SWEEPING_OPT_SW-0&CA_CSIRS_BM_SWEEP_ADPT_SW-0;
//Disabling CSI-RS beam sweeping
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=CARR_CSI_BEAM_SWEEPING_SW-0;
//Disabling SCell deactivation protection
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaCompatibilitySwitch=SCELL_DEACT_PROT_SW-0;
//Disabling SCC PMI-based weight enhancement
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoExtSwitch=SCC_PMI_ENH_SW-0;
//Turning off the CA PDCCH capacity optimization switch
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
CaEnhancedAlgoExtSwitch=CA_PDCCH_CAPACITY_OPT_SWITCH-0;
//Disabling non-CQI-based SCell deactivation for CA UEs performing downlink large-packet services
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=NOT_DEACT_SCC_BASED_CQI_SW-0;
//Disabling consecutive residual block error-triggered SCell deactivation in low-frequency scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaCompatibilitySwitch=BIT_ERROR_SCC_DEACT_SW-0;
//Disabling distributed data volume-triggered SCell deactivation in low-frequency scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
CaEnhancedAlgoExtSwitch=SPLIT_VOL_BASED_SCELL_DEACT_SW-0;
//Disabling delayed SCell deactivation in low-frequency scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=0, ScellDeactvWaitTimer=0;
//Disabling dynamic-codebook-based HARQ feedback over the PUCCH
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoExtSwitch=PUCCH_DYN_HARQ_ACK_CB_SW-0;
//Disabling dynamic PUCCH codebook specification improvement for CA UEs
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=PUCCH_DYN_CB_SPECS_IMP_SW-0;
//Disabling real-time data split by RLC for downlink intra-gNodeB CA
MOD NRDUCELLCARRMGMT: NrDuCellId=0, DlintraGnbCaSplitPolicy=NOT_CONFIG;
//Disabling CA data split stop based on the scheduling capability ratio
MOD NRDUCELLCARRMGMT: NrDuCellId=0, SplitStopSchCapbRatioThld=0;
//Disabling CA UE data split stop optimization
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=SPLIT_STOP_OPT_SW-0;
//Disabling precise SCell activation
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoExtSwitch=PRECISE_SCELL_ACTIVATION_SW-0;
//Turning off the switch for SCell activation and deactivation suspension during DRX wakeup protection
MOD NRDUCELLRSVDEXT01: NrDuCellId=0, RsvdSwParam3=RSVDSWPARAM3_BIT32-0;
//Turning off the switch for SCell activation prohibition under weak coverage
MOD NRDUCELLRSVDEXT01: NrDuCellId=0, RsvdParam15=0;
//Disabling prohibition of SCell configuration inheritance during incoming handovers
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=HO_IN_SCC_CFG_INHERIT_POL_SW-0;
//Disabling prohibition of blind SCell configuration during incoming handovers
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=HO_IN_SCC_BLIND_CONFIG_POL_SW-0;
//Turning off the switch for multi-carrier combination selection based on multiple bandwidth levels
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=MULTI_CARR_MULTI_BWP_OPT_SW-0;
//Disabling smart carrier selection optimization
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=SMART_CARR_SEL_OPT_SW-0;
//Disabling optimized measurement of downlink CA throughput
MOD NRDUCELLLOAMPARAM: NrDuCellId=0,
```

```
StatisticsStrategySwitch=DL_CA_THPT_L_SLOT_STATS_POL_SW-0;  
//Disabling optimized measurement of the numbers of CA-capable UEs in cells  
MOD GNODEBALGO: NrSmartMultiCarrSw=CA_CAP_USER_STATS_POL_SW-0;  
//Disabling CA between low- and high-speed cells  
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=LOW_HIGH_SPEED_CELL_SPT_CA_SW-0;  
//Setting Fr2IntraBandCa200MccMaxNum to DL4CC_UL4CC  
MOD GNODEBCARRMGMT: Fr2IntraBandCa200MccMaxNum=DL4CC_UL4CC;  
//Disabling UE-number-based SCell configuration control  
MOD GNODEBCARRMGMT: CaEnhancedAlgoSwitch=UE_NUM_BASED_SCELL_CONFIG_SW-0;  
MOD NRCELLCAMGMTCONFIG: NrCellId=0, Forbid2ccUeRatioThld=100, Forbid3ccUeRatioThld=100;  
//Turning off the intra-gNodeB C-RNTI conflict detection switch  
MOD GNODEBCARRMGMT: CaEnhancedAlgoSwitch=INTRA_GNB_CRNTI_CON_DETECT_SW-0;  
//Turning off the C-RNTI conflict-triggered reconfiguration switch  
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=CRNTI_CONFLICT_RECONFIG_SW-0;
```

5.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.1.4.2 Activation Verification

Provided that CA UEs are available on the network, downlink intra-band CA has taken effect if any counter in [Table 5-4](#) returns a non-zero value.

Table 5-4 Counters related to downlink intra-band CA

Counter ID	Counter Name
1911816773	N.User.CA.PCell.DL.Avg
1911853842	N.User.CA.PCell.DL.Max
1911816774	N.User.CA.SCell.DL.Avg
1911816900	N.User.CA.PCell.DL.SCellAct.Avg
1911816902	N.User.CA.SCell.DL.Act.Avg
1911827213	N.User.CA.PCell.DL.SCellAct.Max
1911840463	N.CA.PRB.DL.PCell.Used.Avg
1911840465	N.CA.PRB.DL.SCell.Used.Avg

The counters related to the numbers of attempts and successful executions regarding SCell configuration, change, activation, deactivation, and removal are listed in [Table 5-5](#).

Table 5-5 Counters related to SCell management

Counter ID	Counter Name
1911816777	N.CA.SCell.Add.Att
1911816778	N.CA.SCell.Add.Succ
1911816781	N.CA.SCell.Mod.Att
1911816782	N.CA.SCell.Mod.Succ
1911816896	N.CA.SCell.Act.Att
1911816897	N.CA.SCell.Act.Succ
1911816898	N.CA.SCell.Deact.Att
1911816899	N.CA.SCell.Deact.Succ
1911816779	N.CA.SCell.Rmv.Att
1911816780	N.CA.SCell.Rmv.Succ

5.1.4.3 Network Monitoring

Observe the average downlink UE throughput (**User Downlink Average Throughput (DU)**) to check whether the UE data rate increases after this function is enabled.

Observe the average throughput of SCell-configured CA UEs, which is calculated using the following formula: $(N.CA.SccConfigUser.ThpVol.DL - N.CA.SccConfigUser.ThpVol.DL.LastSlot) / N.CA.SccConfigUser.ThpTime.DL.RmvLastSlot$.

Observe the average throughput of SCell-activated CA UEs, which is calculated using the following formula: $(N.CA.SccActiveUser.ThpVol.DL - N.CA.ThpVol.DL.LastSlot) / N.CA.ThpTime.DL.RmvLastSlot$.

NOTE

Optimized measurement of downlink CA throughput can be enabled to improve the accuracy of measuring the average throughput of SCell-activated UEs. For details, see [4.8.1 Optimized Measurement of Downlink CA Throughput](#).

Observe CQI statistics of SCell-activated CA UEs in PCells using the counters from **N.ChMeas.CA.PCell.CQI.0** to **N.ChMeas.CA.PCell.CQI.15** and those in SCells using the counters from **N.ChMeas.CA.SCell.CQI.0** to **N.ChMeas.CA.SCell.CQI.15**.

Observe the numbers of TBs transmitted using different ranks for SCell-activated CA UEs in PCells using the counters from **N.CA.PCell.PDSCH.NbrTbDL.Rank1** to **N.CA.PCell.PDSCH.NbrTbDL.Rank4** and those in SCells using the counters from **N.CA.SCell.PDSCH.NbrTbDL.Rank1** to **N.CA.SCell.PDSCH.NbrTbDL.Rank4**.

Observe MAC statistics related to SCell-activated CA UEs in PCells and SCells using the following counters, which measure the number of downlink TBs initially transmitted and retransmitted to CA UEs:

- **N.DL.SCH.CA.PCell.TB**: measures the number of downlink TBs initially transmitted to CA UEs that treat the local cell as their PCell.
- **N.DL.SCH.CA.SCell.TB**: measures the number of downlink TBs initially transmitted to CA UEs that treat the local cell as their SCell.
- **N.DL.SCH.CA.SCell.ErrTB.Ibler**: measures the number of downlink TBs failing to be initially transmitted to CA UEs that treat the local cell as their SCell.
- **N.DL.SCH.CA.PCell.ErrTB.Ibler**: measures the number of downlink TBs failing to be initially transmitted to CA UEs that treat the local cell as their PCell.
- **N.DL.SCH.CA.PCell.TB.Retrans**: measures the number of downlink TBs retransmitted to CA UEs that treat the local cell as their PCell.
- **N.DL.SCH.CA.SCell.TB.Retrans**: measures the number of downlink TBs retransmitted to CA UEs that treat the local cell as their SCell.
- **N.DL.SCH.CA.PCell.ErrTB.Rbler**: measures the number of downlink TBs failing to be transmitted to CA UEs that treat the local cell as their PCell after the maximum number of retransmissions is reached.
- **N.DL.SCH.CA.SCell.ErrTB.Rbler**: measures the number of downlink TBs failing to be transmitted to CA UEs that treat the local cell as their SCell after the maximum number of retransmissions is reached.

Observe the average MCS index in PDSCH scheduling of SCell-activated CA UEs in PCells, which is calculated using the following formula:

N.ChMeas.CA.PCell.PDSCH.MCS.Sum/N.ChMeas.CA.PCell.PDSCH.

Observe the average MCS index in PDSCH scheduling of SCell-activated CA UEs in SCells, which is calculated using the following formula:

N.ChMeas.CA.SCell.PDSCH.MCS.Sum/N.ChMeas.CA.SCell.PDSCH.

Observe the downlink traffic volume for SCell-activated CA UEs in PCells using the counter **N.CA.PCell.ThpVol.Dl.Cell**.

Observe the downlink traffic volume for CA UEs in SCells using the counter **N.CA.SCell.ThpVol.Dl.Cell**.

5.2 Uplink Intra-Band CA

5.2.1 Principles

5.2.1.1 Overview

With uplink intra-band CA enabled, uplink CA works for a CA UE after an SCell configured for the uplink is activated for the UE. This function can be enabled by turning on the uplink intra-band CA switch, which is the **INTRA_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch.CaAlgoSwitch** parameter. For the band +bandwidth combinations for uplink intra-band CA, see [4.4 CA Band+Bandwidth Combinations](#).

After uplink intra-band CA is enabled, uplink data may be scheduled in an SCell but not in the PCell in a single slot. In such a situation, the gNodeB and UE may make different decisions on whether HARQ feedback is sent together with data on the PUSCH. This will cause incorrect data demodulation on the gNodeB side and

increase the BLER. To prevent this issue, it is recommended that the **CA_PDCCH_CAPACITY_OPT_SWITCH** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch** parameter be selected.

5.2.1.2 CA Uplink Timing

Timing advances (TAs) are used by UEs to adjust the time to transmit uplink data. In a multi-carrier scenario such as CA, a TA is required for each of the carriers. The use of an inappropriate TA affects the uplink demodulation performance and therefore the uplink peak rate.

In intra-band contiguous CA, a CA UE sends data to the PCell and an SCell at nearly the same time through the same RF channel. Therefore, it is acceptable for the UE to use the TA of the PCell in the SCell.

In intra-band non-contiguous CA, however, when a CA UE uses the TA of the PCell in an SCell, uplink demodulation performance is affected. This is because the CA UE is at different distances away from the receive antennas of the PCell and SCell and the UE sends uplink data at different times from the antennas corresponding to the cells. To prevent the impact on uplink demodulation performance, the gNodeB selects a single TA or multiple appropriate TAs from the TAs for the PCell and SCells and delivers the selected TAs to the UE. The number of TAs selected depends on the following functions:

- **Multiple TAs (MTA):** The MTA function takes effect when the gNodeB selects two or more TAs for a UE that has reported the *supportedNumberTAG* IE with a value greater than 1 and its support for *dualPA-Architecture*. TAG is short for TA group. In the CA UE, each CC is assigned to a specific TAG. For all CCs involved in a TAG, the UE uses a same TA to adjust the time to transmit uplink data. The number of TAs selected by the gNodeB is equal to the number of TAGs supported by the UE, which is the value of the *supportedNumberTAG* IE.
 - If the number of TAGs supported by the UE is greater than or equal to the number of CCs, only one CC is assigned to each TAG, and each CC uses its own TA. In other words, all CCs use their own TAs.
 - If the number of TAGs supported by the UE is less than the number of CCs, more than one CC may be assigned to a TAG and share a TA. To select an optimal TA in this case, flexible TA adjustment in uplink multi-CC scenarios is introduced. This function ensures appropriate TA selection and minimizes the impact on uplink demodulation performance. For details about this function, see [Flexible TA Adjustment in Uplink Multi-CC Scenarios](#).
- **Single TA:** The single TA function takes effect when the gNodeB selects a single TA for a UE. This may occur if the UE does not meet requirements or the MTA function is disabled. The UE uses this TA to adjust the time to transmit uplink data on all the CCs. To select an optimal TA in this case, flexible TA adjustment in uplink multi-CC scenarios is introduced, as it is in the MTA case. This function ensures appropriate TA selection and minimizes the impact on uplink demodulation performance. For details about this function, see [Flexible TA Adjustment in Uplink Multi-CC Scenarios](#).

By default, the MTA function takes effect on the gNodeB. To use the single TA function, configure data as described in [MTA Configuration Prohibition for Uplink CA](#).

For details about how the gNodeB obtains the TA of an SCell, see [Procedure for the gNodeB to Obtain the TA of an SCell](#).

MTA Configuration Prohibition for Uplink CA

MTA configuration prohibition for uplink CA is controlled by a cell-level switch and a UE whitelist switch.

- Cell-level switch: If the **UL_CA_MTAG_NOT_CONFIG_SW** option of the **NRDUCellCarrMgmt.CaCompatibilitySwitch** parameter is selected for a cell, the MTA function is disabled for the cell, in which only the single TA function can take effect.

NOTE

If MTA has taken effect for a UE in a cell prior to selecting the **UL_CA_MTAG_NOT_CONFIG_SW** option of the **NRDUCellCarrMgmt.CaCompatibilitySwitch** parameter, the MTA mechanism remains in effect after this option is selected; the UE will switch to the single TA mechanism only after re-accessing the cell.

- UE whitelist switch: Certain UEs may encounter SCell activation failure because of compatibility issues when the MTA function takes effect on them. If the **UL_CA_MTAG_NOT_CONFIG_SW_ON** option of the **gNBUECompatSw.WhitelistCtrlExtSwitch** parameter is selected for the UEs, the MTA function is disabled for these UEs, so that only the single TA function can take effect for them. This option takes effect only when the **UL_CA_MTAG_NOT_CONFIG_SW** option of the **NRDUCellCarrMgmt.CaCompatibilitySwitch** parameter is selected.
 - If the **UL_CA_MTAG_NOT_CONFIG_SW_ON** option is selected, the MTA function does not take effect for specified UEs but takes effect for other UEs.
 - If the **UL_CA_MTAG_NOT_CONFIG_SW_ON** option is deselected, the MTA application status is the same for all UEs. The status is determined by the **UL_CA_MTAG_NOT_CONFIG_SW** option of the **NRDUCellCarrMgmt.CaCompatibilitySwitch** parameter.

NOTE

If MTA has taken effect for a UE in a cell prior to selecting the **UL_CA_MTAG_NOT_CONFIG_SW_ON** option of the **gNBUECompatSw.WhitelistCtrlExtSwitch** parameter, the MTA mechanism remains in effect after this option is selected; the UE will switch to the single TA mechanism only after re-accessing the cell.

Flexible TA Adjustment in Uplink Multi-CC Scenarios

Flexible TA adjustment in uplink multi-CC scenarios is controlled by the **UL_MULTI_CC_TA_FLEX_ADJ_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoSwitch** parameter. With this function enabled, the following TA selection policy applies to all TAGs:

- If the absolute value of the TA difference between any two CCs involved in a TAG is less than the value of the **NRDUCellCarrMgmt.UlMultiCcAbsTaDiffThld** parameter, the TA that has the minimum impact on all the CCs involved in this TAG is selected as the TA for all these CCs.

- If the absolute value of the TA difference between any two CCs involved in a TAG is greater than or equal to the value of the **NRDUCellCarrMgmt.UlMultiCcAbsTaDiffThld** parameter, the TA that has the minimum impact on the PCC is selected as the TA for all the CCs involved in this TAG.

Procedure for the gNodeB to Obtain the TA of an SCell

If an SCell provides uplink resources for a CA UE, the physical random access channel (PRACH) resources are configured for the SCell. After the SCell is activated for the UE, the gNodeB sends a physical downlink control channel (PDCCH) order to the UE in the PCell and, as instructed, the UE initiates a non-contention-based random access procedure in the SCell. After that, the SCell reports the related measurement information to the gNodeB for the gNodeB to obtain the TA of the SCell.

If the number of preamble retransmissions from a CA UE in its SCell reaches the maximum number allowed, the random access procedure fails. The UE stops random access without triggering an RRC connection reestablishment. As a result, the UE is out-of-synchronization in the uplink in the SCell. If the SCell activation conditions are met again and the SCell uplink time alignment timer expires, the PCell sends a PDCCH order to instruct the UE to initiate a random access procedure again. For details about the uplink time alignment timer, see *5G Networking and Signaling*.

5.2.1.3 Optimization Against LO Feedthrough for Uplink CA

After receiving an uplink signal, the gNodeB mixes it with the frequency generated by the oscillator circuit to generate an intermediate frequency (IF) signal at a fixed frequency. The frequency generated by the oscillator circuit is known as the local oscillator (LO) frequency. When LO signals are leaked to an output or input port, this phenomenon is called LO feedthrough. Currently, it is common to see LO feedthrough at UEs of various vendors. LO feedthrough affects the demodulation performance of the gNodeB and therefore causes failures of uplink intra-band CA to reach the expected peak rates.

To resolve this problem, 3GPP Release 16 allows for reporting of the LO locations of UEs through RRC connection reconfiguration messages. Huawei then introduces the function of optimization against LO feedthrough for uplink CA. This function is controlled by the **R16_CA_UE_LO_FEEDTHRU_OPT_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch** parameter.

- If this option is selected, the gNodeB includes the *reportUplinkTxDirectCurrentTwoCarrier-r16* IE in an RRC connection reconfiguration message for SCell configuration and sends the message to a UE that is compatible with the *uplinkTxDC-TwoCarrierReport-r16* IE. Using this message, the gNodeB requests the UE to report its LO locations. During demodulation, the gNodeB sets 0s (no data transmission) at the resource elements (REs) at the LO locations of the UE to reduce the impact of LO interference on demodulation.
- If this option is deselected, the gNodeB does not set 0s at the REs at the LO locations of the UE.

Optimization against LO feedthrough for uplink CA is a trial function. It works only in low frequency bands.

NOTE

Trial functions are functions that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these functions can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial functions shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial functions. Trial functions are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial functions may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial functions but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial functions are free, Huawei is not liable for any trial function malfunctions or any losses incurred by using the trial functions. Huawei does not promise that problems with trial functions will be resolved in the current version. Huawei reserves the rights to convert trial functions into commercial functions in later R/C versions. If trial functions are converted into commercial functions in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial functions. If a customer fails to purchase such a license, the trial function(s) will be invalidated automatically when the product is upgraded.

5.2.1.4 Optimized Fallback upon Uplink CA Exceptions

If an exception (such as an SCell addition failure) occurs in uplink CA, the gNodeB performs fallback by removing the involved SCells, which renders uplink CA ineffective. In some scenarios, the gNodeB may also remove the SCells for downlink CA, causing downlink CA to become ineffective as well.

To address this issue, the 'optimized fallback upon uplink CA exceptions' function has been introduced. This function is controlled by the **UL_CA_FALLBACK_DL_ONLY_CA_SW** option of the **gNodeBCarrMgmt.CaEnhancedAlgoSwitch** parameter. If this option is selected and an exception occurs in uplink CA, the gNodeB removes only uplink SCells but not downlink SCells, ensuring that downlink CA remains unaffected.

This function works only in low frequency bands.

5.2.1.5 SRS and PUSCH/PUCCH Concurrency Compatibility

According to 3GPP specifications, SRS and PUSCH/PUCCH concurrency is supported in intra-band non-contiguous CA, but not in intra-band contiguous CA. In a CA scenario, SRS and PUSCH/PUCCH concurrency means that a PCC and an SCC work in the same symbols where one CC sends SRSs and the other CC sends PUSCH/PUCCH signals.

For intra-band non-contiguous CA, SRS and PUSCH/PUCCH concurrency requires that UEs support this capability. Huawei gNodeBs comply with the protocol specifications to support SRS and PUSCH/PUCCH concurrency, while a UE supports this capability only if it reports the **parallelTxSRS-PUCCH-PUSCH-intraBand-r17** field.

Before the SRS and PUSCH/PUCCH concurrency compatibility function is introduced, the gNodeB by default considers that all UEs support the concurrency. If a UE does not support this capability, the gNodeB demodulates SRSs or PUSCH/PUCCH signals at incorrect time-domain positions, which compromises uplink performance of CA UEs. To address this issue, the SRS and PUSCH/PUCCH

concurrency compatibility function is introduced. This function is controlled by the **SRS_PUSCH_CONCURRENCY_COMPT_SW** option of the **NRDUCellCarrMgmt.CaCompatibilitySwitch** parameter. After the SRS and PUSCH/PUCCH concurrency compatibility function is enabled, the gNodeB checks whether a UE supports this concurrency so that the processing can vary with the UE capability:

- If the UE supports SRS and PUSCH/PUCCH concurrency, the UE can concurrently send SRSs and PUSCH/PUCCH signals.
- If the UE does not support SRS and PUSCH/PUCCH concurrency, the UE does not send SRSs and PUSCH/PUCCH signals concurrently on the PCC and SCCs. Instead, in the corresponding symbols, the UE sends SRSs only but not PUSCH/PUCCH signals.

SRS and PUSCH/PUCCH concurrency compatibility is supported only in low-frequency scenarios.

5.2.2 Network Analysis

5.2.2.1 Benefits

The uplink peak data rates of CA UEs increase.

The peak data rate that a CA UE can achieve using uplink 2CC aggregation in low frequency bands ranges from $(A \times 94\% + B \times 94\%)$ to $(A \times 100\% + B \times 100\%)$.

- A is the uplink data rate that the UE in the single-carrier state can achieve in a cell that is later configured as the PCell for the UE.
- B is the uplink data rate that the UE in the single-carrier state can achieve in a cell that is later configured as an SCell for the UE.

The peak data rate that a CA UE can achieve using uplink 4CC aggregation in high frequency bands ranges from $(A \times 94\% + B \times 94\% + C \times 94\% + D \times 94\%)$ to $(A \times 100\% + B \times 100\% + C \times 100\% + D \times 100\%)$.

- A is the uplink data rate that the UE in the single-carrier state can achieve in a cell that is later configured as the PCell for the UE.
- B , C , and D are the uplink data rates that the UE in the single-carrier state can achieve in individual cells that are later configured as SCells for the UE.

NOTE

The preceding peak data rate ranges can be achieved only if the influencing factors (for example, the UE rank) remain unchanged after CA takes effect.

After flexible TA adjustment in uplink multi-CC scenarios takes effect, the uplink cell throughput may improve. The extent of improvement is influenced by the following factors:

- When there are CCs with negative measured TA values among multiple CCs, the uplink cell throughput improvement is greater compared to scenarios without negative values.
- A larger difference in measured TA values between CCs contributes to greater uplink cell throughput improvement.

- A larger bandwidth difference between CCs contributes to greater uplink cell throughput improvement.
- The impact on the uplink cell throughput depends also on the setting of the **NRDUCellCarrMgmt.UlMultiCcAbsTaDiffThld** parameter. The default value is recommended.
 - A smaller value of this parameter indicates a stricter requirement of TA adjustment to the TA value with a relatively small impact on all CCs. However, if the value is too small, the increase in the uplink cell throughput will be relatively small.
 - A larger value of this parameter indicates a more relaxed requirement. However, if the value is too large, there will be a relatively high probability of a decrease in the uplink throughput.

With optimization against LO feedthrough for uplink CA, the gNodeB sets 0s at the REs at the UE-reported LO locations. This improves uplink demodulation performance and increases the **User Uplink Average Throughput (DU)**.

After optimized fallback upon uplink CA exceptions takes effect, the gNodeB removes only uplink SCells in the event of uplink CA exceptions, ensuring that downlink CA remains unaffected. The detailed impacts are as follows:

- The **User Downlink Average Throughput (DU)** and **Cell Downlink Average Throughput (DU)** increase.
- The **N.User.CA.SCell.DL.Avg** and **N.User.CA.PCell.DL.Avg** increase.
- The **N.CA.SCell.Rmv.Succ** and **N.CA.SCell.Rmv.Att** decrease.

After SCC removal due to consecutive non-split in uplink CA takes effect, the **User Uplink Average Throughput (DU)** of cell-edge UEs increases.

After SRS and PUSCH/PUCCH concurrency compatibility takes effect, the average throughput of SCell-activated uplink CA UEs ($(\text{N.CA.SccActiveUser.ThpVol.Ul} - \text{N.CA.ThpVol.Ul.SmallPkt}) / \text{N.CA.ThpTime.Ul.RmvSmallPkt}$) increases.

If uplink enhancement for smart carrier selection takes effect in suitable scenarios, the average number of uplink CA UEs that treat the local cell as their PCell (**N.User.CA.PCell.Ul.Avg**) increases.

5.2.2.2 Impacts

The impacts between this function and other functions are described in [5.2.3.2.3 Function Impacts](#).

- The smaller the value of **NRDUCellCarrMgmt.CaUlScellActThldSlotNum**, the more likely SCells are activated, and the longer the SCell-activated state lasts. Therefore, a smaller value results in higher UE power consumption and greater values of the following counters:
 - **N.CA.SCell.Act.Att**
 - **N.CA.SCell.Act.Succ**
 - **N.CA.SCell.Deact.Att**
 - **N.CA.SCell.Deact.Succ**
- The smaller the value of **NRDUCellCarrMgmt.CaUlScellDeactThldSlotNum** or **NRDUCellCarrMgmt.RlcUlThroughputThld**, the less likely SCells are deactivated, and the longer the SCell-activated state lasts. Therefore, a

smaller value results in higher UE power consumption and smaller values of the following counters:

- **N.CA.SCell.Deact.Att**
- **N.CA.SCell.Deact.Succ**
- **N.CA.SCell.Act.Att**
- **N.CA.SCell.Act.Succ**
- After uplink intra-band CA is enabled, the CPU usage of the BBP where the PCell is configured increases with the amount of data forwarded from the PCell to SCells at the RLC layer. If the CPU usage is too high, the peak data rate of the UE decreases.
- The data configuration for CA uplink timing has the following impacts:
 - If the **UL_CA_MTAG_NOT_CONFIG_SW** option of the cell-level parameter **NRDUCellCarrMgmt.CaCompatibilitySwitch** is selected, the MTA function does not take effect. In this case, if there is a large TA difference between the PCell and SCell of a CA UE, the uplink throughput of the UE decreases.
 - If the **UL_CA_MTAG_NOT_CONFIG_SW_ON** option of the gNodeB-level parameter **gNBUECompatSw.WhitelistCtrlExtSwitch** is selected, the MTA function does not take effect for specified UEs. In this case, if there is a large TA difference between the PCell and SCell of a CA UE, the uplink throughput of the UE may decrease.
- After SCC removal due to consecutive non-split in uplink CA takes effect, an excessively small value of **NRDUCellCarrMgmt.UlCaContNonSplitSccDelThld** may cause unnecessary SCC removal, decreasing the average throughput of SCell-activated uplink CA UEs (calculated as follows: $(\text{N.CA.SccActiveUser.ThpVol.UL} - \text{N.CA.ThpVol.UL.SmallPkt}) / \text{N.CA.ThpTime.UL.RmvSmallPkt}$).
- After uplink CA data split threshold optimization takes effect, the following indicators change:
 - The total volume of uplink data received at the MAC layer from uplink CA UEs that treat the local cell as their SCell (**N.CA.SCell.ThpVol.UL.Cell**) increases.
 - The total volume of uplink data received at the MAC layer from uplink CA UEs that treat the local cell as their PCell (**N.CA.PCell.ThpVol.UL.Cell**) decreases.
 - The average throughput of SCell-activated uplink CA UEs (calculated as follows: $(\text{N.CA.SccActiveUser.ThpVol.UL} - \text{N.CA.ThpVol.UL.SmallPkt}) / \text{N.CA.ThpTime.UL.RmvSmallPkt}$) varies, influenced by the SCell load. Under low SCell load, this throughput increases.

5.2.3 Requirements

5.2.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-020205	Intra-Band CA	NR0S00IBCA00	per Cell

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.2.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Downlink intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Uplink intra-band CA depends on downlink intra-band CA.
Low-frequency TDD	Uplink intra-band CA	INTRA_BAND_UL_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Optimization against LO feedthrough for uplink CA requires uplink intra-band CA to be enabled.
Low-frequency TDD	MTA configuration prohibition for uplink CA	UL_CA_MTAG_NOT_CONFIG_SW option of the NRDUCellCarrierMgmt.CaCompatibilitySwitch parameter	<i>Carrier Aggregation</i>	The UL_CA_MTAG_NOT_CONFIG_SW_ON option of the gNBUECompatSw.WhitelistCtrlExtSwitch parameter takes effect only when the UL_CA_MTAG_NOT_CONFIG_SW option of the NRDUCellCarrierMgmt.CaCompatibilitySwitch parameter is selected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Flexible TA adjustment in uplink multi-CC scenarios	UL_MULTI_C C_TA_FLEX_ADJ_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgorithmSwitch parameter	<i>Carrier Aggregation</i>	The setting of the NRDUCellCarrierMgmt.UlMultiCcAbsTaDiffThld parameter takes effect only if flexible TA adjustment in uplink multi-CC scenarios is enabled.
Low-frequency TDD	Smart carrier selection	SMART_CARRIER_SEL_SW option of the NRDUCellCollabServ.MultiCarrierMgmtSw parameter	<i>Carrier Aggregation</i>	Uplink enhancement for smart carrier selection takes effect only when both smart carrier selection and service-based optimal carrier selection are enabled.
Low-frequency TDD	Service-based optimal carrier selection	SERVICE_AWARE_CARRIER_SEL_SW option of the NRDUCellCollabServ.MultiCarrierMgmtSw parameter	<i>Carrier Aggregation</i>	
Low-frequency TDD	SRS interference coordination based on self-contained slots	SRS_INTRF_COORD_S_SLOT_SW option of the NRDUCellSrsSrsAlgoExtSwitch parameter	<i>Channel Management</i>	The SRS_INTRF_COORD_S_SLOT_SW option must be selected if the SRS and PUSCH/PUCCH concurrency compatibility function is enabled and the SRS remote interference avoidance switch is turned on.
Low-frequency TDD	SRS remote interference avoidance	SRS_RIM_INTRF_AVOID_SW option of the NRDUCellSrsSrsAlgoExtSwitch parameter	None	

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PUCCH symbol length number increase	PUCCH_SYM_TYPE_NUM_INCR_SW option of the NRDUCellPucch.PucchAlgoSwitch parameter	<i>Channel Management</i>	When the SRS and PUSCH/PUCCH concurrency compatibility function is enabled, the PUCCH_SYM_TYPE_NUM_INCR_SW option must be selected.

5.2.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Compensation for SAR decrease (low-frequency TDD)	REDUCE_SAR_COMPENSATION_SW option of the NRDUCellUeCoop.SpecUeCooperationSwitch parameter	<i>Device-Pipe Synergy</i>	Intra-band CA does not work with compensation for SAR decrease.
Low-frequency TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	Intra-band CA does not work with the function of low latency and high reliability.
Low-frequency TDD	SUL cell	NRDUCell.DuplexMode set to CELL_SUL	<i>Cell Management</i>	Optimization against LO feedthrough for uplink CA does not work in SUL cells.
High-frequency TDD	None	None	None	None

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Adaptive fast processing by the gNodeB	GNB_ADAPT_FAST_PROCESS_SW option of the gNodeBParam.ServiceDiffProcSwitch parameter	None	When uplink intra-band CA is enabled, the GNB_ADAPT_FAST_PROCESS_SW option of the gNodeBParam.ServiceDiffProcSwitch parameter must be deselected.
Low-frequency TDD	Positioning-dedicated SRS transmission in uplink slots	SRS_FOR_POSITION_ON_USLOT_SW option of the NRDUCellPosAlgo.LcsMeasureAlgorithm parameter	<i>Positioning</i>	When the SRS and PUSCH/PUCCH concurrency compatibility function is enabled, positioning-dedicated SRS transmission in uplink slots cannot be enabled together with proactive avoidance of remote interference.
Low-frequency TDD	Proactive avoidance of remote interference	SRS_RIM_INTRF_AVOID_SW and SRS_INTRF_COORD_S_SLOT_SW options of the NRDUCellSrsAlgoExtSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	
Low-frequency TDD	Light-load time sequence optimization	LIGHT_LOAD_TIME_SEQUENCE_OPT_SW option of the NRDUCellAlgoSwitch.LowLatencySwitch parameter	None	When the SRS and PUSCH/PUCCH concurrency compatibility function is enabled, the light-load time sequence optimization function must be disabled.

5.2.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PUSCH MU-MIMO	UL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	If CA UEs do not transmit SRSs in their SCells, they cannot be paired for MU-MIMO in the SCells. As a result, the uplink cell capacity decreases.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Power saving BWP (low-frequency TDD)	BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	BWP2, which provides a narrow bandwidth, is not configured for CA UEs in their SCells. When power saving BWP or "QCI-level threshold control for BWP switching" is enabled together with CA, there are the following impacts: • A larger value of NRDUCellCarrMgmt.CaUIScellActiveThldSlotNum results in a higher traffic volume threshold for UE switching from BWP2 to BWP1 and therefore a lower probability of the switching, which decreases UE power consumption. • A larger value of NRDUCellCarrMgmt.CaUIScellDeactThldSlotNum results in a higher traffic volume threshold for UE switching from BWP1 to BWP2 and therefore a higher probability of the switching, which decreases UE power consumption. • SCells are more likely to be deactivated if the NRDUCellCarrMgmt.RlcUlThroughputThld is set to a higher value.

RAT	Function Name	Function Switch	Reference	Description
				<i>hputThld</i> parameter value is less than the NRDUCellUePwrSaving.Bwp1ToBwp2UlThptThld or NRDUCellQciBearer.QciBwp1ToBwp2UlThptThld parameter value. If NRDUCellCarrMgmt.RlcUlThroughputThld is set to 0, SCells will not be deactivated and UEs will not switch to BWP2.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Time-domain power saving BWP (low-frequency TDD)	TDM_BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	BWP2 is not configured for CA UEs in their SCells. When CA and time-domain power saving BWP are both enabled, there are the following impacts: • A larger value of NRDUCellCarrMgmt.CaUIScellActivateThldSlotNum results in a higher traffic volume threshold for UE switching from BWP2 to BWP1 and therefore a lower probability of the switching, which decreases UE power consumption. • A larger value of NRDUCellCarrMgmt.CaUIScellDeactivateThldSlotNum results in a higher traffic volume threshold for UE switching from BWP1 to BWP2 and therefore a higher probability of the switching, which decreases UE power consumption. • SCells are more likely to be deactivated if the NRDUCellCarrMgmt.RlCUIThroughputThld parameter value is less than the NRDUCellUePwr

RAT	Function Name	Function Switch	Reference	Description
				Saving.Bwp1ToBwp2UlThptThld parameter value. If NRDUCellCarrMgmt.RlcUlThroughputThld is set to 0 , SCells will not be deactivated and UEs will not switch to BWP2.
Low-frequency TDD	Uplink multi-frequency coordination	MULTI_FREQ_UL_COORD_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter	<i>Multi-Frequency Smart Selection</i>	The gNodeB does not select UEs for which uplink CA has taken effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PUSCH scheduling based on SRS interference avoidance	SRS_INTRF_AVOID_PUSCH_SCH_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter	None	When the uplink intra-band CA switch is on, the setting of the SRS_INTRF_AVOID_PUSCH_SCH_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter must be the same for the PCell and the SCell. If it is not the same for the PCell and the SCell, the cells differ in their scheduling in self-contained slots in the uplink, affecting the reporting of the power headroom in the PCell and SCell. As a result, adjustments for transmit power control (TPC) are performed frequently and the average uplink throughput of the uplink CA UE changes greatly.
Low-frequency TDD	Multi-frequency beam coordination	MULTI_FREQ_BEAM_COORD_SW option of the NRDUCellFeatureSwitch.MultiCarrierSwitch parameter	<i>Multi-Frequency Smart Selection</i>	The gNodeB does not select uplink CA UEs that treat the local cell as their PCell or SCell.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CCE-based data split stop for CA UEs	CCE_BASED_CA_SPLIT_STOP_SW option of the NRDUCellCarrierMgmt.CAEnhancedAlgorithmExtSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	CCE-based data split stop for CA UEs does not take effect for uplink CA UEs.
Low-frequency TDD	PDCCH skipping	PDCCH_SKIP_SW option of the NRDUCellUePwrSaving.SchPwrSavingSw parameter	<i>UE Power Saving</i>	With both uplink intra-band CA and PDCCH skipping enabled: - If the CA_UE_PDCCH_SKIP_SW option of the NRDUCellUePwrSaving.SchPwrSavingSw parameter is deselected, PDCCH skipping does not take effect for UEs with SCells configured. - If the CA_UE_PDCCH_SKIP_SW option of the NRDUCellUePwrSaving.SchPwrSavingSw parameter is selected, PDCCH skipping can take effect for CA UEs with SCells not activated, but not for CA UEs with SCells activated.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CA SRS carrier switching	SRS_CARRIER_SWITCHING_SW option of the NRDUCellCarrierMgmt.CAEnhancedAlgorithmSwitch parameter	<i>CA Experience Improvement</i>	When SRS carrier switching and uplink intra-band CA are both enabled, only one of them can take effect at a time. Which of them takes effect is dependent on the multi-carrier capability combination selection policy, which is specified by the NRCellCarrierMgmtConfig.CABandCombSelectionStrategy parameter.
Low-frequency TDD	CA SRS carrier switching enhancement	SRS_CARRIER_SWITCHING_ENH_SW option of the NRDUCellCarrierMgmt.CAEnhancedAlgorithmSwitch parameter	<i>CA Experience Improvement</i>	When SRS carrier switching enhancement and uplink intra-band CA are both enabled, only one of them can take effect at a time. Which of them takes effect is dependent on the multi-carrier capability combination selection policy, which is specified by the NRCellCarrierMgmtConfig.CABandCombSelectionStrategy parameter.
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode	<i>Hyper Cell</i>	Uplink intra-band CA cannot be enabled in hyper cells, where NRDUCell.NrDuCellNetworkingMode is set to HYPER_CELL .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTrp.DataTransMode set to BASIC_MODE or DPS_MODE	<i>Cell Combination</i>	Uplink intra-band CA cannot be enabled in combined cells.
Low-frequency TDD	Joint Transmission Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE and NRDUCellMMultiTrp.DataTransMode set to JOINT_MODE	<i>Cell Combination</i>	Uplink intra-band CA cannot be enabled in combined cells in joint transmission mode.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Experience-based smart carrier selection	MULTI_FREQ_SMART_SEL_SW option of the NRCellAlgoSwitch.MultiFreqAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	Assume that both service-aware carrier selection (controlled by the SERVICE_AWARE_CARR_SEL_SW option of the NRDUCellCollabServ.MultiCarrMgmtSw parameter) and experience-based smart carrier selection are enabled. Experience-based smart carrier selection can take effect to trigger carrier changes for a UE only if the UE has not been considered as a target UE for service-aware carrier selection.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Proactive avoidance of remote interference	SRS_RIM_INTRF_AVOID_SW and SRS_INTRF_COORD_S_SLOT_SW options of the NRDUCellSrsSrsAlgoExtSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	- When the uplink CA switch is on, the SRS_INTRF_AVOID_PUSCH_SCH_SW setting of the NRDUCellSrsSrsAlgoExtSwitch parameter must be the same for the PCell and the SCell. Otherwise, the cells differ in their scheduling in self-contained slots in the uplink, affecting the reporting of the power headroom in the PCell and SCell. As a result, adjustments for TPC are performed frequently and the average uplink throughput of uplink CA UEs changes greatly. - It is good practice to keep the "proactive avoidance of remote interference" switch status consistent between the PCell and the SCell. Otherwise, the bit errors of some UEs may increase due to inconsistent PUSCH scheduling status of the two cells in the same slot. In

RAT	Function Name	Function Switch	Reference	Description
				the event of inconsistent switch status between the two cells, it is good practice to enable the "SRS and PUSCH/PUCCH concurrency compatibility" function to reduce bit errors. For details about the "SRS and PUSCH/PUCCH concurrency compatibility" function, see <i>Carrier Aggregation</i> .
Low-frequency TDD	Device-pipe synergy preallocation	UL_PREFERENTIAL_SWITCH option of the NRDUCellQciBearer.ServiceDiffAlgoSwitch parameter	<i>FWA</i>	For UEs in the CA state, device-pipe synergy preallocation takes effect in their PCells but not SCells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-FR CA	INTER_FR_F R1TOFR2_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	For a UE in the intra-band CA state: • If the band +bandwidth combination for CA at the UE meets the band +bandwidth requirements of inter-FR CA, inter-FR CA can simultaneously take effect for the UE. If this is the case, downlink CA instead of uplink CA can take effect between the two low-frequency cells. • Otherwise, inter-FR CA cannot simultaneously take effect for the UE.
Low-frequency TDD	3GPP R15-compliant uplink slot aggregation	VONR_UL_SLOT_AGGREGATION_SW option of the NRDUCellUISch.VoiceUISchSwitch parameter	<i>VoNR</i>	If uplink intra-band CA has taken effect for UEs with VoNR enabled, the UEs cannot enter the uplink slot aggregation state. If the UEs have been in the uplink slot aggregation state and the conditions for uplink intra-band CA to take effect are met, they will exit the uplink slot aggregation state.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	3GPP R16-compliant uplink slot aggregation	VOICE_R16_UL_SLOT_AGG_SW option of the NRDUCellUlschSwitch parameter	<i>VoNR</i>	If uplink intra-band CA has taken effect for UEs with VoNR enabled, the UEs cannot enter the uplink slot aggregation state. If the UEs have been in the uplink slot aggregation state and the conditions for uplink intra-band CA to take effect are met, they will exit the uplink slot aggregation state.
Low-frequency TDD	Single DCI	DL_SINGLE_DCI_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgorithmSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	If uplink intra-band CA has taken effect for a UE, single DCI will not take effect for the UE.
Low-frequency TDD	MTA configuration prohibition for uplink CA	UL_CA_MTAG_NOT_CONFIG_SW option of the NRDUCellCarrierMgmt.CaCompatibilitySwitch parameter	<i>Carrier Aggregation</i>	The UL_CA_MTAG_NOT_CONFIG_SW_ON option of the gNBUECompatSw.WhitelistCtrlExtSwitch parameter takes effect only when the UL_CA_MTAG_NOT_CONFIG_SW option of the NRDUCellCarrierMgmt.CaCompatibilitySwitch parameter is selected. With this UE whitelist switch turned on, the MTA function takes effect for all UEs except the specified UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE-number-based connected mode MLB	• INTER_FREQ_CONNECTED_MLB_SWITCH option of the NRCellInfo.MlbAlgorithm parameter • NRCellInfo.MlbTriggerMode set to CELL_USER_NUM	<i>Mobility Load Balancing in Connected Mode</i>	When selecting UEs for handovers, the gNodeB excludes those with uplink intra-band CA in effect.
Low-frequency TDD	Spectral-efficiency-based connected mode MLB	• INTER_FREQ_CONNECTED_MLB_SWITCH option of the NRCellInfo.MlbAlgorithm parameter • NRCellInfo.MlbTriggerMode set to SPEC_EFF_BASED_USER_NUM	<i>Mobility Load Balancing in Connected Mode</i>	When selecting UEs for handovers, the gNodeB excludes those with uplink intra-band CA in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	User-rate-based dynamic network slice resource management	DYNAMIC_NETWORKSLICE_SW option of the NRDUCellCommonSw.ServiceDiffAlgorithmExtSwitch parameter	<i>Network Slicing</i>	When user-rate-based dynamic network slice resource management is enabled, the NRDUCellQciBearer.DynResULTargetRate parameter value configured for the PCell also applies to the SCells.
Low-frequency TDD	User-rate-based dynamic QCI resource management	DYNAMIC_QCI_GROUP_SW option of the NRDUCellFeatureSw.QciBasedDiffServiceExpSw parameter	<i>QCI-based Resource Control</i>	When user-rate-based dynamic QCI resource management is enabled, the NRDUCellQciBearer.DynResULTargetRate parameter value configured for the PCell also applies to the SCells.
High-frequency TDD	None	None	None	None

 NOTE

Apart from what are listed above, uplink intra-band CA shares the function impacts of downlink intra-band CA as described in [5.1.3.2.3 Function Impacts](#).

5.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

 NOTE

One prerequisite for CA is that the corresponding cells have been set up. The following requirements for base station models, boards, and RF modules are based on this prerequisite.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

In low frequency bands, CA can be performed in one baseband processing unit or between baseband processing units. In high frequency bands, CA can be performed only in one baseband processing unit.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.2.3.4 Networking

Frequency

There must be at least two frequencies available on the live network.

(Low-frequency scenarios) The **NRCellFreqRelation** MO must be already configured.

The spacing between the center frequencies of CCs for contiguous intra-band CA must meet the following requirements:

For low-frequency intra-band contiguous CA to take effect, the spacing must be less than or equal to the nominal channel spacing required by section 5.4A.1 "Channel spacing for CA" in 3GPP TS 38.101-1 V15.6.0. Otherwise, the CCs are aggregated for intra-band non-contiguous CA. When being less than the nominal channel spacing, the center frequency spacing must be a multiple of the least common multiple of the channel raster and SCS. The nominal channel spacing is calculated using different formulas depending on the channel raster (defined for each frequency band in Table 5.4.2.3-1 "Applicable NR-ARFCN per operating band" in 3GPP TS 38.101-1 V15.6.0):

- When the channel raster for the frequency band is 100 kHz, the nominal channel spacing is calculated using the following formula 1:

$$\text{Nominal channel spacing} = \left\lceil \frac{BW_{\text{Channel}(1)} + BW_{\text{Channel}(2)} - 2 \lfloor GB_{\text{Channel}(1)} - GB_{\text{Channel}(2)} \rfloor}{0.6} \right\rceil \times 0.3$$

- When the channel raster for the frequency band is not 100 kHz, the nominal channel spacing is calculated using the following formula 2:

$$\text{Nominal channel spacing} = \left\lceil \frac{BW_{\text{Channel}(1)} + BW_{\text{Channel}(2)} - 2 \lfloor GB_{\text{Channel}(1)} - GB_{\text{Channel}(2)} \rfloor}{0.015 \times 2^{n+1}} \right\rceil \times 0.015 \times 2^n, n = \mu_0$$

where

- $BW_{\text{Channel}(1)}$ and $BW_{\text{Channel}(2)}$ are the bandwidths of two CCs, CC 1 and CC 2, respectively.

- μ_0 is the μ value corresponding to the maximum subcarrier spacing (SCS) that is supported by both CC 1 and CC 2.
- $GB_{\text{Channel}(1)}$ and $GB_{\text{Channel}(2)}$ are the minimum guard bands corresponding to μ_0 for CC 1 and CC 2, respectively.

Assume that CC 1 (bandwidth: 60 MHz) and CC 2 (bandwidth: 100 MHz) belong to n41. Their nominal channel spacing is calculated as follows:

1. Determine the formula to be used: Formula 2 is to be used because the channel raster for n41 is not 100 kHz, according to Table 5.4.2.3-1 "Applicable NR-ARFCN per operating band" in 3GPP TS 38.101-1 V15.6.0.
2. Determine μ_0 : Both CC 1 and CC 2 support 30 kHz SCS and 60 kHz SCS according to **Figure 5-5**, which lists the SCSs supported by each bandwidth in n41 (the figure is adapted from Table 5.3.5-1 "Channel bandwidths for each NR band" in 3GPP TS 38.101-1 V15.6.0). The maximum SCS that both CCs support is 60 kHz, which corresponds to the μ value of 2. Therefore, μ_0 equals 2. For the μ value corresponding to each SCS, see Table 4.2-1 "Supported transmission numerologies" in 3GPP TS 38.211 V15.6.0.

Figure 5-5 SCSs for n41

NR Band	SCS (kHz)	UE Channel Bandwidth													
		5 MHz	10 MHz	15 MHz	20 MHz	25 MHz	30 MHz	40 MHz	50 MHz	60 MHz	70 MHz	80 MHz	90 MHz	100 MHz	
n41	15	N/A	Yes	Yes	Yes	N/A	Yes	Yes	Yes	N/A	N/A	N/A	N/A	N/A	
n41	30	N/A	Yes	Yes	Yes	N/A	Yes	Yes	Yes	Yes	N/A	Yes	Yes	Yes	
n41	60	N/A	Yes	Yes	Yes	N/A	Yes	Yes	Yes	Yes	N/A	Yes	Yes	Yes	

3. Determine $GB_{\text{Channel}(1)}$ and $GB_{\text{Channel}(2)}$: $GB_{\text{Channel}(1)}$ equals 1530 kHz and $GB_{\text{Channel}(2)}$ equals 1370 kHz according to **Figure 5-6**, which is adapted from Table 5.3.3-1 "Minimum guardband for each UE channel bandwidth and SCS (kHz)" in 3GPP TS 38.101-1 V15.6.0. (In the case of formula 1, the minimum guard bands for two CCs are also determined using the maximum one of the SCSs supported by both CCs.)

Figure 5-6 Minimum guardband for each UE channel bandwidth and SCS (kHz)

SCS (kHz)	UE Channel Bandwidth													
	5 MHz	10 MHz	15 MHz	20 MHz	25 MHz	30 MHz	40 MHz	50 MHz	60 MHz	70 MHz	80 MHz	90 MHz	100 MHz	
15	242.5	312.5	382.5	452.5	522.5	592.5	552.5	692.5	N/A	N/A	N/A	N/A	N/A	
30	505	665	645	805	785	945	905	1045	825	965	925	885	845	
60	N/A	1010	990	1330	1310	1290	1610	1570	1530	1490	1450	1410	1370	

4. Calculate the nominal channel spacing using the formula with $BW_{\text{Channel}(1)}$, $BW_{\text{Channel}(2)}$, $GB_{\text{Channel}(1)}$, $GB_{\text{Channel}(2)}$, and μ_0 determined. The result is 79.8 MHz.

In high-frequency NR TDD, the spacing between the center frequencies of CCs must be less than or equal to the nominal channel spacing required by section 5.4A.1 "Channel spacing for CA" in 3GPP TS 38.101-2 V15.7.0. Otherwise, the CCs are aggregated for intra-band non-contiguous CA. When being less than the nominal channel spacing, the center frequency spacing must be a multiple of the least common multiple of the channel raster and SCS. Use the same method to determine μ_0 for calculating the nominal channel spacing as in low frequency bands.

Coverage

The coverage areas provided by the cells involved in CA must overlap. The smaller the overlapping range, the smaller the area where intra-band CA takes effect.

Cell

The timing advance (TA) offset of the PCell and SCells (specified by the **NRDUCell.TaOffset** parameter) must be the same. Otherwise:

- Uplink throughput is affected in the case of MTA being in effect.
- Uplink CA fails to take effect in the case of MTA being not in effect.

For details about the MTA function, see [5.2.1.2 CA Uplink Timing](#).

5.2.3.5 Others

- For this function to deliver a theoretical peak data rate to SA CA UEs, the maximum bit rate that each CA UE subscribes to in the 5GC cannot be lower than this theoretical value.
- For this NR CA function to deliver a theoretical peak data rate to NSA UEs, the maximum bit rate that each CA UE subscribes to in the EPC cannot be lower than this theoretical value.
- Uplink intra-band CA requires that UEs support CA. During initial access, an incoming handover, or an incoming RRC connection reestablishment, a UE reports its CA capabilities to the gNodeB.
- Optimization against LO feedthrough for uplink CA requires that UEs be compatible with the *uplinkTxDC-TwoCarrierReport-r16* IE.
- UEs must support semi-static codebooks (as indicated by *semiStaticHARQ-ACK-Codebook* in the *phy-Parameters* IE) for feedback over the PUCCH.
- For MTA, UEs must report the *supportedNumberTAG* IE with values greater than 1 and support *dualPA-Architecture*.

5.2.4 Operation and Maintenance

5.2.4.1 Data Configuration

5.2.4.1.1 Data Preparation

Before activating this function in low-frequency scenarios, ensure that the **NRCellRelation** and **NRCellFreqRelation** MOs have been configured. For details about the configuration, see *SA Mobility Management in Connected Mode* and *NSA Mobility Management*.

[Table 5-6](#) and [Table 5-7](#) describe the parameters used for function activation and optimization, respectively. The following parameters do not take effect for SUL cells, for which the **NRDUCell.DuplexMode** parameter is set to **CELL_SUL**.

Table 5-6 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
CA Algorithm Switch	NRDUCellAlgoSwitch.CaAlgoSwitch	INTRA_BAND_UL_CA_SW	Select this option.
CA Activate and BWP Switch UL Thld Slot Num	NRDUCellCarrMgmt.CaUIScellDeactThldSlotNum	N/A	The value 4 is recommended for low-frequency cells. The value 16 is recommended for high-frequency cells. Reduce the NRDUCellCarrMgmt.CaUIScellDeactThldSlotNum parameter value to ensure SCell activation for NR CA when dynamic uplink PDCP data split is used in low-frequency NSA networking (that is, the NsaDcQciParamGroup.UIDataSplitThreshold parameter is set to a value other than INFINITY in Option 3 or the gNBPDcpParamGroup.UIDataSplitThreshold parameter is set to a value other than INFINITY in Option 3x). A larger value of this parameter results in a higher uplink threshold for SCell activation and a lower probability of SCell activation. The value of this parameter must be greater than one half of the NRDUCellCarrMgmt.CaUIScellDeactThldSlotNum parameter value.
CA Deactivate and BWP Switch UL Thld Slot Num	NRDUCellCarrMgmt.CaUIScellDeactThldSlotNum	N/A	The value 1 is recommended for low-frequency cells. The value 4 is recommended for high-frequency cells. A larger value of this parameter results in a higher uplink threshold for SCell deactivation and a higher probability of SCell deactivation.

Parameter Name	Parameter ID	Option	Setting Notes
RLC Uplink Throughput Threshold	NRDUCellCar rMgmt.RlcUl ThroughputThld	N/A	The value 500 is recommended. A larger value of this parameter results in a higher uplink threshold for SCell deactivation and a higher probability of SCell deactivation.
UL CA A2 RSRP Threshold	NRCeCellCaMg mtConfig.UlC aA2RsrpThld	N/A	To trigger SCell removals only when the SCells provide poorer service quality, reduce the value of this parameter. You are advised to set this parameter to a value larger than the value of NRCeCellCaMgmtCon- fig.CaA2RsrpThld .
CA Enhanced Algorithm Extended Switch	NRDUCellCar rMgmt.CaEn hancedAlgoE xtSwitch	R16_CA_UE_L O_FEEDTHRU _OPT_SW	If there are UEs capable of the <i>uplinkTxDC-TwoCarrierReport-r16</i> IE in a cell, it is recommended that this option be selected in uplink large-packet UE tests or peak rate tests.
CA Compatibility Switch	NRDUCellCar rMgmt.CaCo mpatibilityS witch	UL_CA_MTAG _NOT_CONFI G_SW	Select this option if the single TA function is required in a cell.
Parameter Control Type	gNBUEComp atSw.ParamC trlType	N/A	Set this parameter based on the network plan.
Whitelist Control Extension Switch	gNBUEComp atSw.Whiteli stCtrlExtSwit ch	UL_CA_MTAG _NOT_CONFI G_SW_ON	It is recommended that this option be selected when there are UE compatibility issues.
CA Enhanced Algorithm Switch	NRDUCellCar rMgmt.CaEn hancedAlgoS witch	UL_MULTI_CC _TA_FLEX_ADJ _SW	You are advised to select this option.
UL Multi-Carrier Absolute TA Diff Thld	NRDUCellCar rMgmt.UlMul tiCcAbsTaDiff Thld	N/A	You are advised to retain the default value.

Parameter Name	Parameter ID	Option	Setting Notes
CA Enhanced Algorithm Switch	gNodeBCarrMgmt.CaEnhancedAlgoSwitch	UL_CA_FALLBACK_DL_ONLY_CA_SW	You are advised to select this option.
UL CA SCC Split SINR Threshold	NRDUCellCarMgmt.UlCaSccSplitSinrThld	N/A	Use the recommended value. A smaller value of this parameter results in a higher probability that SCCs are qualified for uplink data split. A larger value of this parameter produces the opposite effects.
UL CA Continuous Non Split SCC Delete Threshold	NRDUCellCarMgmt.UlCaContNonSplitSccDelThld	N/A	Use the recommended value. If this parameter is set to a value other than 255 , a smaller value of this parameter results in a higher probability of SCC removal for uplink CA in scenarios with consecutive non-split events and a lower probability of limited UE rate on the PCC in such scenarios. However, unnecessary SCell removal may occur, decreasing uplink UE throughput. A larger value of this parameter results in a lower probability of SCC removal for uplink CA in scenarios with consecutive non-split events and a higher probability of limited UE rate on the PCC in such scenarios, but a lower probability of unnecessary SCell removal.
CA Enhanced Algorithm Switch	NRDUCellCarMgmt.CaEnhancedAlgoSwitch	UL_CA_SPLIT_THLD_OPT_SW	Select this option if the accuracy of data split to SCells needs to be improved.
UL Power Control Algorithm Extension Switch	NRDUCellUIPcConfig.UlPwrCtrlAlgoExtSwitch	MF_ACTUAL_PUSCH_ILPC_SW	It is good practice to select this option in both the PCell and SCells to improve the accuracy of data volume distributed to SCells in the uplink.

Parameter Name	Parameter ID	Option	Setting Notes
CA Compatibility Switch	NRDUCellCar rMgmt.CaCo mpatibilityS witch	SRS_PUSCH_C ONCURRENC Y_COMPT_SW	You are advised to select this option.
Multi-Carrier Management Switch	NRDUCellCol labServ.Multi CarrMgmtSw	SMART_CARR _SEL_UL_ENH _SW	Select this option if preference based on the uplink peak rate is expected to take effect in the CA capability combination selection policy in scenarios where the optimal uplink and downlink CA combinations are not in the same combination during a traffic volume model change.

Table 5-7 Parameters used for optimization

Parameter Name	Parameter ID	Option	Setting Notes
Multi-Carrier Capb Comb Select Strat	NRCeCellCaMg mtConfig.Ca BandCombSe lectionStrat	N/A	Set this parameter as required.
Multi-Carrier Compatibility Switch	NRCeCellCaMg mtConfig.Ca Compatibility Switch	SRS_WT_BASE D_CARR_SEL_ SW	It is recommended that this option be selected if UEs are capable of uplink CA.
Multi-Carrier Management Switch	NRDUCellCol labServ.Multi CarrMgmtSw	SMART_CARR _SEL_SW	You are advised to select this option if the number of an operator's frequencies is greater than the number of frequencies that can be aggregated for a UE.
Multi-Carrier Management Switch	NRDUCellCol labServ.Multi CarrMgmtSw	SERVICE_AWA RE_CARR_SEL _SW	You are advised to select this option if UEs in a cell have various requirements and the number of an operator's frequencies is greater than the number of frequencies that can be aggregated for a UE.

5.2.4.1.2 Using MML Commands

Before using MML commands, refer to [5.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the uplink intra-band CA algorithm switch in low-frequency scenarios
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CaAlgoSwitch=INTRA_BAND_UL_CA_SW-1;
//In high-frequency scenarios, the uplink intra-band CA algorithm switch has been turned on when
downlink intra-band CA is enabled. Therefore, you do not need to turn on this switch again when enabling
uplink intra-band CA.
//Setting the uplink traffic volume thresholds for SCell activation and deactivation in low-frequency
scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaUlScellActThldSlotNum=4, CaUlScellDeactThldSlotNum=1;
//Setting the uplink traffic volume thresholds for SCell activation and deactivation in high-frequency
scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaUlScellActThldSlotNum=16, CaUlScellDeactThldSlotNum=4;
//Setting the uplink throughput threshold for SCell deactivation
MOD NRDUCELLCARRMGMT: NrDuCellId=0, RlcUlThroughputThld=500;
//Setting the RSRP threshold for uplink CA event A2
MOD NRCELLCAMGMTCONFIG: NrCellId=0, UlCaA2RsrpThld=-109;
//Enabling optimization against LO feedthrough for uplink CA in low-frequency scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
CaEnhancedAlgoExtSwitch=R16_CA_UE_LO_FEEDTHRU_OPT_SW-1;
//(Optional) Turning on UL_CA_MTAG_NOT_CONFIG_SW if the single TA function is desired in low-
frequency scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaCompatibilitySwitch=UL_CA_MTAG_NOT_CONFIG_SW-1;
//Enabling whitelist control over MTA configuration prohibition for uplink CA for specified UEs with
compatibility issues in low-frequency scenarios
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
WhitelistCtrlExtSwitch=UL_CA_MTAG_NOT_CONFIG_SW_ON-1;
//(Low frequency) Enabling flexible TA adjustment in uplink multi-CC scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=UL_MULTI_CC_TA_FLEX_ADJ_SW-1;
//(Optional) Setting the UlMultiCcAbsTaDiffThld parameter in low-frequency scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=0, UlMultiCcAbsTaDiffThld=16;
//(Optional) Enabling optimized fallback upon uplink CA exceptions in low-frequency scenarios
MOD GNODEBCARRMGMT: CaEnhancedAlgoSwitch=UL_CA_FALLBACK_DL_ONLY_CA_SW-1;
//(Optional) Setting the UlCaSccSplitSinrThld and UlCaContNonSplitSccDelThld parameters
MOD NRDUCELLCARRMGMT: NrDuCellId=0, UlCaSccSplitSinrThld=-2, UlCaContNonSplitSccDelThld=3;
//(Optional) Turning on UL_CA_SPLIT_THLD_OPT_SW if the accuracy of data split to SCells needs to be
improved
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=UL_CA_SPLIT_THLD_OPT_SW-1;
//(Optional) Turning on MF_ACTUAL_PUSCH_ILPC_SW in both the PCell and SCells (recommended with an
aim to improve the accuracy of data volume distributed to SCells in the uplink)
MOD NRDUCELLULPCCONFIG: NrDuCellId=0, UlPwrCtrlAlgoExtSwitch=MF_ACTUAL_PUSCH_ILPC_SW-1;
//(Optional) Turning on SRS_PUSCH_CONCURRENCY_COMPT_SW
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
CaCompatibilitySwitch=SRS_PUSCH_CONCURRENCY_COMPT_SW-1;
//(Optional) Turning on SMART_CARR_SEL_UL_ENH_SW if preference based on the uplink peak rate is
expected to take effect in the CA capability combination selection policy in scenarios where the optimal
uplink and downlink CA combinations are not in the same combination during a traffic volume model
change
MOD NRDUCELLCOLLABSERV: NrDuCellId=1, MultiCarrMgmtSw=SMART_CARR_SEL_UL_ENH_SW-1;
```

Optimization Command Examples

```
//Enabling the gNodeB to preferentially select the UE-reported band combination that offers the highest
peak data rate
```

```
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaBandCombSelectionStrat=PEAK_RATE_FIRST;  
//Enabling CA capability combination selection with SRS-based weights considered  
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaCompatibilitySwitch=SRS_WT_BASED_CARR_SEL_SW-1;  
//Turning on the smart carrier selection switch  
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, MultiCarrMgmtSw=SMART_CARR_SEL_SW-1;  
//Turning on the service-aware carrier selection switch  
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, MultiCarrMgmtSw=SERVICE_AWARE_CARR_SEL_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the SRS and PUSCH concurrency compatibility switch  
MOD NRDUCELLCARRMGMT: NrDuCellId=0,  
CaCompatibilitySwitch=SRS_PUSCH_CONCURRENCY_COMPT_SW-0;  
//Disabling whitelist control over MTA configuration prohibition for uplink CA for specified UEs in low-  
frequency scenarios  
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,  
WhitelistCtrlExtSwitch=UL_CA_MTAG_NOT_CONFIG_SW_ON-0;  
//Disabling flexible TA adjustment in uplink multi-CC scenarios  
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=UL_MULTI_CC_TA_FLEX_ADJ_SW-0;  
//Disabling MTA configuration prohibition for uplink CA in low-frequency scenarios  
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaCompatibilitySwitch=UL_CA_MTAG_NOT_CONFIG_SW-0;  
//Disabling optimization against LO feedthrough for uplink CA in low-frequency scenarios  
MOD NRDUCELLCARRMGMT: NrDuCellId=0,  
CaEnhancedAlgoExtSwitch=R16_CA_UE_LO_FEEDTHRU_OPT_SW-0;  
//Turning off the uplink intra-band CA algorithm switch in low-frequency scenarios  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CaAlgoSwitch=INTRA_BAND_UL_CA_SW-0;  
//Turning off both the uplink intra-band CA algorithm switch and the downlink intra-band CA algorithm  
switch in high-frequency scenarios  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
CaAlgoSwitch=INTRA_BAND_CA_SW-0&INTRA_BAND_UL_CA_SW-0;  
//Disabling optimized fallback upon uplink CA exceptions in low-frequency scenarios  
MOD GNODEBCARRMGMT: CaEnhancedAlgoSwitch=UL_CA_FALLBACK_DL_ONLY_CA_SW-0;  
//Turning off UL_CA_SPLIT_THLD_OPT_SW  
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=UL_CA_SPLIT_THLD_OPT_SW-0;  
//Turning off MF_ACTUAL_PUSCH_ILPC_SW  
MOD NRDUCELLULPCCONFIG: NrDuCellId=0, ULPwrCtrlAlgoExtSwitch=MF_ACTUAL_PUSCH_ILPC_SW-0;  
//Turning off SMART_CARR_SEL_UL_ENH_SW  
MOD NRDUCELLCOLLABSERV: NrDuCellId=1, MultiCarrMgmtSw=SMART_CARR_SEL_UL_ENH_SW-0;
```

5.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.2.4.2 Activation Verification

Provided that CA UEs are available on the network, uplink intra-band CA has taken effect if any counter in [Table 5-8](#) returns a non-zero value.

Table 5-8 Counters related to uplink intra-band CA

Counter ID	Counter Name
1911816775	N.User.CA.PCell.UL.Avg

Counter ID	Counter Name
1911853846	N.User.CA.PCell.UL.Max
1911816776	N.User.CA.SCell.UL.Avg
1911816901	N.User.CA.PCell.UL.SCellAct.Avg
1911816903	N.User.CA.SCell.UL.Act.Avg
1911827212	N.User.CA.PCell.UL.SCellAct.Max
1911840467	N.CA.PCell.ThpVol.UL.Cell
1911840469	N.CA.SCell.ThpVol.UL.Cell
1911840459	N.CA.PR.B.UL.PCell.Used.Avg
1911840461	N.CA.PR.B.UL.SCell.Used.Avg

The counters related to the numbers of attempts and successful executions regarding SCell configuration, change, activation, deactivation, and removal are listed in [Table 5-9](#).

Table 5-9 Counters related to SCell management

Counter ID	Counter Name
1911816777	N.CA.SCell.Add.Att
1911816778	N.CA.SCell.Add.Succ
1911853943	N.UL.CA.SCell.Add.Att
1911853944	N.UL.CA.SCell.Add.Succ
1911816781	N.CA.SCell.Mod.Att
1911816782	N.CA.SCell.Mod.Succ
1911816896	N.CA.SCell.Act.Att
1911816897	N.CA.SCell.Act.Succ
1911816898	N.CA.SCell.Deact.Att
1911816899	N.CA.SCell.Deact.Succ
1911816779	N.CA.SCell.Rmv.Att
1911816780	N.CA.SCell.Rmv.Succ

5.2.4.3 Network Monitoring

Observe **User Uplink Average Throughput (DU)** to verify the increase in the UE data rate after the function is enabled.

Observe the average throughput of SCell-activated uplink CA UEs using the following calculation: $(N.CA.SccActiveUser.ThpVol.UL - N.CA.ThpVol.UL.SmallPkt) / N.CA.ThpTime.UL.RmvSmallPkt$.

Observe the uplink traffic volume for SCell-activated CA UEs in PCells using the counter **N.CA.PCell.ThpVol.UL.Cell**.

Observe the uplink traffic volume for CA UEs in SCells using the counter **N.CA.SCell.ThpVol.UL.Cell**.

5.3 Flexible Carrier Management (High-Frequency TDD)

5.3.1 Principles

A CA UE with SCells configured consumes the same number of sales units of the license for the number of RRC_CONNECTED UEs as it does before the SCells are configured. However, the CA UE consumes a hardware resource unit in each of its serving cells. With 2CC aggregation for all UEs on the network, the maximum allowed number of RRC_CONNECTED UEs is halved. In high-frequency scenarios where more carriers can be aggregated, this number is proportionally lower. To reduce hardware resource consumption by CA UEs and increase the maximum allowed number of RRC_CONNECTED UEs on the network, flexible carrier management has been introduced for CA in high frequency bands.

Flexible carrier management is implemented through the traffic-volume-based carrier management policy for high-frequency CA by setting the **NRDUCellCarrMgmt.Fr2CaCarrMgmtPolicy** parameter to **USER_TRAFFIC_BASED**. (This parameter specifies the carrier management policy for high-frequency CA and is set to **USER_EXP_BASED** by default.) This function works for UEs in the high-frequency contiguous CA state in common cells (with the **NRDUCell.SupplementaryCellInd** parameter set to **NORMAL_CELL**). It is recommended that flexible carrier management be enabled when cells are moderately- or heavily-loaded.

The user-experience-based policy and the traffic-volume-based policy differ in SCell configuration and removal, as described in the following table.

Table 5-10 Differences between the carrier management policies for high-frequency CA in SCell configuration and removal

Carrier Management Policy for High-Frequency CA	SCell Configuration	SCell Removal
User-experience-based policy	The full SCell configuration policy is used. That is, SCells are configured based on the capabilities of the UE and gNodeB.	SCells are removed if they become unavailable.
Traffic-volume-based policy	SCells are configured dynamically. - If the number of hardware resource units consumed by RRC_CONNECTED UEs on a baseband processing unit or in a sector is higher than hardware resource threshold 1, an SCell is configured for 2CC aggregation initially. Then, the gNodeB checks every 100 ms whether the buffered traffic volume of the UE meets the traffic volume-triggered SCell activation conditions described in 4.6.3 SCell Activation. If the conditions are met three consecutive times, the gNodeB triggers SCell configuration until the number of successfully configured SCells reaches the maximum. - If the number of hardware resource units consumed by RRC_CONNECTED UEs on a baseband processing unit or in a sector is lower than hardware resource threshold 2, the full configuration policy is used.	- If the number of hardware resource units consumed by RRC_CONNECTED UEs on a baseband processing unit or in a sector is higher than hardware resource threshold 1: - SCells are removed dynamically: An SCell is removed if it has been in the deactivated state for more than 5s. - SCells are removed if they become unavailable. - If the number of hardware resource units consumed by RRC_CONNECTED UEs on a baseband processing unit or in a sector is lower than hardware resource threshold 2, SCells are removed if they become unavailable.

 NOTE

The number of hardware resource units consumed by RRC_CONNECTED UEs on a baseband processing unit or in a sector equals $\text{Sum}_{\text{All cells on the baseband processing unit or in the sector}}\{\text{N.User.RRConn.Avg} + \text{N.User.CA.SCell.DL.Avg}\}$.

For a sector, hardware resource thresholds 1 and 2 equal 300 and 200, respectively. For a baseband processing unit, hardware resource thresholds 1 and 2 equal 600 and 400, respectively.

If the number of hardware resource units consumed by RRC_CONNECTED UEs on a baseband processing unit or in a sector changes to be within the range of [Threshold 2, Threshold 1], the processing is the same as that before the change.

In NSA networking, you are advised to set the **NrScgFreqConfig.ScgDLArfcnPriority** parameter to the same value for all frequencies on which common cells in a high-frequency sector operate.

Before enabling this function, you are advised to enable high-frequency UE-number-based idle mode MLB (controlled by the **INTER_FREQ_IDLE_MLB_SW** option of the **NRCCellAlgoSwitch.MlbAlgoSwitch** parameter) in SA networking and to enable UE-number-based MLB in frequency rotation mode (controlled by the **FR1_TO_FR2_FREQ_RR** option of the **NRCCellAlgoSwitch.InterFreqHoSwitch** parameter) in high- and low-frequency hybrid networking. For details about high-frequency UE-number-based idle mode MLB and UE-number-based MLB in frequency rotation mode, see *Mobility Load Balancing in Idle Mode* and *Mobility Load Balancing in Connected Mode*, respectively.

5.3.2 Network Analysis

5.3.2.1 Benefits

After flexible carrier management is enabled, the gNodeB configures and removes SCells based on UE traffic requirements. This reduces the average number of hardware resource units consumed by RRC_CONNECTED UEs on a baseband processing unit or in a sector and increases the maximum allowed number of RRC_CONNECTED UEs.

 NOTE

The increase in the maximum allowed number of RRC_CONNECTED UEs is related to UE location distribution, traffic model, and the usage of other network resources (such as PRB resources). With all UEs supporting *n*CC aggregation on the network, this maximum number can increase by *n* times in extreme cases.

5.3.2.2 Impacts

The impacts between this function and other functions are described in [5.3.3.2.3 Function Impacts](#).

When this function takes effect, the number of RRC connection reconfigurations over the air interface increases and the average number of SCells configured for CA UEs decreases. In addition, the following indicator changes are incurred (you are advised to observe the overall changes in all cells of a sector as cell-level KPIs or indicators fluctuate due to changes in inter-cell UE scheduling in the sector):

- The proportion of CA UEs may decrease, which is indicated by the following counters:
 - **N.User.CA.PCell.DL.Avg**, which measures the average number of downlink CA UEs that treat the local cell as their PCell
 - **N.User.CA.SCell.DL.Avg**, which measures the average number of downlink CA UEs that treat the local cell as their SCell
 - **N.User.CA.PCell.UL.Avg**, which measures the average number of uplink CA UEs that treat the local cell as their PCell
 - **N.User.CA.SCell.UL.Avg**, which measures the average number of uplink CA UEs that treat the local cell as their SCell
- The measured values of the following indicators may change:
 - Average uplink and downlink cell throughputs
 Average uplink cell throughput = **Cell Uplink Average Throughput (DU)**
 Average downlink cell throughput = **Cell Downlink Average Throughput (DU)**
 - Average uplink and downlink UE throughputs
 Average uplink UE throughput = **User Uplink Average Throughput (DU)**
 Average downlink UE throughput = **User Downlink Average Throughput (DU)**
 - Delay
 First Packet Delay in RLC Empty Buffer = **N.Traffic.DL.RlcFirstPktDelay.Time/N.QoS.DL.RlcFirstPktDelay.Num**
 Packet Delay in MAC = **N.QoS.DL.PktDelayAirInterface.Time/N.QoS.DL.PktDelayAirInterface.Num**
 Packet Delay in RLC = **N.QoS.DL.PktDelaygNBdu.Time/N.QoS.DL.PktDelaygNBdu.Num**
 Packet Delay in DU = **N.QoS.DL.PktDelayAirInterface.Time/N.QoS.DL.PktDelayAirInterface.Num + N.QoS.DL.PktDelaygNBdu.Time/N.QoS.DL.PktDelaygNBdu.Num**
 - Average number of active UEs in a cell
 This is measured using the **N.User.RRConn.Active.Avg** counter.
 - Average MCS indexes for uplink and downlink scheduling
 Average MCS index for uplink scheduling (PUSCH Scheduling MCS) = $(0 \times \text{N.ChMeas.PUSCH.MCS.0} + 1 \times \text{N.ChMeas.PUSCH.MCS.1} + 2 \times \text{N.ChMeas.PUSCH.MCS.2} + 3 \times \text{N.ChMeas.PUSCH.MCS.3} + \dots + 26 \times \text{N.ChMeas.PUSCH.MCS.26} + 27 \times \text{N.ChMeas.PUSCH.MCS.27} + 28 \times \text{N.ChMeas.PUSCH.MCS.28}) / (\text{N.ChMeas.PUSCH.MCS.0} + \text{N.ChMeas.PUSCH.MCS.1} + \text{N.ChMeas.PUSCH.MCS.2} + \text{N.ChMeas.PUSCH.MCS.3} + \dots + \text{N.ChMeas.PUSCH.MCS.26} + \text{N.ChMeas.PUSCH.MCS.27} + \text{N.ChMeas.PUSCH.MCS.28})$
 Average MCS index for downlink scheduling (PDSCH Scheduling MCS) = $(0 \times \text{N.ChMeas.PDSCH.MCS.0} + 1 \times \text{N.ChMeas.PDSCH.MCS.1} + 2 \times \text{N.ChMeas.PDSCH.MCS.2} + 3 \times \text{N.ChMeas.PDSCH.MCS.3} + \dots + 26 \times \text{N.ChMeas.PDSCH.MCS.26} + 27 \times \text{N.ChMeas.PDSCH.MCS.27} + 28 \times \text{N.ChMeas.PDSCH.MCS.28}) / (\text{N.ChMeas.PDSCH.MCS.0} + \text{N.ChMeas.PDSCH.MCS.1} + \text{N.ChMeas.PDSCH.MCS.2} + \dots + \text{N.ChMeas.PDSCH.MCS.26} + \text{N.ChMeas.PDSCH.MCS.27} + \text{N.ChMeas.PDSCH.MCS.28})$

- $N.ChMeas.PDSCH.MCS.3 + \dots + N.ChMeas.PDSCH.MCS.26 + N.ChMeas.PDSCH.MCS.27 + N.ChMeas.PDSCH.MCS.28$**
- Average uplink and downlink ranks
 Average uplink rank (AvgULRank) = $(1 \times N.PUSCH.InitTbUL.Rank1 + 2 \times N.PUSCH.InitTbUL.Rank2) / (N.PUSCH.InitTbUL.Rank1 + N.PUSCH.InitTbUL.Rank2)$
 Average downlink rank (AvgDLRank) = $(1 \times N.PDSCH.InitTbDL.Rank1 + 2 \times N.PDSCH.InitTbDL.Rank2 + 3 \times N.PDSCH.InitTbDL.Rank3 + 4 \times N.PDSCH.InitTbDL.Rank4) / (N.PDSCH.InitTbDL.Rank1 + N.PDSCH.InitTbDL.Rank2 + N.PDSCH.InitTbDL.Rank3 + N.PDSCH.InitTbDL.Rank4)$
 - CCE usage
 CCE usage = $N.CCE.Used.Avg / N.CCE.Avail.Avg \times 100\%$
 - Average uplink and downlink RB usages in a cell
 Average uplink RB usage in a cell = **Uplink Resource Block Utilizing Rate (DU)**
 Average downlink RB usage in a cell = **Downlink Resource Block Utilizing Rate (DU)**
 - Uplink and downlink HARQ retransmission rates
 Uplink HARQ retransmission rate (Uplink HARQ Retrans Rate) = $(N.UL.SCH.HalfPiBPSK.TB.Retrans + N.UL.SCH.QPSK.TB.Retrans + N.UL.SCH.16QAM.TB.Retrans + N.UL.SCH.64QAM.TB.Retrans + N.UL.SCH.256QAM.TB.Retrans) / (N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB + N.UL.SCH.HalfPiBPSK.TB.Retrans + N.UL.SCH.QPSK.TB.Retrans + N.UL.SCH.16QAM.TB.Retrans + N.UL.SCH.64QAM.TB.Retrans + N.UL.SCH.256QAM.TB.Retrans) \times 100\%$
 Downlink HARQ retransmission rate (Downlink HARQ Retrans Rate) = $(N.DL.SCH.QPSK.TB.Retrans + N.DL.SCH.16QAM.TB.Retrans + N.DL.SCH.64QAM.TB.Retrans + N.DL.SCH.256QAM.TB.Retrans) / (N.DL.SCH.QPSK.TB + N.DL.SCH.16QAM.TB + N.DL.SCH.64QAM.TB + N.DL.SCH.256QAM.TB + N.DL.SCH.QPSK.TB.Retrans + N.DL.SCH.16QAM.TB.Retrans + N.DL.SCH.64QAM.TB.Retrans + N.DL.SCH.256QAM.TB.Retrans) \times 100\%$
 - Uplink and downlink IBLERs/RBLERs
 Uplink IBLER (Uplink IBLER Rate) = $(N.UL.SCH.HalfPiBPSK.ErrTB.Ibler + N.UL.SCH.QPSK.ErrTB.Ibler + N.UL.SCH.16QAM.ErrTB.Ibler + N.UL.SCH.64QAM.ErrTB.Ibler + N.UL.SCH.256QAM.ErrTB.Ibler) / (N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB)$
 Downlink IBLER (Downlink IBLER Rate) = $(N.DL.SCH.QPSK.ErrTB.Ibler + N.DL.SCH.16QAM.ErrTB.Ibler + N.DL.SCH.64QAM.ErrTB.Ibler + N.DL.SCH.256QAM.ErrTB.Ibler) / (N.DL.SCH.QPSK.TB + N.DL.SCH.16QAM.TB + N.DL.SCH.64QAM.TB + N.DL.SCH.256QAM.TB)$
 Uplink RBLER (Uplink MAC TB BLER) = $(N.UL.SCH.HalfPiBPSK.ErrTB.Rbler + N.UL.SCH.QPSK.ErrTB.Rbler + N.UL.SCH.16QAM.ErrTB.Rbler + N.UL.SCH.64QAM.ErrTB.Rbler + N.UL.SCH.256QAM.ErrTB.Rbler) / (N.UL.SCH.HalfPiBPSK.TB +$

$$\text{N.UL.SCH.QPSK.TB} + \text{N.UL.SCH.16QAM.TB} + \text{N.UL.SCH.64QAM.TB} + \text{N.UL.SCH.256QAM.TB}) \times 100\%$$

$$\begin{aligned} \text{Downlink RBLER (Downlink MAC TB BLER)} = \\ (\text{N.DL.SCH.QPSK.ErrTB.Rbler} + \text{N.DL.SCH.16QAM.ErrTB.Rbler} + \\ \text{N.DL.SCH.64QAM.ErrTB.Rbler} + \text{N.DL.SCH.256QAM.ErrTB.Rbler}) / \\ (\text{N.DL.SCH.QPSK.TB} + \text{N.DL.SCH.16QAM.TB} + \text{N.DL.SCH.64QAM.TB} + \\ \text{N.DL.SCH.256QAM.TB}) \times 100\% \end{aligned}$$

- Average CQI

$$\begin{aligned} \text{Average CQI (AvgCqiSingleCw)} = (0 \times \text{N.ChMeas.CQI.SingleCW.0} + 1 \times \\ \text{N.ChMeas.CQI.SingleCW.1} + 2 \times \text{N.ChMeas.CQI.SingleCW.2} + 3 \times \\ \text{N.ChMeas.CQI.SingleCW.3} + \dots + 13 \times \text{N.ChMeas.CQI.SingleCW.13} + 14 \\ \times \text{N.ChMeas.CQI.SingleCW.14} + 15 \times \text{N.ChMeas.CQI.SingleCW.15}) / \\ (\text{N.ChMeas.CQI.SingleCW.0} + \text{N.ChMeas.CQI.SingleCW.1} + \\ \text{N.ChMeas.CQI.SingleCW.2} + \text{N.ChMeas.CQI.SingleCW.3} + \dots + \\ \text{N.ChMeas.CQI.SingleCW.13} + \text{N.ChMeas.CQI.SingleCW.14} + \\ \text{N.ChMeas.CQI.SingleCW.15}) \end{aligned}$$

- Uplink and downlink CCE aggregation levels

$$\begin{aligned} \text{Uplink CCE aggregation level (UL CCE AGG Level)} = \\ (\text{N.CCE.UL.AggLvl16Num} \times 16 + \text{N.CCE.UL.AggLvl8Num} \times 8 + \\ \text{N.CCE.UL.AggLvl4Num} \times 4 + \text{N.CCE.UL.AggLvl2Num} \times 2 + \\ \text{N.CCE.UL.AggLvl1Num}) / (\text{N.CCE.UL.AggLvl16Num} + \\ \text{N.CCE.UL.AggLvl8Num} + \text{N.CCE.UL.AggLvl4Num} + \\ \text{N.CCE.UL.AggLvl2Num} + \text{N.CCE.UL.AggLvl1Num}) \end{aligned}$$

$$\begin{aligned} \text{Downlink CCE aggregation level (DL CCE AGG Level)} = \\ (\text{N.CCE.DL.AggLvl16Num} \times 16 + \text{N.CCE.DL.AggLvl8Num} \times 8 + \\ \text{N.CCE.DL.AggLvl4Num} \times 4 + \text{N.CCE.DL.AggLvl2Num} \times 2 + \\ \text{N.CCE.DL.AggLvl1Num}) / (\text{N.CCE.DL.AggLvl16Num} + \\ \text{N.CCE.DL.AggLvl8Num} + \text{N.CCE.DL.AggLvl4Num} + \\ \text{N.CCE.DL.AggLvl2Num} + \text{N.CCE.DL.AggLvl1Num}) \end{aligned}$$

- Interference and noise received per PRB in the uplink

This is measured using counters **N.UL.NI.Avg**, **N.UL.NI.Max**, and **N.UL.NI.Min**.

NOTE

If fewer SCells are configured for a UE, the power consumption of the UE decreases.

5.3.3 Requirements

5.3.3.1 Licenses

None

5.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference
High-frequency TDD	Intra-band CA	INTRA_BAND_CA_SW and INTRA_BAND_UL_CA_SW options of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>

5.3.3.2.2 Mutually Exclusive Functions

None

5.3.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Reducing the number of SCCs	OVERHEATING_ASSISTANCE_INFO_SW and UE_SCC_DEACTIVATE_SW options of the NRCCellAlgoSwitch.UeAssistanceInfoSwitch parameter	<i>UE Power Saving</i>	• If an SCell is deactivated for overheating protection or energy saving for more than 5s, dynamic SCell removal will not be triggered. • The gNodeB does not trigger dynamic SCell configuration for UEs in the overheated or energy saving state.

5.3.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

5900 series base stations (macro base stations)

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

5.3.3.4 Cells

If **NRDUCell.SlotAssignment** is set to **2_3_DSUUU**,
NRDUCellCarrMgmt.Fr2CaCarrMgmtPolicy must be set to **USER_EXP_BASED**.

5.3.4 Operation and Maintenance

5.3.4.1 Data Configuration

5.3.4.1.1 Data Preparation

Table 5-11 describes the parameters used for function activation.

Table 5-11 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
FR2 CA Carrier Management Policy	NRDUCellCarrMgmt.Fr2CaCarrMgmtPolicy	It is recommended that this parameter be set to USER_TRAFFIC_BASED to enable flexible carrier management when the cell is moderately or heavily loaded.

5.3.4.1.2 Using MML Commands

Before using MML commands, refer to **5.3.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling flexible carrier management for all high-frequency normal cells  
MOD NRDUCELLCARRMGMT: NrDuCellId=0, Fr2CaCarrMgmtPolicy=USER_TRAFFIC_BASED;  
MOD NRDUCELLCARRMGMT: NrDuCellId=1, Fr2CaCarrMgmtPolicy=USER_TRAFFIC_BASED;
```

Deactivation Command Examples

```
//Disabling flexible carrier management for all high-frequency normal cells  
MOD NRDUCELLCARRMGMT: NrDuCellId=0, Fr2CaCarrMgmtPolicy=USER_EXP_BASED;  
MOD NRDUCELLCARRMGMT: NrDuCellId=1, Fr2CaCarrMgmtPolicy=USER_EXP_BASED;
```

5.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.3.4.2 Activation Verification

Flexible carrier management has taken effect if the value of any of the following indicators increases after this function is enabled:

- **N.CA.SCell.Rmv.Att**, which measures the number of SCell removal attempts for CA UEs
- **N.CA.SCell.Rmv.Succ**, which measures the number of successful SCell removals for CA UEs

5.3.4.3 Network Monitoring

Evaluate the benefits of this function by checking how much the average number of hardware resource units consumed by RRC_CONNECTED UEs on a baseband processing unit or in a sector decreases. (This average number is calculated using the formula described in [5.3.2.1 Benefits](#).) Other indicators also change after this function takes effect. For details, see [5.3.2.2 Impacts](#).

6 Intra-FR Inter-Band CA

6.1 Downlink Intra-FR Inter-Band CA

6.1.1 Principles

6.1.1.1 Overview

Currently, downlink intra-FR inter-band CA is supported in FR1 but not in FR2.

Downlink intra-FR inter-band CA can be enabled by selecting the **INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch.CaAlgoSwitch** parameter. With this function enabled, inter-band SCell configuration is triggered when a CA UE initiates RRC connection setup during initial access, an incoming handover, or an incoming RRC connection reestablishment. After SCells are configured and activated, the CA UE can benefit from CA. 2CC aggregation is supported by default. Aggregation of more than two carriers requires certain other parameter settings. For details about the configuration, see *CA Experience Improvement*.

To learn the band+bandwidth combinations for downlink intra-FR inter-band CA, see [4.4 CA Band+Bandwidth Combinations](#).

6.1.1.2 SCell CSI-RS Beam Sweeping

Basic Principles

For downlink CA to take effect, the gNodeB must obtain the directions of CSI-RS beams for a UE in the PCell and SCells.

In asymmetric CA, some SCells are configured only for the downlink and therefore the gNodeB cannot obtain the beam directions for the UE in these SCells based on the uplink. As a result, these SCells cannot be used. When all the following conditions are met, however, the beam directions for the UE in the PCell and SCells are the same, and the gNodeB can prevent the preceding issue with the use of the CSI-RS beam direction in the PCell:

- The PCell and SCells operate in the same band.
- The PCell and SCells are served by the same RF module.
- The spacing between the center frequencies of the PCell and each SCell is less than 500 MHz. The center frequencies of these cells are specified by the **NRDUCell.UINarfcn** and **NRDUCell.DINarfcn** parameters.

If not all the preceding conditions are met, the beam direction in the PCell cannot be used for SCells. As a result, the SCells cannot be used.

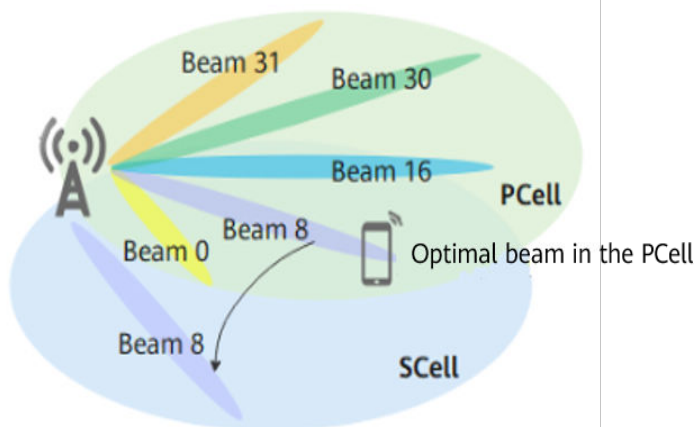
To address this issue, the SCell CSI-RS beam sweeping function is introduced. This function is controlled by the **CARR_CSI_BEAM_SWEEPING_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoSwitch** parameter. This function must be enabled when the beam directions in the PCell and SCells differ. Otherwise, SCells cannot be configured and UE throughput in the PCell decreases due to frequent gap-assisted measurements for SCell configuration.

Figure 6-1 illustrates the SCell CSI-RS beam sweeping function, which works as follows:

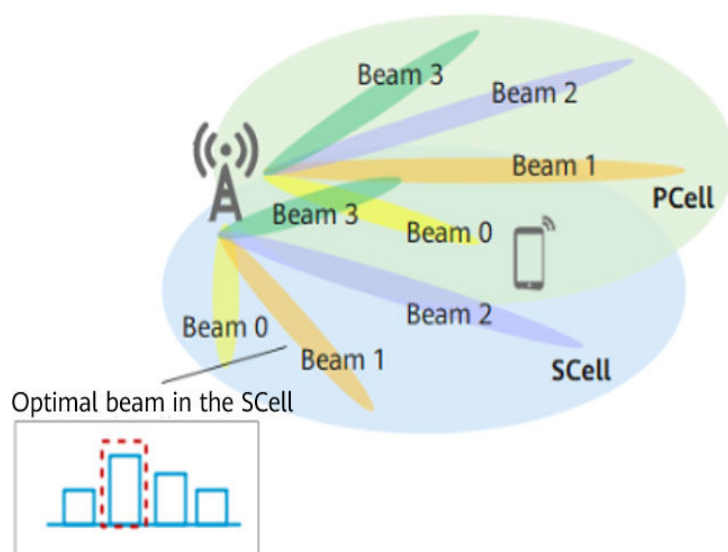
1. The gNodeB sends CSI-RS for beam measurement (CSI-RS for BM) in an SCell on CSI-RS beams in different directions in turn, so as to obtain the optimal beam for the UE in the SCell. Then the gNodeB sends CSI-RS for channel measurement (CSI-RS for CM) on this optimal beam to the UE.
2. The UE performs measurements for CQI, PMI, and rank indication (RI) based on the CSI-RS for CM sent by the SCell.
3. The UE reports the CQI, RI, and PMI measurement results of the SCell on the PUCCH or PUSCH in the PCell.

Figure 6-1 SCell CSI-RS beam sweeping

CSI-RS beam directions are the same in the PCell and SCell.



CSI-RS beam directions are different in the PCell and SCell.



NOTE

In symmetric CA scenarios, the gNodeB obtains the CSI-RS beam directions for a UE in SCells based on the uplink. The SCell CSI-RS beam sweeping function does not take effect even if it is enabled.

SCell CSI-RS beam sweeping needs to be enabled only when the SCell has multiple CSI-RS beams. It does not need to be enabled when the SCell has only one CSI-RS beam. For details about the number of CSI-RS beams in a cell, see *Beam Management*.

CSI-RS Beam Sweeping Optimization

CSI-RS beam sweeping described in [Basic Principles](#) is performed in either periodic or aperiodic mode. After the sweeping succeeds, the UE reports the optimal beam in an SCell. Resources must be reserved for such information

reporting in both periodic and aperiodic sweeping scenarios (or only the supported sweeping scenario if the UE does not support both). The resources available for UEs to report information are limited. If resources are insufficient, other features that require UE information reporting for CA carriers may fail to take effect. Considering that periodic CSI-RS beam sweeping has a high success rate, CSI-RS beam sweeping optimization is introduced to reduce the resource consumption by UE information reporting. CSI-RS beam sweeping optimization is controlled by the **CSI_BEAM_SWEEPING_OPT_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoSwitch** parameter.

- If this option is selected:
 - If a UE supports periodic sweeping, such sweeping is performed and resources are reserved only for UE information reporting for periodic sweeping.
 - If a UE does not support periodic sweeping, aperiodic sweeping is performed and resources are reserved only for UE information reporting for aperiodic sweeping.
- If this option is deselected, CSI-RS beam sweeping optimization does not take effect.

Adaptive CSI-RS Beam Sweeping

The **maxNumberSSB-CSI-RS-ResourceOneTx** field is carried in the capability information that a UE reports to the gNodeB. This field indicates a maximum total quantity of specific reference signal resources (including CSI-RS signal resources) that can be used by the UE to simultaneously measure L1-RSRP in all serving cells in a slot. Without adaptive CSI-RS beam sweeping, the gNodeB does not check whether the total number of CSI-RS resources required by all CCs exceeds the total number of CSI-RS resources supported by a UE during CA configuration. As a result, the total number of CSI-RS resources required by all CCs may exceed the UE capability, and some UEs may experience RRC connection reestablishments or service drops. As such, adaptive CSI-RS beam sweeping is introduced, which is controlled by the **CA_CSIRS_BM_SWEEP_ADPT_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoSwitch** parameter. After this function takes effect, the gNodeB checks whether the CSI-RS resources required by all CCs exceed the total number of CSI-RS resources supported by a UE. If this is true, the gNodeB either avoids SCell configuration or reduces the number of SCCs to prevent issues such as reestablishment or service drops of UEs.

6.1.1.3 CA PDCCH Capacity Optimization

Assume that a UE receives an uplink grant and then a downlink grant in a single slot and sends HARQ feedback on the PUSCH. For certain such UEs, the downlink block error rate (BLER) increases. This issue can be prevented when the start PDCCH symbol for CA UEs is **fixed** at the first available PDCCH symbol in the cells serving these UEs. However, PDCCH resources in the cells are allocated less flexibly for CA UEs using this method. Data transmission for CA UEs may even be affected in the case of insufficient PDCCH resources in heavy-load scenarios.

CA PDCCH capacity optimization is therefore introduced to improve data transmission for CA UEs in heavy-load scenarios. This function allows the start symbol of the PDCCH to be flexibly selected for CA UEs. In addition, the scheduling timing is internally optimized to prevent the downlink BLER from

increasing. This function is controlled by the **CA_PDCCH_CAPACITY_OPT_SWITCH** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch** parameter. This function is supported only in low frequency bands, not in high frequency bands.

6.1.1.4 32-Port CSI-RS Measurement on SCCs

If a UE performs 8-port CSI-RS measurement in a low-frequency cell with 32 or more transmit channels, this downlink channel measurement result is not accurate enough for CA to offer the optimal user experience. To improve the channel measurement accuracy, when such a cell serves as an SCell for a UE, the UE can now perform 32-port CSI-RS measurements on the SCC. This increases the average downlink UE throughput and average downlink cell throughput. This function is controlled by the **CA_SCC_CSIRS_32PORT_SW** option of the **NRDUCellCsirs.CsiSwitch** parameter. It works in low frequency bands, not in high frequency bands.

6.1.1.5 CA PDCCH Aggregation Level Optimization

CA PDCCH aggregation level optimization has been introduced to improve the downlink data transmission performance of CA UEs. This function applies an upper limit (specified by the **NRDUCellPdcchAlgo.PdcchOuterLoopAdjUpLimit** parameter) to the accumulated outer-loop adjustment amount in PDCCH aggregation level adaptation for CA UEs. Outer-loop adjustment for the PDCCH can be performed when the accumulated outer-loop adjustment amount is lower than the upper limit, but it can no longer be performed when this accumulated amount is greater than or equal to the upper limit. A smaller value of the **NRDUCellPdcchAlgo.PdcchOuterLoopAdjUpLimit** parameter indicates a lower upper limit and results in a higher PDCCH aggregation level that is adaptively selected. A larger value results in a lower PDCCH aggregation level. This function is supported in low frequency bands, but not in high frequency bands.

6.1.1.6 Dynamic-Codebook-based HARQ Feedback over the PUCCH

3GPP TS 38.331 has defined two HARQ-ACK codebook types: type 1 and type 2.

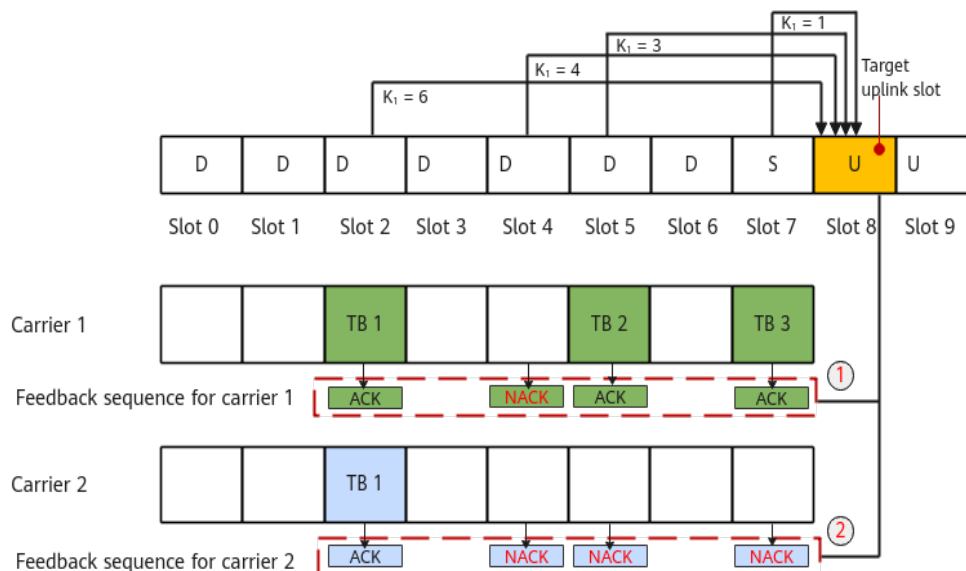
- Type-1 codebooks are semi-static HARQ-ACK codebooks. Once K_1 is configured by the higher layer, the UE provides, in each uplink slot, the HARQ feedback about all the slots that are K_1 slots prior to the current feedback slot. In those slots, data transmission is possible.
- Type-2 codebooks are dynamic HARQ-ACK codebooks. Unlike a UE using a semi-static codebook, a UE using a dynamic codebook sends HARQ feedback only about the resources where data was transmitted.

Figure 6-2 provides an example. In this example, the UE can simultaneously receive data over at most two carriers, and it correctly receives data over both carriers. K_1 received by the UE is {1, 3, 4, 6}, which is configured in the PCell in multi-carrier cases.

- If a semi-static HARQ-ACK codebook is used, the UE sends the HARQ feedback about slots 2, 4, 5, and 7 of both carriers in the target HARQ feedback slot 8. In this situation, eight bits are required for the feedback, regardless of whether the UE has received data in those slots.
- If a dynamic HARQ-ACK codebook is used, the UE sends the HARQ feedback only about slots with data transmission (that is, slots 2, 5, and 7 of carrier 1

and slot 2 of carrier 2) in the target HARQ feedback slot 8. In this situation, only four bits are required for the feedback.

Figure 6-2 Example of data transmission in the 2CC aggregation state



As shown in this example, the use of a semi-static HARQ-ACK codebook consumes more resources than the use of a dynamic HARQ-ACK codebook. Currently, semi-static HARQ-ACK codebooks are always used for the PUCCH, resulting in a limited number of CA UEs that can be scheduled in a cell. To address this issue, dynamic HARQ-ACK codebooks can now be used for the PUCCH. This usage is controlled by the **PUCCH_DYN_HARQ_ACK_CB_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch** parameter.

- If this option is selected and all the following conditions are met, the gNodeB sets the *PDSCH-HARQ-ACK-Codebook* field to **dynamic** when starting SCell configuration for a CA UE during initial access, an incoming handover, or an incoming RRC connection reestablishment, and the UE then uses dynamic HARQ-ACK codebooks for the PUCCH:
 - The number of SCell-configured UEs is greater than the value of the **NRDUCellCarrMgmt.DynCodebookCaUeNumThld** parameter, or the number of RRC_CONNECTED UEs in the PCell is greater than 800.
 - The UE is not an inter-gNodeB CA UE, for which inter-gNodeB SCells have been configured.
- If this option is deselected, UEs use semi-static HARQ-ACK codebooks.

In medium- and heavy-load scenarios (where the PRB usage is greater than 70%), there will be limited specifications of dynamic HARQ-ACK codebooks over the PUCCH, leading to a limited number of CA UEs that can be scheduled in a cell. As such, dynamic PUCCH codebook specification improvement for CA UEs is introduced. This function is controlled by the **PUCCH_DYN_CB_SPECS_IMP_SW** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter. After this function is enabled, there are more available dynamic HARQ-ACK codebooks over the PUCCH for CA UEs, and the number of CA UEs that can be scheduled in a cell increases.

6.1.2 Network Analysis

6.1.2.1 Benefits

The downlink peak data rates of CA UEs increase.

Table 6-1 lists the ranges of peak data rates that a CA UE can achieve using downlink 2CC aggregation in low-speed scenarios.

Table 6-1 Ranges of peak data rates achieved using downlink 2CC aggregation in low-speed scenarios

SCell Weight Type	Peak Data Rate Range
SRS-based	$(A \times 98\% + B \times 98\%)$ to $(A \times 100\% + B \times 100\%)$
PMI-based	$(A \times 98\% + B \times 95\%)$ to $(A \times 100\% + B \times 100\%)$
Note 1: The SCell weight type depends on CA UE capabilities. If a CA UE transmits SRSs in an SCell, SRS-based weights are used in the SCell. Otherwise, PMI-based weights are used in the SCell. Note 2: A is the downlink data rate that the UE in the single-carrier state can achieve in a cell that is later configured as the PCell for the UE. Note 3: B is the downlink data rate that the UE in the single-carrier state can achieve in a cell that is later configured as an SCell and uses SRS- or PMI-based weights.	

In high-speed scenarios, a CA UE in the downlink 2CC aggregation state can reach a peak data rate in the range of $(A \times 95\% + B \times 85\%)$ to $(A \times 98\% + B \times 95\%)$. In this notation, A is the downlink data rate that the UE in the single-carrier state can achieve in a cell that is later configured as the PCell for the UE. B is the downlink data rate that the UE in the single-carrier state can achieve in a cell that is later configured as an SCell for the UE.

 NOTE

The range of peak data rates that CA UEs can reach in high-speed scenarios is lower than that in low-speed scenarios. That is because in high-speed scenarios, the SCC performance gain decreases as affected by Doppler shift and significant channel condition fluctuation.

For the ranges of peak data rates that a CA UE can achieve using aggregation of more than two CCs in the downlink, see *CA Experience Improvement*.

 NOTE

If the PCell of a UE uses the dual-period 8:2 slot configuration and an SCell uses the single-period 8:2 slot configuration or the other way round, one slot in the SCell is unavailable for scheduling due to the restriction that no more than eight k1 values can be configured for a single UE in section 6.3.2 "Radio resource control information elements" of 3GPP TS 38.331. This results in a 1/8 loss in the downlink peak rate of the UE in this SCell, without affecting the downlink peak rate in the PCell. It also affects the overall downlink peak rate of the UE. The amount of the impact varies depending on the PCell and SCell bandwidths. For example, if the two cells have the same bandwidth, there will be a 1/16 loss. The loss in the overall downlink peak rate of the UE is even greater when there are UE compatibility issues. UEs cannot achieve the preceding maximum downlink peak data rates because section 4.1 "Supported max data rate" in 3GPP TS 38.306 V15.6.0 places a limit on the maximum downlink peak data rates of UEs. The following methods can be used to remove the limit:

- For certain UEs powered by chips with strong processing capabilities: Use device-pipe synergy for gNodeBs to identify UEs that support this synergy function. After such UEs are identified, the gNodeBs will remove their limit. In this way, these UEs can achieve higher downlink peak data rates as their chips allow for rates beyond the limit. To enable this function, select both the **SPEC_UE_IDENTIFY_SW** and **FR1_DL_TBS_LIMIT_CANCEL_SW** options of the **NRDUCellUeCoop.SpecUeCooperationSwitch** parameter.
- For all UEs: Reduce the value of the parameter *OH* (indicating overhead) used in the formula for calculating the downlink TBS. The parameter *OH* is specified by the **NRDUCellCarrMgmt.Fr1CaDlTbsParamOh** parameter. A larger value of this parameter results in a lower upper limit to the TBS and a lower upper limit to the downlink peak rate of a UE. A smaller value of this parameter yields the opposite impacts.

After CA is enabled, downlink UE throughput increases. In addition, first packet delay decreases, which equals **N.Traffic.DL.RlcFirstPktDelay.Time** divided by **N.Qos.DL.RlcFirstPktDelay.Num**.

The CA enhancements provide the following benefits:

- SCell CSI-RS beam sweeping can increase the **User Downlink Average Throughput (DU)** for CA UEs in asymmetric inter-band CA scenarios. For the definitions of symmetric and asymmetric CA, see [4.1 Related Concepts](#). CSI-RS beam sweeping optimization in SCell CSI-RS beam sweeping further optimizes the allocation of CA UE reporting capabilities among multiple carriers, improving CA UE performance. After adaptive CSI-RS beam sweeping takes effect, the values of the following indicators decrease if the number of CSI resources for UEs on the network is less than the number of CSI resources required by all CCs: (1) **Service Call Drop Rate (CU, Inactive)**; (2) RRC reestablishment rate ($(N.RRC.ReEst.Att / (N.RRC.SetupReq.Succ + N.RRC.ResumeReq.Succ + N.HO.InterRAT.E2N.ExecSuccIn)) \times 100\%$).
- CA PDCCH capacity optimization, when enabled in heavy-load scenarios, increase both the **User Downlink Average Throughput (DU)** and **Cell Downlink Average Throughput (DU)**.
- 32-port CSI-RS measurement on SCCs increases both the **User Downlink Average Throughput (DU)** and **Cell Downlink Average Throughput (DU)**.
- Dynamic-codebook-based HARQ feedback over the PUCCH reduces the number of bits used for HARQ feedback and increases the probability of using PUCCH format 1 (with two or fewer bits for HARQ feedback), improving PUCCH resource efficiency and increasing the number of CA UEs that can be scheduled in a cell.
- Dynamic PUCCH codebook specification improvement for CA UEs further increases the number of CA UEs that can be scheduled in a cell. This function

improves the downlink throughput of CA UEs in medium- and heavy-load scenarios (where the PRB usage is greater than 70%).

- CA data split stop based on the scheduling capability ratio enables data distribution to the carriers with lower scheduling capabilities to be stopped. This improves CA UE experience and possibly increases the **User Downlink Average Throughput (DU)** when some carriers serving individual CA UEs suffer severe interference.
- CA UE data split stop optimization may shorten the data scheduling delay of CA UEs for which data split is stopped.
- C-RNTI conflict detection prevents interference caused by C-RNTI conflicts, thereby increasing the average throughput of SCell-activated CA UEs in scenarios (such as heavy-load scenarios) where C-RNTI conflicts are likely to occur.
 - Average throughput of SCell-activated downlink CA UEs =
$$\frac{(N.CA.SccActiveUser.ThpVol.DL - N.CA.ThpVol.DL.LastSlot)}{N.CA.ThpTime.DL.RmvLastSlot}$$
 - Average throughput of SCell-activated uplink CA UEs =
$$\frac{(N.CA.SccActiveUser.ThpVol.UL - N.CA.ThpVol.UL.SmallPkt)}{N.CA.ThpTime.UL.RmvSmallPkt}$$
- C-RNTI conflict-triggered reconfiguration reduces the risk of SCell configuration failures due to C-RNTI conflicts, thereby increasing the average throughput of SCell-activated CA UEs in scenarios (such as heavy-load scenarios) where C-RNTI conflicts are likely to occur.
 - Average throughput of SCell-activated downlink CA UEs =
$$\frac{(N.CA.SccActiveUser.ThpVol.DL - N.CA.ThpVol.DL.LastSlot)}{N.CA.ThpTime.DL.RmvLastSlot}$$
 - Average throughput of SCell-activated uplink CA UEs =
$$\frac{(N.CA.SccActiveUser.ThpVol.UL - N.CA.ThpVol.UL.SmallPkt)}{N.CA.ThpTime.UL.RmvSmallPkt}$$
- UE-number-based SCell configuration control limits the use of CA to fewer RRC_CONNECTED UEs in scenarios where the number of RRC_CONNECTED UEs may reach the upper limit (for example, during major sports events). This ensures that more UEs can access the network.
- Non-CQI-based SCell deactivation for CA UEs performing downlink large-packet services increases the average throughput of SCell-activated downlink CA UEs, which is calculated using the following formula:
$$\frac{(N.CA.SccActiveUser.ThpVol.DL - N.CA.ThpVol.DL.LastSlot)}{N.CA.ThpTime.DL.RmvLastSlot}.$$
- Delayed SCell deactivation:
 - A larger value of the SCell deactivation waiting timer results in a longer time for UEs to stay in the SCell-activated state and therefore higher average throughput of SCell-activated CA UEs, but it also leads to higher UE power consumption. If the timer value is too large, flow control may be triggered, affecting user experience.
 - A smaller value of the SCell deactivation waiting timer results in a shorter time for UEs to stay in the SCell-activated state and therefore lower average throughput of SCell-activated CA UEs, but it also leads to lower UE power consumption.

Average throughput of SCell-activated CA UEs =

$$\frac{(N.CA.SccActiveUser.ThpVol.DL - N.CA.ThpVol.DL.LastSlot)}{N.CA.ThpTime.DL.RmvLastSlot}$$

- SCell deactivation protection increases the values of the following indicators in the case of inconsistent SCell deactivation status between the base station and UEs:
 - Average throughput of CA UEs configured with SCells

$$\frac{(N.CA.SccConfigUser.ThpVol.DL - N.CA.SccConfigUser.ThpVol.DL.LastSlot)}{N.CA.SccConfigUser.ThpTime.DL.RmvLastSlot}$$
 - **User Downlink Average Throughput (DU)**
 - **User Uplink Average Throughput (DU)**
- Precise SCell activation enables data to be distributed to SCells earlier, increasing the average throughput of CA UEs with the SCells activated. The more frequently SCells are activated and deactivated, the higher the gains. This function does not offer benefits when SCells remain in the activated or deactivated state without changing their activation states.
- SCell activation and deactivation suspension during DRX wakeup protection: After this function takes effect, the number of UEs for which SCells can be activated and deactivated may decrease in the PCell, the SCell activation success rate ($N.CA.SCell.Act.Succ/N.CA.SCell.Act.Att$) and SCell deactivation success rate ($N.CA.SCell.Deact.Succ/N.CA.SCell.Deact.Att$) may increase, but the number of SCell activation times ($N.CA.SCell.Act.Att$) and the number of SCell deactivation times ($N.CA.SCell.Deact.Att$) may decrease.
- SCell activation prohibition under weak coverage: When this function is in effect, the following impacts may occur depending on the value of **NRDUCellRsvdExt01.RsvdParam15**:
 - A smaller value of **NRDUCellRsvdExt01.RsvdParam15** results in more UEs for which SCells can be activated in the PCell, a lower SCell activation success rate ($N.CA.SCell.Act.Succ/N.CA.SCell.Act.Att$), more SCell activation times ($N.CA.SCell.Act.Att$), and a higher average throughput of SCell-configured CA UEs ($(N.CA.SccConfigUser.ThpVol.DL - N.CA.SccConfigUser.ThpVol.DL.LastSlot)/N.CA.SccConfigUser.ThpTime.DL.RmvLastSlot$).
 - A larger value of **NRDUCellRsvdExt01.RsvdParam15** results in fewer UEs for which SCells can be activated in the PCell, a higher SCell activation success rate ($N.CA.SCell.Act.Succ/N.CA.SCell.Act.Att$), fewer SCell activation times ($N.CA.SCell.Act.Att$), and a lower average throughput of SCell-configured CA UEs ($(N.CA.SccConfigUser.ThpVol.DL - N.CA.SccConfigUser.ThpVol.DL.LastSlot)/N.CA.SccConfigUser.ThpTime.DL.RmvLastSlot$).
- Prohibition of blind SCell configuration during incoming handovers, a sub-function of optimized SCell configuration during PCell handovers, allows cells measured in the source cell of a handover to be preferentially configured as SCells when the SCell coverage on the target cell side is poorer.
- Multi-carrier combination selection based on multiple bandwidth levels increases **User Downlink Average Throughput (DU)** and **User Uplink Average Throughput (DU)** of UEs if a relatively large number of UEs on the network exhibit better CA capabilities (for example, more layers) on the minimum carrier bandwidths.

6.1.2.2 Impacts

The impacts between this function and other functions are described in [6.1.3.2.3 Function Impacts](#).

- System capacity
A CA UE with SCells configured has only one RRC connection to the network. Such a UE consumes one sales unit of the license for the number of RRC_CONNECTED UEs. However, the CA UE consumes one hardware resource unit in the PCell and SCells separately. With n CC aggregation for all UEs on the network, the maximum allowed number of RRC_CONNECTED UEs decreases to $1/n$ (n is an integer).
- Resource usage
 - Overall PRB usage of the network
After CA is enabled, cell load can be rapidly balanced through carrier management and scheduling to help utilize idle resources and increase the overall PRB usage of the network.
 - PUCCH and PUSCH overheads
With CA enabled, each CA UE sends the ACK/NACK and CSI related to its SCells in its PCell, where the PUCCH is available. When the PUSCH is not scheduled, the UE sends the information over the PUCCH, and therefore the PUCCH overheads increase. In addition, fewer PUSCH resources may be available for UEs running uplink full-buffer services, and therefore the throughput of these UEs decreases. When the PUSCH is scheduled, the UE sends the information over the PUSCH, and therefore the signaling overheads on the PUSCH increase.
 - CPU usage of BBPs
After downlink intra-FR inter-band CA is enabled, the CPU usage of the BBP where the PCell is configured increases with the amount of data forwarded from the PCell to SCells at the RLC layer. If the CPU usage is too high, the peak data rate of the UE decreases.
- Overall throughput in the entire network
CA does not directly affect network capacity. However, when the resources on a network have not been exhausted, CA increases the overall resource usage and network throughput.
- KPI fluctuation
 - Channel quality of UEs that treat a cell as their PCell differs from that of UEs that treat the cell as their SCell. Therefore, CA will cause CQIs and MCS indexes to change in a cell configured as an SCell for UEs.
 - After CA is enabled, PUCCH format 3 and format 4 are used more often than before. Therefore, the average value of counters **N.UL.SINR.PUCCH.Index0** to **N.UL.SINR.PUCCH.Index7** decreases slightly.
 - Before CA is enabled, all UEs are in the single-carrier state, have SRS resources, and can use SRS-based weights. After CA is enabled, some UEs have to use only PMI-based weights in their SCells where no SRS resources are available. PMI-based weights result in lower MCS indexes being selected for UEs than SRS-based weights. Therefore, with all other factors constant, the average of counters **N.ChMeas.PDSCH.MCS.0**

through **N.ChMeas.PDSCH.MCS.28** will decrease and the usage of control channel elements (CCEs) on the PDCCH will increase in an SCell after CA is enabled if UEs have no SRS resources in this SCell.

- After CA is enabled, SCells are not activated if the network traffic is light, which is indicated by the value of the **N.User.CA.PCell.DL.SCellAct.Avg** counter being 0. However, the PDCCH CCE usage on each carrier decreases. The reason is that although SCells are not activated for a CA UE, the gNodeB has configured the semi-static HARQ-ACK codebook for the UE, which leads to a lower proportion of discontinuous transmissions (DTXs) than that when the dynamic HARQ-ACK codebook was used before the SCells were configured.
- When CA is in effect, the PUCCH exists only in the PCell. As a result, fewer DTXs are detected, the DTX-based PDCCH aggregation level adjustment in the PCell and SCells is performed less frequently, and the PDCCH aggregation level adjustment slows down. Consequently, the BLER (including the **IBLER** and **RBLER**) may increase.
- After CA is enabled in NSA networking, the values of the **N.NsaDc.SgNB.Mod.Req.Succ** and **N.NsaDc.SgNB.Mod.Req.Att** counters increase because the gNodeB needs to configure SCells through RRCReconfiguration messages.
- After CA is enabled in NSA networking, the values of the **N.NsaDc.SgNB.Mod.Required.Succ** and **N.NsaDc.SgNB.Mod.Required.Att** counters increase because the gNodeB needs to deliver measurement configurations to CA UEs through RRCReconfiguration messages.
- If no cell measured by a UE at the cell edge meets the threshold for event A5, the gNodeB cannot configure an SCell for the UE. In this case, the gNodeB delivers event A5 measurement configurations to the UE every 15s if the UE traffic volume is constantly high. Since these periodic inter-frequency measurements require measurement gaps, the downlink throughput of the UE in the PCell decreases.
- Initial scheduling begins after an SCell is activated. This scheduling affects the average user-perceived data rates of CA UEs to an extent depending on the scheduling duration. (The duration is related to factors such as target weights, rank adjustment period, and PCell air interface quality.) As a result, the average user-perceived data rates of CA UEs decrease in non-full buffer scenarios with frequent SCell deactivation and activation.
- After flow control against multi-carrier operations at the gNodeB is triggered, SCells will no longer be configured for newly admitted UEs. As a result, the **User Uplink Average Throughput (DU)** and **User Downlink Average Throughput (DU)** may decrease. In particular, if this level of flow control is high (indicated by the value **HIGH** of the **gNodeBAlgo.MultiCarrierFlowCtrlLevel** parameter), the **Service Call Drop Rate (CU, Inactive)** in the cell may rise.
- As data is distributed to SCells, scheduling capabilities for CA UEs are stronger. As a result, the proportion of the last slots during each of which the downlink buffer becomes empty in the total data transmission duration may increase and the average user-perceived rate may decrease. (The proportion equals $(N.ThpTime.DL - N.ThpTime.DL.RmvLastSlot) / N.ThpTime.DL$).

- During optimal SCC frequency combination selection, event A5 inter-frequency measurement configurations are delivered to obtain the RSRP of each frequency. As inter-frequency measurement requires measurement gaps, the downlink throughput of UEs in their PCells decreases.
- After CA is enabled, especially when the **SCC_BLIND_CONFIG_SW** option of the **NRCellCaMgmtConfig.CaStrategySwitch** parameter is selected, cell-edge user experience may improve. Cell-edge UEs that would otherwise enter idle or inactive mode due to poor experience may remain in connected mode. As a result, cell-edge UEs may stay online for a longer time, and the proportion of cell-edge UEs may increase. Consequently, the service drop rate of PCells sometimes rises, as cell-edge UEs are more likely to experience service drops.
- As inter-frequency measurement configuration is required for CA SCell configuration, the number of RRC reconfigurations increases and cell-edge UEs are more likely to experience service drops.
- After CA is enabled, certain UEs with compatibility issues in NSA networking may encounter an increase in their service drop rate.
- Parameters related to SCell activation and deactivation have the following impacts on the network:
 - With the **NRDUCellCarrMgmt.CaDiScellActThldSlotNum** parameter set to a smaller value, the following impacts are more likely to occur:
 - There will be a higher probability of SCell activation, a longer SCell-activated duration, higher average throughput of SCell-activated CA UEs, and higher UE power consumption.
 - The values of the following counters increase:
 - **N.CA.SCell.Act.Att**
 - **N.CA.SCell.Act.Succ**
 - **N.CA.SCell.Deact.Att**
 - **N.CA.SCell.Deact.Succ**
 - With the **NRDUCellCarrMgmt.CaDiScellDeactThldSlotNum** or **NRDUCellCarrMgmt.RlcDiThroughputThld** parameter set to a smaller value, the following impacts are more likely to occur:
 - There will be a lower probability of SCell deactivation, a longer SCell-activated duration, higher average throughput of SCell-activated CA UEs, and higher UE power consumption.
 - The values of the following counters decrease:
 - **N.CA.SCell.Deact.Att**
 - **N.CA.SCell.Deact.Succ**
 - **N.CA.SCell.Act.Att**
 - **N.CA.SCell.Act.Succ**
 - Reconfiguring **NRDUCellCarrMgmt.DiStateTransitBufVolThld** or **NRDUCellQciBearer.QciDiStTransitBufVolThld**, which controls both SCell activation and SCell deactivation, causes the values of the following counters to fluctuate (the fluctuation is dependent on the threshold setting and the traffic volume of CA UEs on the live network):

- **N.User.CA.PCell.DL.SCellAct.Avg**
 - **N.CA.PCell.ThpVol.DL.Cell**
 - **N.CA.SCell.ThpVol.DL.Cell**
 - **N.CA.SCell.Deact.Att**
 - **N.CA.SCell.Deact.Succ**
 - **N.CA.SCell.Act.Att**
 - **N.CA.SCell.Act.Succ**
 - With channel quality-triggered SCell deactivation, consecutive residual block error-triggered SCell deactivation, and distributed data volume-triggered SCell deactivation enabled:
 - SCell deactivation is triggered in more scenarios. This results in a shorter SCell-activated duration, lower average throughput of SCell-activated CA UEs, and lower UE power consumption.
 - The values of the following counters increase:
 - **N.CA.SCell.Deact.Att**
 - **N.CA.SCell.Deact.Succ**
 - **N.CA.SCell.Act.Att**
 - **N.CA.SCell.Act.Succ**
 - SCell deactivation protection will change performance counters as follows:
 - The value of the **N.CA.SCell.Deact.Att** counter increases.
 - The value of the **N.CA.SCell.Deact.Succ** counter increases.
 - The value of the **N.CA.SCell.Act.Att** counter changes.
 - The value of the **N.CA.SCell.Act.Succ** counter changes.
- Average throughput of SCell-activated CA UEs =

$$\frac{(\text{N.CA.SccActiveUser.ThpVol.DL} - \text{N.CA.ThpVol.DL.LastSlot})}{\text{N.CA.ThpTime.DL.RmvLastSlot}}$$
- Certain functions have the following additional network impacts:
 - Precise SCell activation: The downlink IBLER increases during the period in which the gNodeB checks SCell availability. During this period as well as the period from when the gNodeB starts data split to when it receives CSI-RS measurement results, the average downlink MCS index and average downlink rank may decrease, and the **Downlink Resource Block Utilizing Rate (DU)** and average downlink CCE aggregation level increase.
 - Early data split: SCells are occasionally activated for CA to take effect for CA UEs in advance. As a result, the CQIs and MCS indexes of the cells acting as SCells fluctuate.
 - SCell CSI-RS beam sweeping: After its subfunction "adaptive CSI-RS beam sweeping" takes effect, the values of counters related to the number of

CA UEs (for example, **N.User.CA.SCell.DL.Avg**) may decrease if the number of CSI resources for UEs on the network is less than the number of CSI resources required by all CCs.

- CA PDCCH aggregation level optimization: A smaller value of the **NRDUCellPdcchAlgo.PdcchOuterLoopAdjUpLimit** parameter results in a better PDCCH demodulation capability but possibly a smaller number of UEs that can be scheduled in the cell.
- Dynamic-codebook-based HARQ feedback over the PUCCH: This function causes the downlink IBLER to increase, because of its handling on missing DCI detection.

Downlink IBLER = $(N.DL.SCH.QPSK.ErrTB.Ibler + N.DL.SCH.16QAM.ErrTB.Ibler + N.DL.SCH.64QAM.ErrTB.Ibler + N.DL.SCH.256QAM.ErrTB.Ibler) / (N.DL.SCH.QPSK.TB + N.DL.SCH.16QAM.TB + N.DL.SCH.64QAM.TB + N.DL.SCH.256QAM.TB) \times 100\%$

- Real-time data split by RLC for downlink intra-gNodeB CA: The **User Downlink Average Throughput (DU)** fluctuates when the load is extremely heavy.
- Prohibition of acting as the PCell: CA will not take effect for UEs that newly access the cell or UEs camping on the cell but not in the CA state. These UEs cannot benefit from CA and therefore optimal user experience cannot be reached. For UEs in the CA state with the cell acting as their PCell, selecting the function option does not affect their existing SCells; the gNodeB can remove the SCells but cannot add new SCells or change SCells for these UEs. The UEs mentioned here refer to those with the "prohibition of acting as the PCell" function in effect. Whether this function is in effect for UEs in different networking depends not only on the cell's function switch setting but also on NSA/SA-specific PCell management policy configuration (detailed in [NSA/SA-specific PCell Management Policy Configuration](#)).
- CA UE data split stop optimization: If data split is stopped for some CA UEs, more data is scheduled on the PCC due to the optimized PCC data split weight. This may decrease the user-perceived rate of non-CA UEs on the PCC.
- C-RNTI conflict-triggered reconfiguration: The RRC reconfiguration times and service drop rate vary, depending on the number of CA capability combinations selected during SCell configuration. Selecting more CA capability combinations increases the likelihood of fewer RRC reconfiguration times and a lower service drop rate. Selecting fewer CA capability combinations increases the likelihood of more RRC reconfiguration times and a higher service drop rate. In addition, this function increases the value of the following counters:
 - **N.CA.SCell.Add.Att**
 - **N.CA.SCell.Add.Succ**
 - **N.CA.SCell.Mod.Att**
 - **N.CA.SCell.Mod.Succ**
- UE-number-based SCell configuration: The proportion of UEs for which CA takes effect and the network-wide UE throughput (including the

uplink and downlink throughputs) change, depending on the configured thresholds expressed as the proportion of UEs for prohibiting 2CC or 3CC aggregation. Higher thresholds indicate higher proportions of CA UEs and higher network-wide UE throughput. Lower thresholds indicate lower proportions of CA UEs and lower network-wide UE throughput. In addition, after the **UE_NUM_BASED_SCELL_CONFIG_SW** option of the **gNodeBCarrMgmt.CaEnhancedAlgoSwitch** parameter is selected, regardless of whether UE-number-based 2CC aggregation prohibition and UE-number-based 3CC aggregation prohibition take effect, more cell load information will be exchanged over the eXn interface, leading to increased length of a single message and possible more message fragments in some scenarios. As a result, the values of counters related to receive and transmit traffic in the SCTP Link Measurement subset may increase.

- Prohibition of SCell configuration inheritance during incoming handovers, a sub-function of optimized SCell configuration during PCell handovers, shortens the handover delay for CA UEs but results in a period of time in which SCell configuration is interrupted. Therefore, the uplink and downlink throughputs of the CA UEs decrease before new SCells are configured for the UEs.
- Multi-carrier combination selection based on multiple bandwidth levels may increase the CPU usage of the main control board, which may decrease the number of CA UEs in heavy-load scenarios. The following counters are involved:
 - **N.User.CA.PCell.DL.Avg** (average number of downlink CA UEs that use the local cell as their PCells)
 - **N.User.CA.SCell.DL.Avg** (average number of downlink CA UEs that use the local cell as their SCells)
 - **N.User.CA.PCell.UL.Avg** (average number of uplink CA UEs that use the local cell as their PCells)
 - **N.User.CA.SCell.UL.Avg** (average number of uplink CA UEs that use the local cell as their SCells)

6.1.3 Requirements

6.1.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit	NE
FOFD-031201	Intra-FR Inter-Band CA	NR0SINFRIBC A	per Cell	Macro gNodeB
FOFD-031201	Intra-FR Inter-Band CA	NR0TSIFIBC01	per Carrier per pRRU	LampSite gNodeB

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

6.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.1.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Configuration of the maximum port number used for CSI-RS for CM	NRDUCellCsirs.FR1MaxCellCsirsPortNum	<i>Channel Management</i>	32-port CSI-RS measurement on SCCs can be enabled only when the NRDUCellCsirs.FR1MaxCellCsirsPortNum parameter is set to 32PORT .
Low-frequency TDD	Downlink intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	32-port CSI-RS measurement on SCCs can be enabled only when downlink intra-band CA or downlink intra-FR inter-band CA has been enabled.
Low-frequency TDD	Downlink intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	32-port CSI-RS measurement on SCCs can be enabled only when downlink intra-band CA or downlink intra-FR inter-band CA has been enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE-specific PDCCH Aggregation Level	NRDUCellPdcch.UeSpecificPdcchAggLvl	None	CA PDCCH aggregation level optimization takes effect only when the NRDUCellPdcch.UeSpecificPdcchAggLvl parameter is set to ADAPTIVE .
Low-frequency TDD	Smart carrier selection	SMART_CARRIER_SEL_SW option of the NRDUCellCollabServ.MultiCarrierMgmtSw parameter	<i>Carrier Aggregation</i>	Smart carrier selection must be enabled before smart carrier selection optimization can be enabled.
Low-frequency TDD	Prohibition of acting as the PCell	PCELL_PROHIBIT_SW option of the NRCeCellCaMgmtConfig.CarrierStrategySwitch parameter	<i>Carrier Aggregation</i>	The NSA/SA-specific PCell management policy configuration function takes effect only when the "prohibition of acting as the PCell" function is enabled.
Low-frequency TDD	Dynamic-codebook-based HARQ feedback over the PUCCH	PUCCH_DYNAMIC_HARQ_ACK_CB_SW option of the NRDUCellCarrierMgmt.CarrierEnhancedAlgorithmExtSwitch parameter	<i>Carrier Aggregation</i>	The "dynamic PUCCH codebook specification improvement for CA UEs" function takes effect only when the "dynamic-codebook-based HARQ feedback over the PUCCH" function is enabled.
Low-frequency TDD	Downlink Minimum Carrier Bandwidth	NRDUCellBwp.DIMinCarrierBw	<i>Scalable Bandwidth</i>	Multi-carrier combination selection based on multiple bandwidth levels requires that the NRDUCellBwp.DIMinCarrierBw parameter be set to a value other than NOT_CONFIG .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink Minimum Carrier Bandwidth	NRDUCellBwp.UlMinCarrierBw	<i>Scalable Bandwidth</i>	Multi-carrier combination selection based on multiple bandwidth levels requires that NRDUCellBwp.UlMinCarrierBw be set to a value other than NOT_CONFIG .
Low-frequency TDD	Smart carrier selection	SMART_CARRIER_SEL_SW option of the NRDUCellCoIlabServ.MultiCarrierMgmtSw parameter	<i>Carrier Aggregation</i>	Multi-carrier combination selection based on multiple bandwidth levels takes effect only when smart carrier selection is enabled.

6.1.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Compensation for SAR decrease (low-frequency TDD)	REDUCE_SAR_COMPENSATION_SW option of the NRDUCellUeCoop.SpecUeCooperationSwitch parameter	<i>Device-Pipe Synergy</i>	Intra-FR inter-band CA does not work with compensation for SAR decrease.
Low-frequency TDD	Optimization of non-contention-based RA beyond cell radius	CONTENTION_FREE_RA_OPT_SW option of the NRDUCellPrach.RachAlgoOptSwitch parameter	<i>Random Access Control</i>	Downlink intra-FR inter-band CA does not work with optimization of non-contention-based RA beyond cell radius.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Amount of CSI-RS resources in a cell	NRDUCellCsirs.CsirsCellResourceNum	<i>Channel Management</i>	When the NRDUCellCsirs.CsirsCellResourceNum parameter is set to 1_RESOURCE , the CARR_CSI_BEAM_SWEEEPING_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter cannot be selected.
Low-frequency TDD	CSI-RS measurement period	NRDUCellCsirs.CsiPeriod	<i>Channel Management</i>	32-port CSI-RS measurement on SCCs cannot be enabled when NRDUCellCsirs.CsiPeriod is set to SLOT10 .
Low-frequency TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode	<i>Cell Combination</i>	32-port CSI-RS measurement on SCCs cannot be enabled in combined cells whose NRDUCell.NrDuCellNetworkingMode is set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTrp.DataTransMode is set to BASIC_MODE or DPS_MODE .
Low-frequency TDD	R16 type 2	R16_TYPE2_SW option of the NRDUCellPdschPrecode.DlCodebookTypeSwitch parameter	None	32-port CSI-RS measurement on SCCs cannot be enabled when the R16_TYPE2_SW option of NRDUCellPdschPrecode.DlCodebookTypeSwitch is selected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-FR CA	INTER_FR_FR1TOFR2_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	When service-aware carrier selection is enabled (by selecting the SERVICE_AWARE_CARR_SEL_SW option of the NRDUCellCollabServ.v.MultiCarrMgmtSw parameter), inter-FR CA cannot be enabled.
Low-frequency TDD	gNodeB Adaptive Fast Processing Switch	GNB_ADAPT_FAST_PROCESS_SW option of the gNodeBParam.ServiceDiffProcSwitch parameter	None	Dynamic-codebook-based HARQ feedback over the PUCCH (controlled by the PUCCH_DYN_HARQ_ACK_CB_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter) cannot be enabled when the gNodeB adaptive fast processing switch is on. This switch cannot be turned on when this function is enabled.
Low-frequency TDD	Downlink 8-layer transmission for a single FWA user	DL_SU_EIGHT_LAYER_SW option of the NRDUCellDLRank.DlRankSelAlgoSw parameter	<i>FWA</i>	Dynamic-codebook-based HARQ feedback over the PUCCH does not work with downlink 8-layer transmission for a single FWA user.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Distributed massive MIMO cell	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAIgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	SCell CSI-RS beam sweeping cannot be enabled in distributed massive MIMO cells.
Low-frequency TDD	One piece of cell resource for CSI-RS	NRDUCellCsirs.CellResourceNum set to 1_RESOURCE	<i>Beam Management</i>	SCell CSI-RS beam sweeping must be disabled in a cell where one piece of cell resource is used for CSI-RS and either of the following conditions is not met: - The NRDUCellTrp.TxRxMode parameter is set to 64T64R or 128T128R in at least one NRDUCellTrp MO associated with this cell through the NRDUCellCsirs.NrDuCellId parameter. - The NRDUCellCsirs.FR1MaxCellCsirsPortNum parameter is set to 32PORT.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Experience-based smart carrier selection	MULTI_FREQ_SMART_SEL_SW option of the NRCellAlgoSwitch.MultiFreqAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	If the NRCellCaMgmtConfig.CaBandCombSelectionStrat parameter (specifying the policy on multi-carrier capability combination selection) is set to DEFAULT , experience-based smart carrier selection cannot be enabled.
Low-frequency TDD	Cell deployment in IAB mode	NRDUCell.DeploymentPosition set to DEPLOY_IAB	None	NRDUCell.DeploymentPosition can be set to DEPLOY_IAB only when downlink intra-FR inter-band CA is disabled.
Low-frequency TDD	80 MHz UE access	80M_UE_ACCESS_SW option of the NRDUCellUeCoop.UeAccessSwitch parameter	None	When the 80M_UE_ACCESS_SW option of the NRDUCellUeCoop.UeAccessSwitch parameter is selected, downlink intra-FR inter-band CA cannot be enabled.

6.1.3.2.3 Function Impacts

- Impact relationships with basic functions under RAN functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Slot Configuration (TDD)	None	<i>Standards Compliance</i>	Downlink intra-FR inter-band CA can take effect only when the absolute value of the frame offset difference between the PCell and an SCell is less than or equal to 625 Ts. As such, the frame offset modified due to Slot Configuration (TDD) must meet this requirement to prevent downlink intra-FR inter-band CA from not taking effect. For details about the frame offset difference between the PCell and SCells in downlink intra-FR inter-band CA, see 6.1.3.4 Networking.
Low-frequency TDD	Cell-level PUCCH time-frequency format	None	<i>Channel Management</i>	The NRDUCellPucch.StructureType parameter must be set to LONG_STRUCTURE. If it is set to SHORT_STRUCTURE, intra-FR inter-band CA does not take effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS period	NRDUCellSrs.SrsPeriod	<i>Channel Management</i>	When the SRS period (specified by the NRDUCellSrs.SrsPeriod parameter) in the PCell of an antenna-selection UE is set to 5 ms or 10 ms, SRS resources may fail to be allocated to the UE in its SCells. If this is the case, user experience is affected.
Low-frequency TDD	PDCCH rate matching	PDCCH_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	PDCCH rate matching does not take effect when SCells of CA UEs are transmitting the common PDCCH. It is recommended that common PDCCH rate matching in SCells (controlled by the SCC_COMMON_PDCCH_RM_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgorithmExtSwitch parameter) be enabled, to increase the peak data rates in the SCells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink enhanced scheduling of aperiodic CSI reports	APER_CSI_RPT_ENHANCE_SW option of the NRDUCellConfig.CsiSwitch parameter	<i>Uplink Scheduling</i>	The status (enabled or disabled) of uplink enhanced scheduling of aperiodic CSI reports must be consistent between the PCell and all SCells of a CA UE. Status inconsistency affects the downlink throughput of the UE.

- Impact relationships with functions related to UCN under RAN functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UL CoMP	INTRA_GNB_UL_COMP_SW option of the NRDUCellConfig.CompSwitch parameter	<i>CoMP</i>	UL CoMP cannot be performed for CA UEs in their SCells.
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellConfig.CompSwitch parameter	<i>CoMP</i>	When CA is enabled, intra-base-station joint transmission cannot be performed for CA UEs in their SCells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Joint Transmission Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTrp.DataTransMode set to JOINT_MODE	<i>Cell Combination</i>	In low-speed scenarios, downlink intra-FR inter-band 2CC aggregation is supported by combined cells in joint transmission mode (that is, cells with Joint Transmission Cell Combination enabled) that are served by UBBP _g or UBBP _{fw} series baseband processing units. If downlink intra-FR inter-band CA is enabled for combined cells in joint transmission mode, candidate SCells must also be combined cells in joint transmission mode.
Low-frequency TDD	Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	When downlink intra-FR inter-band CA is enabled, Co-MM cannot be performed for CA UEs in their SCells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworking Mode	<i>Hyper Cell</i>	• If CA flexible networking (controlled by the CA_FLEX_NETWORKING_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter) is disabled, downlink intra-FR inter-band CA does not work in hyper cells. • If CA flexible networking is enabled, downlink intra-FR inter-band CA works in hyper cells.
Low-frequency TDD	Cell Combination	NRDUCell.NrDuCellNetworking Mode	<i>Cell Combination</i>	• If CA flexible networking (controlled by the CA_FLEX_NETWORKING_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter) is disabled, downlink intra-FR inter-band CA does not work in combined cells. • If CA flexible networking is enabled, downlink intra-FR inter-band CA works in combined cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Hyper Cell Cell Combination	NRDUCell.NrDuCellNetworkingMode	<i>Hyper Cell Cell Combination</i>	The cells involved in downlink intra-FR inter-band CA cannot include both hyper cells and combined cells. (Combined cells are those with NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTrp.DataTransMode set to BASIC_MODE or DPS_MODE .)

- Impact relationships with functions related to energy saving

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Power saving BWP	BWP2_SWITCH option of the NRDUCellUePwrSaving . BwpPwrSavingSw parameter	<i>UE Power Saving</i>	BWP2, which provides a narrow bandwidth, is not configured for CA UEs in their SCells. When power saving BWP or "QCI-level threshold control for BWP switching" is enabled together with CA, there are the following impacts: • A larger value of NRDUCellCarrMgmt.CaDisCellActivateThldSlotNum results in a higher traffic volume threshold for UE switching from BWP2 to BWP1 and therefore a lower probability of the switching, which decreases UE power consumption. • A larger value of NRDUCellCarrMgmt.CaDisCellDeactivateThldSlotNum results in a higher traffic volume threshold for UE switching from BWP1 to BWP2 and therefore a higher probability of the switching, which decreases UE power consumption. • A larger value of NRDUCellCarrM

RAT	Function Name	Function Switch	Reference	Description
				gmt.DlStateTransitionBufVolThld or NRDUCellQciBearer.QciDlStateTransitionBufVolThld results in a lower probability of UE switching from BWP2 to BWP1 and a higher probability of UE switching from BWP1 to BWP2. As a result, UEs stay in BWP2 for a longer time and their power consumption decreases. • A smaller value of NRDUCellCarrierMgmt.DlStateTransitionBufVolThld or NRDUCellQciBearer.QciDlStateTransitionBufVolThld results in a higher probability of UE switching from BWP2 to BWP1 and a lower probability of UE switching from BWP1 to BWP2. As a result, UEs stay in BWP2 for a shorter time and their power consumption increases. BWP switching and RRC reconfiguration may occur more

RAT	Function Name	Function Switch	Reference	Description
				frequently. For cell-edge UEs under weak coverage, the service drop rate may increase. • SCells are more likely to be deactivated if the NRDUCellCarrMgmt.RlcDLThroughputThld parameter value is less than the NRDUCellUePwrSaving.Bwp1ToBwp2DLThptThld or NRDUCellQciBearer.QciBwp1ToBwp2DLThptThld parameter value. If NRDUCellCarrMgmt.RlcDLThroughputThld is set to 0, SCells will not be deactivated and UEs will not switch to BWP2.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Time-domain power saving BWP (low-frequency TDD)	TDM_BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	BWP2 is not configured for CA UEs in their SCells. When CA and time-domain power saving BWP are both enabled, there are the following impacts: • A larger value of NRDUCellCarrMgmt.CaDisCellActThldSlotNum results in a higher traffic volume threshold for UE switching from BWP2 to BWP1 and therefore a lower probability of the switching, which decreases UE power consumption. • A larger value of NRDUCellCarrMgmt.CaDisCellDeactThldSlotNum results in a higher traffic volume threshold for UE switching from BWP1 to BWP2 and therefore a higher probability of the switching, which decreases UE power consumption. • A larger value of NRDUCellCarrMgmt.DlStateTransitBufVolThld results in a lower probability

RAT	Function Name	Function Switch	Reference	Description
				of UE switching from BWP2 to BWP1 and a higher probability of UE switching from BWP1 to BWP2. As a result, UEs stay in BWP2 for a longer time and their power consumption decreases. • A smaller value of NRDUCellCarrMgmt.DlStateTransitBufVolThld results in a higher probability of UE switching from BWP2 to BWP1 and a lower probability of UE switching from BWP1 to BWP2. As a result, UEs stay in BWP2 for a shorter time and their power consumption increases. BWP switching and RRC reconfiguration may occur more frequently. For cell-edge UEs under weak coverage, the service drop rate may increase. • SCells are more likely to be deactivated if the NRDUCellCarrM

RAT	Function Name	Function Switch	Reference	Description
				gmt.RlcDlThroughputThld parameter value is less than the NRDUCellUePowerSaving.Bwp1ToBwp2DlThptThld parameter value. If NRDUCellCarrMgmt.RlcDlThroughputThld is set to 0 , SCells will not be deactivated and UEs will not switch to BWP2.
Low-frequency TDD	Intelligent carrier shutdown	INTRA_GNB_MULTI_CA_RR_SD_SW or INTER_GNB_MULTI_CA_RR_SD_SW option of the NRDUCellAgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When intelligent carrier shutdown takes effect, the capacity-layer cells that have been shut down cannot be configured as SCells for UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DRX	NRDUCellUePwrSaving.NrDuCellDRXAlgoSwitch	<i>DRX</i>	Data split for a UE is affected if the UE is in different DRX states in its PCell and SCells. When intra-FR inter-band CA and DRX are both enabled, it is recommended that collaboration between CA and DRX be enabled. The collaboration is controlled by the CA_AND_DRX_STATUS_COLLAB_SWITCH option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter.
Low-frequency TDD	Maximum-bandwidth reduction	OVERHEATING_ASSISTANCE_INFO_SW and UE_BW_REDUCE_SW options of the NRCCellAlgoSwitch.UeAssistanceInfoSwitch parameter	<i>UE Power Saving</i>	When a UE requests maximum-bandwidth reduction, the base station deactivates all SCells of the UE in multi-carrier scenarios.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH skipping	PDCCH_SKIP_SW option of the NRDUCellUePwrSaving.SchPwrSavingSw parameter	<i>UE Power Saving</i>	With both downlink intra-FR inter-band CA and PDCCH skipping enabled: • If the CA_UE_PDCCH_SKIP_SW option of the NRDUCellUePwrSaving.SchPwrSavingSw parameter is deselected, PDCCH skipping does not take effect for UEs with SCells configured. • If the CA_UE_PDCCH_SKIP_SW option of the NRDUCellUePwrSaving.SchPwrSavingSw parameter is selected, PDCCH skipping can take effect for CA UEs with SCells not activated, but not for CA UEs with SCells activated.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SSSG switching	UE_PWR_SAVING_ENHANCE_SW and SSSG_SWITCHING_SW options of the NRDUCellUePwrSaving . <i>UePowerSavingSwitch</i> parameter	<i>UE Power Saving</i>	With both downlink intra-FR inter-band CA and SSSG switching enabled, a sparse SSSG can be configured for PCCs but not for SCCs. In this case, SCell activation for CA requires switching from a sparse SSSG to a dense SSSG and therefore is delayed.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast carrier shutdown	Capacity-layer cells: • INTRA_GNB_MULTICARRIER_SD_SWITCH option (intra-base-station multi-carrier scenarios) or INTER_GNB_MULTICARRIER_SD_SWITCH option (inter-base-station multi-carrier scenarios) of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter selected • FAST_CARRIER_SHUTDOWN_SWITCH option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter Basic-layer cells:	<i>Energy Conservation and Emission Reduction</i>	After fast carrier shutdown takes effect, the capacity-layer cells that have been shut down cannot be configured as SCells for UEs.

RAT	Function Name	Function Switch	Reference	Description
		FAST_CARR_SHUT_SERV_ASSUR_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter		
Low-frequency TDD	Symbol power saving	NRDUCellAlgoSwitch.SymbolShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	After dynamic PUCCH codebook specification improvement for CA UEs takes effect, the number of scheduling times for CA UEs increases. This shortens the symbol power saving duration of a cell (N.PowerSaving.SymbolShutdown.Duration).
Low-frequency TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATS parameter	<i>Multi-RAT Coordinated Energy Saving</i>	After RF module fast deep dormancy takes effect, capacity-layer cells that have been shut down cannot be configured as SCells for UEs.

- Impact relationships with functions related to RAN services

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CA prohibition for VoNR UEs	VONR_NOT_CONFIG_CA_SW option of the NRCellCaMgmtConfig.CaStrategySwitch parameter	<i>VoNR</i>	When CA prohibition for VoNR UEs is enabled, CA does not take effect for VoNR UEs.
Low-frequency TDD	SRS field strength-based positioning	• SRS_RSRP_LOCATION_SW option of the NRCellAIgoSwitch.LcsSwitch parameter • gNB Rfsp Config. UserServiceType set to LCS	<i>Positioning</i>	Downlink intra-FR inter-band CA does not take effect for UEs whose UE service type is LCS when SRS field strength-based positioning is enabled and the COMMON_USER_LCS_SW option of the NRDUCellPosAlgo.LcsMeasureAlgorithm parameter is deselected.
Low-frequency TDD	Precise positioning (LampSite NR)	• UTDOA_SW option of the NRCellAIgoSwitch.LcsSwitch parameter • gNB Rfsp Config. UserServiceType set to LCS	<i>Positioning</i>	Downlink intra-FR inter-band CA does not take effect for UEs whose UE service type is LCS when precise positioning (LampSite NR) is enabled and the COMMON_USER_LCS_SW option of the NRDUCellPosAlgo.LcsMeasureAlgorithm parameter is deselected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Precise positioning (NR)	• AOA_UT DOA_HY BRID_LOCATION_SW option of the NRCellAlgSwitch.LcsSwitch parameter • gNBRfsp Config.UserServiceType set to LCS	<i>Positioning</i>	Downlink intra-FR inter-band CA does not take effect for UEs whose UE service type is LCS when precise positioning (NR) is enabled and the COMMON_USER_LCS_SW option of the NRDUCellPosAlgo.LcsMeasureAlgoSw parameter is deselected.
Low-frequency TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	Downlink intra-FR inter-band CA does not take effect for UEs for which the low latency and high reliability function has taken effect. This constraint does not apply to UEs performing low latency (4 ms) and high reliability services.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-base-station CA downlink data split for Cloud X services	CLOUD_X_INTRA_GNB_CA_ENH_S W option of the NRDUCellA lgoSwitch.C loudVrSchS witch parameter	<i>Ultimate Cloud X Experience</i>	When both intra-base-station CA downlink data split for Cloud X services and distributed data volume-triggered SCell deactivation (controlled by the SPLIT_VOL_BASED_SCELL_DEACT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter) are enabled: - A larger value of the gNBQciBearer.CloudServSplitCoefficient parameter results in a higher CA downlink data split threshold for Cloud X UEs. If the buffered data volume is lower than the data split threshold for a long time, SCells will be deactivated and the values of the N.CA.SCell.Deactivate.Att and N.CA.SCell.Deactivate.Succ counters will increase. - A smaller value of the gNBQciBearer.CloudServSplitCoefficient parameter results in a lower CA

RAT	Function Name	Function Switch	Reference	Description
				downlink data split threshold for Cloud X UEs, a longer SCell-activated duration, and higher UE power consumption.

- Impact relationships with functions related to carrier management under RAN functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Experience-based smart carrier selection	MULTI_FREQ_SMART_SEL_SW option of the NRCellAlgoSwitch.MultiFreqAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	If both smart carrier selection (controlled by the SMART_CARR_SEL_SW option of the NRDUCellCollabServ.MultiCarrMgmtSw parameter) and experience-based smart carrier selection are enabled, inter-frequency measurement configurations related to event A5 for experience-based smart carrier selection are preferentially delivered.
Low-frequency TDD	Uplink multi-frequency coordination	MULTI_FREQ_UL_COORD_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter	<i>Multi-Frequency Smart Selection</i>	The gNodeB does not select UEs for which downlink CA has taken effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-FR CA	INTER_FR_FR1TOFR2_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	For a UE in the intra-FR inter-band CA state: • If the band +bandwidth combination for CA at the UE meets the band +bandwidth requirements of inter-FR CA, inter-FR CA can simultaneously take effect for the UE. If this is the case, downlink CA instead of uplink CA can take effect between the two low-frequency cells. • Otherwise, inter-FR CA cannot simultaneously take effect for the UE.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CSI-RS beam sweeping	CARR_CSI_BEAM_SWEEPING_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoSwitch parameter	<i>Carrier Aggregation</i>	In low-frequency scenarios with a 32T32R RF module and 32 CSI-RS ports, 32-port CSI-RS measurement on SCCs can take effect even if CSI-RS beam sweeping is disabled. In low-frequency NR TDD scenarios with a 64T64R RF module and 32 CSI-RS ports, 32-port CSI-RS measurement on SCCs can take effect only when CSI-RS beam sweeping is enabled.
Low-frequency TDD	SRS carrier switching	SRS_CARRIER_SWITCHING_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoSwitch parameter	<i>CA Experience Improvement</i>	32-port CSI-RS measurement on SCCs does not take effect for CA UEs for which SRS carrier switching is in effect.
Low-frequency TDD	SRS carrier switching	SRS_CARRIER_SWITCHING_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoSwitch parameter	<i>CA Experience Improvement</i>	Dynamic-codebook-based HARQ feedback over the PUCCH does not take effect for UEs for which SRS carrier switching has taken effect and from which <i>switchingTimeUL</i> with a value other than 0 μ s is reported.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	3CC aggregation and massive CA	NRDUCellCarrierMgmt.CaDLMaxCcNum set to DL3CC , DL4CC , or DL5CC	<i>CA Experience Improvement</i>	32-port CSI-RS measurement on SCCs does not take effect for CA UEs for which 3CC aggregation or massive CA is in effect.
Low-frequency TDD	Massive CA	NRDUCellCarrierMgmt.CaDLMaxCcNum set to DL4CC or DL5CC	<i>CA Experience Improvement</i>	Dynamic-codebook-based HARQ feedback over the PUCCH does not take effect for UEs in the massive CA state.
Low-frequency TDD	Multi-frequency beam coordination	MULTI_FREQ_BEAM_COORD_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter	<i>Multi-Frequency Smart Selection</i>	The gNodeB does not select downlink CA UEs whose SCells perform SRS measurement.
Low-frequency TDD	CCE-based data split stop for CA UEs	CCE_BASED_CA_SPLIT_STOP_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoExtSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	CCE-based data split stop for CA UEs takes effect only for UEs in the downlink CA state. After this function takes effect, data scheduling for CA UEs with SCells activated may accelerate and traffic volume-triggered SCell deactivation is more likely to take effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-FR CA coverage enhancement	PUCCH_GROUP_ADAPT_SW option of the NRDUCellCarrierMgmt.InterFrCaEnhSw parameter	<i>Carrier Aggregation</i>	Dynamic-codebook-based HARQ feedback over the PUCCH does not take effect for inter-FR CA UEs for each of which a single set of PUCCH-based feedback mechanisms is used.
Low-frequency TDD	Experience-based smart carrier selection	MULTI_FREQ_SMART_SELECTOR_SW option of the NRCCellAlgoSwitch.MultiFreqAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	Assume that both service-aware carrier selection (controlled by the SERVICE_AWARE_CARRIER_SELECTOR_SW option of the NRDUCellCollabService.MultiCarrierMgmtSw parameter) and experience-based smart carrier selection are enabled. Experience-based smart carrier selection can take effect to trigger carrier changes for a UE only if the UE has not been considered as a target UE for service-aware carrier selection.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SCC coverage expansion	- SA networking: SA_SCC_COV_THLD_ADAPT_SW option of the NRCellAlgSwitch.MultiFreqReqAlgoSwitch parameter - NSA networking: NSA_SCC_COV_THLD_ADAPT_SW option of the NRCellAlgSwitch.MultiFreqReqAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	For CA UEs for which SCell coverage expansion has taken effect, the threshold for event A2 related to SCell coverage expansion is used in A2-triggered SCell removal.
Low-frequency TDD	CA and DRX status collaboration	CA_AND_DRX_STATUS_COLLAB_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoExtSwitch parameter	<i>Carrier Aggregation</i>	If CA and DRX status collaboration is disabled for an SCell, the SCell may be in sleep time, so that precise SCell activation may not take effect. It is recommended that CA and DRX status collaboration be enabled when precise SCell activation has been enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Early data split	EARLY_SPLIT_SW option of the NRDUCellCarrierMgmt.CaCompatibilitySwitch parameter	<i>Carrier Aggregation</i>	When early data split and precise SCell activation are both enabled, the gNodeB now considers that an SCell has been successfully activated if one of the following conditions is met: • The SCell receives valid CQI reports. • The SCell receives valid CSI-RS beam sweeping reports. • The SCell is available according to the proactive check by the gNodeB.
Low-frequency TDD	RSRP prediction model optimization	NR_TO_NR_COV_GRID_SW option of the gNodeBAlgo.NrSmartMultiCarrierSw parameter	<i>Virtual Grid-based Multi-Frequency Coordination</i>	If the NR_TO_NR_COV_GRID_SW option of the gNodeBAlgo.NrSmartMultiCarrierSw parameter is selected, data distribution to SCells is delayed. Therefore, the average downlink throughput of CA UEs, which is equal to $(N.CA.SccActiveUser.ThpVol.DL - N.CA.ThpVol.DL.LastSlot) / N.CA.ThpTime.DL.RmvLastSlot$, slightly decreases in burst service scenarios.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Single-period 8:2 and dual-period 8:2 slot configuration combinations	CA_SLOT_CONFIG_EXT_SW option of the NRCellCaMgmtConfig.CaStrategySwitch parameter	<i>Carrier Aggregation</i>	When the CA_SLOT_CONFIG_EXT_SW option of the NRCellCaMgmtConfig.CaStrategySwitch parameter is selected, CA PDCCH capacity optimization takes effect by default, regardless of whether the switch controlling this function is on or not.

- Impact relationships with functions related to mobility management under RAN functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE-number-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SWITCH option of the NRCellAIgoSwitch.MlbAlgorithmSwitch parameter NRCellMlb.MlbTriggerMode set to CELL_USER_NUM	<i>Mobility Load Balancing in Connected Mode</i>	During the procedure of UE-number-based connected mode MLB, the gNodeB can select CA UEs as the target UEs to be handed over only when NRCellMlb.MlbTransCA is set to DL_CA , and can only transfer downlink CA UEs for the purpose of uplink MLB. After a CA UE is handed over to the target cell, the CA configuration may no longer be applicable, resulting in reduced uplink and downlink user-perceived rates of the UE.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Spectral-efficiency-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SWITCH option of the NRCellInfo.MlbAlgorithmSwitch parameter NRCellInfo.MlbTriggerMode set to SPEC EFF_BASED_USER_NUMBER	<i>Mobility Load Balancing in Connected Mode</i>	During the procedure of spectral-efficiency-based connected mode MLB, the gNodeB can select CA UEs as the target UEs to be handed over only when NRCellInfo.MlbTransferCA is set to DL_CA , and can only transfer downlink CA UEs for the purpose of uplink MLB. After a CA UE is handed over to the target cell, the CA configuration may no longer be applicable, resulting in reduced uplink and downlink user-perceived rates of the UE.
Low-frequency TDD	5QI-specific UE-number-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SWITCH option of the NRCellInfo.MlbAlgorithmSwitch parameter NRCellInfo.MlbTriggerMode set to SPECIFIED_5QI_USER_NUMBER	<i>Mobility Load Balancing in Connected Mode</i>	When selecting UEs to be handed over, the gNodeB does not select UEs on which CA has taken effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Radio-resource-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_Switch.MlbAlgoSwitch parameter selected, and NRCellMlb.MlbTriggerMode set to RADIO_RESOURCE	<i>Mobility Load Balancing in Connected Mode</i>	The gNodeB does not select UEs that treat the local cell as their PCell or SCell for handovers.
Low-frequency TDD	Service-based handover to SCC frequencies	SERV_BASED_HO_TO_SCC_FREQ_Switch option of the gNBMobilityCommParam.MobilityAlgoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	CA UEs are allowed to be handed over to cells on their SCC frequencies through service-based inter-frequency handovers only when service-based handover to SCC frequencies is enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NSA/SA selection based on downlink traffic volume	• MOBILITY_TO_EUTRAN_SWITCH option of the NRCellAIgoSwitch.InterRatServiceMobilitySwitch parameter • NSA_SA_DL_SELECTION_OPTION_SWITCH option of the gNodeBParam.NRworkingOptionOptSwitch parameter	<i>NSA-SA Selection Based on User Experience</i>	The gNodeB includes the selected EN-DC combination in the <i>selectedbandCombinationInfoEN-DC-v1540</i> IE of the handover request message. The eNodeB then configures the EN-DC combination for NSA networking. This EN-DC combination remains unchanged before the SCG added by NSA/SA selection based on downlink traffic volume is released. Therefore, the eNodeB cannot add SCells based on the selection by downlink CA.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NSA/SA selection based on uplink coverage	MOBILITY_TO_EUTRAN_SW option of the NRCellAIgoSwitch.InterRatServiceMobilitySw parameter NSA_SA_UL_SEL_OPT_SW option of the gNodeBParam.NetworkinOptionOptSw parameter	<i>NSA-SA Selection Based on User Experience</i>	The gNodeB includes the selected EN-DC combination in the <i>selectedbandCombinationInfoEN-DC-v1540</i> IE of the handover request message. The eNodeB then configures the EN-DC combination for NSA DC. This EN-DC combination remains unchanged before the SCG added by NSA/SA selection based on uplink coverage is released. Therefore, the eNodeB cannot add SCells based on the selection by downlink CA.

- Impact relationships with functions related to interference mitigation under RAN functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Enhanced intra-frequency SSB interference avoidance	INTRAFREQSSBAVOIDENH_SW option of the NRDUCellAIgoSwitch.CommChnInterfAvoidSwitch parameter	<i>3D Networking Experience Improvement</i>	When downlink intra-FR inter-band CA is enabled, enhanced intra-frequency SSB interference avoidance does not take effect for CA UEs in their SCells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-frequency CSI-RS interference avoidance	INTRA_FREQ_CSIRS_AVOID_SW option of the NRDUCellAsgoSwitch.CommChnlInterfAvoidSwitch parameter	<i>3D Networking Experience Improvement</i>	When downlink intra-FR inter-band CA is enabled, intra-frequency CSI-RS interference avoidance does not take effect for CA UEs in their SCells.
Low-frequency TDD	Enhanced inter-frequency SSB interference avoidance	INTER_FREQ_SSB_AVOID_ENH_SW option of the NRDUCellAsgoSwitch.CommChnlInterfAvoidSwitch parameter	<i>3D Networking Experience Improvement</i>	When downlink intra-FR inter-band CA is enabled, enhanced inter-frequency SSB interference avoidance does not take effect for CA UEs in their SCells.
Low-frequency TDD	Inter-frequency CSI-RS interference avoidance	INTER_FREQ_CSIRS_AVOID_SW option of the NRDUCellAsgoSwitch.CommChnlInterfAvoidSwitch parameter	<i>3D Networking Experience Improvement</i>	When downlink intra-FR inter-band CA is enabled, inter-frequency CSI-RS interference avoidance does not take effect for CA UEs in their SCells.
Low-frequency TDD	Proactive avoidance of remote interference	SRS_RIM_INTERF_AVOID_SW and SRS_INTERF_COORD_S_SLOT_SW options of the NRDUCellSrsSrsAlgoExtSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	2CC aggregation takes effect for fewer UEs when proactive avoidance of remote interference is enabled.

- Impact relationships with functions related to MIMO under RAN functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDSCH MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellA lgoSwitch . MuMimoSwitch parameter	<i>MIMO (TDD)</i>	If CA UEs do not transmit SRSs in their SCells, they cannot be paired for MU-MIMO in the SCells. As a result, the downlink cell capacity decreases.
Low-frequency TDD	Service-based adaptive scheduling	NRDUCellPdsch.DLSchOptTimeThld	<i>MIMO (TDD)</i> <i>Massive MIMO AHR (Low-Frequency TDD)</i>	For UEs in their SCells of downlink intra-FR inter-band CA, the activation of service-based adaptive scheduling may be affected by the mechanism of data distribution to the SCells.

- Impact relationships with functions related to NSA networking under RAN functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	TDM power control	- NR: NSA_DC_ENH_UL_POWER_CONTROL_SW option of the NRCellNsaDcConfig.NsaDcAlgoSwitch parameter - LTE: NSA_DC_ENH_UL_POWER_CONTROL_SW and TDM_SWITCH options of the NsaDcMgmtConfig.NsaDcAlgoSwitch parameter	<i>EPC-based NSA Performance Enhancement</i>	TDM power control allows for the LTE-NR relative frame offset of 3 ms. If the frame offset is changed for an NR cell with the downlink intra-FR inter-band CA function enabled, the frame offset of the primary secondary cell (PSCell) must also be changed because this function requires that the absolute value of the frame offset difference between NR carriers be less than or equal to 625 Ts. For details about the frame offset difference between the PCell and SCells in downlink intra-FR inter-band CA, see 6.1.3.4 Networking .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Secondary intermodulation interference avoidance	- NR: LNR_RELATIVE_FRAME_OFFSET_ADAPT_SWITCH option of the gNodeBParam.NsaDcOptSwitch parameter - LTE: LNR_RELATIVE_FRAME_OFFSET_ADAPT_SWITCH option of the NsaDcAlgoParam.NsaDcAlgoSwitch parameter	<i>EPC-based NSA Basics</i>	Secondary intermodulation interference avoidance allows for the LTE-NR relative frame offset of 3 ms. If the frame offset is changed for an NR cell with the downlink intra-FR inter-band CA function enabled, the frame offset of the PCell must also be changed because this function requires that the absolute value of the frame offset difference between NR carriers be less than or equal to 625 Ts. For details about the frame offset difference between the PCell and SCells in downlink intra-FR inter-band CA, see 6.1.3.4 Networking.

- Impact relationships with other functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-RAT ANR	NR_NR_ANR_SW option of the NRCellAlgoSwitch.AnrSwitch parameter	<i>ANR</i>	If intra-FR inter-band CA is disabled, measurement-based SCell configuration does not take effect; therefore, intra-RAT ANR will not be triggered by CA. If intra-band CA is enabled but SCells are configured in a blind manner, intra-RAT ANR will not be triggered in this scenario either.
Low-frequency TDD	Light-load time sequence optimization	LIGHT_LOAD_TIME_SEQUENCE_OPT_SW option of the NRDUCellAlgoSwitch.LowLatencySwitch parameter	None	Light-load time sequence optimization does not take effect for CA UEs.
Low-frequency TDD	Network slicing	Multiple parameters are required for network slicing deployment. For details, see <i>Network Slicing</i> .	<i>Network Slicing</i>	The network slices configured for an SCell must include UEs' network slices.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	User-rate-based dynamic network slice resource management	DYNAMIC_NETWORKS_LICE_SW option of the NRDUCellCommonSw.ServiceDiffAlgoExtSwitch parameter	<i>Network Slicing</i>	When user-rate-based dynamic network slice resource management is enabled, the NRDUCellQciBearer.DynResDlTargetRate parameter value configured for the PCell also applies to the SCells.
Low-frequency TDD	User-rate-based dynamic QCI resource management	DYNAMIC_QCI_GROUP_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter	<i>QCI-based Resource Control</i>	When user-rate-based dynamic QCI resource management is enabled, the NRDUCellQciBearer.DynResDlTargetRate parameter value configured for the PCell also applies to the SCells.
Low-frequency TDD	QCI-based uplink basic preallocation	UL_PREALLOCATION_QCI_SW option of the gNBDUSchParamGroup.ScheduleOptSwitch parameter	<i>Service-differentiated Experience Guarantee</i>	For CA UEs, QCI-based uplink preallocation is performed in their PCells but not SCells.
Low-frequency TDD	Remote Interference Avoidance for SRS Sw	SRS_RIM_INTERF_AVOID_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter	None	2CC aggregation takes effect for fewer UEs when SRS remote interference avoidance is enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	32-port CSI-RS measurement by single-carrier UEs	SGL_OR_PC_C_CSIRS_32_PORT_SW option of the NRDUCellC sirs.CsiSwitch parameter	None	- When 32-port CSI-RS measurement by single-carrier UEs is enabled, it is recommended that 32-port CSI-RS measurement on SCCs be enabled. In this case, no extra resources are consumed in the cell, and the downlink cell throughput increases with the number of UEs for which 32-port CSI-RS measurement on SCCs takes effect. - When 32-port CSI-RS measurement by single-carrier UEs is disabled, 32-port CSI-RS measurement on SCCs results in extra resource consumption in the cell. Therefore, 32-port CSI-RS measurement on SCCs is recommended only when the percentage of UEs that support this function exceeds 50% and the traffic volume meets the data split threshold. The

RAT	Function Name	Function Switch	Reference	Description
				downlink cell throughput increases with the number of UEs for which 32-port CSI-RS measurement on SCCs takes effect.
Low-frequency TDD	UE bandwidth adaptation	UE_BW_ADAPTIVE_SWITCH option of the NRDUCellBwp.BwpConfigSwitch parameter	<i>Scalable Bandwidth</i>	Consider a scenario where UE bandwidth adaptation is enabled and minimum carrier bandwidths (NRDUCellBwp.DLMinCarrierBw and NRDUCellBwp.ULMinCarrierBw) are configured. After multi-carrier combination selection based on multiple bandwidth levels takes effect, if the bandwidth of the PCell in the optimal combination selected based on the CA capability combination policy differs from the bandwidth in use for a UE, the gNodeB instructs the UE to use the optimal bandwidth.
Low-frequency TDD	PDCP Duplication Indicator	gNBpdcuParamGroup.PdcpDuplicationInd	None	Precise SCell activation does not take effect if the gNBpdcuParamGroup.PdcpDuplicationInd parameter is set to BOTH_SUPPORT or DL_SUPPORT .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Directional outmigration of cell-center UEs from an extended-range cell	VIRTUAL_D EDICATED_ NET_SW option of the NRCellAlgo Switch . <i>ExtendedCellRangePolSw</i> parameter	<i>Extended Cell Range</i>	Directional outmigration of cell-center UEs from an extended-range cell can take effect for CA PCells but not SCells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-base-station CA downlink data split	CLOUD_X_INTRA_GNB_CA_ENH_SWITCH option of the NRDUCellAsgoSwitch.CloudVrSwitch parameter	<i>Layered QoS</i>	When both intra-base-station CA downlink data split and distributed data volume-triggered SCell deactivation are enabled: - A larger value of the gNBQciBearer.CloudServSplitCoefficient parameter results in a higher CA downlink data split threshold for R18 PDU Set UEs. If the buffered data volume is lower than the data split threshold for a long time, SCells will be deactivated and the values of the N.CA.SCell.Deactivate.Att and N.CA.SCell.Deactivate.Succ counters will increase. - A smaller value of the gNBQciBearer.CloudServSplitCoefficient parameter results in a lower CA downlink data split threshold for R18 PDU Set UEs, a longer SCell-activated duration, and higher UE power consumption.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Flexible experience differentiation	FLEX_EXP_DIFF_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter	<i>Service-differentiated Experience Guarantee</i>	After both "CA" and "flexible experience differentiation" are enabled, the base station estimates the uplink or downlink peak data rate of each carrier of a UE. After both "CA" and "flexible experience differentiation" are enabled, the following rules also apply: (1) If the SCell activation threshold is set to a small value, the estimated uplink or downlink peak UE data rate is more accurate, but the UE power consumption increases. (2) If an SCell cannot be activated because the SCell activation threshold is set to a large value, the estimated uplink or downlink peak UE data rate may be inaccurate. It is recommended that the NRDUCellCarrMgmt.CaDIScellActThldSlotNum and NRDUCellCarrMgmt.CaUIScellActThldSlotNum parameters be set to 0 in this scenario.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UL and DL Decoupling	NRDUCellA lgoSwitch.U lDlDecouplingSwitch	<i>UL and DL Decoupling</i>	For UEs that support both UL and DL Decoupling and intra-FR inter-band CA, either UL and DL Decoupling or intra-FR inter-band CA can take effect at a time.

6.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

NOTE

One prerequisite for CA is that the corresponding cells have been set up. The following requirements for base station models, boards, and RF modules are based on this prerequisite.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

For downlink intra-FR inter-band CA:

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

CA can be performed in one baseband processing unit or between baseband processing units.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

RF module requirement of 32-port CSI-RS measurement on SCCs:

Low-frequency RF modules with at least 32T32R support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

RF module requirement of the SCell CSI-RS beam scheme:

Low-frequency RF modules with at least 16T16R support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

6.1.3.4 Networking

Frequency

There must be at least two frequencies available on the live network.

Coverage

The coverage areas provided by the cells involved in CA must overlap. The smaller the overlapping range, the smaller the area where intra-FR inter-band CA takes effect.

Cell

The inter-frequency NR cell relationships have been configured (using the **NRCellRelation** MO).

For intra-frequency cells, the network must be planned to prevent physical cell identifier (PCI) conflicts.

The absolute value of the frame offset difference between the PCell and an SCell must be less than or equal to 625 Ts. Otherwise, intra-FR inter-band CA does not take effect. The frame offset is specified by the **gNBFreqBandConfig.FrameOffset** or **gNodeBParam.FrameOffset** parameter. The value of **gNBFreqBandConfig.FrameOffset** takes precedence. If this parameter is either not configured or set to **4294967295**, the value of **gNodeBParam.FrameOffset** applies.

Intra-FR inter-band CA does not take effect in NR SUL cells, for which the **NRDUCell.DuplexMode** parameter is set to **CELL_SUL**.

The **NRDUCell.HighSpeedFlag** parameter, if configured, of the cells to be engaged in downlink intra-FR inter-band CA must meet one of the following requirements before downlink intra-FR inter-band CA can take effect:

- If the **LOW_HIGH_SPEED_CELL_SPT_CA_SW** option of the **NRCellCaMgmtConfig.CaStrategySwitch** parameter is deselected, the **NRDUCell.HighSpeedFlag** parameter must be set to the same value for all these cells.
- If the **LOW_HIGH_SPEED_CELL_SPT_CA_SW** option of the **NRCellCaMgmtConfig.CaStrategySwitch** parameter is selected, the **NRDUCell.HighSpeedFlag** parameter can be set to either the same value or different values for these cells. If the values are different between the cells, the value must be **LOW_SPEED** for the PCell, whereas the value **HIGH_SPEED** can be set for some or all SCells. After this option is selected, the number of

CA UEs increases, and the **User Downlink Average Throughput (DU)** increases.

Intra-FR inter-band CA does not take effect for non-NR-SUL cells configured with short PUCCH in low-frequency scenarios.

Dynamic-codebook-based HARQ feedback over the PUCCH requires that the PCell be a low-frequency NR TDD cell.

In inter-gNodeB CA scenarios, aperiodic CSI reporting requires that the inter-gNodeB transmission delay be less than or equal to 4 ms.

SCell CSI-RS beam sweeping is supported only when the **NRDUCELL.DuplexMode** parameter is set to **CELL_TDD**.

SCell CSI-RS beam sweeping is supported only when the **NRDUCellTrp.TxRxMode** parameter is set to **16T16R** or higher.

6.1.3.5 Others

- For this function to deliver a theoretical peak data rate to SA CA UEs, the maximum bit rate that each CA UE subscribes to in the 5GC cannot be lower than this theoretical value.
- For this NR CA function to deliver a theoretical peak data rate to NSA UEs, the maximum bit rate that each CA UE subscribes to in the EPC cannot be lower than this theoretical value.
- The requirements for UEs are as follows:
 - UEs must support CA. During initial access, an incoming handover, or an incoming RRC reestablishment, a UE reports its CA capabilities to the gNodeB.
 - The UE must support 32-port CSI-RS measurement on SCCs. That is, the UE must report the **maxNumberTxPortsPerResource** IE whose values include p32.
 - UEs must support semi-static codebooks (as indicated by *semiStaticHARQ-ACK-Codebook* in the *phy-Parameters* IE) for feedback over the PUCCH.

6.1.4 Operation and Maintenance

6.1.4.1 Data Configuration

6.1.4.1.1 Data Preparation

Before activating this function, ensure that the **NRCeIlRelatIon** and **NRCeIlFreqRelatIon** MOs have been configured. For details about the configuration, see *SA Mobility Management in Connected Mode* and *NSA Mobility Management*.

Table 6-2 and **Table 6-3** describe the parameters used for function activation and optimization, respectively. The following parameters do not take effect for SUL cells, for which the **NRDUCell.DuplexMode** parameter is set to **CELL_SUL**.

Table 6-2 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
CA Algorithm Switch	NRDUCellAlgoSwitch.CaAlgoSwitch	INTRA_FR_INTER_BAND_CA_SW	Select this option.
NR Cell ID	NRCeIlCaMgmtConfig.NrCellId	N/A	Set this parameter based on the network plan.
CA Strategy Switch	NRCeIlCaMgmtConfig.CaStrategySwitch	SCC_BLIND_CONFIG_SW	Select this option if blind SCell configuration is required.
CA Frequency Group ID	gNBcAFrequency.CaFreqGroupId	N/A	Set this parameter based on the network plan. It is recommended that all frequencies planned for CA be configured in the same frequency group.
CA Enhanced Algorithm Switch	NRDUCellCarrierMgmt.CaEnhancedAlgorithmSwitch	CARR_CSI_BEAM_SWEEPING_SW	Select this option for cells served by low-frequency RF modules with 16T16R or higher.
CA Enhanced Algorithm Switch	NRDUCellCarrierMgmt.CaEnhancedAlgorithmSwitch	CSI_BEAM_SWEEPING_OPTION_SW	You are advised to select this option.
CA Enhanced Algorithm Switch	NRDUCellCarrierMgmt.CaEnhancedAlgorithmSwitch	CA_CSIRS_BEAM_SWEEP_ADAPT_SW	You are advised to select this option.
Downlink NARFCN	gNBcAFrequency.DlNarfcn	N/A	Set this parameter based on the network plan.
SSB Frequency Position	gNBcAFrequency.SsbFreqPositions	N/A	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
Operator ID List	gNBCaFrequency.OperatorIdList	N/A	Set this parameter based on the network plan. The value of this parameter is expressed in decimal format. The gNodeB converts the decimal number into a 32-bit binary number. If the number is less than 32 bits, the most significant bits are 0s by default. The bits from right to left have a one-to-one relationship with the local operator indexes (specified by the gNBOperator.OperatorId parameter) in ascending order. For example, bit 0 at the rightmost place corresponds to the operator with the gNBOperator.OperatorId parameter set to 0. If a bit is set to 1, the corresponding operator can use the frequency. If a bit is set to 0, the corresponding operator cannot use the frequency. Take the value 5 of the gNBCaFrequency.OperatorIdList parameter as an example. The binary format of this value is 101, indicating that the operators with gNBOperator.OperatorId set to 0 and 2 can use the CA frequency and the operators with gNBOperator.OperatorId set to 1 and values greater than 2 cannot use the CA frequency.
CA A5 RSRP Threshold 2	NRCaMgmtConfig.CaA5RsrpThld2	N/A	To trigger SCell configuration only when the SCells provide better service quality, increase the value of this parameter. The value of this parameter must be greater than that of NRCaMgmtConfig.CaA2RsrpThld.

Parameter Name	Parameter ID	Option	Setting Notes
CA SCC A5 RSRP Threshold 2 Offset	gNB CaFrequency.CaScCA5RsrpThld2Offset	N/A	To adjust the CA event A5 threshold for different SCCs, set this parameter according to operators' policies.
CA A5 Time-to-Trigger	NR CellCaMgmtConfig.CaA5TimeToTrigger	N/A	Set this parameter based on the network plan. A smaller value of this parameter results in a shorter time required before event A5 can be reported and a greater impact of RSRP fluctuations after the SCell coverage meets the entering conditions for event A5. A larger value of this parameter results in the opposite effects.
CA A6 Time-to-Trigger	NR CellCaMgmtConfig.CaA6TimeToTrigger	N/A	Set this parameter based on the network plan. A smaller value of this parameter results in a shorter time required before intra-frequency event A6 can be reported and a greater impact of RSRP fluctuations after the intra-frequency neighboring cell meets the entering conditions for event A6. A larger value of this parameter results in the opposite effects.

Parameter Name	Parameter ID	Option	Setting Notes
Multi-Carrier Measurement Wait Timer	gNodeBAlgo. MultiCarrier MeasWaitTi mer	N/A	Retain the default value. The smaller the value of this parameter, the shorter the measurement waiting time. In scenarios with a large number of candidate frequencies, however, if the value is too small, SCells may not be configured because no measurement reports are received, or better SCells may not be configured because no measurement reports on better candidates are received. As a result, the user-perceived rates of CA UEs decrease. A larger value of this parameter indicates a longer measurement waiting time and results in a higher probability of configuring better SCells but a longer time taken for SCell configuration.
CA A6 Offset	NRCeIlCaMg mtConfig.Ca A6Offset	N/A	To trigger SCell changes only when the neighboring cells of the SCells provide better service quality, increase the value of this parameter.
SCC Deactivation CQI Threshold	NRDUCellCar rMgmt.SccDe actvCqiThld	N/A	It is recommended that this parameter be set to 5. To ensure that SCells are activated only when the channel quality of CA UEs is high, increase the value of this parameter.
CA A2 RSRP Threshold	NRCeIlCaMg mtConfig.Ca A2RsrpThld	N/A	To trigger SCell removals only when the SCells provide poorer service quality, reduce the value of this parameter.
CA A2 RSRP Threshold Offset	gNBcAFreque ncy.CaA2Rsrp ThldOffset	N/A	To adjust the CA event A2 threshold for different SCCs, set this parameter according to operators' policies.

Parameter Name	Parameter ID	Option	Setting Notes
Multi-carrier Flow Control Level	gNodeBAlgo.MultiCarrierFlowCtrlLevel	N/A	Use the recommended value.
Intra-gNodeB CA DL Data Split Threshold	NRDUCellCarrierMgmt.IntraGnbCaDLSplitThld	N/A	The value ADAPTIVE is recommended. Set this parameter to a large value if it is expected that data is distributed to SCells only when the volume of buffered RLC data is large.
DL Intra-gNodeB CA Split Policy	NRDUCellCarrierMgmt.DLIntraGnbCaSplitPolicy	N/A	The value REAL_TIME is recommended.
Split Stop Sch Capb Ratio Threshold	NRDUCellCarrierMgmt.SplitStopSchCapbRatioThld	N/A	The value 50 is recommended.
CA Enhanced Algorithm Switch	NRDUCellCarrierMgmt.CaEnhancedAlgoSwitch	SPLIT_STOP_OPT_SW	You are advised to select this option.
CA Strategy Switch	NRCeCellCaMgmtConfig.CaStrategySwitch	CA_SLOT_CONFIG_EXT_SW	Select this option if compatibility with single-period 8:2 and dual-period 8:2 slot configuration combinations is required.
Customized CA Band Combination ID	gNBCustCaBandComb.CustCaBandCombId	N/A	This parameter identifies an operator-defined CA band +bandwidth combination configured on the base station. Set this parameter as required.
Bandwidth Combination Set	gNBCustCaBandComb.BandwidthCombSet	N/A	Set this parameter as required.
Frequency Band 1	gNBCustCaBandComb.FrequencyBand1	N/A	This parameter specifies CA band 1. Set this parameter as required.

Parameter Name	Parameter ID	Option	Setting Notes
Frequency Bandwidth 1	gNBCustCaB andComb.FrequencyBandwidth1	N/A	This parameter specifies the bandwidth of CA band 1. Set this parameter as required.
Frequency Band 2	gNBCustCaB andComb.FrequencyBand2	N/A	This parameter specifies CA band 2. Set this parameter as required.
Frequency Bandwidth 2	gNBCustCaB andComb.FrequencyBandwidth2	N/A	This parameter specifies the bandwidth of CA band 2. Set this parameter as required.
FR1 CA Downlink TBS Parameter OH	NRDUCellCar rMgmt.Fr1Ca DLTbsParam Oh	N/A	The value 14 is recommended. To increase the maximum downlink TBS, set this parameter to a smaller value.

Parameter Name	Parameter ID	Option	Setting Notes
CA Activate and BWP Switch DL Thld Slot Num	NRDUCellCarrMgmt.CaDIScellActThldSlotNum	N/A	The value 16 is recommended for low-frequency NR TDD cells. A larger value of this parameter results in a higher downlink threshold for SCell activation. The value of this parameter must be greater than one half of the NRDUCellCarrMgmt.CaDIScellDeactThldSlotNum parameter value. In NSA networking, to ensure NR SCell activation when downlink dynamic data split mode is used at the PDCP layer, reduce the value of the NRDUCellCarrMgmt.CaDIScellActThldSlotNum parameter as follows: In the case of the PCell being an NR TDD cell, set the NRDUCellCarrMgmt.CaDIScellActThldSlotNum parameter to a value less than or equal to 20. NOTE Downlink dynamic data split mode is used at the PDCP layer when data is configured as follows: - In Option 3, the NsaDcQciParamGroup.DlDataPdcpsplitMode parameter is set to SCG_AND_MCG, SCG_PREFERRED, or ADAPT_PRIMARY_PATH. - In Option 3x, the gNBpdcpsplitMode parameter is set to SCG_AND_MCG, MCG_PREFERRED, or ADAPT_PRIMARY_PATH.
CA Deactivate and BWP Switch DL Thld Slot Num	NRDUCellCarrMgmt.CaDIScellDeactThldSlotNum	N/A	The value 4 is recommended. A larger value of this parameter results in a higher downlink threshold for SCell deactivation.

Parameter Name	Parameter ID	Option	Setting Notes
DL State Transition Buffer Volume Threshold	NRDUCellCar rMgmt.DlSta teTransitBuf VolThld	N/A	- Regardless of whether NSA or SA networking is used, when the CA activation rate in NSA networking is not required to be as high as that in SA networking: The value 20 is recommended for TDD cells. - For the CA activation rate in NSA networking to reach that in SA networking: A value less than 20 is recommended for TDD cells.
Service Differentiation Algorithm Switch	NRDUCellQci Bearer.Servic eDiffAlgoSwi tch	QCI_CA_BWP_ SWITCH_THL D_CTRL_SW	Select this option if the buffered traffic volume threshold for downlink state transitions needs to be set on a per QCI basis.
QCI DL State Transit Buffer Volume Threshold	NRDUCellQci Bearer.QciDl StTransitBuf VolThld	N/A	- Regardless of whether NSA or SA networking is used, when the CA activation rate in NSA networking is not required to be as high as that in SA networking: The value 20 is recommended for TDD cells. - For the CA activation rate in NSA networking to reach that in SA networking: A value less than 20 is recommended for TDD cells.
RLC Downlink Throughput Threshold	NRDUCellCar rMgmt.RlcDl ThroughputT hld	N/A	The value 2000 is recommended. A larger value of this parameter results in a higher downlink threshold for SCell deactivation.
RLC Buffer Filtering Coefficient	NRDUCellCar rMgmt.RlcBuf ferFilterCoeff	N/A	Use the recommended value. A larger value of this parameter results in a higher level of sensitivity of CA UEs to the instantaneous change in the buffered data volume.

Parameter Name	Parameter ID	Option	Setting Notes
CA SCell Config PRB Usage Threshold	NRCeCellCaMgmtConfig.CaScellConfigPrbUsageThld	N/A	To prevent cells with high PRB usages from being configured as SCells for CA, reduce the value of this parameter.
CA Enhanced Algorithm Extended Switch	NRDUCellCarMgmt.CaEnhancedAlgorithmExtendedSwitch	CA_PDCCH_CAPACITY_OPT_SWITCH	Use the recommended value.
CA Enhanced Algorithm Extended Switch	NRDUCellCarMgmt.CaEnhancedAlgorithmExtendedSwitch	SPLIT_VOL_BASED_SCELL_DEACT_SW	To further reduce UE power consumption in low-frequency scenarios, it is recommended that this option be selected.
CSI Switch	NRDUCellCarMgmt.CsiSwitch	CA_SCC_CSIRS_32PORT_SW	You are advised to select this option.
CA Enhanced Algorithm Switch	NRDUCellCarMgmt.CaEnhancedAlgorithmSwitch	NOT_DEACT_SCC_BASED_CQI_SW	You are advised to select this option.
CA Compatibility Switch	NRDUCellCarMgmt.CaCompatibilitySwitch	BIT_ERROR_SCC_DEACTV_SW	It is recommended that this option be selected in low-frequency scenarios.
PDCCH Outer-Loop Adjustment Upper Limit	NRDUCellPdcchAlgo.PdcchOuterLoopAdjustUpperLimit	None	Use the recommended value. When this parameter is set, the NRDUCellPdcch.UeSpecificPdcchAggLvl parameter must be set to ADAPTIVE .
SCC Configuration Interval	NRDUCellCarMgmt.SccConfigInterval	None	You are advised to use the recommended value.
Dynamic Codebook CA UE Number Threshold	NRDUCellCarMgmt.DynCodebookCAUENumThld	None	Set this parameter as required.
CA Enhanced Algorithm Extended Switch	NRDUCellCarMgmt.CaEnhancedAlgorithmExtendedSwitch	PUCCH_DYN_HARQ_ACK_CB_SW	You are advised to select this option.

Parameter Name	Parameter ID	Option	Setting Notes
PUCCH Algorithm Switch	NRDUCellPucch.PucchAlgoSwitch	PUCCH_DYN_CB_SPECS_IM_P_SW	You are advised to select this option in medium- and heavy-load scenarios.
SCell Deactivation Wait Timer	NRDUCellCarMgmt.ScellDeactvWaitTimer	None	Set this parameter as required.
CA Compatibility Switch	NRDUCellCarMgmt.CaCompatibilitySwitch	SCELL_DEACT_PROT_SW	Select this option if SCell deactivation protection is required.
CA Strategy Switch	NRCeCellCaMgmtConfig.CaStrategySwitch	HO_IN_SCC_CFG_INHERIT_POL_SW	Set this option as required.
CA Strategy Switch	NRCeCellCaMgmtConfig.CaStrategySwitch	HO_IN_SCC_B_LIND_CONFIG_POL_SW	Set this option as required.
CA Strategy Switch	NRCeCellCaMgmtConfig.CaStrategySwitch	MULTI_CARR_MULTI_BWP_OPT_SW	You are advised to select this option if a relatively large number of UEs on the network exhibit better CA capabilities (for example, more layers) on the minimum carrier bandwidths. It is recommended that this option be deselected in other scenarios.
CA Enhanced Algorithm Extended Switch	NRDUCellCarMgmt.CaEnhancedAlgorithmSwitch	PRECISE_SCELL_ACTIVATION_SW	You are advised to select this option.
Reserved Switch Parameter 3	NRDUCellRsvdExt01.RsvdSwitchParam3	RSVDSWPARAM3_BIT32	Select this option as required if DRX is enabled and the SCell activation or deactivation success rate is relatively low.
Reserved Parameter 15	NRDUCellRsvdExt01.RsvdParam15	None	Set this parameter as required when a large number of UEs on the network are under weak coverage and the SCell activation success rate is relatively low.

Parameter Name	Parameter ID	Option	Setting Notes
CA Strategy Switch	NRCellCaMgmtConfig.CaStrategySwitch	SMART_CARR_SEL_OPT_SW	Retain the default value.
Statistics Strategy Switch	NRDUCellOamParam.StatisticsStrategySwitch	DL_CA_THPT_SLOT_STAT_POL_SW	Select this option if the average throughput of SCell-activated UEs is expected to more accurately reflect the actual air interface capability of CA UEs.
NR Smart Multi-Carrier Switch	gNodeBAlgo.NrSmartMultiCarrSw	CA_CAP_USER_STATS_POL_SW	Select this option if it is expected that CA-capable UE counters measure only the numbers of UEs that can actually work in the capability-indicated CA state with the present cell configurations.
CA Strategy Switch	NRCellCaMgmtConfig.CaStrategySwitch	LOW_HIGH_SPEED_CELL_SELECT_CA_SW	You are advised to select this option.
CA Enhanced Algorithm Switch	gNodeBCarrMgmt.CaEnhancedAlgoSwitch	UE_NUM_BASED_SCELL_CONFIG_SW	Select this option if the number of UEs that can access the network may be limited (for example, during major sports events).
Forbid 2CC UE Ratio Threshold	NRCellCaMgmtConfig.Forbid2ccUeRatioThld	None	Set this parameter as required. If the number of UEs that can access the network may be limited (for example, during major sports events), this parameter can be set to a value other than 100 . If both NRCellCaMgmtConfig.Forbid2ccUeRatioThld and NRCellCaMgmtConfig.Forbid3ccUeRatioThld are not set to 100 , the value of NRCellCaMgmtConfig.Forbid2ccUeRatioThld must be greater than that of NRCellCaMgmtConfig.Forbid3ccUeRatioThld .

Parameter Name	Parameter ID	Option	Setting Notes
Forbid 3CC UE Ratio Threshold	NRCeMgmtConfig.Forbid3ccUeRatioThld	None	Set this parameter as required. If the number of UEs that can access the network may be limited (for example, during major sports events), this parameter can be set to a value other than 100 . If both NRCeMgmtConfig.Forbid2ccUeRatioThld and NRCeMgmtConfig.Forbid3ccUeRatioThld are not set to 100 , the value of NRCeMgmtConfig.Forbid2ccUeRatioThld must be greater than that of NRCeMgmtConfig.Forbid3ccUeRatioThld .
CA Enhanced Algorithm Switch	gNodeBCarrMgmt.CaEnhancedAlgorithmSwitch	INTRA_GNB_CRNTI_CONFLICT_SW	You are advised to select this option.
CA Enhanced Algorithm Switch	NRDUCellCarrierMgmt.CaEnhancedAlgorithmSwitch	CRNTI_CONFLICT_RECONFIG_SW	You are advised to select this option.

Table 6-3 Parameters used for optimization

Parameter Name	Parameter ID	Option	Setting Notes
Specified UE Cooperation Switch	NRDUCellUeCoop.SpecUeCooperationSwitch	SPEC_UE_IDENTITY_SW	Set this option as required.
Specified UE Cooperation Switch	NRDUCellUeCoop.SpecUeCooperationSwitch	FR1_DL_TBS_LIMIT_CANCEL_SW	Set this option as required.
CA Enhanced Algorithm Extended Switch	NRDUCellCarrierMgmt.CaEnhancedAlgorithmExtendedSwitch	CA_AND_DRX_STATUS_COELAB_SW	It is recommended that this option be selected when both intra-FR inter-band CA and DRX are enabled.

Parameter Name	Parameter ID	Option	Setting Notes
Multi-Carrier Capb Comb Select Strat	NRCeIlCaMg mtConfig.Ca BandCombSe lectionStrat	N/A	Set this parameter as required.
Multi-Carrier Compatibility Switch	NRCeIlCaMg mtConfig.Ca Compatibility Switch	SRS_WT_BASE D_CARR_SEL_ SW	It is recommended that this option be selected only when a UE reports its capability of multiple CA band+bandwidth combinations that differ only in the SRS carrier switching capability.
Multi-Carrier Management Switch	NRDUCellCol labServ.Multi CarrMgmtSw	SMART_CARR _SEL_SW	You are advised to select this option if the number of an operator's frequencies is greater than the number of frequencies that can be aggregated for a UE.
Multi-Carrier Management Switch	NRDUCellCol labServ.Multi CarrMgmtSw	SERVICE_AWA RE_CARR_SEL _SW	You are advised to select this option if UEs in a cell have various requirements and the number of an operator's frequencies is greater than the number of frequencies that can be aggregated for a UE.
CA Compatibility Switch	NRDUCellCar rMgmt.CaCo mpatibilityS witch	EARLY_SPLIT_ SW	You are advised to select this option.
CA Enhanced Algorithm Extended Switch	NRDUCellCar rMgmt.CaEn hancedAlgoE xtSwitch	SCC_COMMO N_PDCCH_RM _SW	It is recommended that this option be selected when both intra-FR inter-band CA and PDCCH rate matching (controlled by the PDCCH_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchS witch parameter) are enabled in low-frequency scenarios.
CA Strategy Switch	NRCeIlCaMg mtConfig.CaS trategySwitc h	PCELL_PROHI BIT_SW	You are advised to use the recommended value.

Parameter Name	Parameter ID	Option	Setting Notes
Specific UE PCell Policy	NRCeLLCaMg mtConfig.Spe cUePcellPolic y	None	Set this parameter based on the network plan.

6.1.4.1.2 Using MML Commands

Before using MML commands, refer to [6.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling downlink intra-FR inter-band CA
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CaAlgoSwitch=INTRA_FR_INTER_BAND_CA_SW-1;
//Enabling SCell CSI-RS beam sweeping (The SCells cannot be used if the function is disabled.)
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=CARR_CSI_BEAM_SWEEPING_SW-1;
//((Optional) Enabling CSI-RS beam sweeping optimization and adaptive CSI-RS beam sweeping
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
CaEnhancedAlgoSwitch=CSI_BEAM_SWEEPING_OPT_SW-1&CA_CSIRS_BM_SWEEP_ADPT_SW-1;
//((NR TDD) Configuring the CA frequency group and its frequencies, the offsets for event A2 and event A5,
and the operators allowed to use the frequencies (all operators in this example). The DInarfcn and
SsbFreqPos parameter values must be the same as those in the NRDUCell MO.
ADD GNB CAFREQUENCY: CaFreqGroupId=0, DInarfcn=630000, SsbFreqPos=7812, CaA2RsrpThldOffset=0,
CaSccA5RsrpThld2Offset=0, OperatorIdList=4294967295;
//Setting the measurement waiting timer for multi-carrier operations
MOD GNODEBALGO: MultiCarrierMeasWaitTimer=30;
//Setting CA-related parameters
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaA2RsrpThld=-109, CaA5RsrpThld2=-105,
CaA5TimeToTrig=640MS, CaA6Offset=2, CaA6TimeToTrig=640MS;
//((Optional) Setting parameters related to blind SCell configuration when required
//Turning on the switch of blind SCell configuration
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=SCC_BLIND_CONFIG_SW-1;
//Setting BlindScellConfigFlag to TRUE
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1234, CellId=1,
BlindScellConfigFlag=TRUE;
//Setting the SCC configuration interval
MOD NRDUCELLCARRMGMT: NrDuCellId=0, SccCfgInterval=8;
//Setting the level of flow control against multi-carrier operations at the gNodeB
MOD GNODEBALGO: MultiCarrierFlowCtrlLevel=MEDIUM;
//Setting the CQI threshold for SCell deactivation
MOD NRDUCELLCARRMGMT: NrDuCellId=0, SccDeactvCqiThld=5;
//Setting the downlink data split threshold for intra-gNodeB CA
MOD NRDUCELLCARRMGMT: NrDuCellId=0, IntraGnbCaDlSplitThld=ADAPTIVE;
//Enabling real-time data split by RLC for downlink intra-gNodeB CA
MOD NRDUCELLCARRMGMT: NrDuCellId=0, DlintraGnbCaSplitPolicy=REAL_TIME;
//Enabling CA data split stop based on the scheduling capability ratio
MOD NRDUCELLCARRMGMT: NrDuCellId=0, SplitStopSchCapbRatioThld=50;
//((Optional) Enabling CA UE data split stop optimization
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=SPLIT_STOP_OPT_SW-1;
//((Optional) Turning on CA_SLOT_CONFIG_EXT_SW if compatibility with single-period 8:2 and dual-period
```

8:2 slot configuration combinations is required

MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=CA_SLOT_CONFIG_EXT_SW-1;
 //(Optional; TDD) Configuring an operator-defined CA band+bandwidth combination (taking 2CC aggregation as an example)

ADD GNBCUSTCABANDCOMB: CustCaBandCombId=1, BandwidthCombSet=0, FrequencyBand1=N41, FrequencyBandwidth1=CELL_BW_90M, FrequencyBand2=N78, FrequencyBandwidth2=CELL_BW_100M;
 //(Optional) Enabling non-CQI-based SCell deactivation for CA UEs performing downlink large-packet services

MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=NOT_DEACT_SCC_BASED_CQI_SW-1;
 //Enabling consecutive residual block error-triggered SCell deactivation in low-frequency scenarios

MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaCompatibilitySwitch=BIT_ERROR_SCC_DEACTV_SW-1;
 //Enabling distributed data volume-triggered SCell deactivation in low-frequency scenarios

MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoExtSwitch=SPLIT_VOL_BASED_SCELL_DEACT_SW-1;
 //Setting the SCell deactivation waiting timer in low-frequency scenarios

MOD NRDUCELLCARRMGMT: NrDuCellId=0, SCellDeactvWaitTimer=10;
 //(Optional) Enabling SCell deactivation protection in low-frequency scenarios

MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaCompatibilitySwitch=SCELL_DEACT_PROT_SW-1;
 //Setting the downlink traffic volume thresholds for SCell activation and deactivation

MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaDiScellActThldSlotNum=16, CaDiScellDeactThldSlotNum=4;
 //Setting the cell-level threshold of buffered traffic volume for downlink state transitions related to CA SCells in NR TDD scenarios

MOD NRDUCELLCARRMGMT: NrDuCellId=0, DIStateTransitBufVolThld=20;
 //Turning on QCI_CA_BWP_SWITCH_THLD_CTRL_SW in low-frequency NR TDD scenarios if the buffered traffic volume threshold for downlink state transitions needs to be set on a per QCI basis

MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=80, ServiceDiffAlgoSwitch=QCI_CA_BWP_SWITCH_THLD_CTRL_SW-1;
 //Setting the QCI-level threshold of buffered traffic volume for downlink state transitions related to CA SCells in low-frequency NR TDD scenarios

MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=80, QciDIStTransitBufVolThld=20;
 //Setting the RlcBufferFilterCoeff parameter

MOD NRDUCELLCARRMGMT: NrDuCellId=0, RlcBufferFilterCoeff=1;
 //Setting the downlink throughput threshold for SCell deactivation

MOD NRDUCELLCARRMGMT: NrDuCellId=0, RlcDIThroughputThld=2000;
 //Setting the parameter OH for downlink TBS in CA

MOD NRDUCELLCARRMGMT: NrDuCellId=0, Fr1CaDIbtsParamOh=14;
 //Setting the PRB usage threshold for CA SCell configuration

MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaScellConfigPrbUsageThld=100;
 //Turning on the CA PDCCH capacity optimization switch

MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoExtSwitch=CA_PDCCH_CAPACITY_OPT_SWITCH-1;
 //Enabling 32-port CSI-RS measurement on SCCs

MOD NRDUCELLCSIRS: NrDuCellId=0, CsiSwitch=CA_SCC_CSIRS_32PORT_SW-1;
 //Setting the PdcchOuterLoopAdjUpLimit parameter

MOD NRDUCELLPDCCH: NrDuCellId=0, UeSpecificPdcchAggLvl=ADAPTIVE;
 MOD NRDUCELLPDCCHALGO: NrDuCellId=0, PdcchOuterLoopAdjUpLimit=40;
 //Setting DynCodebookCaUeNumThld (take 120 as an example)

MOD NRDUCELLCARRMGMT: NrDuCellId=0, DynCodebookCaUeNumThld=120;
 //Enabling dynamic-codebook-based HARQ feedback over the PUCCH

MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoExtSwitch=PUCCH_DYN_HARQ_ACK_CB_SW-1;
 //(Optional; recommended in medium- or heavy-load scenarios) Enabling dynamic PUCCH codebook specification improvement for CA UEs

MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=PUCCH_DYN_CB_SPECS_IMP_SW-1;
 //Enabling precise SCell activation (The switch must be turned on for both the PCell and SCells.)

MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoExtSwitch=PRECISE_SCELL_ACTIVATION_SW-1;
 //(Optional) Turning on the switch for SCell activation and deactivation suspension during DRX wakeup protection if the SCell activation and deactivation success rates are relatively low when DRX is enabled

MOD NRDUCELLRSVDEXT01: NrDuCellId=0, RsvdSwParam3=RSVDSWPARAM3_BIT32-1;
 //(Optional) Turning on the switch for SCell activation prohibition under weak coverage if a large number of UEs on the network are under weak coverage and the SCell activation success rate is relatively low

MOD NRDUCELLRSVDEXT01: NrDuCellId=0, RsvdParam15=27;
 //(Optional) Turning on HO_IN_SCC_CFG_INHERIT_POL_SW to enable prohibition of SCell configuration inheritance during incoming handovers

MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=HO_IN_SCC_CFG_INHERIT_POL_SW-1;
 //(Optional) Turning on HO_IN_SCC_BLIND_CONFIG_POL_SW to enable prohibition of blind SCell configuration during incoming handovers

MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=HO_IN_SCC_BLIND_CONFIG_POL_SW-1;
 //(Optional) Turning on SMART_CARR_SEL_OPT_SW to enable smart carrier selection optimization

MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=SMART_CARR_SEL_OPT_SW-1;

```

//(Optional) Selecting the MULTI_CARR_MULTI_BWP_OPT_SW option (recommended only when a relatively
large number of UEs on the network exhibit better CA capabilities, for example, more layers, on the
minimum carrier bandwidths; not recommended in other scenarios)
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=MULTI_CARR_MULTI_BWP_OPT_SW-1;
//(Optional) Enabling optimized measurement of downlink CA throughput so that the average throughput
of SCell-activated UEs can more accurately reflect the actual air interface capability of CA UEs
MOD NRDUCELLLOAMPARAM: NrDuCellId=0,
StatisticsStrategySwitch=DL_CA_THPT_L_SLOT_STATS_POL_SW-1;
//(Optional) Enabling optimized measurement of the numbers of CA-capable UEs in cells so that CA-
capable UE counters measure only the numbers of UEs that can actually work in the capability-indicated
CA state with the present cell configurations
MOD GNODEBALGO: NrSmartMultiCarrSw=CA_CAP_USER_STATS_POL_SW-1;
//(Optional; recommended) Enabling CA between low- and high-speed cells
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=LOW_HIGH_SPEED_CELL_SPT_CA_SW-1;
//(Optional) Enabling UE-number-based SCell configuration control if the number of UEs that can access
the network may be limited (for example, during major sports events)
MOD GNODEBCARRMGMT:CaEnhancedAlgoSwitch=UE_NUM_BASED_SCELL_CONFIG_SW-1;
MOD NRCELLCAMGMTCONFIG: NrCellId=0, Forbid2ccUeRatioThld=80, Forbid3ccUeRatioThld=70;
//(Optional; recommended) Turning on the intra-gNodeB C-RNTI conflict detection switch
MOD GNODEBCARRMGMT: CaEnhancedAlgoSwitch=INTRA_GNB_CRNTI_CON_DETECT_SW-1;
//(Optional) Turning on the C-RNTI conflict-triggered reconfiguration switch
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=CRNTI_CONFLICT_RECONFIG_SW-1;

```

Optimization Command Examples

```

//Enabling device-pipe synergy to remove the limit on the maximum downlink peak data rates of UEs
MOD NRDUCELLUECOOP: NrDuCellId=0,
SpecUeCooperationSwitch=SPEC_UE_IDENTIFY_SW-1&FR1_DL_TBS_LIMIT_CANCEL_SW-1;
//Enabling collaboration between CA and DRX when both intra-FR inter-band CA and DRX are enabled
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
CaEnhancedAlgoExtSwitch=CA_AND_DRX_STATUS_COLLAB_SW-1;
//Enabling preferential selection of a UE-reported CA capability combination based on the peak data rate
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaBandCombSelectionStrat=PEAK_RATE_FIRST;
//Enabling CA capability combination selection with SRS-based weights considered
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaCompatibilitySwitch=SRS_WT_BASED_CARR_SEL_SW-1;
//Turning on the smart carrier selection switch
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, MultiCarrMgmtSw=SMART_CARR_SEL_SW-1;
//Turning on the service-aware carrier selection switch
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, MultiCarrMgmtSw=SERVICE_AWARE_CARR_SEL_SW-1;
//Turning on the early data split switch
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaCompatibilitySwitch=EARLY_SPLIT_SW-1;
//Turning on the switch for common PDCCH rate matching in SCells in low-frequency scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoExtSwitch=SCC_COMMON_PDCCH_RM_SW-1;
//Turning on PCELL_PROHIBIT_SW and configuring the UE-specific PCell policy (If PCELL_PROHIBIT_SW is
turned on for a cell, the cell cannot act as the PCell for CA. Determine the switch setting based on the
network plan.)
MOD NRCELLCAMGMTCONFIG: NrCellId=0,
CaStrategySwitch=PCELL_PROHIBIT_SW-1,SpecUePcellPolicy=NSA_SA;

```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```

//Disabling downlink intra-FR inter-band CA
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CaAlgoSwitch= INTRA_FR_INTER_BAND_CA_SW-0;
//Disabling CSI-RS beam sweeping optimization and adaptive CSI-RS beam sweeping
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
CaEnhancedAlgoSwitch=CSI_BEAM_SWEEPING_OPT_SW-0&CA_CSIRS_BM_SWEEP_ADPT_SW-0;
//Disabling SCell CSI-RS beam sweeping
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=CARR_CSI_BEAM_SWEEPING_SW-0;
//Turning off the blind SCell configuration switch
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=SCC_BLIND_CONFIG_SW-0;
//Disabling SCell deactivation protection
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaCompatibilitySwitch=SCELL_DEACT_PROT_SW-0;
//Turning off the CA PDCCH capacity optimization switch
MOD NRDUCELLCARRMGMT: NrDuCellId=0,

```

```

CaEnhancedAlgoExtSwitch=CA_PDCCH_CAPACITY_OPT_SWITCH-0;
//Disabling 32-port CSI-RS measurement on SCCs
MOD NRDUCELLCSIRS: NrDuCellId=0, CsiSwitch=CA_SCC_CSIRS_32PORT_SW-0;
//Disabling non-CQI-based SCell deactivation for CA UEs performing downlink large-packet services
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=NOT_DEACT_SCC_BASED_CQI_SW-0;
//Disabling consecutive residual block error-triggered SCell deactivation in low-frequency scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaCompatibilitySwitch=BIT_ERROR_SCC_DEACTV_SW-0;
//Disabling distributed data volume-triggered SCell deactivation in low-frequency scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
CaEnhancedAlgoExtSwitch=SPLIT_VOL_BASED_SCELL_DEACT_SW-0;
//Disabling delayed SCell deactivation in low-frequency scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=0, ScellDeactvWaitTimer=0;
//Disabling dynamic-codebook-based HARQ feedback over the PUCCH
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoExtSwitch=PUCCH_DYN_HARQ_ACK_CB_SW-0;
//Disabling dynamic PUCCH codebook specification improvement for CA UEs
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=PUCCH_DYN_CB_SPECS_IMP_SW-0;
//Disabling real-time data split by RLC for downlink intra-gNodeB CA
MOD NRDUCELLCARRMGMT: NrDuCellId=0, DlintraGnbCaSplitPolicy=NOT_CONFIG;
//Disabling CA data split stop based on the scheduling capability ratio
MOD NRDUCELLCARRMGMT: NrDuCellId=0, SplitStopSchCapbRatioThld=0;
//Disabling CA UE data split stop optimization
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=SPLIT_STOP_OPT_SW-0;
//Turning off CA_SLOT_CONFIG_EXT_SW
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=CA_SLOT_CONFIG_EXT_SW-0;
//Disabling precise SCell activation
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoExtSwitch=PRECISE_SCELL_ACTIVATION_SW-0;
//Turning off the switch for SCell activation and deactivation suspension during DRX wakeup protection
MOD NRDUCELLRSVDEXT01: NrDuCellId=0, RsvdSwParam3=RSVDSWPARAM3_BIT32-0;
//Turning off the switch for SCell activation prohibition under weak coverage
MOD NRDUCELLRSVDEXT01: NrDuCellId=0, RsvdParam15=0;
//Disabling prohibition of SCell configuration inheritance during incoming handovers
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=HO_IN_SCC_CFG_INHERIT_POL_SW-0;
//Disabling prohibition of blind SCell configuration during incoming handovers
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=HO_IN_SCC_BLIND_CONFIG_POL_SW-0;
//Turning off the switch for multi-carrier combination selection based on multiple bandwidth levels
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=MULTI_CARR_MULTI_BWP_OPT_SW-0;
//Disabling smart carrier selection optimization
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=SMART_CARR_SEL_OPT_SW-0;
//Disabling optimized measurement of downlink CA throughput
MOD NRDUCELLLOAMPARAM: NrDuCellId=0,
StatisticsStrategySwitch=DL_CA_THPT_L_SLOT_STATS_POL_SW-0;
//Disabling optimized measurement of the numbers of CA-capable UEs in cells
MOD GNODEBALGO: NrSmartMultiCarrSw=CA_CAP_USER_STATS_POL_SW-0;
//Disabling CA between low- and high-speed cells
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=LOW_HIGH_SPEED_CELL_SPT_CA_SW-0;
//Disabling UE-number-based SCell configuration control
MOD GNODEBCARRMGMT: CaEnhancedAlgoSwitch=UE_NUM_BASED_SCELL_CONFIG_SW-0;
MOD NRCELLCAMGMTCONFIG: NrCellId=0, Forbid2ccUeRatioThld=100, Forbid3ccUeRatioThld=100;
//Turning off the intra-gNodeB C-RNTI conflict detection switch
MOD GNODEBCARRMGMT: CaEnhancedAlgoSwitch=INTRA_GNB_CRNTI_CON_DETECT_SW-0;
//Turning off the C-RNTI conflict-triggered reconfiguration switch
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=CRNTI_CONFLICT_RECONFIG_SW-0;

```

6.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.1.4.2 Activation Verification

Provided that CA UEs are available on the network, downlink intra-FR inter-band CA has taken effect if any counter in [Table 6-4](#) returns a non-zero value.

Table 6-4 Counters related to downlink intra-FR inter-band CA

Counter ID	Counter Name
1911816773	N.User.CA.PCell.DL.Avg
1911853842	N.User.CA.PCell.DL.Max
1911816774	N.User.CA.SCell.DL.Avg
1911816900	N.User.CA.PCell.DL.SCellAct.Avg
1911816902	N.User.CA.SCell.DL.Act.Avg
1911827213	N.User.CA.PCell.DL.SCellAct.Max
1911840463	N.CA.PR.B.DL.PCell.Used.Avg
1911840465	N.CA.PR.B.DL.SCell.Used.Avg

The counters related to the numbers of attempts and successful executions regarding SCell configuration, change, activation, deactivation, and removal are listed in [Table 6-5](#).

Table 6-5 Counters related to SCell management

Counter ID	Counter Name
1911816777	N.CA.SCell.Add.Att
1911816778	N.CA.SCell.Add.Succ
1911816781	N.CA.SCell.Mod.Att
1911816782	N.CA.SCell.Mod.Succ
1911816896	N.CA.SCell.Act.Att
1911816897	N.CA.SCell.Act.Succ
1911816898	N.CA.SCell.Deact.Att
1911816899	N.CA.SCell.Deact.Succ
1911816779	N.CA.SCell.Rmv.Att
1911816780	N.CA.SCell.Rmv.Succ

6.1.4.3 Network Monitoring

Observe the average downlink UE throughput (**User Downlink Average Throughput (DU)**) to check whether the UE data rate increases after this function is enabled.

Observe the average throughput of SCell-configured CA UEs, which is calculated using the following formula: **(N.CA.SccConfigUser.ThpVol.DL –**

**N.CA.SccConfigUser.ThpVol.DL.LastSlot)/
N.CA.SccConfigUser.ThpTime.DL.RmvLastSlot.**

Observe the average throughput of SCell-activated CA UEs, which is calculated using the following formula: **(N.CA.SccActiveUser.ThpVol.DL – N.CA.ThpVol.DL.LastSlot)/N.CA.ThpTime.DL.RmvLastSlot.**

 NOTE

Optimized measurement of downlink CA throughput can be enabled to improve the accuracy of measuring the average throughput of SCell-activated UEs. For details, see [4.8.1 Optimized Measurement of Downlink CA Throughput](#).

Observe the CQI statistics of SCell-activated CA UEs in PCells using the counters **N.ChMeas.CA.PCell.CQI.0** to **N.ChMeas.CA.PCell.CQI.15** and those in SCells using the counters **N.ChMeas.CA.SCell.CQI.0** to **N.ChMeas.CA.SCell.CQI.15**.

Observe the numbers of TBs transmitted using different ranks for SCell-activated CA UEs in PCells using the counters **N.CA.PCell.PDSCH.NbrTbDL.Rank1** to **N.CA.PCell.PDSCH.NbrTbDL.Rank4** and those in SCells using the counters **N.CA.SCell.PDSCH.NbrTbDL.Rank1** to **N.CA.SCell.PDSCH.NbrTbDL.Rank4**.

Observe MAC statistics related to SCell-activated CA UEs in PCells and SCells using the following counters, which measure the number of downlink TBs initially transmitted and retransmitted to CA UEs:

- **N.DL.SCH.CA.PCell.TB**: measures the number of downlink TBs initially transmitted to CA UEs that treat the local cell as their PCell.
- **N.DL.SCH.CA.SCell.TB**: measures the number of downlink TBs initially transmitted to CA UEs that treat the local cell as their SCell.
- **N.DL.SCH.CA.SCell.ErrTB.Ibler**: measures the number of downlink TBs failing to be initially transmitted to CA UEs that treat the local cell as their SCell.
- **N.DL.SCH.CA.PCell.ErrTB.Ibler**: measures the number of downlink TBs failing to be initially transmitted to CA UEs that treat the local cell as their PCell.
- **N.DL.SCH.CA.PCell.TB.Retrans**: measures the number of downlink TBs retransmitted to CA UEs that treat the local cell as their PCell.
- **N.DL.SCH.CA.SCell.TB.Retrans**: measures the number of downlink TBs retransmitted to CA UEs that treat the local cell as their SCell.
- **N.DL.SCH.CA.PCell.ErrTB.Rbler**: measures the number of downlink TBs failing to be transmitted to CA UEs that treat the local cell as their PCell after the maximum number of retransmissions is reached.
- **N.DL.SCH.CA.SCell.ErrTB.Rbler**: measures the number of downlink TBs failing to be transmitted to CA UEs that treat the local cell as their SCell after the maximum number of retransmissions is reached.

Observe the average MCS index in PDSCH scheduling of SCell-activated CA UEs in PCells, which is calculated using the following formula:

N.ChMeas.CA.PCell.PDSCH.MCS.Sum/N.ChMeas.CA.PCell.PDSCH.

Observe the average MCS index in PDSCH scheduling of SCell-activated CA UEs in SCells, which is calculated using the following formula:

N.ChMeas.CA.SCell.PDSCH.MCS.Sum/N.ChMeas.CA.SCell.PDSCH.

Observe the downlink traffic volume for SCell-activated CA UEs in PCells using the counter **N.CA.PCell.ThpVol.DL.Cell**.

Observe the downlink traffic volume for CA UEs in SCells using the counter **N.CA.SCell.ThpVol.DL.Cell**.

6.2 Uplink Intra-FR Inter-Band CA

6.2.1 Principles

6.2.1.1 Overview

Currently, uplink intra-FR inter-band CA is supported only in FR1 rather than FR2.

To fully utilize PCell and SCell resources, uplink intra-FR inter-band CA requires that uplink data be transmitted simultaneously in the PCell and SCells, achieving higher uplink peak throughput. This function can be enabled by selecting the **INTRA_FR_INTER_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch.CaAlgoSwitch** parameter. With this function enabled, uplink inter-band SCell configuration is triggered when a CA UE initiates RRC setup during initial access, an incoming handover, or an incoming RRC connection reestablishment. After SCells are configured and activated, the CA UE can benefit from uplink CA. For the band+bandwidth combinations for uplink intra-FR inter-band CA, see [4.4 CA Band+Bandwidth Combinations](#).

After uplink intra-FR inter-band CA is enabled, uplink data may be scheduled in an SCell but not in the PCell in a single slot. In such a situation, the gNodeB and UE may make different decisions on whether HARQ feedback is sent together with data on the PUSCH. This will cause incorrect data demodulation on the gNodeB side and increase the BLER. To prevent this issue, it is recommended that the **CA_PDCCH_CAPACITY_OPT_SWITCH** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch** parameter be selected.

NOTE

Uplink intra-FR inter-band CA works only in TDD-only scenarios.

6.2.1.2 CA Uplink Timing

Timing advances (TAs) are used by UEs to adjust the time to transmit uplink data. In a multi-carrier scenario such as CA, a TA is required for each of the carriers. The use of an inappropriate TA affects the uplink demodulation performance and therefore the uplink peak rate.

In uplink intra-FR inter-band CA, however, when a CA UE uses the TA of the PCell in an SCell, uplink demodulation performance is affected. This is because the CA UE is at different distances away from the receive antennas of the PCell and SCell and the UE sends uplink data at different times from the antennas corresponding to the cells. To prevent the impact on uplink demodulation performance, the gNodeB selects a single TA or multiple appropriate TAs from the TAs for the PCell and SCells and delivers the selected TAs to the UE. The number of TAs selected depends on the following functions:

- Multiple TAs (MTA): The MTA function takes effect when the gNodeB selects two or more TAs for a UE that has reported the *supportedNumberTAG* IE with

a value greater than 1 and its support for *dualPA-Architecture*. TAG is short for TA group. In the CA UE, each CC is assigned to a specific TAG. For all CCs involved in a TAG, the UE uses a same TA to adjust the time to transmit uplink data. The number of TAs selected by the gNodeB is equal to the number of TAGs supported by the UE, which is the value of the *supportedNumberTAG* IE.

- If the number of TAGs supported by the UE is greater than or equal to the number of CCs, only one CC is assigned to each TAG, and each CC uses its own TA. In other words, all CCs use their own TAs.
- If the number of TAGs supported by the UE is less than the number of CCs, more than one CC may be assigned to a TAG and share a TA. To select an optimal TA in this case, flexible TA adjustment in uplink multi-CC scenarios is introduced. This function ensures appropriate TA selection and minimizes the impact on uplink demodulation performance. For details about this function, see [Flexible TA Adjustment in Uplink Multi-CC Scenarios](#).
- Single TA: The single TA function takes effect when the gNodeB selects a single TA for a UE. This may occur if the UE does not meet requirements or the MTA function is disabled. The UE uses this TA to adjust the time to transmit uplink data on all the CCs. To select an optimal TA in this case, flexible TA adjustment in uplink multi-CC scenarios is introduced, as it is in the MTA case. This function ensures appropriate TA selection and minimizes the impact on uplink demodulation performance. For details about this function, see [Flexible TA Adjustment in Uplink Multi-CC Scenarios](#).

By default, the MTA function takes effect on the gNodeB. To use the single TA function, configure data as described in [MTA Configuration Prohibition for Uplink CA](#).

For details about how the gNodeB obtains the TA of an SCell, see [Procedure for the gNodeB to Obtain the TA of an SCell](#).

MTA Configuration Prohibition for Uplink CA

MTA configuration prohibition for uplink CA is controlled by a cell-level switch and a UE whitelist switch.

- Cell-level switch: If the **UL_CA_MTAG_NOT_CONFIG_SW** option of the **NRDUCellCarrMgmt.CaCompatibilitySwitch** parameter is selected for a cell, the MTA function is disabled for the cell, in which only the single TA function can take effect.

NOTE

If MTA has taken effect for a UE in a cell prior to selecting the **UL_CA_MTAG_NOT_CONFIG_SW** option of the **NRDUCellCarrMgmt.CaCompatibilitySwitch** parameter, the MTA mechanism remains in effect after this option is selected; the UE will switch to the single TA mechanism only after re-accessing the cell.

- UE whitelist switch: Certain UEs may encounter SCell activation failure because of compatibility issues when the MTA function takes effect on them. If the **UL_CA_MTAG_NOT_CONFIG_SW_ON** option of the **gNBUECompatSw.WhitelistCtrlExtSwitch** parameter is selected for the UEs, the MTA function is disabled for these UEs, so that only the single TA function

can take effect for them. This option takes effect only when the **UL_CA_MTAG_NOT_CONFIG_SW** option of the **NRDUCellCarrMgmt.CaCompatibilitySwitch** parameter is selected.

- If the **UL_CA_MTAG_NOT_CONFIG_SW_ON** option is selected, the MTA function does not take effect for specified UEs but takes effect for other UEs.
- If the **UL_CA_MTAG_NOT_CONFIG_SW_ON** option is deselected, the MTA application status is the same for all UEs. The status is determined by the **UL_CA_MTAG_NOT_CONFIG_SW** option of the **NRDUCellCarrMgmt.CaCompatibilitySwitch** parameter.

 NOTE

If MTA has taken effect for a UE in a cell prior to selecting the **UL_CA_MTAG_NOT_CONFIG_SW_ON** option of the **gNBUECompatSw.WhitelistCtrlExtSwitch** parameter, the MTA mechanism remains in effect after this option is selected; the UE will switch to the single TA mechanism only after re-accessing the cell.

Flexible TA Adjustment in Uplink Multi-CC Scenarios

Flexible TA adjustment in uplink multi-CC scenarios is controlled by the **UL_MULTI_CC_TA_FLEX_ADJ_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoSwitch** parameter. With this function enabled, the following TA selection policy applies to all TAGs:

- If the absolute value of the TA difference between any two CCs involved in a TAG is less than the value of the **NRDUCellCarrMgmt.UlMultiCcAbsTaDiffThld** parameter, the TA that has the minimum impact on all the CCs involved in this TAG is selected as the TA for all these CCs.
- If the absolute value of the TA difference between any two CCs involved in a TAG is greater than or equal to the value of the **NRDUCellCarrMgmt.UlMultiCcAbsTaDiffThld** parameter, the TA that has the minimum impact on the PCC is selected as the TA for all the CCs involved in this TAG.

Procedure for the gNodeB to Obtain the TA of an SCell

If an SCell provides uplink resources for a CA UE, the physical random access channel (PRACH) resources are configured for the SCell. After the SCell is activated for the UE, the gNodeB sends a physical downlink control channel (PDCCH) order to the UE in the PCell and, as instructed, the UE initiates a non-contention-based random access procedure in the SCell. After that, the SCell reports the related measurement information to the gNodeB for the gNodeB to obtain the TA of the SCell.

If the number of preamble retransmissions from a CA UE in its SCell reaches the maximum number allowed, the random access procedure fails. The UE stops random access without triggering an RRC connection reestablishment. As a result, the UE is out-of-synchronization in the uplink in the SCell. If the SCell activation conditions are met again and the SCell uplink time alignment timer expires, the PCell sends a PDCCH order to instruct the UE to initiate a random access procedure again. For details about the uplink time alignment timer, see *5G Networking and Signaling*.

6.2.1.3 Optimized Fallback upon Uplink CA Exceptions

If an exception (such as an SCell addition failure) occurs in uplink CA, the gNodeB performs fallback by removing the involved SCells, which renders uplink CA ineffective. In some scenarios, the gNodeB may also remove the SCells for downlink CA, causing downlink CA to become ineffective as well.

To address this issue, the 'optimized fallback upon uplink CA exceptions' function has been introduced. This function is controlled by the **UL_CA_FALLBACK_DL_ONLY_CA_SW** option of the **gNodeBCarrMgmt.CaEnhancedAlgoSwitch** parameter. If this option is selected and an exception occurs in uplink CA, the gNodeB removes only uplink SCells but not downlink SCells, ensuring that downlink CA remains unaffected.

This function works only in low frequency bands.

6.2.1.4 SRS and PUSCH/PUCCH Concurrency Compatibility

As defined in 3GPP specifications, SRS and PUSCH/PUCCH concurrency is supported in intra-FR inter-band CA. In a CA scenario, SRS and PUSCH/PUCCH concurrency means that a PCC and an SCC work in the same symbols where one CC sends SRSs and the other CC sends PUSCH/PUCCH signals.

For intra-FR inter-band CA, SRS and PUSCH/PUCCH concurrency requires that UEs support this capability. Huawei gNodeBs comply with the protocol specifications to support SRS and PUSCH/PUCCH concurrency, while a UE supports this capability only if it reports the **parallelTxSRS-PUSCH-PUSCH** field.

Before the SRS and PUSCH/PUCCH concurrency compatibility function is introduced, the gNodeB by default considers that all UEs support the concurrency. If a UE does not support this capability, the gNodeB demodulates SRSs or PUSCH/PUCCH signals at incorrect time-domain positions, which compromises uplink performance of CA UEs. To address this issue, the SRS and PUSCH/PUCCH concurrency compatibility function is introduced. This function is controlled by the **SRS_PUSCH_CONCURRENCY_COMPT_SW** option of the **NRDUCellCarrMgmt.CaCompatibilitySwitch** parameter. After the SRS and PUSCH/PUCCH concurrency compatibility function is enabled, the gNodeB checks whether a UE supports this concurrency so that the processing can vary with the UE capability:

- If the UE supports SRS and PUSCH/PUCCH concurrency, the UE can concurrently send SRSs and PUSCH/PUCCH signals.
- If the UE does not support SRS and PUSCH/PUCCH concurrency, the UE does not send SRSs and PUSCH/PUCCH signals concurrently on the PCC and SCCs. Instead, in the corresponding symbols, the UE sends SRSs only but not PUSCH/PUCCH signals.

SRS and PUSCH/PUCCH concurrency compatibility is supported only in low-frequency scenarios.

6.2.2 Network Analysis

6.2.2.1 Benefits

The uplink peak data rates of CA UEs increase.

The peak data rate that a CA UE can achieve using uplink 2CC aggregation ranges from $(A \times 94\% + B \times 94\%)$ to $(A \times 100\% + B \times 100\%)$.

In this expression:

- A is the uplink data rate that the UE in the single-carrier state can achieve in a cell that is later configured as the PCell for the UE.
- B is the uplink data rate that the UE in the single-carrier state can achieve in a cell that is later configured as an SCell for the UE.

 NOTE

- The preceding peak data rate ranges can be achieved only if the influencing factors (for example, the UE rank) remain unchanged after CA takes effect.
- Uplink CA increases the uplink data rate only if the UE uses more transmit antennas in the uplink CA state than in the single-carrier state. In other situations, the uplink data rate may not increase despite greater uplink bandwidth due to CA, because the number of layers for uplink multiple-input multiple-output (MIMO) on each carrier decreases.
- UE capabilities affect the uplink peak data rate. For example, some UEs reduce their transmit power after CA takes effect, which affects the uplink peak data rate.

After flexible TA adjustment in uplink multi-CC scenarios takes effect, the uplink cell throughput may improve. The extent of improvement is influenced by the following factors:

- When there are CCs with negative measured TA values among multiple CCs, the uplink cell throughput improvement is greater compared to scenarios without negative values.
- A larger difference in measured TA values between CCs contributes to greater uplink cell throughput improvement.
- A larger bandwidth difference between CCs contributes to greater uplink cell throughput improvement.
- The impact on the uplink cell throughput depends also on the setting of the **NRDUCellCarrMgmt.UlMultiCcAbsTaDiffThld** parameter. The default value is recommended.
 - A smaller value of this parameter indicates a stricter requirement of TA adjustment to the TA value with a relatively small impact on all CCs. However, if the value is too small, the increase in the uplink cell throughput will be relatively small.
 - A larger value of this parameter indicates a more relaxed requirement. However, if the value is too large, there will be a relatively high probability of a decrease in the uplink throughput.

With optimized fallback upon uplink CA exceptions, the gNodeB removes only uplink SCells in the event of uplink CA exceptions, ensuring that downlink CA remains unaffected. The detailed impacts are as follows:

- The **User Downlink Average Throughput (DU)** and **Cell Downlink Average Throughput (DU)** increase.
- The **N.User.CA.SCell.DL.Avg** and **N.User.CA.PCell.DL.Avg** increase.
- The **N.CA.SCell.Rmv.Succ** and **N.CA.SCell.Rmv.Att** decrease.

After SCC removal due to consecutive non-split in uplink CA takes effect, the **User Uplink Average Throughput (DU)** of cell-edge UEs increases.

After SRS and PUSCH/PUCCH concurrency compatibility takes effect, the average throughput of SCell-activated uplink CA UEs ($(N.CA.SccActiveUser.ThpVol.UL - N.CA.ThpVol.UL.SmallPkt)/N.CA.ThpTime.UL.RmvSmallPkt$) increases.

If uplink enhancement for smart carrier selection takes effect in suitable scenarios, the average number of uplink CA UEs that treat the local cell as their PCell ($N.User.CA.PCell.UL.Avg$) increases.

6.2.2.2 Impacts

The impacts between this function and other functions are described in [6.2.3.2.3 Function Impacts](#).

- The smaller the value of **NRDUCellCarrMgmt.CaUlScellActThldSlotNum**, the more likely SCells are activated, and the longer the SCell-activated state lasts. Therefore, a smaller value results in higher UE power consumption and greater values of the following counters:
 - **N.CA.SCell.Act.Att**
 - **N.CA.SCell.Act.Succ**
 - **N.CA.SCell.Deact.Att**
 - **N.CA.SCell.Deact.Succ**
- The smaller the value of **NRDUCellCarrMgmt.CaUlScellDeactThldSlotNum** or **NRDUCellCarrMgmt.RlcUlThroughputThld**, the less likely SCells are deactivated, and the longer the SCell-activated state lasts. Therefore, a smaller value results in higher UE power consumption and smaller values of the following counters:
 - **N.CA.SCell.Deact.Att**
 - **N.CA.SCell.Deact.Succ**
 - **N.CA.SCell.Act.Att**
 - **N.CA.SCell.Act.Succ**
- After uplink intra-FR inter-band CA is enabled, the CPU usage of the BBP where the PCell is configured increases with the amount of data forwarded from the PCell to SCells at the RLC layer. If the CPU usage is too high, the peak data rate of the UE decreases.
- The data configuration for CA uplink timing has the following impacts:
 - If the **UL_CA_MTAG_NOT_CONFIG_SW** option of the cell-level parameter **NRDUCellCarrMgmt.CaCompatibilitySwitch** is selected, the MTA function does not take effect. In this case, if there is a large TA difference between the PCell and SCell of a CA UE, the uplink throughput of the UE decreases.
 - If the **UL_CA_MTAG_NOT_CONFIG_SW_ON** option of the gNodeB-level parameter **gNBUECompatSw.WhitelistCtrlExtSwitch** is selected, the MTA function does not take effect for specified UEs. In this case, if there is a large TA difference between the PCell and SCell of a CA UE, the uplink throughput of the UE may decrease.
- After SCC removal due to consecutive non-split in uplink CA takes effect, an excessively small value of **NRDUCellCarrMgmt.UlCaContNonSplitSccDelThld** may cause unnecessary SCC removal, decreasing the average throughput of SCell-activated uplink CA

UEs (calculated as follows: $(N.CA.SccActiveUser.ThpVol.UL - N.CA.ThpVol.UL.SmallPkt) / N.CA.ThpTime.UL.RmvSmallPkt$).

- After uplink CA data split threshold optimization takes effect, the following indicators change:
 - The total volume of uplink data received at the MAC layer from uplink CA UEs that treat the local cell as their SCell ($N.CA.SCell.ThpVol.UL.Cell$) increases.
 - The total volume of uplink data received at the MAC layer from uplink CA UEs that treat the local cell as their PCell ($N.CA.PCell.ThpVol.UL.Cell$) decreases.
 - The average throughput of SCell-activated uplink CA UEs (calculated as follows: $(N.CA.SccActiveUser.ThpVol.UL - N.CA.ThpVol.UL.SmallPkt) / N.CA.ThpTime.UL.RmvSmallPkt$) varies, influenced by the SCell load. Under low SCell load, this throughput increases.

6.2.3 Requirements

6.2.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit	NE
FOFD-031201	Intra-FR Inter-Band CA	NR0SINFRIBC A	per Cell	Macro gNodeB
FOFD-031201	Intra-FR Inter-Band CA	NR0TSIFIBC01	per Carrier per pRRU	LampSite gNodeB

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

6.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.2.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Uplink intra-FR inter-band CA (controlled by the INTRA_FR_INTER_BAND_UL_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter) requires that downlink intra-FR inter-band CA be enabled.
Low-frequency TDD	MTA configuration prohibition for uplink CA	UL_CA_MTAG_NOT_CONFIG_SW option of the NRDUCellCarrMgmt.CaCompatibilitySwitch parameter	<i>Carrier Aggregation</i>	The UL_CA_MTAG_NOT_CONFIG_SW_ON option of the gNBUECompatSw.WhitelistCtrlExtSwitch parameter takes effect only when the UL_CA_MTAG_NOT_CONFIG_SW option of the NRDUCellCarrMgmt.CaCompatibilitySwitch parameter is selected.
Low-frequency TDD	Flexible TA adjustment in uplink multi-CC scenarios	UL_MULTI_CC_TA_FLEX_ADJ_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter	<i>Carrier Aggregation</i>	The setting of the NRDUCellCarrMgmt.UlMultiCcAbsTaDiffThld parameter takes effect only if flexible TA adjustment in uplink multi-CC scenarios is enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Smart carrier selection	SMART_CARR_SEL_SW option of the NRDUCellConfigCommon.SmartCarrierSelectionSwitch parameter	<i>Carrier Aggregation</i>	Uplink enhancement for smart carrier selection takes effect only when both smart carrier selection and service-based optimal carrier selection are enabled.
Low-frequency TDD	Service-based optimal carrier selection	SERVICE_BASED_CARR_SEL_SW option of the NRDUCellConfigCommon.SmartCarrierSelectionSwitch parameter	<i>Carrier Aggregation</i>	
Low-frequency TDD	SRS interference coordination based on self-contained slots	SRS_INTRFCOORD_SLOT_SW option of the NRDUCellConfigCommon.SrsInterferenceCoordinationSwitch parameter	<i>Channel Management</i>	The SRS_INTRFCOORD_SLOT_SW option must be selected if the SRS and PUSCH/PUCCH concurrency compatibility function is enabled and the SRS remote interference avoidance switch is turned on.
Low-frequency TDD	SRS remote interference avoidance	SRS_REMOTE_INTERF_AVOID_SW option of the NRDUCellConfigCommon.SrsInterferenceCoordinationSwitch parameter	None	
Low-frequency TDD	PUCCH symbol length number increase	PUCCH_SYM_LEN_NUM_INCR_SW option of the NRDUCellConfigCommon.PucchSymbolLengthIncreaseSwitch parameter	<i>Channel Management</i>	When the SRS and PUSCH/PUCCH concurrency compatibility function is enabled, the PUCCH_SYM_LEN_NUM_INCR_SW option must be selected.

6.2.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode	<i>Hyper Cell</i>	Uplink intra-FR inter-band CA cannot be enabled in hyper cells, where NRDUCell.NrDuCellNetworkingMode is set to HYPER_CELL .
Low-frequency TDD	Adaptive fast processing by the gNodeB	GNB_ADAPT_FAST_PROCESS_SW option of the gNodeBParam.ServiceDiffProcSwitch parameter	None	When uplink intra-FR inter-band CA is enabled by selecting the INTRA_FR_INTER_BAND_UL_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter, the GNB_ADAPT_FAST_PROCESS_SW option of the gNodeBParam.ServiceDiffProcSwitch parameter must be deselected.
Low-frequency TDD	Positioning-dedicated SRS transmission in uplink slots	SRS_FOR_POSITION_ON_USLOT_SW option of the NRDUCellPosAlgo.LcsMeasureAlgorithm parameter	<i>Positioning</i>	When the SRS and PUSCH/PUCCH concurrency compatibility function is enabled, positioning-dedicated SRS transmission in uplink slots cannot be enabled together with proactive avoidance of remote interference.
Low-frequency TDD	Proactive avoidance of remote interference	SRS_RIM_INTRF_AVOID_SW and SRS_INTRF_COORD_S_SLOT_SW options of the NRDUCellSrsSrsAlgoExtSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Light-load time sequence optimization	NRDUCellAIgoSwitch.LIGHT_LOAD_TIME_SEQ_OPT_SW option of the LowLatencySwitch parameter	None	When the SRS and PUSCH/PUCCH concurrency compatibility function is enabled, the light-load time sequence optimization function must be disabled.

 NOTE

Apart from what are listed above, uplink intra-FR inter-band CA shares the mutually exclusive functions of downlink intra-FR inter-band CA as described in [6.1.3.2.2 Mutually Exclusive Functions](#).

6.2.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Remote interference adaptive avoidance	RIM_ADAPT_AVOID_SW option of the NRDUCellAIgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	When remote interference adaptive avoidance takes effect, the MTA function does not take effect for UEs in the uplink CA state.
Low-frequency TDD	DL Blanking	NRDUCell.PeerCellSlotAssignment	<i>Standards Compliance</i>	If DL Blanking is enabled in the PCell, only the single TA function, but not the function of MTA, can be used for UEs, affecting uplink peak data rates of CA UEs.
Low-frequency TDD	PUSCH MU-MIMO	UL_MU_MIMO_SW option of the NRDUCellAIgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	If CA UEs do not transmit SRSs in their SCells, they cannot be paired for MU-MIMO in the SCells. As a result, the uplink cell capacity decreases.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Power saving BWP (low-frequency TDD)	BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	BWP2, which provides a narrow bandwidth, is not configured for CA UEs in their SCells. When power saving BWP or "QCI-level threshold control for BWP switching" is enabled together with CA, there are the following impacts: • A larger value of NRDUCellCarrMgmt.CaUIScellActiveThldSlotNum results in a higher traffic volume threshold for UE switching from BWP2 to BWP1 and therefore a lower probability of the switching, which decreases UE power consumption. • A larger value of NRDUCellCarrMgmt.CaUIScellDeactThldSlotNum results in a higher traffic volume threshold for UE switching from BWP1 to BWP2 and therefore a higher probability of the switching, which decreases UE power consumption. • SCells are more likely to be deactivated if the NRDUCellCarrMgmt.RlcUlThroug

RAT	Function Name	Function Switch	Reference	Description
				<i>hputThld</i> parameter value is less than the <i>NRDUCellUePwrSaving.Bwp1ToBwp2UlThptThld</i> or <i>NRDUCellQciBearer.QciBwp1ToBwp2UlThptThld</i> parameter value. If <i>NRDUCellCarrMgmt.RlcUlThroughputThld</i> is set to 0, SCells will not be deactivated and UEs will not switch to BWP2.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Time-domain power saving BWP (low-frequency TDD)	TDM_BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	BWP2 is not configured for CA UEs in their SCells. When CA and time-domain power saving BWP are both enabled, there are the following impacts: • A larger value of NRDUCellCarrMgmt.CaUIScellActivateThldSlotNum results in a higher traffic volume threshold for UE switching from BWP2 to BWP1 and therefore a lower probability of the switching, which decreases UE power consumption. • A larger value of NRDUCellCarrMgmt.CaUIScellDeactivateThldSlotNum results in a higher traffic volume threshold for UE switching from BWP1 to BWP2 and therefore a higher probability of the switching, which decreases UE power consumption. • SCells are more likely to be deactivated if the NRDUCellCarrMgmt.RlcUIThroughputThld parameter value is less than the NRDUCellUePwr

RAT	Function Name	Function Switch	Reference	Description
				Saving.Bwp1ToBwp2UlThptThld parameter value. If NRDUCellCarrierMgmt.RlcUlThroughputThld is set to 0 , SCells will not be deactivated and UEs will not switch to BWP2.
Low-frequency TDD	Uplink multi-frequency coordination	MULTI_FREQ_UL_COORD_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter	<i>Multi-Frequency Smart Selection</i>	The gNodeB does not select UEs for which uplink CA has taken effect.
Low-frequency TDD	Multi-frequency beam coordination	MULTI_FREQ_BEAM_COORD_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter	<i>Multi-Frequency Smart Selection</i>	The gNodeB does not select uplink CA UEs that treat the local cell as their PCell or SCell.
Low-frequency TDD	CCE-based data split stop for CA UEs	CCE_BASED_CA_SPLIT_STOP_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgorithmExtSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	CCE-based data split stop for CA UEs does not take effect for uplink CA UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH skipping	PDCCH_SKIP_SW option of the NRDUCellUePwrSaving.SchPwrSavingSw parameter	<i>UE Power Saving</i>	With both uplink intra-FR inter-band CA and PDCCH skipping enabled: • If the CA_UE_PDCCH_SKIP_SW option of the NRDUCellUePwrSaving.SchPwrSavingSw parameter is deselected, PDCCH skipping does not take effect for UEs with SCells configured. • If the CA_UE_PDCCH_SKIP_SW option of the NRDUCellUePwrSaving.SchPwrSavingSw parameter is selected, PDCCH skipping can take effect for CA UEs with SCells not activated, but not for CA UEs with SCells activated.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CA SRS carrier switching	SRS_CARRIER_SWITCHING_SW option of the NRDUCellCarrierMgmt.CAEnhancedAlgorithmSwitch parameter	<i>CA Experience Improvement</i>	When SRS carrier switching and uplink intra-FR inter-band CA are both enabled, only one of them can take effect at a time. Which of them takes effect is dependent on the multi-carrier capability combination selection policy, which is specified by the NRCellCarrierMgmtConfig.CABandCombSelectionStrategy parameter.
Low-frequency TDD	CA SRS carrier switching enhancement	SRS_CARRIER_SWITCHING_ENH_SW option of the NRDUCellCarrierMgmt.CAEnhancedAlgorithmSwitch parameter	<i>CA Experience Improvement</i>	When SRS carrier switching enhancement and uplink intra-FR inter-band CA are both enabled, only one of them can take effect at a time. Which of them takes effect is dependent on the multi-carrier capability combination selection policy, which is specified by the NRCellCarrierMgmtConfig.CABandCombSelectionStrategy parameter.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Proactive avoidance of remote interference	SRS_RIM_INTRF_AVOID_SW and SRS_INTRF_COORD_S_SLOT_SW options of the NRDUCellSrsSrsAlgoExtSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	- When the uplink CA switch is on, the SRS_INTRF_AVOID_PUSCH_SCH_SW setting of the NRDUCellSrsSrsAlgoExtSwitch parameter must be the same for the PCell and the SCell. Otherwise, the cells differ in their scheduling in self-contained slots in the uplink, affecting the reporting of the power headroom in the PCell and SCell. As a result, adjustments for TPC are performed frequently and the average uplink throughput of uplink CA UEs changes greatly. - It is good practice to keep the "proactive avoidance of remote interference" switch status consistent between the PCell and the SCell. Otherwise, within the same symbols, SRSs may be transmitted on one carrier but PUSCH data on the other. This increases bit

RAT	Function Name	Function Switch	Reference	Description
				errors of some UEs. In the event of inconsistent switch status between the two cells, it is good practice to enable the "SRS and PUSCH/PUCCH concurrency compatibility" function to avoid bit errors. For details about the "SRS and PUSCH/PUCCH concurrency compatibility" function, see <i>Carrier Aggregation</i> .
Low-frequency TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTrp.DataTransMode set to BASIC_MODE or DPS_MODE	<i>Cell Combination</i>	Uplink intra-FR inter-band CA cannot be enabled in combined cells.
Low-frequency TDD	Joint Transmission Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTrp.DataTransMode set to JOINT_MODE	<i>Cell Combination</i>	Uplink intra-FR inter-band CA cannot be enabled in combined cells in joint transmission mode (that is, cells with Joint Transmission Cell Combination enabled).

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Device-pipe synergy preallocation	UL_PREFERE NTIAL_SCH_ SW option of the NRDUCellQc iBearer.Servi ceDiffAlgoS witch parameter	<i>FWA</i>	For UEs in the CA state, device-pipe synergy preallocation takes effect in their PCells but not SCells.
Low-frequency TDD	Inter-FR CA	INTER_FR_F R1TOFR2_C A_SW option of the NRDUCellAl goSwitch.Ca AlgoSwitch parameter	<i>Carrier Aggregation</i>	For a UE in the intra-FR inter-band CA state: • If the band +bandwidth combination for CA at the UE meets the band +bandwidth requirements of inter-FR CA, inter-FR CA can simultaneously take effect for the UE. If this is the case, downlink CA instead of uplink CA can take effect between the two low-frequency cells. • Otherwise, inter-FR CA cannot simultaneously take effect for the UE.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	3GPP R15-compliant uplink slot aggregation	VONR_UL_SLOT_AGGREGATION_SW option of the NRDUCellULSch.VoiceULSchSwitch parameter	<i>VoNR</i>	If uplink intra-FR inter-band CA has taken effect for UEs with VoNR enabled, the UEs cannot enter the uplink slot aggregation state. If the UEs have been in the uplink slot aggregation state and the conditions for uplink intra-FR inter-band CA to take effect are met, they will exit the uplink slot aggregation state.
Low-frequency TDD	3GPP R16-compliant uplink slot aggregation	VOICE_R16_UL_SLOT_AGG_SW option of the NRDUCellULSch.VoiceULSchSwitch parameter	<i>VoNR</i>	If uplink intra-FR inter-band CA has taken effect for UEs with VoNR enabled, the UEs cannot enter the uplink slot aggregation state. If the UEs have been in the uplink slot aggregation state and the conditions for uplink intra-FR inter-band CA to take effect are met, they will exit the uplink slot aggregation state.
Low-frequency TDD	Single DCI	DL_SINGLE_DCI_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgorithmSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	If uplink intra-FR inter-band CA has taken effect for a UE, single DCI will not take effect for the UE.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	MTA configuration prohibition for uplink CA	UL_CA_MTAG_NOT_CONFIG_SW option of the NRDUCellCarrierMgmt.CaCompatibilitySwitch parameter	<i>Carrier Aggregation</i>	The UL_CA_MTAG_NOT_CONFIG_SW_ON option of the gNBUECompatSw.WhitelistCtrlExtSwitch parameter takes effect only when the UL_CA_MTAG_NOT_CONFIG_SW option of the NRDUCellCarrierMgmt.CaCompatibilitySwitch parameter is selected. With this UE whitelist switch turned on, the MTA function takes effect for all UEs except the specified UEs.
Low-frequency TDD	UE-number-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SW option of the NRCellAccessSwitch.MlbAlgorithmSwitch parameter NRCellMLB.MlbTriggerMode set to CELL_UER_NUM	<i>Mobility Load Balancing in Connected Mode</i>	When selecting UEs for handovers, the gNodeB excludes those with uplink intra-FR inter-band CA in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Spectral-efficiency-based connected mode MLB	INTER_FR_EQ_CONNECTED_MLB_SW option of the NRCellAIgoSwitch.MlbAlgoSwitch parameter NRCellMlb.MlbTriggerMode set to SPEC_EFF_BASED_USER_NUM	<i>Mobility Load Balancing in Connected Mode</i>	When selecting UEs for handovers, the gNodeB excludes those with uplink intra-FR inter-band CA in effect.
Low-frequency TDD	User-rate-based dynamic network slice resource management	DYNAMIC_NETWORK_SLICE_SW option of the NRDUCellCommonSw.ServiceDiffAlgExtSwitch parameter	<i>Network Slicing</i>	When user-rate-based dynamic network slice resource management is enabled, the NRDUCellQciBearer.DynResULTargetRate parameter value configured for the PCell also applies to the SCells.
Low-frequency TDD	User-rate-based dynamic QCI resource management	DYNAMIC_QCI_GROUP_SW option of the NRDUCellFeatureSw.QciBasedDiffServiceExpSw parameter	<i>QCI-based Resource Control</i>	When user-rate-based dynamic QCI resource management is enabled, the NRDUCellQciBearer.DynResULTargetRate parameter value configured for the PCell also applies to the SCells.

NOTE

Apart from what are listed above, uplink intra-FR inter-band CA shares the function impacts of downlink intra-FR inter-band CA as described in [6.1.3.2.3 Function Impacts](#).

6.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

NOTE

One prerequisite for CA is that the corresponding cells have been set up. The following requirements for base station models, boards, and RF modules are based on this prerequisite.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

CA can be performed in one baseband processing unit or between baseband processing units.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

6.2.3.4 Networking

The timing advance (TA) offset of the PCell and SCells (specified by the **NRDUCell.TaOffset** parameter) must be the same. Otherwise:

- Uplink throughput is affected in the case of MTA being in effect.
- Uplink CA fails to take effect in the case of MTA being not in effect.

For details about the MTA function, see [6.2.1.2 CA Uplink Timing](#).

According to section 4.2 "Transmission timing adjustments" in 3GPP TS 38.213 V17.3.0, the TA offset values for the PCell and SCells in the same TA group must be the same. If the effective TA offset values for the cells involved in CA are different, uplink CA cannot be configured.

6.2.3.5 Others

- For this function to deliver a theoretical peak data rate to SA CA UEs, the maximum bit rate that each CA UE subscribes to in the 5GC cannot be lower than this theoretical value.

- For this NR CA function to deliver a theoretical peak data rate to NSA UEs, the maximum bit rate that each CA UE subscribes to in the EPC cannot be lower than this theoretical value.
- UEs
 - UEs must support CA. During initial access, an incoming handover, or an incoming RRC connection reestablishment, a UE reports its CA capabilities to the gNodeB.
 - UEs must support semi-static codebooks (as indicated by *semiStaticHARQ-ACK-Codebook* in the *phy-Parameters* IE) for feedback over the PUCCH.
 - For MTA, UEs must report the *supportedNumberTAG* IE with values greater than 1 and support *dualPA-Architecture*.

6.2.4 Operation and Maintenance

6.2.4.1 Data Configuration

6.2.4.1.1 Data Preparation

Before activating this function, ensure that the **NRCellRelation** and **NRCellFreqRelation** MOs have been configured. For details about the configuration, see *SA Mobility Management in Connected Mode* and *NSA Mobility Management*.

Table 6-6 and **Table 6-7** describe the parameters used for function activation and optimization, respectively. The following parameters do not take effect for SUL cells, for which the **NRDUCell.DuplexMode** parameter is set to **CELL_SUL**.

Table 6-6 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
CA Algorithm Switch	NRDUCellAlgoSwitch.CaAlgorithmSwitch	INTRA_FR_INTER_BAND_UL_CA_SW	Select this option.

Parameter Name	Parameter ID	Option	Setting Notes
CA Activate and BWP Switch UL Thld Slot Num	NRDUCellCarrMgmt.CaUIScellActThldSlotNum	N/A	The value 4 is recommended. A larger value of this parameter results in a higher uplink threshold for SCell activation. Reduce the NRDUCellCarrMgmt.CaUIScellDeactThldSlotNum parameter value as required to ensure SCell activation for NR CA when dynamic uplink PDCP data split is used in NSA networking (that is, the NsaDcQciParamGroup.UIDataSplitThreshold parameter is set to a value other than INFINITY in Option 3 or the gNBPDcpParamGroup.UIDataSplitThreshold parameter is set to a value other than INFINITY in Option 3x). A larger value of this parameter results in a higher uplink threshold for SCell activation and a lower probability of SCell activation. The value of this parameter must be greater than one half of the NRDUCellCarrMgmt.CaUIScellDeactThldSlotNum parameter value.
CA Deactivate and BWP Switch UL Thld Slot Num	NRDUCellCarrMgmt.CaUIScellDeactThldSlotNum	N/A	The value 1 is recommended. A larger value of this parameter results in a higher uplink threshold for SCell deactivation.
RLC Uplink Throughput Threshold	NRDUCellCarrMgmt.RlcULThroughputThld	N/A	The value 500 is recommended. A larger value of this parameter results in a higher uplink threshold for SCell deactivation.

Parameter Name	Parameter ID	Option	Setting Notes
UL CA A2 RSRP Threshold	NRCeCellCaMgmtConfig.UlCaA2RsrpThld	N/A	To trigger SCell removals only when the SCells provide poorer service quality, reduce the value of this parameter. You are advised to set this parameter to a value larger than the value of NRCeCellCaMgmtConfig.CaA2RsrpThld .
CA Compatibility Switch	NRDUCellCarMgmt.CaCompatibilitySwitch	UL_CA_MTAG_NOT_CONFIG_SW	Select this option if the single TA function is required in a cell.
Parameter Control Type	gNBUECompatibilityParamCtrlType	N/A	Set this parameter based on the network plan.
Whitelist Control Extension Switch	gNBUECompatibilityParamCtrlTypeWhitelistCtrlExtSwitch	UL_CA_MTAG_NOT_CONFIG_SW_ON	It is recommended that this option be selected when there are UE compatibility issues.
CA Enhanced Algorithm Switch	NRDUCellCarMgmt.CaEnhancedAlgoSwitch	UL_MULTI_CC_TA_FLEX_ADJ_SW	You are advised to select this option.
UL Multi-Carrier Absolute TA Diff Thld	NRDUCellCarMgmt.UlMultiCcAbsTaDiffThld	N/A	You are advised to retain the default value.
CA Enhanced Algorithm Switch	gNodeBCarMgmt.CaEnhancedAlgoSwitch	UL_CA_FALLBACK_DL_ONLY_CA_SW	You are advised to select this option.
UL CA SCC Split SINR Threshold	NRDUCellCarMgmt.UlCaSccSplitSinrThld	N/A	Use the recommended value. A smaller value of this parameter results in a higher probability that SCCs are qualified for uplink data split. A larger value of this parameter produces the opposite effects.

Parameter Name	Parameter ID	Option	Setting Notes
UL CA Continuous Non Split SCC Delete Threshold	NRDUCellCar rMgmt.UlCaC ontNonSplitS ccDelThld	N/A	Use the recommended value. If this parameter is set to a value other than 255 , a smaller value of this parameter results in a higher probability of SCC removal for uplink CA in scenarios with consecutive non-split events and a lower probability of limited UE rate on the PCC in such scenarios. However, unnecessary SCell removal may occur, decreasing uplink UE throughput. A larger value of this parameter results in a lower probability of SCC removal for uplink CA in scenarios with consecutive non-split events and a higher probability of limited UE rate on the PCC in such scenarios, but a lower probability of unnecessary SCell removal.
CA Enhanced Algorithm Switch	NRDUCellCar rMgmt.CaEn hancedAlgoS witch	UL_CA_SPLIT_THLD_OPT_SW	Select this option if the accuracy of data split to SCells needs to be improved.
UL Power Control Algorithm Extension Switch	NRDUCellUIP cConfig.UlPw rCtrlAlgoExtS witch	MF_ACTUAL_PUSCH_ILPC_SW	It is good practice to select this option in both the PCell and SCells to improve the accuracy of data volume distributed to SCells in the uplink.
CA Compatibility Switch	NRDUCellCar rMgmt.CaCo mpatibilityS witch	SRS_PUSCH_CONCURRENCY_COMPT_SW	You are advised to select this option.

Parameter Name	Parameter ID	Option	Setting Notes
Multi-Carrier Management Switch	NRDUCellCol labServ.Multi CarrMgmtSw	SMART_CARR _SEL_UL_ENH _SW	Select this option if preference based on the uplink peak rate is expected to take effect in the CA capability combination selection policy in scenarios where the optimal uplink and downlink CA combinations are not in the same combination during a traffic volume model change.

Table 6-7 Parameters used for optimization

Parameter Name	Parameter ID	Option	Setting Notes
Multi-Carrier Capb Comb Select Strat	NRCeCellCaMg mtConfig.Ca BandCombSe lectionStrat	N/A	Set this parameter as required.
Multi-Carrier Compatibility Switch	NRCeCellCaMg mtConfig.Ca Compatibility Switch	SRS_WT_BASE D_CARR_SEL_ SW	It is recommended that this option be selected if UEs are capable of uplink CA.
Multi-Carrier Management Switch	NRDUCellCol labServ.Multi CarrMgmtSw	SMART_CARR _SEL_SW	You are advised to select this option if the number of an operator's frequencies is greater than the number of frequencies that can be aggregated for a UE.
Multi-Carrier Management Switch	NRDUCellCol labServ.Multi CarrMgmtSw	SERVICE_AWA RE_CARR_SEL _SW	You are advised to select this option if UEs in a cell have various requirements and the number of an operator's frequencies is greater than the number of frequencies that can be aggregated for a UE.

6.2.4.1.2 Using MML Commands

Before using MML commands, refer to [6.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling uplink intra-FR inter-band CA
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CaAlgoSwitch=INTRA_FR_INTER_BAND_UL_CA_SW-1;
//Setting the uplink traffic volume thresholds for SCell activation and deactivation
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaUlScellActThldSlotNum=4, CaUlScellDeactThldSlotNum=1;
//Setting the RlcBufferFilterCoeff parameter
MOD NRDUCELLCARRMGMT: NrDuCellId=0, RlcBufferFilterCoeff=1;
//Setting the uplink RLC throughput threshold
MOD NRDUCELLCARRMGMT: NrDuCellId=0, RlcUlThroughputThld=500;
//Setting the RSRP threshold for uplink CA event A2
MOD NRCELLCAMGMTCONFIG: NrCellId=0, UlCaA2RsrpThld=-109;
//Optional Turning on UL_CA_MTAG_NOT_CONFIG_SW if the single TA function is desired in low-frequency scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaCompatibilitySwitch=UL_CA_MTAG_NOT_CONFIG_SW-1;
//Enabling whitelist control over MTA configuration prohibition for uplink CA for specified UEs with compatibility issues in low-frequency scenarios
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
WhitelistCtrlExtSwitch=UL_CA_MTAG_NOT_CONFIG_SW_ON-1;
//Enabling flexible TA adjustment in uplink multi-CC scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=UL_MULTI_CC_TA_FLEX_ADJ_SW-1;
//Optional Setting the UlMultiCcAbsTaDiffThld parameter
MOD NRDUCELLCARRMGMT: NrDuCellId=0, UlMultiCcAbsTaDiffThld=16;
//Optional Enabling optimized fallback upon uplink CA exceptions in low-frequency scenarios
MOD GNODEBCARRMGMT: CaEnhancedAlgoSwitch=UL_CA_FALLBACK_DL_ONLY_CA_SW-1;
//Optional Setting the UlCaSccSplitSinrThld and UlCaContNonSplitSccDelThld parameters
MOD NRDUCELLCARRMGMT: NrDuCellId=0, UlCaSccSplitSinrThld=-2, UlCaContNonSplitSccDelThld=3;
//Optional Turning on UL_CA_SPLIT_THLD_OPT_SW if the accuracy of data split to SCells needs to be improved
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=UL_CA_SPLIT_THLD_OPT_SW-1;
//Optional Turning on MF_ACTUAL_PUSCH_ILPC_SW in both the PCell and SCells (recommended with an aim to improve the accuracy of data volume distributed to SCells in the uplink)
MOD NRDUCELLULPCCONFIG: NrDuCellId=0, UlPwrCtrlAlgoExtSwitch=MF_ACTUAL_PUSCH_ILPC_SW-1;
//Optional Turning on SRS_PUSCH_CONCURRENCY_COMPT_SW
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
CaCompatibilitySwitch=SRS_PUSCH_CONCURRENCY_COMPT_SW-1;
//Optional Turning on SMART_CARR_SEL_UL_ENH_SW if preference based on the uplink peak rate is expected to take effect in the CA capability combination selection policy in scenarios where the optimal uplink and downlink CA combinations are not in the same combination during a traffic volume model change
MOD NRDUCELLCOLLABSERV: NrDuCellId=1, MultiCarrMgmtSw=SMART_CARR_SEL_UL_ENH_SW-1;
```

Optimization Command Examples

```
//Enabling the gNodeB to preferentially select the UE-reported band combination that offers the highest peak data rate
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaBandCombSelectionStrat=PEAK_RATE_FIRST;
//Enabling CA band combination selection with SRS-based weights considered
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaCompatibilitySwitch=SRS_WT_BASED_CARR_SEL_SW-1;
//Turning on the smart carrier selection switch
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, MultiCarrMgmtSw=SMART_CARR_SEL_SW-1;
//Turning on the service-aware carrier selection switch
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, MultiCarrMgmtSw=SERVICE_AWARE_CARR_SEL_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the SRS and PUSCH concurrency compatibility switch
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
```

```

CaCompatibilitySwitch=SRS_PUSCH_CONCURRENCY_COMPT_SW-0;
//Disabling whitelist control over MTA configuration prohibition for uplink CA for specified UEs in low-
frequency scenarios
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
WhitelistCtrlExtSwitch=UL_CA_MTAG_NOT_CONFIG_SW_ON-0;
//Disabling flexible TA adjustment in uplink multi-CC scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=UL_MULTI_CC_TA_FLEX_ADJ_SW-0;
//Disabling MTA configuration prohibition for uplink CA in low-frequency scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaCompatibilitySwitch=UL_CA_MTAG_NOT_CONFIG_SW-0;
//Disabling uplink intra-FR inter-band CA
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CaAlgoSwitch=INTRA_FR_INTER_BAND_UL_CA_SW-0;
//Disabling optimized fallback upon uplink CA exceptions in low-frequency scenarios
MOD GNODEBCARRMGMT: CaEnhancedAlgoSwitch=UL_CA_FALLBACK_DL_ONLY_CA_SW-0;
//Turning off UL_CA_SPLIT_THLD_OPT_SW
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=UL_CA_SPLIT_THLD_OPT_SW-0;
//Turning off MF_ACTUAL_PUSCH_ILPC_SW
MOD NRDUCELLULPCONFIG: NrDuCellId=0, ULPwrCtrlAlgoExtSwitch=MF_ACTUAL_PUSCH_ILPC_SW-0;
//Turning off SMART_CARR_SEL_UL_ENH_SW
MOD NRDUCELLCOLLABSERV: NrDuCellId=1, MultiCarrMgmtSw=SMART_CARR_SEL_UL_ENH_SW-0;

```

6.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.2.4.2 Activation Verification

Provided that CA UEs are available on the network, uplink intra-FR inter-band CA has taken effect if any counter in [Table 6-8](#) returns a non-zero value.

Table 6-8 Counters related to uplink intra-FR inter-band CA

Counter ID	Counter Name
1911816775	N.User.CA.PCell.UL.Avg
1911853846	N.User.CA.PCell.UL.Max
1911816776	N.User.CA.SCell.UL.Avg
1911816901	N.User.CA.PCell.UL.SCellAct.Avg
1911816903	N.User.CA.SCell.UL.Act.Avg
1911827212	N.User.CA.PCell.UL.SCellAct.Max
1911840467	N.CA.PCell.ThpVol.UL.Cell
1911840469	N.CA.SCell.ThpVol.UL.Cell
1911840459	N.CA.PRB.UL.PCell.Used.Avg
1911840461	N.CA.PRB.UL.SCell.Used.Avg

The counters related to the numbers of attempts and successful executions regarding SCell configuration, change, activation, deactivation, and removal are listed in [Table 6-9](#).

Table 6-9 Counters related to SCell management

Counter ID	Counter Name
1911816777	N.CA.SCell.Add.Att
1911816778	N.CA.SCell.Add.Succ
1911853943	N.UL.CA.SCell.Add.Att
1911853944	N.UL.CA.SCell.Add.Succ
1911816781	N.CA.SCell.Mod.Att
1911816782	N.CA.SCell.Mod.Succ
1911816896	N.CA.SCell.Act.Att
1911816897	N.CA.SCell.Act.Succ
1911816898	N.CA.SCell.Deact.Att
1911816899	N.CA.SCell.Deact.Succ
1911816779	N.CA.SCell.Rmv.Att
1911816780	N.CA.SCell.Rmv.Succ

6.2.4.3 Network Monitoring

Observe **User Uplink Average Throughput (DU)** to verify the increase in the UE data rate after the function is enabled.

Observe the average uplink throughput of SCell-activated CA UEs using the following calculation: $(N.CA.SccActiveUser.ThpVol.UL - N.CA.ThpVol.UL.SmallPkt)/N.CA.ThpTime.UL.RmvSmallPkt$.

Observe the uplink traffic volume for SCell-activated CA UEs in PCells using the counter **N.CA.PCell.ThpVol.UL.Cell**.

Observe the uplink traffic volume for CA UEs in SCells using the counter **N.CA.SCell.ThpVol.UL.Cell**.

7 Inter-FR Inter-Band CA

Inter-FR inter-band CA, also known as inter-FR CA, allows for aggregation of high- and low-frequency cells. Inter-FR CA is a trial feature, and all of its functions are trial functions.

NOTE

Trial features are features that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these features can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial features shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial features. Trial features are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial features may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial features but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial features are free, Huawei is not liable for any trial feature malfunctions or any losses incurred by using the trial features. Huawei does not promise that problems with trial features will be resolved in the current version. Huawei reserves the rights to convert trial features into commercial features in later R/C versions. If trial features are converted into commercial features in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial features. If a customer fails to purchase such a license, the trial feature(s) will be invalidated automatically when the product is upgraded.

7.1 Principles

7.1.1 Inter-FR CA (Trial)

Inter-FR CA allows for aggregation of CCs in the high and low frequency ranges (FRs) to achieve a larger bandwidth. It is controlled by the **INTER_FR_FR1TOFR2_CA_SW** option of the **NRDUCellAlgoSwitch.CaAlgoSwitch** parameter.

Inter-FR CA may involve different numbers of downlink low-frequency CCs. The maximum number of such CCs is determined by the **NRDUCellCarrMgmt.InterFrCaFr1DlMaxCcNum** parameter. Based on the number of low-frequency CCs, inter-FR CA is classified into the following types:

- Inter-FR CA with one low-frequency CC: The one low-frequency CC acts as the PCC, and all high-frequency CCs act as SCCs. This type of CA works in both the uplink and the downlink.
- Inter-FR CA with two low-frequency CCs: The two low-frequency CCs are a PCC and an SCC, and all high-frequency CCs act as SCCs. This type of CA works in both the uplink and the downlink. However, uplink CA does not work between the two low-frequency CCs.

Inter-FR CA works only in NR TDD.

For the band+bandwidth combinations compatible with inter-FR CA, see [4.4 CA Band+Bandwidth Combinations](#).

For details about HARQ feedback that works for inter-FR CA, see [HARQ Feedback for Inter-FR CA](#).

For how inter-FR CA collaborates with low-frequency CA, see [Collaboration Between Inter-FR CA and Low-Frequency CA](#).

In the case of inter-FR CA, measurement gap patterns can be configured independently for FR1 and FR2 carriers. This is known as independent measurement gap pattern configuration. For details, see [Independent Measurement Gap Pattern Configuration](#).

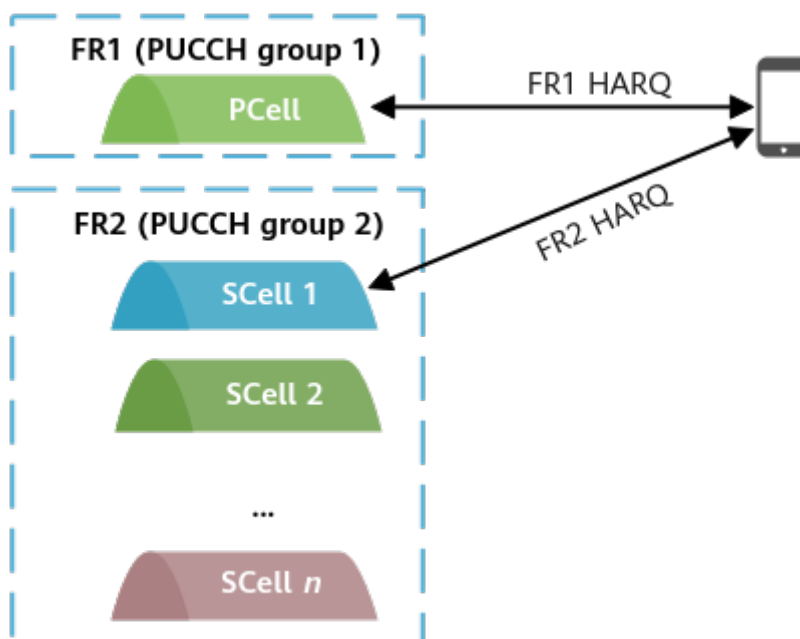
For whether inter-FR CA works in inter-gNodeB CA scenarios, see *CA Experience Improvement*.

HARQ Feedback for Inter-FR CA

For inter-FR CA, SCell management (configuration, change, activation, deactivation, and removal) messages are transmitted to and from the PCell. HARQ feedback is transmitted to the high- and low-frequency cells over separate PUCCHs to reduce signaling exchanges between the high-frequency SCells and the low-frequency PCell. That means two sets of PUCCH-based feedback mechanisms are used. Accordingly, the cells engaged in CA are divided into two groups: PUCCH group 1 (low-frequency) and PUCCH group 2 (high-frequency). The HARQ feedback procedure differs as follows:

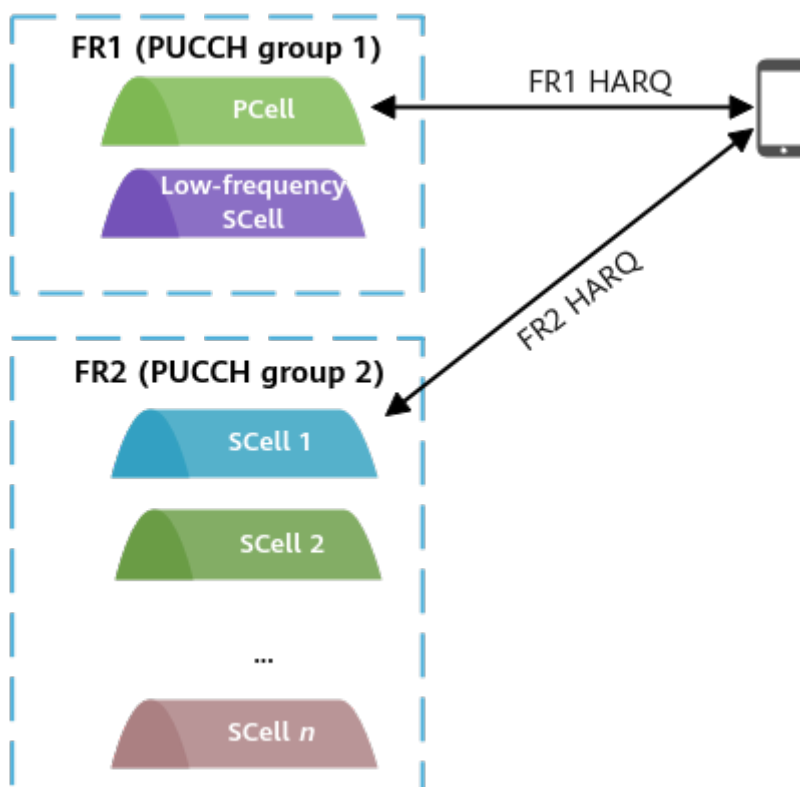
- Inter-FR CA with one low-frequency CC: See [Figure 7-1](#). The low-frequency PCell uses a PUCCH alone for HARQ feedback, but the high-frequency SCells share a PUCCH. For all these cells, the codebook format is determined by *pdsch-HARQ-ACK-Codebook*, and dynamic codebooks take effect.

Figure 7-1 HARQ feedback in the case of inter-FR CA with one low-frequency CC



- Inter-FR CA with two low-frequency CCs: See [Figure 7-2](#).
 - Low-frequency cells: The low-frequency PCell and SCell share a PUCCH for HARQ feedback. For these cells, the codebook format is determined by *pdsch-HARQ-ACK-Codebook*, and dynamic codebooks or semi-static codebooks can take effect. For the requirements that must be met for dynamic codebooks to take effect, see [5.1.1.6 Dynamic-Codebook-based HARQ Feedback over the PUCCH](#).
 - High-frequency cells: The high-frequency SCells share a PUCCH for HARQ feedback. The PUCCH to be shared is selected by the UE from the PUCCHs in the high-frequency cells acting as the SCells, depending on UE capabilities. Different UEs sometimes select different PUCCHs. The SCell that provides the shared PUCCH is then called a PUCCH-SCell (PU-SCell). For the high-frequency cells, the codebook format is determined by *pdsch-HARQ-ACK-Codebook-secondaryPUCCHgroup*, and dynamic codebooks always take effect.

Figure 7-2 HARQ feedback in the case of inter-FR CA with two low-frequency CCs



NOTE

An uplink and downlink supplementary cell, whose **NRDUCell.SupplementaryCellInd** is set to **ULDL_SUPPLEMENTARY_CELL**, in the high FR cannot act as a PU-SCell for inter-FR CA.

Collaboration Between Inter-FR CA and Low-Frequency CA

Assume that a UE supports the CA capability combinations for low-frequency CA (either intra-band CA or intra-FR inter-band CA) and those for inter-FR CA. Upon function activation for inter-FR CA and low-frequency CA to take effect simultaneously:

- If the UE is accessing its serving cell during initial access, a handover, RRC reestablishment, or RRC resume and CA has not taken effect for the UE, the CA capability combination selected for the UE is dependent on the inter-FR carrier selection policy (controlled by the **NRCeIlCaMgmtConfig.InterFrCarrSelStrat** parameter) specified for the serving cell.
 - If this parameter is set to **DEFAULT**, the CA capability combination is selected based on coverage. In case the network provides high-frequency coverage, a CA capability combination for inter-FR CA is preferentially selected for the UE. In case the network does not provide high-frequency coverage, a CA capability combination for low-frequency CA can be selected for the UE.
 - If this parameter is set to **FR2_PRIORITY**, only a CA capability combination for inter-FR CA or a subset of such a combination can be selected. In case the network provides high-frequency coverage, high-

frequency SCells can be added based on measurements so that inter-FR CA can quickly take effect.

- If CA has taken effect for the UE, the following situations are possible:
 - If low-frequency CA has taken effect for the UE, the following rule applies: If the UE is in the intra-base-station CA state and the band +bandwidth combination engaged in the low-frequency CA meets the requirement of inter-FR CA with two low-frequency CCs, then inter-FR CA with two low-frequency CCs can adaptively take effect for the UE. Otherwise, inter-FR CA cannot take effect for the UE.
 - If inter-FR CA with one low-frequency CC has taken effect for the UE, only this type of CA can take effect for the UE in the serving cell. After the low-frequency PCell changes, low-frequency CA or inter-FR CA will take effect depending on the network conditions.
 - If inter-FR CA with two low-frequency CCs has taken effect for the UE, inter-FR CA with one low-frequency CC or low-frequency CA takes effect adaptively depending on coverage changes.

 NOTE

When uplink low-frequency CA (either uplink intra-band CA or uplink intra-FR inter-band CA) is enabled, it may or may not take effect for a UE capable of this type of CA. If the UE is in the low-frequency CA state, uplink CA takes effect for the UE. If the UE is in the state of inter-FR CA with two low-frequency CCs, uplink CA does not take effect between the two low-frequency CCs for the UE; in this case, only downlink CA takes effect between the two CCs.

Independent Measurement Gap Pattern Configuration

A UE supports independent measurement gap pattern configuration if it meets one of the following conditions:

- The UE supports *independentGapConfig*.
- The UE supports *independentGapConfig-maxCC-r17*, and the number of CCs aggregated for the UE is less than or equal to the value of this IE reported by the UE. (This IE specifies the maximum number of CCs to which independent measurement gap patterns apply.) In addition, the **INDEP_GAP_CONFIG_MAXCC_SW** option of the **NRDUCellMeas.MeasAlgoSwitch** parameter is selected.

Measurement gap patterns are configured as follows:

- For a UE that does not support independent measurement gap pattern configuration, the per-UE measurement gap pattern is maintained.
- For a UE that supports independent measurement gap pattern configuration, independent measurement gap patterns for FR1 and FR2 can be configured for the UE. Specifically:
 - If only low-frequency cells are serving the UE and the target frequencies in the A5 measurement configuration all reside in FR2, measurement gaps are not required for the low-frequency serving cells. This does not affect low-frequency data transmission.
 - If the UE stays in the inter-FR CA state and the target frequencies in the A5 measurement configuration are all non-serving intra-RAT frequencies in FR1, it is allowed to deliver only the per-FR1 measurement gap pattern to the UE. This does not affect high-frequency data transmission.

- In other scenarios, measurement gap pattern configuration works as described in Table 9.1.2-3 of 3GPP TS 38.133 V18.3.0.

7.1.2 Inter-FR CA Coverage Enhancement

In high frequency bands, the PUCCH covers limited areas because of limited UE transmit power. Therefore, inter-FR CA can take effect only in relatively small coverage areas. This hinders inter-FR CA from offering the optimal user experience. To improve user experience delivered by inter-FR CA, inter-FR CA coverage enhancement is introduced. It includes two functions:

- PUCCH grouping adaptation for inter-FR CA, which is controlled by the **PUCCH_GROUP_ADAPT_SW** option of the **NRDUCellCarrMgmt.InterFrCaEnhSw** parameter
- PUCCH grouping adaptation enhancement for inter-FR CA, which is controlled by the **PUCCH_GROUP_ADAPT_ENH_SW** option of the **NRDUCellCarrMgmt.InterFrCaEnhSw** parameter

PUCCH grouping adaptation enhancement for inter-FR CA requires PUCCH grouping adaptation for inter-FR CA to be enabled. These functions work as follows:

- With PUCCH grouping adaptation for inter-FR CA enabled:
 - When the high-frequency PU-SCell of an inter-FR CA UE is not prone to limited uplink coverage, the high-frequency serving cells and low-frequency serving cells of the UE use separate PUCCHs for HARQ feedback transmission. That means two sets of PUCCH-based feedback mechanisms are used.
 - When the high-frequency PU-SCell of an inter-FR CA UE is prone to limited uplink coverage, the high- and low-frequency serving cells of the UE all use the PUCCH of the low-frequency PCell for HARQ feedback transmission. That means a single set of PUCCH-based feedback mechanisms is used. In this situation, the HARQ feedback for the high-frequency cells is transmitted in the low-frequency PCell with better coverage, so that inter-FR CA takes effect in a larger area and delivers a better user experience. A high-frequency SCell may or may not have uplink physical channels, depending on PUCCH grouping adaptation enhancement for inter-FR CA.
 - If PUCCH grouping adaptation enhancement for inter-FR CA is enabled, the high-frequency cell has downlink physical channels but no uplink physical channels. In this case, downlink super HARQ for inter-FR CA can be used to further improve downlink user experience. For details about this function, see [Downlink Super HARQ for Inter-FR CA](#).
 - If PUCCH grouping adaptation enhancement for inter-FR CA is disabled, the high-frequency cell has both uplink and downlink physical channels.

When PUCCH grouping adaptation for inter-FR CA is enabled, the **gNBCaFrequency.CaSccA5RsrpThld2Offset** and **gNBCaFrequency.CaA2RsrpThldOffset** parameters must be set to **-2** and **-11**, respectively, for high-frequency cell configuration and removal on the serving gNodeB of the low-frequency PCell.

- With PUCCH grouping adaptation for inter-FR CA disabled, the high-frequency serving cells and low-frequency serving cells of a UE use separate PUCCHs for HARQ feedback transmission. That is, two sets of PUCCH-based feedback mechanisms are used. For details, see [7.1.1 Inter-FR CA \(Trial\)](#).

Inter-FR CA coverage enhancement works with macro base stations but not LampSite base stations.

For whether inter-FR CA coverage enhancement works in inter-gNodeB CA scenarios, see *CA Experience Improvement*.

Downlink Super HARQ for Inter-FR CA

When PUCCH grouping adaptation enhancement for inter-FR CA has taken effect for a UE, downlink super HARQ for inter-FR CA can take effect for the UE to further increase the downlink UE throughput, if both of the following conditions are met:

- Device-pipe synergy is enabled for both the low-frequency PCell and the high-frequency PU-SCell. This function is controlled by the **SPEC_UE_IDENTIFY_SW** option of the **NRDUCellUeCoop.SpecUeCooperationSwitch** parameter. For details about device-pipe synergy, see *Device-Pipe Synergy*.
- Downlink super HARQ for inter-FR CA is enabled for the high-frequency PU-SCell. This function is controlled by the **INTER_FR_CA_DL_SUPER_HARQ_SW** option of the **NRDUCellUeCoop.SpecUeCooperationSwitch** parameter.

NOTE

Downlink super HARQ for inter-FR CA does not work in the case of an NG-based handover from the low-frequency PCell of a UE for which PUCCH grouping adaptation enhancement for inter-FR CA has taken effect.

7.1.3 Fast SCell Configuration for Inter-FR CA

SCell configuration for inter-FR CA takes a long time because UE measurement of high-frequency cells requires much time. To shorten the time for SCell configuration, fast SCell configuration for inter-FR CA is introduced. This function is controlled by the **FR2_IDLE_INACTIVE_MEAS_SW** option of the **NRCeMgmtConfig.CaStrategySwitch** parameter.

- If this option is selected for a low-frequency cell, high-frequency cell measurement configuration is initiated upon UE release to idle or inactive mode in the low-frequency cell. (This type of measurement is also known as early measurement.) If a UE in idle or inactive mode has detected a target high-frequency cell and if the measurement is not stopped when the UE accesses the low-frequency cell, the UE directly reports the measurement result to the low-frequency cell. This way, a high-frequency SCell can be quickly configured. This process saves the time required for high-frequency cell measurement, whose configuration used to be delivered after the UE accesses the low-frequency cell.

 NOTE

- In the low-frequency cell, high-frequency cell measurement configurations for idle/inactive-mode UEs are delivered only to UEs capable of inter-FR CA.
- Fast SCell configuration based on idle/inactive-mode UE measurement does not work if a UE in idle or inactive mode does not detect a target high-frequency cell that meets the requirements or if the measurement has been stopped before the UE accesses the low-frequency cell.
- If this option is deselected for a low-frequency cell, inter-frequency measurement configuration of high-frequency cells is initiated after UEs access the low-frequency cell. When a UE in connected mode detects a target high-frequency cell, it reports the measurement result to the low-frequency cell. Then, the high-frequency cell is configured as an SCell.

7.1.4 FR2 Primary-SCell-Inclusive Activation and Deactivation

In the case of FR2 SCell deactivation for inter-FR CA, the primary FR2 SCell stays activated to prevent FR2 UEs from being out of synchronization. This leads to rapid drain of UE power. To address this, the "FR2 primary-SCell-inclusive activation and deactivation" function is introduced to allow the primary FR2 SCell to be deactivated and activated again. This function includes the following sub-functions:

- FR2 SCell activation and deactivation: controlled by the **FR2_SCELL_ACTV_DEACTV_SW** option of the **NRDUCellCarrMgmt.InterFrCaEnhSw** parameter.
- FR2 SCell deactivation and fast activation: controlled by the **FR2_SCELL_FAST_ACTV_DEACTV_SW** option of the **NRDUCellCarrMgmt.InterFrCaEnhSw** parameter.

The two sub-functions differ in support for fast activation. With "FR2 SCell deactivation and fast activation", fast activation is supported, whereby the gNodeB directly sends a MAC CE for activating the primary FR2 SCell to a UE and then the UE sends a related measurement report to fast trigger such activation. With "FR2 SCell activation and deactivation", fast activation is not supported. Instead, the gNodeB first sends an RRC reconfiguration message to instruct a UE to measure the primary FR2 SCell; after the UE sends a related measurement report, the gNodeB sends the MAC CE for activating the primary FR2 SCell.

The switches of the preceding two sub-functions are configured on the PCell.

Table 7-1 describes how the "FR2 primary-SCell-inclusive activation and deactivation" function works in different scenarios.

Table 7-1 "FR2 primary-SCell-inclusive activation and deactivation" in different scenarios

FR2_SCELL_ACTV_DEACTV_SW Option	FR2_SCELL_FAST_ACTV_DEACTV_SW Option	Implementation of "FR2 Primary-SCell-Inclusive Activation and Deactivation"
Deselected	Deselected	The "FR2 primary-SCell-inclusive activation and deactivation" function does not take effect.

FR2_SCELL_ACTIVATION_DEACTIVATION_OPTION	FR2_SCELL_FAST_ACTIVATION_DEACTIVATION_OPTION	Implementation of "FR2 Primary-SCell-Inclusive Activation and Deactivation"
Selected	Deselected	The primary FR2 SCell can be deactivated or activated again based on traffic volume ^a .
Selected	Selected	- For UEs^b that support measurement reporting for unknown SCell activation as defined in 3GPP Release 18, their primary FR2 SCells can be deactivated or fast activated again based on traffic volume. - For other UEs, their primary FR2 SCells can be deactivated or activated based on traffic volume (fast activation is not supported).
Deselected	Selected	- For UEs^b that support measurement reporting for unknown SCell activation as defined in 3GPP Release 18, their primary FR2 SCells can be deactivated or fast activated again based on traffic volume. - For other UEs, the "FR2 primary-SCell-inclusive activation and deactivation" function does not take effect.
a: For details about traffic volume-based deactivation and activation, see Traffic Volume-triggered SCell Deactivation and SCell Activation Through a MAC CE. b: The l3-MeasUnknownSCellActivation-r18 field in the UE-reported capability information indicates whether the UE supports measurement reporting for unknown SCell activation, as detailed in section 4.2.9 in 3GPP TS 38.306 V18.3.0.		

The "FR2 primary-SCell-inclusive activation and deactivation" function is recommended only when power saving is required for UEs. It is not recommended when power saving is not strongly required, such as in FWA or toB scenarios.

A certain time interval must be maintained between the activation and deactivation of a primary FR2 SCell to prevent ping-pong issues.

- After being activated, the primary FR2 SCell can be deactivated based on traffic volume only when the timer specified by **NRDUCellCarrMgmt.Fr2ScellDeactvProhibitTmr** expires.
- After being deactivated, the primary FR2 SCell cell can be activated or fast activated again based on traffic volume only after the interval specified by **NRDUCellCarrMgmt.Fr2ScellActvInterval** elapses.

Both **NRDUCellCarrMgmt.Fr2ScellDeactvProhibitTmr** and **NRDUCellCarrMgmt.Fr2ScellActvInterval** are configured on the PCell.

7.2 Network Analysis

7.2.1 Benefits

Inter-FR CA

- Inter-FR CA increases the uplink or downlink peak data rates of CA UEs. The uplink or downlink peak data rate that a UE can achieve ranges from $(A \times 95\% + B \times 95\%)$ to $(A \times 100\% + B \times 100\%)$. A is the sum of the uplink/downlink data rates that the UE can achieve in all the low-frequency cells. (If there is only one low-frequency CC, A is the uplink/downlink data rate that the UE can achieve in the single low-frequency cell.) B is the sum of the uplink/downlink data rates that the UE can achieve in all the high-frequency cells.

NOTE

The preceding peak data rate ranges can be achieved only if the influencing factors (for example, the UE rank) remain unchanged after CA takes effect.

- Independent measurement gap pattern configuration reduces service interruption caused by gap pattern configurations, improving user experience.

Inter-FR CA Coverage Enhancement

- PUCCH grouping adaptation for inter-FR CA and PUCCH grouping adaptation enhancement for inter-FR CA change two sets of PUCCH-based feedback mechanisms into one set for inter-FR CA UEs at the edges of their SCells. These functions enable downlink data transmission to continue in the SCells and therefore increase the **User Downlink Average Throughput (DU)**.
- Downlink super HARQ for inter-FR CA further increases the **User Downlink Average Throughput (DU)**.

Fast SCell Configuration for Inter-FR CA

Fast SCell configuration for inter-FR CA saves the time required for high-frequency cell measurement, because the gNodeB no longer has to wait to deliver high-frequency cell measurement configurations until UEs access low-frequency cells. The saved time can reach 0.5s to 1s in the following typical scenario:

A UE in idle or inactive mode detects a target high-frequency cell that meets requirements. In addition, the duration for the UE to stay in idle or inactive mode is less than the measurement duration (300s), which means the measurement has not been stopped when the UE accesses a low-frequency cell.

FR2 Primary-SCell-Inclusive Activation and Deactivation

The "FR2 primary-SCell-inclusive activation and deactivation" function shortens the time for the primary FR2 SCell to stay activated, reducing UE power consumption.

7.2.2 Impacts

The impacts between this function and other functions are described in [7.3.2.3 Function Impacts](#).

Inter-FR CA

This function has the following network impacts:

- Resource usage
Due to data distribution from the PCell to SCells, the online duration of UEs decreases in the PCell and increases in the SCells. As a result, resource indicators such as the number of RRC_CONNECTED UEs and PRB usage may decrease in the PCell and may increase in the SCells.
- Other performance indicators
 - After inter-FR CA is enabled, the transmission capabilities of inter-FR CA UEs change, affecting the number of online UEs in their PCells and SCells. As a result, the CQIs and MCS indexes in these cells alter, and related performance indicators (such as the cell throughput and initial block error rate [IBLER]) vary.
 - If no cell measured by a UE at the cell edge meets the threshold for event A5, the gNodeB cannot configure SCells for inter-FR CA for the UE. In this case, the gNodeB delivers event A5 measurement configurations to the UE every 15s if the UE traffic volume is constantly high. Since these periodic inter-frequency measurements require measurement gaps, the downlink throughput of the UE in the PCell decreases.
- The **FR2_PRIORITY** setting of the inter-FR carrier selection policy enables inter-FR CA to take precedence. It increases the PRB usage of high-frequency carriers but reduces the number of UEs for which low-frequency CA and SUL take effect.

Inter-FR CA Coverage Enhancement

This function has the following network impacts:

- Accessibility-related performance indicators
 - With PUCCH grouping adaptation for inter-FR CA and PUCCH grouping adaptation enhancement for inter-FR CA: Switching between one set and two sets of PUCCH-based feedback mechanisms for an inter-FR CA UE causes SCells to be removed and configured again. In addition, the switching usually occurs in weak uplink coverage areas of SCells. As a result, the service drop rate and the probability of RRC reestablishment in the PCell may increase.
 - With PUCCH grouping adaptation for inter-FR CA: The non-contention-based random access success rate in SCells decreases.
- Other performance indicators
With PUCCH grouping adaptation for inter-FR CA and PUCCH grouping adaptation enhancement for inter-FR CA:
 - The downlink coverage areas of SCells are expanded through the PCell. This increases the number of cell-edge UEs and traffic volume in the SCells. As a result, the average MCS index, average CQI, and average rank of the PCell and SCells change, causing variations in related performance

counters, such as cell throughput and IBLER. In addition, the PDCCH CCE aggregation level increases in the SCells.

- Switching between one set and two sets of PUCCH-based feedback mechanisms for an inter-FR CA UE causes SCells to be removed and configured again and also causes PUCCH resources to be reconfigured. The UE throughput decreases during this process.
- After two sets of PUCCH-based feedback mechanisms are changed into one set for an inter-FR CA UE, channel information and HARQ feedback for the PCell and SCells are all transmitted over the PUCCH in the PCell. As a result, the PUCCH bit rate is lifted, causing increases in the downlink IBLER, residual block error rate (RBLER), and retransmission probability in the PCell. When the HARQ feedback is sent together with data over the PUSCH in the PCell, some of the HARQ feedback may be discarded if the PCell coverage is poor. The downlink IBLER and RBLER in the SCells of the UE further rise, so that the IBLER cannot converge on the target IBLER (specified by the **NRDUCellPdsch.DlTargetIbler** parameter). The average downlink throughput of the UE in the low-frequency cell may decrease.
- After two sets of PUCCH-based feedback mechanisms are changed into one set for an inter-FR CA UE running downlink large-packet services in a lightly loaded cell, **User Downlink Average Throughput (DU)** may decrease if channel conditions are unstable.

Fast SCell Configuration for Inter-FR CA

This function has the following network impacts:

- Accessibility-related performance indicators
In the case of fast configuration of a high-frequency cell as an SCell based on the measurement results reported by an idle- or inactive-mode UE in a low-frequency cell, the high-frequency SCell may fail to be configured or activated because the measurement results are invalid. As a result, the non-contention-based random access success rate in the high-frequency cell decreases, and the service drop rate and the probability of RRC connection reestablishment in the low-frequency cell even rise.
- Other performance indicators
In the case of fast configuration of a high-frequency cell as an SCell based on the measurement results reported by an idle- or inactive-mode UE in a low-frequency cell, the high-frequency SCell may fail to be configured or activated because the measurement results are invalid. The low-frequency cell delivers the inter-frequency measurement configurations related to high frequencies to the UE in connected mode again. This process introduces an extra delay, during which the UE performs service transmission only in the low-frequency cell.

FR2 Primary-SCell-Inclusive Activation and Deactivation

This function has the following network impacts:

- The uplink and downlink traffic volumes carried by FR1 and FR2 cells change, causing the average uplink and downlink throughputs of UEs and PCells to change.
 - Average uplink UE throughput: **User Uplink Average Throughput (DU)**

- Average downlink UE throughput: **User Downlink Average Throughput (DU)**
- Average uplink cell throughput: **Cell Uplink Average Throughput (DU)**
- Average downlink cell throughput: **Cell Downlink Average Throughput (DU)**
- The uplink and downlink traffic volumes carried by FR1 and FR2 cells change, causing the air interface scheduling indicators of these cells to fluctuate, including the PRB usage, CCE usage, IBLER, RBLE, MCS, and rank.
- The number of RRC reconfiguration messages increases, which may increase the service drop rate and the probability of RRC connection reestablishment.
- The number of MAC CEs delivered in FR1 cells increases. When there are a large number of inter-FR CA UEs, the average uplink and downlink throughputs of UEs in the FR1 cells decrease.
- Uplink random access increases in FR2 cells. When there are a large number of inter-FR CA UEs, uplink access to FR2 cells is delayed or even fails. This causes removal of FR2 SCells, reducing the number of inter-FR CA UEs. Counters related to the number of inter-FR CA UEs:
 - **N.User.InterFRCA.PCell.Avg**
 - **N.User.InterFRCA.PCell.Max**
 - **N.User.InterFRCA.PUCCHSCell.Avg**
 - **N.User.InterFRCA.PUCCHSCell.Max**
- As there are more SCell activation and deactivation scenarios, the values of the following counters increase:
 - **N.CA.SCell.Deact.Att**
 - **N.CA.SCell.Deact.Succ**
 - **N.CA.SCell.Act.Att**
 - **N.CA.SCell.Act.Succ**
- In inter-gNodeB scenarios, gNodeBs need to exchange configuration information related to this function through the eXn interface. As such, a single message becomes longer and, in certain scenarios, may be broken into more fragments. This may increase the values of counters related to the received and transmitted traffic volumes in the **SCTP Link Measurement** function subset.

7.3 Requirements

7.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-070206	Inter-FR CA	NR0S00FRCA00	per Cell

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Inter-FR CA	INTER_FR_FR1TOFR2_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	• PUCCH grouping adaptation for inter-FR CA requires the inter-FR CA switch to be turned on. • Fast SCell configuration for inter-FR CA requires the inter-FR CA switch to be turned on. • If inter-FR CA is disabled, the NRCellCaMgmtConfig.InterFrCarrSelStrat parameter can only be set to DEFAULT. • The NRDUCellCarrMgmt.InterFrCaFr1DlMaxCcNum parameter can be set to DL2CC only if inter-FR CA is enabled. • If inter-FR CA is disabled, the NRDUCellCarrMgmt.InterFrCaFr1DlMaxCcNum parameter must be set to NOT_CONFIG.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	If the two low-frequency CCs in inter-FR CA are downlink intra-band carriers, this type of inter-FR CA can take effect only if downlink intra-band CA is enabled.
Low-frequency TDD	Downlink intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	If the two low-frequency CCs in inter-FR CA are downlink intra-FR inter-band carriers, this type of inter-FR CA can take effect only if downlink intra-FR inter-band CA is enabled.
High-frequency TDD	Narrow-beam probing	NARROW_BEAM_PROBE_SW option of the NRDUCellAlgoSwitch.FullChannelCovEnhSwitch parameter	<i>Random Access Control</i>	PUCCH grouping adaptation for inter-FR CA requires the narrow-beam probing switch to be turned on.
Low-frequency TDD High-frequency TDD	PUCCH grouping adaptation for inter-FR CA	PUCCH_GROUP_ADAPT_SW option of the NRDUCellCarrierMgmt.InterFrCaEnhSw parameter	<i>Carrier Aggregation</i>	In inter-FR CA scenarios, PUCCH grouping adaptation enhancement for inter-FR CA requires PUCCH grouping adaptation for inter-FR CA to be enabled.
High-frequency TDD	Device-pipe synergy	SPEC_UE_IDENTITY_SW option of the NRDUCellUeCoop.SpecUeCooperationSwitch parameter	<i>Device-Pipe Synergy</i>	Downlink super HARQ for inter-FR CA (controlled by the INTER_FR_CA_DL_SUPER_HARQ_SW option of the NRDUCellUeCoop.SpecUeCooperationSwitch parameter) can be enabled only if device-pipe synergy has been enabled.
High-frequency TDD	TRS Beam Type	NRDUCellCsirs.TrsBeamType	None	Inter-FR CA takes effect only when the type is set to NARROW_BEAM .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS carrier switching	SRS_CARRIER_SWITCHING_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter	<i>CA Experience Improvement</i>	Inter-FR CA with two low-frequency CCs can take effect for a UE only when SRS carrier switching is enabled and takes effect for the UE.
Low-frequency TDD High-frequency TDD	FR2 primary-SCell-inclusive activation and deactivation	Either the FR2_SCELL_ACTIV_DEACTIVATION_SW or FR2_SCELL_FAST_ACTIV_DEACTIVATION_SW option of the NRDUCellCarrMgmt.InterFrCaEnhSw parameter selected	<i>Carrier Aggregation</i>	The NRDUCellCarrMgmt.Fr2ScellDeactivProhibitTmr and NRDUCellCarrMgmt.Fr2ScellActivInterval parameters take effect only when the "FR2 primary-SCell-inclusive activation and deactivation" function is enabled.

7.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell-level PUCCH time-frequency format	NRDUCellPucch.StructureType	<i>Channel Management</i>	- When inter-FR CA is enabled, the NRDUCellPucch.StructureType parameter cannot be set to SHORT_STRUCTURE. - When PUCCH grouping adaptation for inter-FR CA is enabled, the NRDUCellPucch.StructureType parameter cannot be set to SHORT_STRUCTURE.
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode	<i>Hyper Cell</i>	Inter-FR CA cannot be enabled in hyper cells, where NRDUCell.NrDuCellNetworkingMode is set to HYPER_CELL .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode	<i>Cell Combination</i>	Inter-FR CA cannot be enabled in combined cells whose NRDUCell.NrDuCellNetworkingMode is set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTRP.DataTransMode is set to BASIC_MODE or DPS_MODE .
Low-frequency TDD	Joint Transmission Cell Combination	NRDUCell.NrDuCellNetworkingMode and NRDUCellMultiTRP.DataTransMode	<i>Cell Combination</i>	Inter-FR CA cannot be enabled in a cell where Joint Transmission Cell Combination has been enabled (by setting NRDUCell.NrDuCellNetworkingMode to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTRP.DataTransMode to JOINT_MODE).
Low-frequency TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	Inter-FR CA does not work with the low latency and high reliability function. They cannot be both enabled.
Low-frequency TDD	Optimization of non-contention-based RA beyond cell radius	CONTENTION_FREE_RA_OPT_SW option of the NRDUCellPrach.RachAlgoOptSwitch parameter	<i>Random Access Control</i>	Inter-FR CA does not work with optimization of non-contention-based RA beyond cell radius. They cannot be both enabled.
Low-frequency TDD	Service-aware carrier selection	SERVICE_AWARE_CARRIER_SELECTION_SW option of the NRDUCellCollabServ.MultiCarrierMgmtSw parameter	<i>Carrier Aggregation</i>	Inter-FR CA cannot be enabled when service-aware carrier selection is enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	Inter-FR CA cannot be enabled in high-speed cells, where NRDUCell.HighSpeedFlag is set to HIGH_SPEED .
Low-frequency TDD	SSB beam switching	NRDUCellCsirs.SsbBeamSwitchingMode	<i>Channel Management</i>	PUCCH grouping adaptation for inter-FR CA does not work with SSB beam switching. They cannot be both enabled. The NRDUCellCsirs.SsbBeamSwitchingMode parameter must be set to NO_SWITCHING before PUCCH grouping adaptation for inter-FR CA can be enabled.
Low-frequency TDD	SRS-based positioning	SRS_FOR_POSITION_ON_USLOT_SW option of the NRDUCellPosAlgo.LcsMeasureAlgoSw parameter	None	PUCCH grouping adaptation for inter-FR CA does not work with SRS-based positioning. They cannot be both enabled.
Low-frequency TDD	SRS period shortening for downlink 1T4R UEs	DL_1T4R_SRS_PERIOD_SHORTEN_SW option of the NRDUCellPdschPrecode.SumimoSrsPeriodShortSw parameter	None	PUCCH grouping adaptation for inter-FR CA does not work with SRS period shortening for downlink 1T4R UEs. They cannot be both enabled.
Low-frequency TDD	SRS period shortening for downlink 2T4R UEs	DL_2T4R_SRS_PERIOD_SHORTEN_SW option of the NRDUCellPdschPrecode.SumimoSrsPeriodShortSw parameter	None	PUCCH grouping adaptation for inter-FR CA does not work with SRS period shortening for downlink 2T4R UEs. They cannot be both enabled.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Full-buffer UE-specific CSI-RS for BM	USER_BM_CSI_RS_SW option of the NRDUCellCsirs.CsiSwitch parameter	<i>Channel Management</i>	Full-buffer UE-specific CSI-RS for BM cannot be enabled when PUCCH grouping adaptation for inter-FR CA is enabled.
High-frequency TDD	Uplink experience-based scheduling	UL_EXP_BASED_SCH_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	<i>FWA</i>	In high-frequency TDD, uplink experience-based scheduling does not work with inter-FR CA.
High-frequency TDD	Downlink experience-based scheduling	DL_EXP_BASED_SCH_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	<i>FWA</i>	In high-frequency TDD, downlink experience-based scheduling does not work with inter-FR CA.
Low-frequency TDD High-frequency TDD	mmWave 3D coverage pattern	NRDUCellTrpMmwavBeam.CoverageScenario	<i>mmWave Beam Management (High-Frequency TDD)</i>	Inter-FR CA does not take effect when the NRDUCellTrpMmwavBeam.CoverageScenario parameter is set to SCENARIO_103(Single Beam Scenario) .
Low-frequency TDD	DL Blanking	NRDUCell.PeerCellSlotAssignment	<i>Standards Compliance</i>	When DL Blanking is enabled (by setting the NRDUCell.PeerCellSlotAssignment parameter to a value other than NOT_CONFIG), inter-FR CA cannot be enabled.
Low-frequency TDD	Cell deployment in IAB mode	NRDUCell.DeployPosition set to DEPLOY_IAB	None	NRDUCell.DeployPosition can be set to DEPLOY_IAB only when inter-FR CA is disabled.

7.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Power saving BWP	BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	Power saving BWP does not take effect for inter-FR CA UEs.
Low-frequency TDD	Time-domain power saving BWP (low-frequency TDD)	TDM_BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	Time-domain power saving BWP does not take effect for inter-FR CA UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Light-load time sequence optimization	LIGHT_LOAD_TIME_SEQUENCE_OPT_SW option of the NRDUCellIgoSwitch.LowLatencySwitch parameter	None	In the case of inter-FR CA with one low-frequency CC where the NRDUCell.SlotAssignment parameter is set to 4_1_DDSU for the low-frequency cell, light-load time sequence optimization does not take effect for UEs for which PUCCH grouping adaptation for inter-FR CA has taken effect. In the case of inter-FR CA with two low-frequency CCs, the two low-frequency CCs work in the form of intra-band CA or intra-FR inter-band CA. However, as described in 5.1.2.2 Impacts and 6.1.2.2 Impacts, light-load time sequence optimization does not take effect for the UE with intra-band CA or intra-FR inter-band CA in effect. Therefore, light-load time sequence optimization does not take effect in the case of inter-FR CA with two low-frequency CCs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	DRX	BASIC_DRX_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch parameter	<i>DRX</i>	DRX can take effect for an inter-FR CA UE only when DRX is enabled for all the serving cells of the UE. The DRX settings for the low-frequency PCell apply to the UE.
Low-frequency TDD	Intra-band CA	INTRA_BAND_CA_SW or INTRA_BAND_UL_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	For UEs that support the CA capability combinations for low-frequency CA and those for inter-FR CA, see Collaboration Between Inter-FR CA and Low-Frequency CA for the collaboration between inter-FR CA and low-frequency CA.
Low-frequency TDD	Intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW or INTRA_FR_INTER_BAND_UL_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	For UEs that support the CA capability combinations for low-frequency CA and those for inter-FR CA, see Collaboration Between Inter-FR CA and Low-Frequency CA for the collaboration between inter-FR CA and low-frequency CA.
Low-frequency TDD	Power saving scheduling	POWER_SAVING_SCHEDULE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When inter-FR CA is enabled, data convergence in symbols does not take effect in the slots where RAR messages are sent.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When RF channel intelligent shutdown takes effect in high-frequency cells, some of the cells no longer send SSBs and may fail to be configured for inter-FR CA. To increase the success rate of high-frequency cell configuration for inter-FR CA, full-frequency measurement (controlled by the FULL_FREQ_MEAS_SW option of the NRCellAlgoSwitch.MeasPolicySwitch parameter) can be enabled for low-frequency cells.
Low-frequency TDD High-frequency TDD	Channel quality-triggered SCell deactivation	NRDUCellCarrierMgmt.SccDeactivCqiThreshold	<i>Carrier Aggregation</i>	Channel quality-triggered SCell deactivation does not take effect for UEs for which PUCCH grouping adaptation for inter-FR CA has taken effect.
High-frequency TDD	PRACH configuration index	NRDUCellPrach.PrachConfigurationIndex	<i>Random Access Control</i>	When inter-FR CA is enabled, it is recommended that the PRACH configuration index be set to 188 or 65535 .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CA prohibition for VoNR UEs	VONR_NOT_CONFIG_CA_SW option of the NRCellCaMgmtConfig.CaStrategySwitch parameter	<i>VoNR</i>	When CA prohibition for VoNR UEs is enabled, CA does not take effect for VoNR UEs.
Low-frequency TDD	Maximum-bandwidth reduction	OVERHEATING_ASSISTANCE_INFO_SW and UE_BW_REDUCE_SW options of the NRCellAlgoSwitch.UeAssistanceInfoSwitch parameter	<i>UE Power Saving</i>	- If the OVERHEATING_ASSISTANCE_INFO_SW option of the NRCellAlgoSwitch.UeAssistanceInfoSwitch parameter is selected, the base station removes high-frequency SCells for a UE when the UE requests to reduce the bandwidth due to overheating. - If the UE_BW_REDUCE_SW option of the NRCellAlgoSwitch.UeAssistanceInfoSwitch parameter is selected, the function of maximum-bandwidth reduction does not take effect for an inter-FR CA UE when the UE requests to reduce the bandwidth only due to power saving.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Reducing the number of MIMO layers	OVERHEATING_ASSISTANCE_INFO_SW and UE_MIMO_LAYER_REDUCE_SW options of the NRCellAlgoSwitch.UeAssistanceInfoSwitch parameter	<i>UE Power Saving</i>	• If the OVERHEATING_ASSISTANCE_INFO_SW option of the NRCellAlgoSwitch.UeAssistanceInfoSwitch parameter is selected, the base station removes high-frequency SCells for a UE when the UE requests to reduce the number of MIMO layers due to overheating. • If the UE_MIMO_LAYER_REDUCE_SW option of the NRCellAlgoSwitch.UeAssistanceInfoSwitch parameter is selected, the function of reducing the number of MIMO layers does not take effect for an inter-FR CA UE when the UE requests to reduce the number of MIMO layers only due to power saving.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Reducing the number of SCCs	OVERHEATING_ASSISTANCE_INFO_SW and UE_SCC_DEACTIVATE_SW options of the NRCellAlgoSwitch.UeAssistanceInfoSwitch parameter	<i>UE Power Saving</i>	If the function of reducing the number of SCCs is enabled in SA-networking inter-FR CA scenarios, values of the following counters increase when high-frequency SCell removal is triggered for overheating protection or UE power saving: • N.CA.SCell.Add.Att or N.CA.SCell.Add.Succ • N.CA.SCell.Rmv.Att or N.CA.SCell.Rmv.Succ
Low-frequency TDD High-frequency TDD	SSSG switching	UE_PWR_SAVING_ENHANCE_SW and SSSG_SWITCHING_SW options of the NRDUCellUePwrSaving.UePowerSavingSwitch parameter	<i>UE Power Saving</i>	In inter-FR CA, a sparse SSSG can be configured for PCCs but not for SCCs, and the SSSG function cannot be enabled for high-frequency SCCs. In this case, SCell activation for CA requires switching from a sparse SSSG to a dense SSSG and therefore is delayed.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Operator-specific limitation on the maximum number of available RBs	MAX_RB_LIMIT_SW option of the NRDUCellAIgoSwitch.RanSharingAlgoSwitch parameter	<i>Multi-Operator Sharing</i>	In inter-FR CA scenarios, when operator-specific limitation on the maximum number of available RBs takes effect for the PCell, RB resource estimation is inaccurate during uplink data split. As a result, the average uplink throughput of inter-FR CA UEs fluctuates.
Low-frequency TDD	Dynamic-codebook-based HARQ feedback over the PUCCH	PUCCH_DYN_HARQ_ACK_CB_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoExtSwitch parameter	<i>Carrier Aggregation</i>	Dynamic-codebook-based HARQ feedback over the PUCCH does not take effect for inter-FR CA UEs for each of which a single set of PUCCH-based feedback mechanisms is used.
High-frequency TDD	Dynamic beam allocation	DYNAMIC_BEAM_ALLOC_SW option of the NRDuCellIMobileBhAlgo.MobileBackhaulAlgoSw parameter	<i>mmWave Beam Management (High-Frequency TDD)</i>	Dynamic beam allocation does not take effect for UEs for which PUCCH grouping adaptation enhancement for inter-FR CA has taken effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	mmWave 3D coverage pattern	NRDUCellTrpMmwavBeam.CoverageScenario	<i>mmWave Beam Management (High-Frequency TDD)</i>	Inter-FR CA coverage enhancement is not recommended when the NRDUCellTrpMmwavBeam.CoverageScenario parameter is set to SCENARIO_104(Wide Beam Scenario) . That is because inter-FR CA coverage enhancement is used in mmWave eMBB scenarios to address limited uplink coverage caused by limited UE transmit power in high frequency bands. However, the NRDUCellTrpMmwavBeam.CoverageScenario parameter is set to SCENARIO_104(Wide Beam Scenario) when the high-frequency LOS coverage is not limited, in which case the uplink coverage of high frequency bands is not limited. If inter-FR CA coverage enhancement takes effect unexpectedly in this case, the UE throughput decreases.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	2-port CSI-RS coverage enhancement	CSIRS_2PORT_COV_ENH_SW option of the NRDUCellChnCovAlgo.DlCoverageAlgoSwitch parameter	None	Beam-polling-based detection takes place with a 40 ms CSI-RS period for UEs for which PUCCH grouping adaptation enhancement for inter-FR CA has taken effect.
High-frequency TDD	Quick MCS adjustment and VAM measurement period shortening	DL_MCS_AND_VAM_WT_QUICK_ADJ_SW option of the NRDUCellPdsch.DLinkAdaptEnhancementSw parameter	None	Quick MCS adjustment and VAM measurement period shortening do not take effect for UEs for which PUCCH grouping adaptation enhancement for inter-FR CA has taken effect.
High-frequency TDD	SRS SINR measurement optimization	SRS_SINR_MEAS_OPT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter	<i>MIMO (TDD)</i>	SRS SINR measurement optimization does not take effect for UEs for which PUCCH grouping adaptation enhancement for inter-FR CA has taken effect.
Low-frequency TDD High-frequency TDD	mmWave PUSCH MU-MIMO	UL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	mmWave PUSCH MU-MIMO does not take effect for inter-FR CA UEs.
High-frequency TDD	PUCCH receive channel adaptation	NRDUCellPucch.PucchRxAllocMode set to ADAPTIVE	<i>Intelligent Aggregate Scheduling (High-Frequency TDD)</i>	PUCCH receive channel adaptation does not take effect for UEs with PUCCH grouping adaptation enhancement for inter-FR CA in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink 8-layer transmission for a single FWA user	DL_SU_EIGHT_LAYER_SW option of the NRDUCellDLRank.DlRankSelAlgoSw parameter	<i>FWA</i>	PUCCH grouping adaptation for inter-FR CA does not take effect for a UE for which downlink 8-layer transmission for a single FWA user has taken effect.
High-frequency TDD	Intelligent beam selection	INTELLIGENT_BEAM_SELECTION_SW option of the NRDUCellBeamAlgo.BeamOptAlgoSwitch parameter	<i>mmWave Super Coverage (High-Frequency TDD)</i>	Intelligent beam selection does not take effect when PUCCH grouping adaptation for inter-FR CA is in effect, in which case a single set of PUCCH-based feedback mechanisms is used in the PCell and SCells of a UE in the inter-FR CA state.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SCC coverage expansion	- SA networking: SA_SCC_COV_THLD_ADAPT_SW option of the NRCellIgoSwitch.MultiFreqAlgoSwitch parameter - NSA networking: NSA_SCC_COV_THLD_ADAPT_SW option of the NRCellIgoSwitch.MultiFreqAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	SCC coverage expansion does not take effect for UEs engaged in inter-FR CA.
Low-frequency TDD	Independent SUL deployment	SUPER_UL_TDM_SCH_SW option of the NRDUCellIgoSwitch.SuperUplinkSwitch parameter	<i>Super Uplink</i>	Inter-FR CA and super uplink with independent SUL deployment cannot simultaneously take effect for a UE.
Low-frequency TDD	UL and DL Decoupling	NRDUCellIgoSwitch.ULDDecouplingSwitch	<i>UL and DL Decoupling</i>	Inter-FR CA and UL and DL Decoupling cannot simultaneously take effect for a UE.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	BWP spatial adaptation	MIMO_BWP_ADAPT_SW option of the NRDUCellUeCoop.SpecUeCooperationSwitch parameter	None	BWP spatial adaptation does not take effect for UEs for which downlink super HARQ for inter-FR CA has taken effect.
Low-frequency TDD	DRX preference configuration	DRX_PREFERENCE_SW option of the NRDUCellUeCoop.SpecUeCooperationSwitch parameter	None	DRX preference configuration does not take effect for UEs for which downlink super HARQ for inter-FR CA has taken effect.
Low-frequency TDD	CSI-RS beam sweeping	CSIRS_BEAM_SWEEPING_SW option of the NRDUCellCsiRs.CsiSwitch parameter	None	CSI-RS beam sweeping cannot take effect for UEs in the inter-FR CA state.
Low-frequency TDD	Blind SCell configuration	SCC_BLIND_CONFIG_SW option of the NRCeCellCaMgmtConfig.CaStrategySwitch parameter	<i>Carrier Aggregation</i>	After inter-FR CA with one low-frequency CC takes effect for a UE, low-frequency SCells cannot be added based on traffic volume if the PCell remains unchanged. As a result, inter-FR CA with two low-frequency CCs cannot take effect. When the blind SCell configuration switch is off, inter-FR CA with one low-frequency CC is more likely to take effect for UEs. As a result, inter-FR CA with two low-frequency CCs is less likely to take effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	PUCCH grouping adaptation for inter-FR CA	PUCCH_GROUP_ADAPT_SW option of the NRDUCellCarrierMgmt.InterFrCaEnhSw parameter	<i>Carrier Aggregation</i>	When PUCCH grouping adaptation for inter-FR CA is enabled, no more than four high-frequency CCs can be aggregated even if the NRDUCellCarrierMgmt.InterFrCa200MFr2CcMaxNum parameter is set to DL6CC_UL4CC. When two sets of PUCCH-based feedback mechanisms are changed to one set for an inter-FR CA UE with six high-frequency CCs, the maximum high-frequency bandwidth that can be reached in the downlink decreases from (6CC x 200 MHz) to (4CC x 200 MHz). This decreases the downlink data rate of the UE.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Intra-sector UE-number-based connected mode MLB	INTER_FR_EQ_CONNECTED_MLB_SW option of the NRCellIgoSwitch.MlbAlgoSwitch parameter selected NRCellIb.MlbTriggerMode set to INTRA_SECTOR_CELL_USER_NUM NRCellIb.MlbDirection set to UPLINK	<i>Mobility Load Balancing in Connected Mode</i>	Intra-sector UE-number-based connected mode MLB does not take effect for inter-FR CA UEs.
Low-frequency TDD High-frequency TDD	UE-specific TRS resource management	USER_TRS_BEAM_SW option of the NRDUCellCsirs.CsiSwitch parameter	<i>Channel Management</i>	UE-specific TRS resource management does not take effect on high-frequency SCCs of inter-FR CA.
Low-frequency TDD High-frequency TDD	UE bandwidth adaptation	UE_BW_ADAPTIVE_SW option of the NRDUCellBwp.BwpConfigSwitch parameter	<i>Scalable Bandwidth</i>	Inter-FR CA does not take effect for UEs with UE bandwidth adaptation in effect.
Low-frequency TDD	Flexible experience differentiation	FLEX_EXP_DIFF_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter	<i>Service-differentiated Experience Guarantee</i>	Flexible experience differentiation does not take effect for inter-FR CA UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Network slicing	Multiple parameters are required for network slicing deployment. For details, see <i>Network Slicing</i> .	<i>Network Slicing</i>	The network slices configured for an SCell must include UEs' network slices.

7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

NOTE

One prerequisite for inter-FR CA is that the corresponding cells have been set up. The following requirements for base station models, boards, and RF modules are based on this prerequisite.

Base Station Models

The following are base station model requirements for cells that will participate in inter-FR CA:

- For low-frequency cells:
 - Inter-FR CA, fast SCell configuration for inter-FR CA, and FR2 primary-SCell-inclusive activation and deactivation:
 - 3900 series base stations (macro base stations). They must be configured with the BBU3910.
 - 5900 series base stations (macro base stations). They cannot be configured with the BookBBU5901/BookBBU5901a.
 - DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.
 - Inter-FR CA coverage enhancement
 - 3900 series base stations (macro base stations). They must be configured with the BBU3910.
 - 5900 series base stations (macro base stations). They cannot be configured with the BookBBU5901/BookBBU5901a.
- For high-frequency cells: 5900 series base stations (macro base stations). They cannot be configured with the BookBBU5901/BookBBU5901a.

Boards

The following are board requirements for cells that will participate in inter-FR CA:

- For low-frequency cells:
 - Baseband processing unit: UBBPg series and later low-frequency NR TDD-capable baseband processing units support this function.
 - Main control board: All main control boards capable of low-frequency NR TDD support this function. UMPTg series and later main control boards are recommended.
- For high-frequency cells: All main control boards and baseband processing units that are capable of high-frequency NR TDD support this function. If high-frequency cells are configured on UBBPh boards that reside in a different BBU than the serving BBPs of low-frequency cells, these serving BBPs of low-frequency cells must run the version of V100R020C10SPC150 or later.

For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

7.3.4 Networking

The coverage areas provided by the cells involved in inter-FR CA must overlap. The smaller the overlapping range, the smaller the area where inter-FR CA takes effect.

Inter-gNodeB inter-FR CA has additional networking requirements. For details, see the networking requirements of inter-gNodeB CA described in *CA Experience Improvement*.

For inter-FR CA, the high-frequency SCells must be served by the same gNodeB. To be specific, these cells must be configured on the same BBP.

For inter-FR CA with two low-frequency CCs, the two low-frequency cells must be served by the same gNodeB, and the high- and low-frequency cells can be either intra- or inter-gNodeB cells.

7.3.5 Cells

The inter-frequency NR cell relationships have been configured (using the **NRCeLLRelation** MO).

The **NRCeLL.DuplexMode** parameter must be set to **CELL_TDD** for the cells involved.

The frame offsets between the PCell and high-frequency SCells and between the PCell and low-frequency SCells must meet the following requirements. Otherwise, inter-FR CA does not take effect.

- The frame offsets of the PCell and high-frequency SCells must be the same.

- The absolute value of the frame offset difference between the PCell and low-frequency SCells must be less than or equal to 625 Ts.

The frame offset is specified by the **gNBFreqBandConfig.FrameOffset** or **gNodeBParam.FrameOffset** parameter. The value of **gNBFreqBandConfig.FrameOffset** takes precedence. If **gNBFreqBandConfig.FrameOffset** is either not configured or set to **4294967295**, the value of **gNodeBParam.FrameOffset** applies.

7.3.6 Others

- Transmission requirements: When the cells involved in inter-FR CA are served by different gNodeBs, data is transmitted over the eXn interface. The eXn packet loss rate must be less than or equal to 10^{-6} , and the delay variation must be less than or equal to 1 ms. If the eXn packet loss rate is excessively high, sequence numbers (SNs) become out of order, causing the number of intra-cell UE handovers to increase.
- UE requirements:
 - UEs must support inter-FR CA. During initial access, an incoming handover, or an incoming RRC reestablishment, a UE reports its CA capabilities to the gNodeB.
 - UEs must report the *supportedNumberTAG* field with the value of n2, n3, or n4.
 - Each UE must be capable of sending its HARQ feedback to its PCell and SCells over separate PUCCHs. That is, UEs must report the *diffNumerologyAcrossPUCCH-Group* field.
 - Additional requirement of inter-FR CA coverage enhancement: UEs must be capable of sending its HARQ feedback to its PCell and SCells all over the PUCCH of the PCell. That is, UEs must report the *diffNumerologyWithinPUCCH-GroupSmallerSCS* field.
 - Additional requirement of fast SCell configuration for inter-FR CA: UEs must support idle- or inactive-mode measurement. That is, UEs must report the *idleInactiveNR-MeasReport-r16* and *idleInactiveNR-MeasBeamReport-r16* fields of high-frequency cells.
 - UEs must act as follows during SCell selection: If both inter-FR CA and low-frequency intra-FR CA are enabled, inter-FR CA UEs preferentially select high-frequency cells as SCells.
 - UEs must support semi-static codebooks (as indicated by *semiStaticHARQ-ACK-Codebook* in the *phy-Parameters* IE) for feedback over the PUCCH.
 - Inter-FR CA with two low-frequency CCs requires that UEs support *harqACK-CB-SpatialBundlingPUCCH-Group-r16*.
 - UEs must support independent measurement gap pattern configuration. That is, each UE must meet one of the following conditions:
 - The UE supports *independentGapConfig*.
 - The UE supports *independentGapConfig-maxCC-r17*, and the number of CCs aggregated for the UE is less than or equal to the value of this IE reported by the UE. (This IE specifies the maximum number of CCs to which independent measurement gap patterns apply.) In

addition, the **INDEP_GAP_CONFIG_MAXCC_SW** option of the **NRDUCellMeas.MeasAlgoSwitch** parameter is selected.

- As for fast activation of the primary FR2 SCell in the "FR2 primary-SCell-inclusive activation and deactivation" function, UEs must support measurement reporting for unknown SCell activation in accordance with 3GPP Release 18. This capability is indicated by the **I3-MeasUnknownSCellActivation-r18** field in the UE-reported capability information, as detailed in section 4.2.9 in 3GPP TS 38.306 V18.3.0.
- Interference requirements: Inter-FR CA coverage enhancement requires that the value of any counter in the range of **N.UL.NI.Max.PRB0** to **N.UL.NI.Max.PRB272** minus the value of **N.UL.NI.Min** be less than 10 dB in the low-frequency cell (PCell). If this requirement is not met, this function will fail to yield the expected effect of coverage area extension or will cause user experience to deteriorate.

7.4 Operation and Maintenance

7.4.1 Data Configuration

7.4.1.1 Data Preparation

For the parameters and configuration methods for inter-gNodeB inter-FR CA, see *CA Experience Improvement*.

Table 7-2 describes the parameters used for function activation.

Table 7-2 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
CA Algorithm Switch	NRDUCellAlgoSwitch.CaAlgoSwitch	INTER_FR_FR1TOFR2_CA_SW	Select this option for both the PCell and SCells that will participate in inter-FR CA.
CA Algorithm Switch	NRDUCellAlgoSwitch.CaAlgoSwitch	INTER_GNODEB_CA_SW	Select this option for the inter-gNodeB cells that will participate in inter-FR CA. Before selecting this option, set the NRDUCellCarrMgmt.InterGnodebDelay parameter based on the inter-gNodeB transmission delay and set the NRCellRelation.InterGnodebF1lag parameter to define inter-gNodeB neighboring cells for inter-FR CA.

Parameter Name	Parameter ID	Option	Setting Notes
Inter-FR Carrier Aggregation Enhance Switch	NRDUCellCar rMgmt.InterF rCaEnhSw	PUCCH_GRO UP_ADAPT_S W	It is recommended that this option be selected for both the PCell and SCells for inter-FR CA to improve the coverage of inter-FR CA.
Inter-FR Carrier Aggregation Enhance Switch	NRDUCellCar rMgmt.InterF rCaEnhSw	PUCCH_GRO UP_ADAPT_E NH_SW	It is recommended that this option be selected for both the PCell and SCells for inter-FR CA to improve the coverage of inter-FR CA.
CA Strategy Switch	NRCeCellCaMg mtConfig.CaS trategySwitc h	FR2_IDLE_INA CTIVE_MEAS_ SW	Select this option for the low-frequency cells that will participate in inter-FR CA.
Downlink NARFCN	gNBCellCar rMgmt.DlNarfc n	None	Set this parameter based on the network plan. The downlink NARFCN and SSB frequency-domain position must be the same as those in the NRDUCell MO.
SSB Frequency Position	gNBCellCar rMgmt.SsbFreq Pos	None	Set this parameter based on the network plan. The downlink NARFCN and SSB frequency-domain position must be the same as those in the NRDUCell MO.

Parameter Name	Parameter ID	Option	Setting Notes
Operator ID List	gNBCaFrequency.OperatorIdList	N/A	Set this parameter based on the network plan. The value of this parameter is expressed in decimal format. The gNodeB converts the decimal number into a 32-bit binary number. If the number is less than 32 bits, the most significant bits are 0s by default. The bits from right to left have a one-to-one relationship with the local operator indexes (specified by the gNBOperator.OperatorId parameter) in ascending order. For example, bit 0 at the rightmost place corresponds to the operator with the gNBOperator.OperatorId parameter set to 0. If a bit is set to 1, the corresponding operator can use the frequency. If a bit is set to 0, the corresponding operator cannot use the frequency. Take the value 5 of the gNBCaFrequency.OperatorIdList parameter as an example. The binary format of this value is 101, indicating that the operators with gNBOperator.OperatorId set to 0 and 2 can use the CA frequency and the operators with gNBOperator.OperatorId set to 1 and values greater than 2 cannot use the CA frequency.
CA A5 RSRP Threshold 2	NRCellCaMgmtConfig.CaA5RsrpThld2	None	To trigger SCell configuration only when the SCells provide better service quality, increase the value of this parameter. The value of this parameter must be greater than that of NRCellCaMgmtConfig.CaA2RsrpThld. The default value is recommended.

Parameter Name	Parameter ID	Option	Setting Notes
DL State Transition Buffer Volume Threshold	NRDUCellCarrierMgmt.DlStateTransitBufVolThld	N/A	The value 20 is recommended for low-frequency NR TDD cells.
Service Differentiation Algorithm Switch	NRDUCellQciBearer.ServiceDiffAlgoSwitch	QCI_CA_BWP_SWITCH_THL D_CTRL_SW	Select this option if the buffered traffic volume threshold for downlink state transitions needs to be set on a per QCI basis.
QCI DL State Transit Buffer Volume Threshold	NRDUCellQciBearer.QciDlStTransitBufVolThld	N/A	Use the recommended value of this parameter for low-frequency NR TDD cells.
CA SCC A5 RSRP Threshold 2 Offset	gNBCellFrequency.CaSccA5RsrpThld2Offset	None	To adjust the CA event A5 threshold for different SCCs, set this parameter according to policies. The value -2 is recommended for inter-FR CA coverage enhancement, and the default value is recommended in other scenarios.
Multi-Carrier Measurement Wait Timer	gNodeBAlgo.MultiCarrierMeasWaitTimer	N/A	Retain the default value. The smaller the value of this parameter, the shorter the measurement waiting time. In scenarios with a large number of candidate frequencies, however, if the value is too small, SCells may not be configured because no measurement reports are received, or better SCells may not be configured because no measurement reports on better candidates are received. As a result, the user-perceived rates of CA UEs decrease. A larger value of this parameter indicates a longer measurement waiting time and results in a higher probability of configuring better SCells but a longer time taken for SCell configuration.

Parameter Name	Parameter ID	Option	Setting Notes
CA A2 RSRP Threshold	NRCeCellCaMgmtConfig.CaA2RsrpThld	None	To trigger SCell removals only when the SCells provide poorer service quality, reduce the value of this parameter. The default value is recommended.
CA A2 RSRP Threshold Offset	gNBCellFrequency.CaA2RsrpThldOffset	None	To adjust the CA event A2 threshold for different SCCs, set this parameter according to policies. The value -11 is recommended for inter-FR CA coverage enhancement, and the default value is recommended in other scenarios.
No Handover Flag	NRCeCellRelation.NoHoFlag	None	If the network serves UEs that support inter-FR CA but do not support mmWave SA networking, set this parameter to FORBID_HO .
Inter-FR CA FR1 Downlink Max CC Number	NRDUCellCarrierMgmt.InterFrCaFr1DlMaxCcNum	None	It is recommended that this parameter be set to DL2CC when there are two or more low frequencies on the network.
Inter-FR CA 200M FR2 CC Max Num	NRDUCellCarrierMgmt.InterFrCa200MFr2CcMaxNum	None	- When there are four or fewer contiguous high-frequency carriers with a bandwidth of 200 MHz each, it is recommended that this parameter be set to DL4CC_UL4CC. - When there are five or six contiguous high-frequency carriers with a bandwidth of 200 MHz each, it is recommended that this parameter be set to DL6CC_UL4CC. This parameter is configured on the PCell.
Specified UE Cooperation Switch	NRDUCellUeCoop.SpecUeCooperationSwitch	INTER_FR_CA_DL_SUPER_HARQ_SW	To enable downlink super HARQ for inter-FR CA, select this option.

Parameter Name	Parameter ID	Option	Setting Notes
Inter-FR Carrier Selection Strategy	NRCellCaMgmtConfig.InterFrCarrSelStrat	None	Set this parameter based on the network plan.
Measurement Algorithm Switch	NRDUCellMeas.MeasAlgorithmSwitch	INDEP_GAP_CONFIG_MAXC_C_SW	You are advised to select this option.
Inter-FR Carrier Aggregation Enhance Switch	NRDUCellCarMgmt.InterFrCaEnhSw	FR2_SCELL_ACTIV_DEACTV_SW	You are advised to select this option only when power saving is required for UEs and to deselect this option when power saving is not strongly required (such as FWA or toB scenarios).
Inter-FR Carrier Aggregation Enhance Switch	NRDUCellCarMgmt.InterFrCaEnhSw	FR2_SCELL_FAST_ACTIV_DEACTV_SW	You are advised to select this option only when power saving is required for UEs and to deselect this option when power saving is not strongly required (such as FWA or toB scenarios).
FR2 SCell Deactivation Prohibit Timer	NRDUCellCarMgmt.Fr2ScellDeactivProhibitTmr	N/A	Retain the default value. A larger value of this parameter results in a longer duration for inter-FR CA UEs to stay in the FR2 SCell-activated state, higher UE throughput, and higher UE power consumption. A smaller value of this parameter produces the opposite effects. However, if the value of this parameter is too small, FR2 SCells may be frequently activated and deactivated, affecting user experience and increasing the radio signaling overhead.

Parameter Name	Parameter ID	Option	Setting Notes
FR2 SCell Activation Interval	NRDUCellCar rMgmt.Fr2Sc ellActvInterv al	N/A	Retain the default value. A smaller value of this parameter results in more frequent attempts to activate FR2 SCells for inter-FR CA UEs, a shorter time to activate FR2 SCells, and a larger increase in the number of air-interface signaling messages. A larger value of this parameter produces the opposite effects.

7.4.1.2 Using MML Commands

Before using MML commands, refer to [7.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling inter-FR CA for low- and high-frequency NR TDD cells
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CaAlgoSwitch=INTER_FR_FR1TOFR2_CA_SW-1;
MOD NRDUCELLALGOSWITCH: NrDuCellId=10, CaAlgoSwitch=INTER_FR_FR1TOFR2_CA_SW-1;
//Enabling PUCCH grouping adaptation for inter-FR CA and PUCCH grouping adaptation enhancement for
inter-FR CA for the low- and high-frequency NR TDD cells
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
InterFrCaEnhSw=PUCCH_GROUP_ADAPT_SW-1&PUCCH_GROUP_ADAPT_ENH_SW-1;
MOD NRDUCELLCARRMGMT: NrDuCellId=10,
InterFrCaEnhSw=PUCCH_GROUP_ADAPT_SW-1&PUCCH_GROUP_ADAPT_ENH_SW-1;
//Enabling fast SCell configuration for inter-FR CA for the low-frequency NR TDD cell
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=FR2_IDLE_INACTIVE_MEAS_SW-1;
//Configuring the operating frequency of the high-frequency NR TDD cell as a neighboring frequency for
the low-frequency NR TDD cell
ADD NRCELLFREORELATION: NrCellId=0, SsbFreqPos=22395, FrequencyBand=N257,
SubcarrierSpacing=120KHZ;
//Configuring an NR cell relationship for the low-frequency NR TDD cell (Set
NRCellRelation.InterGnodebFlag to TRUE in an inter-gNodeB case and to False in an intra-gNodeB case.)
ADD NRCELLRELATION: NrCellId=0, Mcc="262", Mnc="04", gNBId=1, CellId=10, InterGnodebFlag=TRUE;
//Configuring the CA frequency group and its frequencies on the base station serving the low-frequency
PCell (using inter-FR CA with two carriers as an example) and the operators that are allowed to use the
frequencies (all operators in this example). The downlink frequency and SSB frequency-domain position
must be the same as those in the NRDUCell MO.
ADD GNBCAFREQUENCY: CaFreqGroupId=3, DINarfcn=634050, SsbFreqPos=7854,
OperatorIdList=4294967295;
ADD GNBCAFREQUENCY: CaFreqGroupId=3, DINarfcn=2057499, SsbFreqPos=22395,
OperatorIdList=4294967295;
//Setting CA algorithm parameters for the low-frequency NR TDD cell
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaA2RsrpThld=-109, CaA5RsrpThld2=-105;
```

```
//Setting the cell-level threshold of buffered traffic volume for downlink state transitions related to CA
SCells for the low-frequency NR TDD cell
MOD NRDUCELLCARRMGMT: NrDuCellId=0, DlStateTransitBufVolThld=20;
//Turning on QCI_CA_BWP_SWITCH_THLD_CTRL_SW for the low-frequency NR TDD cell if the buffered
traffic volume threshold for downlink state transitions needs to be set on a per QCI basis
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=80,
ServiceDiffAlgoSwitch=QCI_CA_BWP_SWITCH_THLD_CTRL_SW-1;
//Setting the QCI-level threshold of buffered traffic volume for downlink state transitions related to CA
SCells in low-frequency NR TDD scenarios
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=80, QciDlStTransitBufVolThld=20;
//Setting the offsets for events A2 and A5 relevant to the high frequency for inter-FR CA coverage
enhancement on the serving gNodeB of the low-frequency PCell
MOD GNBCAFREQUENCY: CaFreqGroupId=3, Dlnarfcn=2057499, CaA2RsrpThldOffset=-11,
CaSccA5RsrpThld2Offset=-2;
//Setting the measurement waiting timer for multi-carrier operations
MOD GNODEBALGO: MultiCarrierMeasWaitTimer=30;
//Disabling low-to-high-frequency handover and redirection for the low-frequency NR TDD cell if the
network does not allow standalone access of UEs to the high frequency
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=10, NoHoFlag=FORBID_HO;
//Setting the InterFrCaFr1DlMaxCcNum parameter to DL2CC when there are two or more low frequencies
on the network
MOD NRDUCELLCARRMGMT: NrDuCellId=0, InterFrCaFr1DlMaxCcNum=DL2CC;
//Setting InterFrCa200MFr2CcMaxNum to DL6CC_UL4CC for the PCell if there are five or six contiguous
high-frequency carriers with a bandwidth of 200 MHz each
MOD NRDUCELLCARRMGMT: NrDuCellId=0, InterFrCa200MFr2CcMaxNum=DL6CC_UL4CC;
//Enabling downlink super HARQ for inter-FR CA for the high-frequency PU-SCell to further increase the
downlink throughput of UEs for which PUCCH grouping adaptation enhancement for inter-FR CA has taken
effect
MOD NRDUCELLUECOOP: NrDuCellId=10, SpecUeCooperationSwitch=INTER_FR_CA_DL_SUPER_HARQ_SW-1;
//Setting the inter-FR carrier selection policy for the low-frequency NR TDD PCell when only inter-FR CA is
required
MOD NRCELLCAMGMTCONFIG: NrCellId=0, InterFrCarrSelStrat=FR2_PRIORITY;
//Turning on INDEP_GAP_CONFIG_MAXCC_SW for the low-frequency NR TDD PCell to enable independent
measurement gap pattern configuration for UEs that support independentGapConfig-maxCC-r17
MOD NRDUCELLMEAS: NrDuCellId=0, MeasAlgoSwitch=INDEP_GAP_CONFIG_MAXCC_SW-1;
//((Optional) Enabling "FR2 primary-SCell-inclusive activation and deactivation" for UE power saving
purposes
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
InterFrCaEnhSw=FR2_SCELL_ACTV_DEACTV_SW-1&FR2_SCELL_FAST_ACTV_DEACTV_SW-1;
//((Optional) Setting Fr2ScellDeactvProhibitTmr and Fr2ScellActvInterval
MOD NRDUCELLCARRMGMT: NrDuCellId=0, Fr2ScellDeactvProhibitTmr=30, Fr2ScellActvInterval=5;
```

Deactivation Command Examples

```
//((In the case of inter-FR CA with two low-frequency CCs) Setting InterFrCaFr1DlMaxCcNum to
NOT_CONFIG before disabling inter-FR CA
MOD NRDUCELLCARRMGMT: NrDuCellId=0, InterFrCaFr1DlMaxCcNum=NOT_CONFIG;
//Disabling fast SCell configuration for inter-FR CA for the low-frequency NR TDD cell
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaStrategySwitch=FR2_IDLE_INACTIVE_MEAS_SW-0;
//Disabling downlink super HARQ for the high-frequency NR TDD cell
MOD NRDUCELLUECOOP: NrDuCellId=10, SpecUeCooperationSwitch=INTER_FR_CA_DL_SUPER_HARQ_SW-0;
//Disabling PUCCH grouping adaptation for inter-FR CA and PUCCH grouping adaptation enhancement for
inter-FR CA for the low- and high-frequency NR TDD cells
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
InterFrCaEnhSw=PUCCH_GROUP_ADAPT_SW-0&PUCCH_GROUP_ADAPT_ENH_SW-0;
MOD NRDUCELLCARRMGMT: NrDuCellId=10,
InterFrCaEnhSw=PUCCH_GROUP_ADAPT_SW-0&PUCCH_GROUP_ADAPT_ENH_SW-0;
//Setting InterFrCa200MFr2CcMaxNum to DL4CC_UL4CC
MOD NRDUCELLCARRMGMT: NrDuCellId=0, InterFrCa200MFr2CcMaxNum=DL4CC_UL4CC;
//Disabling "FR2 primary-SCell-inclusive activation and deactivation"
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
InterFrCaEnhSw=FR2_SCELL_ACTV_DEACTV_SW-0&FR2_SCELL_FAST_ACTV_DEACTV_SW-0;
//Disabling inter-FR CA for the low- and high-frequency NR TDD cells
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CaAlgoSwitch=INTER_FR_FR1TOFR2_CA_SW-0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=10, CaAlgoSwitch=INTER_FR_FR1TOFR2_CA_SW-0;
```

7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.4.2 Activation Verification

Provided that CA UEs are available on the network, inter-FR CA has taken effect if any counter in [Table 7-3](#) returns a non-zero value.

Table 7-3 Counters related to inter-FR CA

Counter ID	Counter Name
1911835934	N.User.InterFRCA.PUCCHSCell.Max
1911835933	N.User.InterFRCA.PUCCHSCell.Avg
1911835932	N.User.InterFRCA.PCell.Max
1911835931	N.User.InterFRCA.PCell.Avg

Provided that CA UEs are available on the network, inter-FR CA coverage enhancement has taken effect if any counter in [Table 7-4](#) returns a non-zero value.

Table 7-4 Counters related to inter-FR CA coverage enhancement

Counter ID	Counter Name
1911840395	N.User.InterFRCA.PCell.OnePUCCHGroup.Avg
1911840396	N.User.InterFRCA.PCell.OnePUCCHGroup.Max
1911840397	N.User.InterFRCA.HostSCell.OnePUCCHGroup.Avg
1911840398	N.User.InterFRCA.HostSCell.OnePUCCHGroup.Max

The counters related to the numbers of attempts and successful executions regarding SCell configuration, change, activation, deactivation, and removal are listed in [Table 7-5](#).

Table 7-5 Counters related to SCell management

Counter ID	Counter Name
1911816777	N.CA.SCell.Add.Att
1911816778	N.CA.SCell.Add.Succ
1911816781	N.CA.SCell.Mod.Att
1911816782	N.CA.SCell.Mod.Succ
1911816896	N.CA.SCell.Act.Att
1911816897	N.CA.SCell.Act.Succ
1911816898	N.CA.SCell.Deact.Att
1911816899	N.CA.SCell.Deact.Succ
1911816779	N.CA.SCell.Rmv.Att
1911816780	N.CA.SCell.Rmv.Succ

7.4.3 Network Monitoring

Observe the average downlink UE throughput and average uplink UE throughput to check whether the UE data rate increases after this function is enabled.

The average downlink UE throughput is indicated by **User Downlink Average Throughput (DU)**.

The average uplink UE throughput is indicated by **User Uplink Average Throughput (DU)**.

8 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

9 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

10 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

11 Reference Documents

- 3GPP TS 38.101-1: "NR; User Equipment (UE) radio transmission and reception; Part 1: Range 1 Standalone"
- 3GPP TS 38.101-2: "NR; User Equipment (UE) radio transmission and reception; Part 2: Range 2 Standalone"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.300: "NR; Overall description; Stage-2"
- 3GPP TS 38.306: "NR; User Equipment (UE) radio access capabilities"
- 3GPP TS 38.321: "NR; Medium Access Control (MAC) protocol specification"
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) Protocol specification"
- 3GPP TS 38.133: "NR; Requirements for support of radio resource management"
- 3GPP TS 38.213: "NR; Physical layer procedures for control"
- *CoMP*
- *Extended Cell Range*
- *UE Power Saving*
- *mmWave Beam Management (High-Frequency TDD)*
- *Channel Management*
- *Positioning*
- *Modulation Schemes*
- *Uplink Scheduling*
- *ANR*
- *High Speed Mobility*
- *Hyper Cell*
- *Cell Combination*
- *License Control Item Lists*
- *FWA*
- *Standards Compliance*
- *Massive MIMO AHR (Low-Frequency TDD)*
- *5G Networking and Signaling*

- *SA Mobility Management in Connected Mode*
- *NSA Mobility Management*
- *Energy Conservation and Emission Reduction*
- *Device-Pipe Synergy*
- *VoNR*
- *Network Slicing*
- *Mobility Load Balancing in Connected Mode*
- *Mobility Load Balancing in Idle Mode*
- *Remote Interference Management (Low-Frequency TDD)*
- *URLLC*
- *DRX*
- *EPC-based NSA Performance Enhancement*
- *EPC-based NSA Basics*
- *3D Networking Experience Improvement*
- *Scalable Bandwidth*
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*
- *Multi-Frequency Smart Selection*
- *Multi-Frequency Smart Aggregation*
- *Downlink Scheduling*
- *Beam Management*
- *Co-MM (Low-Frequency TDD)*
- *Virtual Grid-based Multi-Frequency Coordination*
- *MIMO (TDD)*
- *Cell Management*
- *Super Uplink*
- *UL and DL Decoupling*
- *Multi-Operator Sharing*
- *Random Access Control*
- *CA Experience Improvement*
- *NSA-SA Selection Based on User Experience*
- *Distributed Antenna Solutions (Low-Frequency TDD)*
- *Ultimate Video Experience*
- *Ultimate Cloud X Experience*
- *Service-differentiated Experience Guarantee*
- *Beam Management*
- *Layered QoS*
- *Power Control*
- *mmWave Super Coverage (High-Frequency TDD)*
- *Intelligent Aggregate Scheduling (High-Frequency TDD)*

- *Multi-RAT Coordinated Energy Saving*